Quantitative Finance and Systematic Research

Chapters 1–45, Projects 1–6, and Capstone · Active Recall and Complete Answers

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October 17, 2026

Edition and Use Notes

This volume contains active-recall practice and same-chapter complete answers for Chapters 1–45, Projects 1–6, and the Capstone, as a local complete candidate. Each chapter and project gives reading and fill-in material first, then exercises; answers start on a new page within the same chapter or project. The active-recall projections for Projects 1–4 are new in this edition; they follow the accepted exercises and complete answers, keeping original identifiers and meaning, and new questions carry their own identifiers. Constructed numbers are for practicing methods, not market evidence; research statuses follow the main text. The main text is `editions/en/body/main.pdf`.

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Contents

Chapter 1 —Quantitative Finance and the Quant Industry 1

Chapter 2 —Python for Quantitative Research 9

Chapter 3 —Financial Data Engineering 16

Chapter 4 —Put Units Back into Rates 21

Chapter 5 —Read Shapes before Risk 28

Chapter 6 —an alert is not the event 34

Chapter 7 —mean uncertainty and return risk 41

Chapter 8 —Regression 48

Chapter 9 —Optimization 54

Chapter 10 —Financial Markets 61

Chapter 11 —Returns and Compounding 73

Project 1 —Returns, Wealth, and Risk Description 79

Chapter 12 —Financial Statements for Quantitative Research 87

Chapter 13 —Valuation for Quants 97

Chapter 14 —Capital Asset Pricing Model and Asset Pricing (CAPM) 110

Chapter 15 —Factor Models 116

Project 2 —Factor Regressions and Evidentiary Limits 124

Chapter 16 —Time Series I: Dependence and ARIMA 131

Chapter 17 —Time Series II: Unit Roots, Cointegration, and Prediction 137

Chapter 18 —Volatility Modeling and Forecasting 147

Chapter 19 —State-Space Models and the Kalman Filter 153

Chapter 20 —Modern Portfolio Theory 159

20.1 From variance to optimal weights 160

20.2 Cash, tangency, and uncertainty 161

20.3 Moments, constraints, and information clocks 162

Chapter 21 —Practical Portfolio Construction **169**

21.1 Covariance estimation and costs after drift 170

21.2 Views, leverage, and constraints 171

21.3 Research judgment and code diagnostics 172

21.4 Amount ledger and research judgment 173

Project 3 —Rolling Portfolios and Out-of-Sample Evaluation **179**

Chapter 22 —Quantitative Research Methodology **186**

Chapter 23 —Cross-Sectional Research **189**

Chapter 24 —Value Factors **194**

Chapter 25 —Momentum **198**

Chapter 26 —Quality **203**

Chapter 27 —Carry, Defensive Investing, and Low Volatility **207**

Chapter 28 —Multi-Factor Models: Preprocessing, Combination, and Constraints **212**

Chapter 29 —Building a Backtesting Engine **217**

Chapter 30 —Backtesting Biases **221**

Chapter 31 —Financial Model Validation **225**

Chapter 32 —Performance Statistics & Inference **229**

Chapter 33 —Risk Management **234**

Chapter 34 —Market Microstructure **239**

Chapter 35 —Transaction Costs **248**

Chapter 36 —Algorithmic Execution and Trade Scheduling **260**

Project 4 —From Fields to Signals to Execution **271**

Project 5 —Pairs Trading **284**

Chapter 37 —Accuracy, Decisions, and Strength of Evidence **298**

Chapter 38 —Time Contracts, Splitting, and Training-Only Fitting **306**

Chapter 39 —Shrinkage, Thresholding, and Baselines **313**

Chapter 40 —Splitting, Averaging, and Stepwise Correction **320**

Chapter 41 —Variance, Distance, and When Information Is Available **327**

Chapter 42 —Deep Learning Foundations and Quantitative Applications **333**

42.1 Loss Function, Optimizer, and Model Selection 335

42.2 Structural Inductive Bias: CNN, RNN, and Transformer 337

42.3 Attention and the Financial-Research Boundary 338

Chapter 43 —Bayesian Inference, Causal Identification, and Alternative Text Data **344**

Project 6 —Machine-Learning Forecasts, Costs, and Evidence **351**

Chapter 44 —Stochastic Calculus and Derivatives **367**

Chapter 45 —Production Quant Pipelines **385**

Capstone: from admissible input to an interpretable net conclusion **404**

Chapter 1 —Quantitative Finance and the Quant Industry

Data provide features for signals and forecasts. Targets allocate capital. Execution turns targets into orders; fills update positions. Later prices and cash flows determine PnL.

Trigger. Does a good wealth curve use information available before the decision and actual fills?

Definition 1.1 *(Decisions and outcomes).* A target weight is a desired capital fraction. An order requests a trade. A fill records executed quantity, price, and time. Positions update from filled quantities. Record each object separately.

Your turn. **1-L01: decisions and fills (1-E02 and Section 5.3)**
After Signal, fill in the forecast object, capital-allocation decision, submitted instruction, execution record, post-fill quantity, and subsequent result. Explain why a target is not a position.

Your turn. **1-L02: traded notional and fees (Section 6.2, 1-E06)**
Risky weights change from $(0.3, 0.5)$ to $(0.5, 0.3)$ at NAV 200,000. Fee: 5 bps per traded notional. Count purchases plus sales without halving. Find the traded fraction, total notional, and fee. Does the cash-weight change add another trade?

Your turn. 1-L03: future information (1-P02)
A rule sets holdings at t and then measures the next interval. Identify the leaking line, provide runnable corrected code, and compare PnL fractions.

```
past_return = -0.02
next_return = 0.03
weight = float(next_return > 0.0)
pnl_fraction = weight * next_return
print(pnl_fraction)
```

Your turn. 1-L04: industry-return evidence (1-E04, 1-C01)
French ten-industry files supply historical portfolio returns. Name a supported analysis and three evidence categories needed for stock-level execution. Propose a preregistered control that might overturn a momentum explanation.

Definition 1.2 *(A score needs a forecast mapping).* A rank has no return units. A specified mapping supplies a return forecast; evidence is needed to assess that mapping. The realized return belongs to a later interval.

Your turn. 1-L05: scores and forecasts (1-N01, Section 8)
Constructed past prices: A 98 to 100, B 101 to 100. Give the stronger asset score+1 and the weaker score−1. Use $\hat{\mu}_i = 0.01 + 0.005 s_i$, decimal return per holding period. Later returns are A 2%, B−1%. Calculate historical returns, scores, and forecasts. Which columns are known when the signal forms; which must wait?

Your turn. 1-L06: no fills (1-N04)
Targets 0.6/0.2 do not fill; actual weights remain 0.5/0.3, cash 0.2. A/B/cash returns 2%/−1%/0%. Fees 10 bps apply only to actual notional. Gross return is _____, cost fraction _____, net _____. Explain why planned turnover cannot determine actual fees.

Definition 1.3 *(Reconcile actual fills).* A trade changes asset quantity and cash together. A fee reduces NAV. An unfilled target quantity earns no holding PnL. Post-fee weights use post-fee NAV.

Your turn. **1-L07: partial fills (1-B01)**
Start with 500 A,300 B, both priced 100, and cash 20,000. Plan to buy 100 A/sell 100 B, but only 40 A fill and all 100 B sell. Pay 10 bps from cash immediately. Final prices A 102/B 99. Calculate actual notional, fee, cash, quantities, post-fill/final NAV, and net PnL. Why does target 600 A overstate profit?

Chapter 1 solutions: 1-L01 and 1-L02

1-L01

Forecast, Target Weight, Order, Fill, Position, PnL. Forecasts estimate future outcomes; targets decide desired capital; orders request trades; fills record actual execution. No fill, partial fill, or a different price can leave actual holdings different from targets. Subsequent holdings and cash flows generate PnL.

1-L02

$T = |0.5 - 0.3| + |0.3 - 0.5| = 0.4$. Traded notional $200,000(0.4) = 80,000$. Five bps is 0.0005, so fee $80,000(0.0005) = 40$ currency units. This convention counts risky purchases/sales, excluding the balancing cash change.

Chapter 1 solutions: 1-L03 and 1-L04

1-L03

The weight line uses the next return, unknown at decision time. Historical information gives weight 0 and PnL 0; future-informed weight 1 gives 3%, a leaked outcome.

```
past_return = -0.02
next_return = 0.03
weight = float(past_return > 0.0)
pnl_fraction = weight * next_return
assert pnl_fraction == 0.0
print(pnl_fraction)
```

This repairs input timing only. Actual tests also specify availability, fills, return endpoints, and fees.

1-L04

Use the series to describe industry-return differences or practice bounded statistics. Stock-level execution needs historical constituents/quantities, executable prices/depth/times, corporate actions, borrowing/financing, and costs. Freeze a momentum window and out-of-sample period, comparing a preregistered reversal window or equal-weight benchmark. Keep results that do not support the explanation. Return files alone do not establish buying, selling, or borrowing availability.

Chapter 1 solutions: 1-L05 and 1-L06

1-L05

Historical returns are $100/98 - 1 \approx 2.040816\%$ and $100/101 - 1 \approx -0.990099\%$. Scores+1/−1 map to 1.5%/0.5%. The mapping is assumed without fitted evidence. Historical inputs, scores, and forecasts can be recorded at signal time;2%/−1% realizations wait until the holding interval ends. A signal does not establish a forecast's validity or profit.

1-L06

Gross $0.5(0.02)+0.3(-0.01) = 0.007 = 0.7\%$. With no actual trade, the stated fee is 0 and net return 0.7%. Targets remain decisions; fills update quantities. The question excludes opportunity costs and rejection fees.

Chapter 1 solutions: 1-L07

1-L07

A purchase spends 4,000 and the B sale raises 10,000; total notional 14,000 and fee 14. Cash $20,000 - 4,000 + 10,000 - 14 = 25,986$; quantities 540 A/200 B. Post-fill NAV $54,000 + 20,000 + 25,986 = 99,986$. Final NAV $540(102) + 200(99) + 25,986 = 100,866$, net PnL 866 and 0.866% of initial NAV. Price PnL $540(2) + 200(-1) = 880$, less 14, agrees. The unfilled 60 shares would falsely add 120 profit. Post-fee weights divide by 99,986. Keep this event ledger separate from Section 8's terminal-fee simplification.

Chapter 2 —Python for Quantitative Research

Vectorization shortens computation; security labels, holding intervals and units determine what is computed. Three prices, two returns and one product of wealth factors form this chapter's main thread.

Trigger. When a hand calculation disagrees with a DataFrame.

Definition 2.1 *(Prices, intervals and missingness; text §§4,6).* For the same asset, currency and share unit, positive finite prices without omitted investment cash flows give price return $R_t = P_t/P_{t-1} - 1$. Price is currency per unit; return is a dimensionless decimal. T endpoints give at most $T-1$ adjacent returns. Keep the first NaN: there is no predecessor. Total return additionally requires dividends and actions.

Computation contract

`compute_returns` requires an ascending, unique, timezone-aware DatetimeIndex and positive finite nonmissing prices. A missing price removes its own and the next adjacent return. It does not silently sort, fill or invent zeros. In (T, N), time is rows and assets columns; still check label order for $(N,)$ weights.

Example 2.2 *(Constructed teaching prices; text §8).* The three comparable prices are 100, 110 and 99, without dividends, splits, fees or external flows.

Your turn. **2-L01: Calculate and prove; text §8.2.**
The two simple returns are _____ and _____; compounded return is _____. Write the wealth-factor product, then use the log-product rule to prove that the sum of log returns is ln(99/100). Why is the first NaN not a third return?

Chapter 2: Code diagnosis and research judgment

> Alignment can expose a mismatch, but does not choose a sample or execution time. Identify each value's security and interval before checking whether code expresses that relationship.

Your turn. 2-L02: Panel diagnosis; 2-P02.
Two assets alternate by date in a long table. What does global `shift(1)` mispair? Give sorting keys and grouping keys for within-asset shift and same-date rank. How does a (T, N) wide table avoid cross-security predecessors?

Your turn. 2-L03: Missingness and counting; 2-P01.
For 100, NaN, 99, which returns does `compute_returns` leave missing? Why does compounding after `fillna(0)` not establish wealth? How many returns come from four complete endpoints? Should a real 0% holding return be removed?

Your turn. 2-L04: Clocks and scope; 2-C01, 2-P03.
A version is public at 16:05 and ingested at 16:12. Can a 16:10 signal use it? Does shifting a closing signal one row prove execution at that close? Does month-end `last()` prove a trading session? Can French monthly returns be treated as prices and differenced again? State the missing evidence.

Your turn. 2-L05: Two windows; 2-N02.
A prices are 100, 110, 99, 108.9. For rolling(2), min_periods=2, the t2 elements are _____ and mean _____. For expanding, min_periods=1, t2 elements are _____ and mean _____. Give both t3 results and the first NaN position.

Your turn. 2-L06: Whose predecessor? 2-N03, 2-N04.
Rows are t0/A/100, t0/B/50, t1/A/110, t1/B/52. Give the third row's global and within-asset predecessors and returns. At t1, A 10% and B 4%: give average-rank percentiles. What if both returns are 1%?

Definition 2.3 *(Axis meaning before broadcasting).* A one-dimensional vector broadcasts along the last dimension. Equal time/asset lengths do not make them interchangeable. Date factors need a column vector; asset weights need checked labels.

Your turn. **2-L07: Dates and assets; 2-B01.**
Construct $R = \left(\begin{smallmatrix} 0.10 & -0.05 \\ -0.10 & 0.10 \end{smallmatrix}\right)$ with holding intervals in rows and A/B in columns. Baseline weights are $w = (0.2, 0.8)$, risky fractions specified before each interval are $g = (1, 0.5)$; cash earns zero and there are no fees. Calculate $R \odot w$, $(R \odot g_{\text{col}}) \odot w$ and incorrectly column-broadcast $(R \odot g) \odot w$. Give row returns, second-period cash and the $(2, 1)$ shape.

Chapter 2 answers: Check after completing the questions

2-L01. Returns are $110/100 - 1 = 10\%$ and $99/110 - 1 = -10\%$. The product is $1.1(0.9) = 0.99$, cumulative -1%. Using $\ln a + \ln b = \ln(ab)$,

$$\ln(110/100) + \ln(99/110) = \ln[(110/100)(99/110)] = \ln(99/100).$$

The intermediate 110 cancels. Three prices give two intervals; the first NaN means no predecessor, not a third observed 0% return.

2-L02. Global shift can make A use B. Sort by security_id/timestamp and shift within security_id; windows also stay within securities. Rank by timestamp across securities. A wide table shifts each column along time, still requiring ordered unique indices and verified asset labels.

2-L03. All three returns are missing: no predecessor, missing current price, missing predecessor. Zero filling invents known unchanged intervals and cannot restore terminal wealth or drawdown. Four complete endpoints give three returns. Retain initial wealth in wealth/drawdown paths but not in return mean/volatility counts. Real 0% is a valid observation. Dropping interior NaN cannot create a continuous path.

2-L04. At 16:10 the ingestion gate fails. Shift expresses row position; fills need quotes, order deadlines, calculation latency and execution time. A finalized-close signal cannot presume a same-close fill. Month-end comes from bins; check the actual last input and calendar. French columns are already monthly decimal returns, not prices. Current snapshots also do not prove historical availability.

Transfer to the text

Rerun 2-P01–2-P03 using the authorized reader/course route. Record available_at, ingested_at, signal_time, trade_time and return endpoints. Type hints do not replace finite-value checks.

Answers to 2-L05 and 2-L06

2-L05. t2 rolling uses 110,99, mean 104.5; expanding uses 100,110,99, mean 103. t3 rolling is $(99 + 108.9)/2 = 103.95$; expanding is $417.9/4 = 104.475$. Rolling t0 has fewer than two observations and is NaN; expanding starts at 100. Both include the current input without establishing same-close execution.

2-L06. The third row is t1/A/110. Global predecessor 50 (B) gives a wrong 120%; within-asset predecessor 100 (A) gives 10%. Sort time before grouping. At t1, B rank 1 and A rank 2 give 0.5 and 1. Equal returns average ranks to 1.5, giving 0.75 each.

Answer to 2-L07

Baseline contributions are $(0.02, -0.04)$ and $(-0.02, 0.08)$, returns $-2\%, 6\%$. A $(2, 1)$ date column scales the second return row to $(-0.05, 0.05)$; asset weighting then gives $(-0.01, 0.04)$. Row returns are $-2\%, 3\%$. Second-period risky weights $0.10/0.40$ and cash 0.50 sum to 1. Wrong column broadcasting halves B in both periods, giving contributions $(0.02, -0.02)$ and $(-0.02, 0.04)$, returns 0%,2%. Equal lengths do not prove axes; align labels and dimensions explicitly.

Chapter 3 —Financial Data Engineering

> Preserve security identity, units, rights and versions before asking what a signal could see. Actions change units; release and ingestion determine whether a version belongs to a historical decision.

Trigger. When a service shows only current EPS, or prices cross a split.

Definition 3.1 *(Dual visibility; text §4.3).* For a fixed fact key (permanent ID, reporting period and specified concept/unit), both available_at and ingested_at must be no later than signal_time. All clocks must be comparable timezone-aware instants; exclude unproven same-second ordering or retain receipt sequences. Select latest available_at among eligible versions. Later receipt does not mean newer source data; conflicting same-release rows cannot be resolved by file order.

Definition 3.2 *(Rights ledger; text §6.1).* An old quantity N_0 becomes qN_0 new shares. With cash entitlement D per new share and ending price P_1, $W_1 = qN_0(P_1 + D)$ and $W_0 = N_0P_0$. Use one currency, recognize dividend receivables and assume no tax, fees, reinvestment or external deposits. Return is $W_1/W_0 - 1$.

Example 3.3 *(Constructed; text §8).* One old share worth 100 splits 2:1; both new shares close at 51 and have cash entitlement 1 each.

Your turn. **3-L01: Rights calculation; 3-E05.**
Ending stock value _____, receivable _____, wealth _____, total return _____. What is price return in comparable new-share units? Why is $51/100-1$ wrong?

Chapter 3: Filter eligibility, then order source versions

> $\max(available_at, ingested_at)$ answers when both gates are passed, not which source version was released latest.

Your turn. 3-L02: Late older version; 3-C01.
EPS 10 is released April 1 and ingested April 20; restated EPS 8 is released April 10 and ingested April 11. What does an April 21 signal select? Why is ordering by the maximum clock wrong? Independent variation: at April 10 12:00 (all times −04:00), old release April 1 10:00/receipt April 2 09:00, new release April 10 10:00/receipt April 11 09:00. What is lost by selecting the global latest release and then deleting unreceived rows?

Your turn. 3-L03: Code boundaries; 3-P01, 3-P03.
Initial release 16:05, ingestion 16:12, no older version: what do 16:10 and 16:15 select? How should conflicting rows at the same fact key/release be handled? Why does NaN volume not pass a trustworthy $V \geq 0$ check? Does an offset-aware timestamp prove a trading day?

Your turn. 3-L04: Missingness and scope; 3-P02, 3-C02.
D1 price 10, D2 suspended/missing, D3 reopened at 9. Specify interval, session span and D2 status. Why not invent two daily returns? Do today's securities and companyfacts prove a historical executable strategy? Name at least two missing data requirements.

Your turn. 3-L05: Per-version eligibility; 3-N02, text §8.1.
In 2024: v1 released 3/15 16:05, ingested 16:12, value 10; v2 released 4/2 16:05, ingested 16:08, value 8; v_old released 3/20 10:00, ingested 4/20 09:00, value 9.5. All −04:00. At 4/2 16:06, check release/receipt for every version before selection; repeat at 4/21 16:00. Why not globally select v2 before filtering?

Your turn. 3-L06: Cash is not delisted; 3-N03.
After the action, stock is USD 102 and cash USD 2. Stock alone returns −20%, then −50%; cash stays unchanged and unreinvested. Ending stock _____, account wealth _____, account return from 104 _____. Diagnose 104(0.4). Can unknown terminal value become −50%?

Chapter 3 answers: Check after completing the questions

3-L01. Stock value $2(51) = 102$, receivable $2(1) = 2$, wealth 104 and total return 4% from 100. The comparable initial price is $100/2 = 50$; price return $51/50 - 1 = 2\%$. $51/100 - 1 = -49\%$ mixes units; omitting the dividend gives price return, not total return. Retain entitlement/payment dates separately and never add cash again to a dividend-inclusive return series.

3-L02. At April 21 both versions qualify; April 10's EPS 8 is the latest release. The old version's April 20 eligibility is later, but its release is older. The independent April 10 12:00 variation selects old EPS 10 because the new version is unreceived. Selecting the new version first and then deleting it wrongly returns empty. Do not confuse this separate variation with April 21. Filter both gates per signal, then order public versions.

3-L03. 16:10 is empty; 16:15 selects the initial report. Public availability is not receipt. Reject/quarantine ambiguous same-key/release rows and preserve originals. NaN is neither reliable nonnegative volume nor 0; check type, finiteness and missingness before relationships. Offsets enable comparison; venue zones and actual calendars still determine sessions.

3-L04. D2 is unknown, reason=suspended. D3 returns $9/10 - 1 = -10\%$ over D1-close to D3-close across two trading sessions. Unknown intermediate wealth cannot be decomposed into daily observations or prove suspended fills. Current lists condition on later survival; current responses can be restated. Obtain historical membership/permanent mappings, version payloads and clocks, terminal proceeds and executable quotes/costs. Preserve unknowns and limit conclusions where evidence is missing.

Transfer to the text

Run the dual-clock interface and adversarial inputs through the authorized route. Field checks are necessary conditions. NAV ratios require no external flows and a unified ledger for fees, receivables, liabilities and currencies.

Answers to 3-L05 and 3-L06

3-L05. At 4/2 16:06: v1 yes/yes, v2 yes/no, v_old yes/no. Only v1 qualifies, value 10. At 4/21 all pass; latest public v2 gives 8. Retain late older data without letting it win. Selecting v2 before filtering at 4/2 16:06 loses eligible v1. Eligibility and public-version ordering are distinct.

3-L06. Stock $102(0.8)(0.5) = 40.8$; unchanged cash 2 gives 42.8. Account return $42.8/104 - 1 \approx -58.846154\%$, versus stock-only -60%. $104(0.4)$ wrongly applies stock loss to cash. Investigate unknown terminal values; do not substitute the constructed -50%.

Chapter 4: Put Units Back into Rates

A derivative describes a local linear response. Input units and the variables held fixed determine which question it answers. Curvature does not forecast a trend.

Trigger. Why does a one-bp price change differ so much from the derivative itself?

Definition 4.1 *(Local approximation; text §§5–6).* For a constructed annual-compounding zero-coupon bond with face value 100 and one payment in two years, $P(y) = 100/(1+y)^2$. Holding maturity fixed gives $P' = -200/(1+y)^3$ and $P'' = 600/(1+y)^4$. With the required smoothness,

$$P(y+h) - P(y) = P'(y)h + \tfrac{1}{2}P''(y)h^2 + O(h^3).$$

Both y and h are decimal yields; price and cash flow use the same currency.

Example 4.2 *(Constructed inputs).* At $y = .05$, $P = 90.7029478458$, $P' = -172.7675197063$, $P'' = 493.6214848751$. The linear term scales with the shock and curvature with its square.

Your turn. 4-L01: text 4-E02 and 4-E03
One bp corresponds to $h =$ _____, with linear price change about _____. Why does $h = .01$ increase this shock a hundredfold?

Your turn. 4-L02: text 4-E04, extended to maturity
For $P(y(t), \tau(t))$, write the full chain rule as remaining maturity changes with calendar time. Which contribution is omitted by keeping only the yield partial?

Your turn. 4-L03: diagnose text 4-P01

Use 4-P01's function and yield. Diagnose:

```
first = finite_difference_gradient(price, [y],
step=1e-3)[0]
second = finite_difference_hessian(price, [y],
step=1e-6)[0, 0]
linear_1bp = first * 0.01
```

Which two derivative results fail the specified tolerances? Which line uses 100 bp in place of one bp? Restore the specified steps and give the order of magnitude of the one-bp linear price change.

Your turn. 4-L04: text 4-C01, approximation decision

Predeclare absolute linear-approximation error at most .001 price units. For $P(y) = 100/(1 + y)^2$ at .05, compare yield rises of 1, 25 and 100 bp. Which pass? Does this establish the same accuracy for a real bond?

From inputs to a local response

The rule $q(x) = 2x^2 + 1$ maps input x to output $q(x)$. Increment h changes the output by $\Delta q = q(x + h) - q(x)$. For nonzero h, the quotient $\Delta q/h$ is an interval-average rate. The derivative is its limit as h tends to zero.

Your turn. 4-L05: text 4-N01, proof scaffold
Move from $x = 2$ to $2 + h$. Fill $q(2) =$ _____, then complete

$$\frac{q(2+h)-q(2)}{h} = \frac{2[(2+h)^2 - 2^2]}{h} = \frac{2(4h+h^2)}{h} = \underline{\qquad\qquad}.$$

[fill in: Factor and cancel nonzero h before taking its limit. Explain why direct substitution of zero is invalid.] What is the limiting slope?

Your turn. 4-L06: text 4-N02, powers and basis points
Let $u = 1 + y$ in $P = 100u^{-2}$. Fill $du/dy =$ _____ and

$$P'(y) = 100(-2)u^{-3}(du/dy) = \underline{\qquad\qquad},$$
$$P''(y) = (-200)(-3)u^{-4}(du/dy) = \underline{\qquad\qquad}.$$

25 bp = 25× _____ = _____. At .05, the linear price change is about _____ price units.

Your turn. 4-L07: diagnose text 4-N03
A program labels this a price change, with $h = .0001$:

```
reported_price_change = (price(y + h) - price(y)) / h
```

The quotient measures _____, in units _____. For a one-bp linear price change, multiply by _____, giving about _____.

Your turn. 4-L08: assess text 4-N04
A researcher concludes from $P'(.05) = -172.7675$ that yield will rise tomorrow and a real bond will fall. Which two types of evidence are missing? Given a one-bp yield rise, what local response does this formula support?

A direction and a finite step

For the fixed quadratic score with $g = \Sigma w - \mu \neq 0$, the exact update is

$$f(w - \eta g) - f(w) = -\eta g^\top g + \tfrac{1}{2}\eta^2 g^\top \Sigma g.$$

Linear descent and quadratic curvature scale differently. Constraints require a separate check.

Your turn. **4-L09: text 4-B01**
For this fixed positive definite quadratic score, use $w = (.6, .4)^\top$, $g = (-.022, -.038)^\top$, $g^\top g = .001928$ and $g^\top \Sigma g = .00016604$. Derive the strict-descent step interval; compare .1 and 30 and check budgets. Along $d = (-1, 1)^\top$ from $(.6, .4)$ with $s \geq 0$, derive the long-only upper bound. Show intermediate steps.

Chapter 4 Answers: 4-L01–4-L09

4-L01

$h = .0001$, linear change about $-.017276752$ price units. The shock .01 is 100 bp, a hundred times one bp.

4-L02

The full rule is $dP/dt = P_y(dy/dt) + P_\tau(d\tau/dt)$. The omitted term is the response to shrinking remaining maturity. Real holding-period analysis also needs cash flows, accrual and quote conventions.

4-L03

First differences at 10^{-3} have relative error about 1.81×10^{-6}, above 10^{-6}; second differences at 10^{-6} have error about 4.25×10^{-5}, above 2×10^{-6} in the recorded environment. Restore 10^{-4} and 10^{-3} respectively. Multiply by .0001 for one bp, giving about $-.017276752$ price units. Smaller steps need not improve accuracy; cancellation affects the second difference.

4-L04

Absolute linear errors are about .000002468, .001537685 and .024371353 price units. Only one bp passes .001. This conclusion belongs to the constructed single-cash-flow formula. Real bonds need cash-flow, day-count, compounding, quote and market-data checks, plus stability across times or securities.

Answers to 4-L05–4-L08

4-L05

$q(2) = 9$; the square difference is $4h + h^2$, quotient $8 + 2h$, and limiting slope 8.

4-L06

$du/dy = 1$; $P' = -200(1 + y)^{-3}$, $P'' = 600(1 + y)^{-4}$. One bp is .0001, 25 bp .0025. At .05, $P' \approx -172.7675$, yielding about $-.4319$ price units.

4-L07

The quotient is an average rate approximating P', with price/decimal-yield units. Multiply by $h = .0001$ for a linear change of about $-.0172768$ price units. An exact constructed change is $P(y + h) - P(y)$.

4-L08

The derivative contains no forecast of future yield direction; the construction has not established that real quotes follow it. Given a one-bp rise, it supports a linear response about $-.0173$ price units.

Answer to 4-L09

The exact change is $-.001928\eta + .00008302\eta^2$, so strict descent needs $0 < \eta < 23.22331968$. Steps .1 and 30 change the score by $-.0001919698$ and $+.016878$; weights are $(.6022, .4038)$ and $(1.26, 1.54)$, with sums 1.006 and 2.8, both violating the budget. Along $(.6 - s, .4 + s)$, sum one is preserved and nonnegativity requires $0 \leq s \leq .6$. Positive-step descent further needs $s < .016/.055$. A teaching-score improvement is not evidence of future return or actual execution.

Chapter 5: Read Shapes before Risk

> Identify the object on every row and column. Vectors and matrices then give a calculable mapping; risk and PCA interpretations depend on those meanings.

Trigger. Which axis records observations, and which records assets?

□□□□□

Vectors, transpose and inner product

Write weights and same-interval returns as columns. Transpose changes 2×1 into 1×2; an inner product pairs, multiplies and sums to a scalar.

Example 5.1 *(Text 5-N01: one interval).*

$$w=(.6,.4)^\top,\quad R_1=(.02,-.01)^\top,\quad w^\top R_1=.6(.02)+.4(-.01)=.008.$$

Reset weights at each interval's start and ignore fees. Weights/returns are dimensionless. The two-interval A/B panel is $X=\begin{bmatrix}.02 & -.01\\ -.03 & .04\end{bmatrix}$; its first row becomes column R_1.

Your turn. 5-L04: text 5-N01
Give shapes of $w^\top, w^\top R_1, Xw$. Which row supplies R_1? Compute both portfolio returns and identify rows and columns.

A matrix maps inputs to outputs

For fixed A, $x\mapsto Ax$ is linear. Its first column is the image of $(1,0)^\top$, the second of $(0,1)^\top$. Other inputs combine those columns by their coordinates.

Your turn. 5-L05: text 5-N02
Compute every cell as a left-row/right-column dot product:

$$A=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix},\quad B=\begin{bmatrix}5 & 6\\ 7 & 8\end{bmatrix},\quad AB=\begin{bmatrix}_____ & _____\\ _____ & _____\end{bmatrix}.$$

Why common-sample covariance gives no negative variance

Theorem 5.2 *(Text 6.2: common-sample covariance is PSD).* For $n \geq 2$ common complete rows centered by column, $\widehat{\Sigma} = X_c^\top X_c/(n-1)$ satisfies $v^\top \widehat{\Sigma} v \geq 0$ for every v.

Proof.

$$v^\top \widehat{\Sigma} v = \frac{v^\top X_c^\top X_c v}{n-1}.$$

[fill in: Associate the product and identify the numerator as a squared Euclidean norm.] A squared length is nonnegative.

Your turn. 5-L06: text 6.2, proof scaffold

Complete the missing equality. Which sample combination supplies this length? Why does estimating each covariance on different dates prevent reuse of this proof?

Your turn. 5-L07: text 5-N04 and §5.4

Retain directions V_k. Distinguish directions, scores and projected points. Training scores are _____; projected points are _____. Give the held-out row formula using only training parameters. Does high explained variance establish prediction or trading value?

Risk includes joint movement

A matrix handles several assets after observation and security alignment. Covariance records joint changes; separate individual risks cannot replace its cross terms.

Trigger. Why can volatile assets form a less volatile portfolio?

Definition 5.3 *(Portfolio variance).* For column weights w and same-interval $p \times p$ covariance Σ, $\sigma_p^2 = w^\top \Sigma w$. Common centered rows give

$$v^\top \widehat{\Sigma} v = \|X_c v\|^2/(n-1) \geq 0.$$

This is positive semidefiniteness (PSD). Pairwise missingness on different rows may lose the shared Gram structure.

Example 5.4 *(Constructed annual decimal returns).*

$$w = (.6, .4)^\top, \qquad \Sigma = \begin{bmatrix} .04 & .01 \\ .01 & .09 \end{bmatrix}.$$

Your turn. 5-L01: variant of text 5-E05, using §8.1
Use 60/40 weights here and practice variance only; original 5-E05 uses equal weights and also asks expected return and correlation. Fill variance _____ and standard deviation _____. How much variance is omitted if the off-diagonals are ignored?

Your turn. 5-L02: text 5-P03
PCA centers training data. How can adding future rows change the mean, covariance directions and historical coordinates? Even if a direction happens to stay unchanged, why does it breach the cutoff contract?

Your turn. **5-L03: diagnose text 5-P03**

Use these arrays and diagnose the fit:

```
training = np.array([
[.01, -.02], [.02, .01],
[-.01, .03], [0., -.01]])
future = np.array([[.03, .02]])
fit = pca_fit_transform(
np.vstack((training, future)), n_components=2)
score = (future - fit.center) @ fit.components.T
```

Does the future row enter only the transform? Fix the fit input. Replace future by an extreme row: which fitted parameters must stay unchanged, and must scores stay unchanged too? Is feature-count agreement sufficient?

Chapter 5 Answers: 5-L01–5-L07

5-L01

Variance is $.6^2(.04) + 2(.6)(.4)(.01) + .4^2(.09) = .0336$, standard deviation about .183303. Omitting off-diagonals removes .0048 variance. Return to original 5-E05 for equal weights, expected return and correlation.

5-L02

Future rows can change the mean, directions and past coordinates, though a direction may happen not to change in one case. Any future contribution to estimated parameters breaches the past cutoff. Fit past rows only, then freeze and transform holdout observations.

5-L03

The fit includes the future row. Correct it:

```
import numpy as np
from quant_course.math_tools import pca_fit_transform
fit = pca_fit_transform(training, n_components=2)
score = (future - fit.center) @ fit.components.T
```

Compare the same training data with two holdout inputs:

```
def project(train, held):
    fit = pca_fit_transform(train, n_components=2)
    return fit, (held - fit.center) @ fit.components.T
fit_a, score_a = project(training, future)
extreme = np.array([[100., -100.]])
fit_b, score_b = project(training, extreme)
assert np.allclose(fit_a.center, fit_b.center)
assert np.allclose(fit_a.components, fit_b.components)
assert not np.allclose(score_a, score_b)
```

Centers and directions stay the same; scores can change because the observation changed. These positional arrays have no feature names. Production callers also check feature order and training cutoff: equal column counts do not detect a swap.

5-L04

Shapes are 1×2, 1×1 and 2×1. The first panel row supplies R_1. Returns are .008 and $-.002$; rows are intervals and columns A/B assets.

5-L05

$$AB = \begin{bmatrix} 1(5) + 2(7) & 1(6) + 2(8) \\ 3(5) + 4(7) & 3(6) + 4(8) \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}.$$

5-L06

$$v^\top X_c^\top X_c v = (X_c v)^\top (X_c v) = \|X_c v\|_2^2.$$

The weighted sample combination $X_c v$ supplies the squared length. Pairwise samples generally lack one shared Gram matrix, so this guarantee no longer applies.

5-L07

Scores are $T = X_c V_k$, projected points $TV_k^\top$, with the training mean added for uncentered reconstruction. A held-out row has score $(X_h - \bar{x}_{train}^\top)V_k$; multirow arrays broadcast the same training mean across rows. EVR describes training variation without target labels, extrapolation, execution constraints or fees, and cannot establish trading success.

Chapter 6: an alert is not the event

Start with an outcome table: define events, then condition. Map outcomes to random variables and use their distributions to calculate moments. The classical i.i.d. CLT concerns standardized sample means, not single-period returns.

Trigger. Why can an alert remain unreliable with 95% sensitivity?

Definition 6.1 *(Bayes updating and dependence assumptions).* Event E and positive alert $+$ have $p = \Pr(E)$, $a = \Pr(+|E)$ and $b = \Pr(+|E^c)$. Then

$$\Pr(E|+) = \frac{pa}{\qquad\qquad}.$$

Weak white noise has zero mean, finite constant variance and _____ autocovariance at nonzero lags, without requiring independence. Classical i.i.d. CLT additionally requires _____; dependent sequences need other assumptions.

Example 6.2 *(Constructed inputs for 6-L01).* Use $p = .005, a = .95, b = .05$ for 10,000 objects. These are specified teaching parameters, not financial-model validation results.

Your turn. 6-L01 Section 8.1; a 6-E05 method variant

Name the two cells in the positive denominator, then calculate $\Pr(+)$ and $\Pr(E|+)$. Multiply their relevant joint probabilities by 10,000 to obtain expected true- and false-positive counts.

$$\Pr(+) = pa + (1-p)b = _____,$$
$$\Pr(E|+) = \frac{pa}{\qquad\qquad} = _____.$$

Your turn. 6-L02 Derivation: Section 8.2 four-state portfolio

Four equally likely states are $(R_A, R_B) = (.1,.1), (-.1,-.1), (.1,-.1), (-.1,.1)$, with equal-weight return $R_p = (R_A + R_B)/2$. Let C denote the first two aligned states and information $\mathcal{G} = \sigma(\mathbf{1}_C)$.

Fill in four R_p values: _____, _____, _____, _____. Within C, $\mathbb{E}[R_p|\mathcal{G}] =$ _____ and $\mathrm{Var}(R_p|\mathcal{G}) =$ _____; within C^c these are _____ and _____. Recover unconditional moments using $\mathbb{E}[R_p] = \mathbb{E}[\mathbb{E}(R_p|\mathcal{G})]$ and $\mathrm{Var}(R_p) = \mathbb{E}[\mathrm{Var}(R_p|\mathcal{G})] + \mathrm{Var}(\mathbb{E}[R_p|\mathcal{G}])$.

Your turn. 6-L03 Code diagnosis: CLT standardization

Assume *rng* exists. The following code aims to standardize Bernoulli sample means for CLT. What scaling is missing, and what is the wrong statistic ' s variance?

```
import math
p = 0.30
n = 120
means = rng.binomial(n=n, p=p, size=30_000) / n
z = (means - p) / math.sqrt(p * (1 - p))
```

Your turn. 6-L04 Research judgment: correlation and independence

Monthly-return sample autocorrelations are insignificant, and a fixed-seed i.i.d. simulation looks bell-shaped. Does this establish i.i.d. real returns and classical mean intervals? Complete the gaps: failure to reject zero correlation does not establish _____; the simulation checks only _____; a classical interval also needs _____.

Define frequency, information time, tails and moments, structural changes and the out-of-sample window.

Your turn. 6-L05 Identify objects: 6-N01
A bag contains 3 red and 7 blue cards. Replace and mix after each draw. Draw independently twice, encoding red 1 and blue 0; ordered outcomes are $00, 01, 10, 11$. Write Ω and probabilities _____, _____, _____, _____. Set $A = \{11\}$ and $B = \{01, 10, 11\}$, and define X as success count. Map the four outcomes through X and give its support.

Your turn. 6-L06 One calculation: 6-N02
In 6-L05, circle outcomes retained by B and calculate $\Pr(A|B) = \frac{\text{probability of } A \cap B}{\text{probability of } B}$. Identify the rows in numerator and denominator.

Your turn. 6-L07 Complete calculation: 6-N03
Group outcome rows into $\Pr(X = 0) =$ _____, $\Pr(X = 1) =$ _____ and $\Pr(X = 2) =$ _____, then calculate $\mathbb{E}[X]$, $\mathbb{E}[X^2]$ and $\text{Var}(X)$. Show substitution into $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

Your turn. 6-L08 Code diagnosis: 6-N04

Each sample has n Bernoulli(p) observations and statistic $z = (\bar{X} - p)/\sqrt{p(1-p)}$. Complete the derivation: independence removes cross covariances, giving $\mathrm{Var}(\bar{X}) = n^{-2}\sum_{i=1}^{n}\mathrm{Var}(X_i) =$ _____ and hence $\mathrm{Var}(z) =$ _____. Keeping the numerator requires denominator _____; alternatively change the numerator to _____. What can many replications check?

Your turn. 6-L09 Research judgment: 6-N05

Evaluate: "A bell-shaped simulated histogram and failure to reject zero autocorrelation prove real returns are i.i.d. normal, the signal causes losses, and classical CLT intervals are valid." State the definitions or assumptions needed separately for prediction, causality and mean inference.

Your turn. 6-L10 Derivation and diagnosis: 6-B01

The whole sequence shares A, taking 1 or 2 with probability $1/2$ each. ε_t are independent standard-normal innovations, independent of A, and $X_t = A\varepsilon_t$. Complete $\mathbb{E}[A^2] =$ _____ and $\mathbb{E}[A^4] =$ _____. Cross-period linear covariance is _____; squared-observation covariance is _____.

Conditional on shared scale, half the distribution of $Z_n = \sqrt{n}\bar{X}_n/\sqrt{2.5}$ is $N(0,$ _____$)$ and half $N(0,$ _____$)$. Derive its fourth moment and explain why a correct mean-variance formula still does not justify a classical IID normal interval.

Chapter 6 solutions: definitions and 6-L01–6-L10

Definition blanks

The Bayes denominator $pa + (1-p)b$ combines true and false positives. Weak white noise has 0 autocovariance at all nonzero lags. Classical i.i.d. CLT requires independence, identical distributions and finite, strictly positive variance.

6-L01

Positive probability is $.005(.95) + .995(.05) = .0545$, posterior $.00475/.0545 = 19/218 \approx .08715596$, approximately 8.7156%. Expected true positives are $10000(.005)(.95) = 47.5$ and false positives $10000(.995)(.05) = 497.5$.

6-L02

$\Pr(C) = \Pr(C^c) = 1/2$. Within C returns $.1, -.1$ have mean 0 and variance .01; within C^c returns $0, 0$ have both moments 0. Thus $\mathbb{E}[R_p] = \frac{1}{2}(0) + \frac{1}{2}(0) = 0$ and $\mathrm{Var}(R_p) = \mathbb{E}[\mathrm{Var}(R_p|\mathcal{G})] + \mathrm{Var}(\mathbb{E}[R_p|\mathcal{G}]) = \frac{1}{2}(.01) + \frac{1}{2}(0) + 0 = .005$, matching direct computation.

6-L03

Each sample contains $n = 120$ independent Bernoulli(p) observations; mean variance is $p(1-p)/n$. The denominator $\sqrt{p(1-p)}$ gives wrong variance $[p(1-p)/n]/p(1-p) = 1/n = 1/120$ rather than 1. With numerator $\bar{X} - p$, use denominator $\sqrt{p(1-p)/n}$; alternatively use numerator $\sqrt{n}(\bar{X} - p)$ and denominator $\sqrt{p(1-p)}$. This checks constructed-model standardization, not i.i.d. real data.

6-L04

No. Nonrejection of sample autocorrelation does not prove zero population autocovariance at every lag, much less independence. The simulation checks only specified generation. Define sequence, frequency, information and return interval, examine tails, moments, structural changes and dependence, and state the interval' s distribution/independence or dependence-aware assumptions. A dependent CLT or resampling method needs its additional conditions. No single test proves population i.i.d.

6-L05

Sample space is $\Omega = \{00, 01, 10, 11\}$, with independent-trial probabilities .49, .21, .21, .09. Events $A = \{11\}$ and $B = \{01, 10, 11\}$; mapping $X(00) = 0$, $X(01) = X(10) = 1$ and $X(11) = 2$ has support $\{0, 1, 2\}$. The PMF is $\Pr(X = 0) = .49$, $\Pr(X = 1) = .42$ and $\Pr(X = 2) = .09$.

6-L06

B retains $01, 10, 11$ with total $.21 + .21 + .09 = .51$. Intersection $A \cap B$ contains only 11 with probability .09; hence $\Pr(A|B) = .09/.51 = 3/17 \approx .17647059$.

6-L07

Grouping gives $\Pr(X = 0) = .49$, $\Pr(X = 1) = .21 + .21 = .42$ and $\Pr(X = 2) = .09$, followed by $\mathbb{E}[X] = 0(.49) + 1(.42) + 2(.09) = .60$, $\mathbb{E}[X^2] = 0^2(.49) + 1^2(.42) + 2^2(.09) = .78$ and $\mathrm{Var}(X) = .78 - (.60)^2 = .42$. Single-trial variance is $.3(.7) = .21$ and independent variance sum .42.

6-L08

Mean variance is $p(1-p)/n$, so z has variance $[p(1-p)/n]/p(1-p) = 1/n$. Keep the numerator and use $\sqrt{p(1-p)/n}$, or use numerator $\sqrt{n}(\bar{X}-p)$ with denominator $\sqrt{p(1-p)}$. More independent replications check the specified distribution and implementation; they do not prove CLT or i.i.d. real returns.

6-L09

The histogram concerns specified generation. Nonrejection of sample autocorrelation proves neither zero population covariance, independence nor normality. Prediction needs events, windows, information cutoff and out-of-sample assessment; causality needs intervention and randomization or defensible confounder identification. Classical CLT mean intervals need i.i.d. observations with finite positive variance and a centered mean standardized by each sample's n. Sample-standard-deviation inference also needs consistency and appropriate distributional assumptions. Dependent inference has its own dependence and moment conditions.

6-L10

$\mathbb{E}[A^2] = (1+4)/2 = 2.5$ and $\mathbb{E}[A^4] = (1+16)/2 = 8.5$. Linear cross covariance is 0 but squared covariance $8.5 - 2.5^2 = 2.25$, so the shared scale retains dependence. Conditional normal variances $1/2.5 = .4$ and $4/2.5 = 1.6$ each occur with probability one half; the normal fourth-moment identity in Section 6.5a gives $3(.4^2 + 1.6^2)/2 = 4.08$. The mixture is unchanged with n and differs from a standard normal with fourth moment 3, so it has no standard-normal limit. Mean variance $2.5/n$ is only a second-order result; classical IID CLT also needs independence. This diagnoses an argument rather than fitting markets.

Chapter 7: mean uncertainty and return risk

Move from an unknown parameter to a sample statistic and its realized estimate. Inference asks how that estimator varies when the sampling mechanism repeats. Standard error describes this variation; return risk describes single-period dispersion.

Trigger. Why can a positive sample mean still leave insufficient evidence?

Definition 7.1 *(Classical mean interval).* Under IID Gaussian observations with unknown positive variance, the interval is

$$\bar{X} \pm t_{1-\alpha_{\text{sig}}/2,n-1}\frac{S}{\sqrt{n}}.$$

$\bar{X}$ is sample mean, $S^2 = \sum(X_i - \bar{X})^2/(n-1)$, and α_{sig} significance level. This exact conclusion cannot be transferred unconditionally to arbitrary returns.

Example 7.2 *(Inputs for the four-return calculation).* Returns $(-.01, .01, .02, .02)$ have mean .01 and squared-deviation sum .0006; the 95% interval' s t critical value is about 3.182446.

Your turn. 7-L01 Mapping: 7-E02
Find sample variance, mean SE and the 95% interval. Does it contain zero? Explain the different quantities described by sample standard deviation and mean SE.

Your turn. 7-L02 7-C02 method limits; not its full experiment
What temporal information does independent bootstrap destroy? What do blocks preserve, and why can they not automatically repair a structural break?

Your turn. 7-L03 Code diagnosis: sample size for power

Section 6.7 ' s constructed design gives noninteger bound 2473.0229. Does this code achieve target power? Correct rounding and explain the scope.

```
raw_n = 2473.0229
required = int(raw_n)
```

Your turn. 7-L04 Research judgment: dependent power design

For constructed stationary AR(1) monthly returns with $\phi = .7$ and $n = 180$, can an independent-Gaussian sample-size formula guarantee real research power? Propose a dependence-aware assessment and distinguish simulation from market conclusions.

Your turn. 7-L05 Identify objects: 7-N01

Study a fixed rule's mean daily return. Population parameter: _____; day i random return: _____; realized observation: _____. Estimator: _____; estimate: _____; sampling unit: _____. Repeated sampling means _____. With dependence, repetition preserves _____.

Your turn. 7-L06 One test calculation: 7-N02

For a right-sided test, $z = 1.50$ and $\Phi(1.50) = .9332$. Complete $p = 1 - \Phi(z) =$ _____; at $\alpha = .05$, _____. Interpret p as "assuming _____, the probability of a _____ at least this large."

Your turn. 7-L07 Test and interval with mean given: 7-N03

Assume IID Gaussian sampling with known population standard deviation. Each measurement is a sampling unit in common output units. Given $n = 4, \sigma = 2, \bar{x} = 2, H_0 : \mu = 0, H_1 : \mu > 0$, exact critical value $z_{.95}$ is approximately 1.645. Complete

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{2}{\sqrt{4}} = _____,$$
$$z = \frac{\bar{x} - 0}{SE} = _____,$$
$$p = 1 - \Phi(2) = _____.$$

Lower confidence bound is $\bar{x} - 1.645SE =$ _____. Under H_0 and the rounded threshold, $\alpha \approx$ _____. At $\mu_1 = 2.645$, $\beta = \Phi(1.645 - 2.645) \approx$ _____ and power _____. Invert the pivot: $Z = (\bar{X} - \mu)/SE \leq c$ is equivalent to $\mu \geq$ _____, giving coverage $\Phi(c) =$ _____. Show steps and identify the repeated interval-generating process.

Your turn. 7-L08 Code diagnosis: 7-N04

‘p = 2 * (1 - Phi(z))‘ is used for a prespecified right-sided test. For $z \geq 0$ it computes _____; for $z < 0$ it _____. Correct formula: _____. At $z = 1.50$, correct $p =$ _____; if $z < 0$, right-tail p _____.

Your turn. **7-L09 Research judgment: 7-N05**

Gross daily mean .02% minus daily cost .03% gives net estimate _____. The version was selected from 40 candidates, with nominal IID test $p = .03$ and positive serial correlation. Give one needed check each for selection, dependence and economic meaning, and the 40-test Bonferroni threshold.

Chapter 7 solutions: 7-L01–7-L09

7-L01

Unbiased variance is .0002, $s = \sqrt{.0002} \approx .0141421$ and mean SE $s/\sqrt{4} \approx .0070711$. Interval $.01 \pm 3.182446(.0070711) \approx [-.0125033, .0325033]$ contains zero. s describes single-period sample dispersion; $s/\sqrt{n}$ describes sample-mean variability under IID assumptions. They are not interchangeable.

7-L02

Pointwise IID resampling destroys neighboring dependence. Blocks preserve within-block order but cut dependence at joins. Length, stationarity and dependence decay affect inference. A structural break can invalidate a single stationary-population approximation; blocks do not automatically fix it.

7-L03

Rounding down gives 2473, below bound 2473.0229, so target power is not guaranteed. Round up to 2474 for these parameters. Independence, Gaussianity, known standard deviation and prespecified direction underlie that number; it is not a general data threshold.

```
import math
required = math.ceil(raw_n)
assert required == 2474
```

7-L04

At $\phi = .7$, long-run variance ratio $(1 + \phi)/(1 - \phi) = 17/3$ gives SE inflation $\sqrt{17/3} \approx 2.3805$ and approximate mean effective sample size $180/(17/3) \approx 31.76$. Define target and information window, then use justified long-run variance/HAC, dependence-aware bootstrap or power simulation with an explicit AR mechanism. Examine block lengths, dependence parameters and structural changes. This construction neither proves real AR dynamics nor supplies universal degrees of freedom.

7-L05

The unknown parameter is target mean daily return μ. X_i is unit i' s random return and x_i its realization. $\bar{X}$ is the estimator and $\bar{x}$ the realized estimate. A unit is a trading day defined by the rule. Repeated equal-length samples under the same model reveal variation in $\bar{X}$; dependence requires preserving temporal structure.

7-L06

$p = 1 - .9332 = .0668$. Since $p > .05$, do not reject H_0. The p-value is the null-model probability of a right-tail statistic at least as extreme; it is not the probability the null is true.

7-L07

$SE = 2/\sqrt{4} = 1$, $z = 2/1 = 2$ and $p \approx 1 - .9772 = .0228$; lower bound is approximately $2-1.645(1) = .355$. Exact $z_{.95}$ gives Type I error .05; rounded threshold 1.645 gives about $.049985 \approx .05$. Approximate alternative $\mu_1 = 2.645$ with the hand threshold gives $\beta \approx \Phi(-1) = .1587$ and power .8413; exact quantiles give $\beta \approx .15862$. Inversion gives $\mu \geq \bar{X} - cSE$; exact $c = z_{.95}$ yields $\Phi(c) = .95$. Coverage repeats interval generation under the sampling mechanism. The parameter stays fixed; observed endpoints no longer vary.

7-L08

For $z \geq 0$, the code gives two-sided p; for $z < 0$, it exceeds 1 and is not a probability. Correct two-sided expression is $2[1-\Phi(|z|)]$. Right-sided p is $p = 1-\Phi(z)$, giving .0668 at $z = 1.50$. A negative z increases the right-tail probability because it opposes the prespecified positive alternative. Do not change tails after observing the statistic.

7-L09

Net estimate is $.02\% - .03\% = -.01\%$. Disclose and address selection among 40 candidates; assess mean uncertainty with HAC or prespecified dependent bootstrap; use net returns and report risk/capacity for economic evaluation. Bonferroni threshold is $.05/40 = .00125$. Nominal IID p-values correct neither search nor dependence; the numbers do not support a positive net point estimate.

Chapter 8 —Regression

Thread: from market-risk exposure to predictions available in time

> OLS point estimates, coefficient inference and future prediction need different conditions. Robust standard errors do not repair future-information leakage.

Trigger. Why can tests change when HAC leaves coefficients unchanged?

Definition 8.1 *(Point estimates and uncertainty).* With full column rank, _____, the solution $\hat{\beta} = (X^\top X)^{-1} X^\top y$ is unique. HC3 addresses heteroskedasticity and HAC additionally estimates cross-period score dependence. Both change coefficient covariance, leaving OLS estimates unchanged. Reference t tails for dependent heteroskedastic samples have no classical small-sample exactness guarantee.

Example 8.2 *(Constructed inputs and recorded outputs).* Each row is a UTC calendar day; feature and response share that day and daily decimal excess-return units. The first 280 rows train, the last 120 are held out; estimated slope is 1.124669. Classical, HC3 and HAC(5) standard errors are approximately .036144, .052391, .055402. Contemporaneous holdout error is not an available period-start forecast.

Your turn. 8-L01 Hand calculation mapped to 8-P01

Regress $y = (1, 2, 4)^\top$ on $x = (-1, 0, 1)^\top$ with an intercept, complete the derivation step by step. For $\hat{y} = 2$ and $\hat{y} = 2 + x$, first list fits, residuals and RSS. Write $Q(a, b) = \sum_i (y_i - a - bx_i)^2$, differentiate in both parameters and set derivatives to zero.

Here $\sum x_i =$ ____, $\sum y_i =$ ____, $\sum x_i^2 =$ ____ and $\sum x_i y_i =$ ____, hence $a =$ ____ and $b =$ ____. Give $X^\top X$, $X^\top y$, OLS fits, residuals, RSS, *TSS* and R^2, then explain the geometry of $X^\top e = 0$.

This covers 8-P01 ' s hand calculation only, not its normal-equation code, statsmodels cross-check or assertions.

Your turn. **8-L02 Code diagnosis: scaling within training folds, 8-P03**
Although this code receives training data only, each validation fold influences the fitted scaler. How do you fit preprocessing using only each fold's training observations?

```
scaler = StandardScaler().fit(X_train)
X_scaled = scaler.transform(X_train)
cv = TimeSeriesSplit(n_splits=5, test_size=30, gap=5)
scores = cross_val_score(Ridge(alpha=1.0), X_scaled, y_train
   ↪ , cv=cv)
```

Your turn. 8-L03 Inference: 8-P02 and 8-E06

Use slope 1.124669 and the three standard errors to calculate t statistics. Which error structures correspond to classical, HC3 and HAC covariance? Why not choose the most significant result afterward? Explain why these t calculations do not replace P02's full report of all three $\hat{\alpha}$, $\hat{\beta}$, standard errors, p-values and common point estimates.

Your turn. 8-L04 Research judgment: contemporaneous fit and prediction, 8-C02

Market return x_t is known after the close and y_t is that day's asset return. Why does low holdout error fail to establish prediction at period start? State next-period timing, training, inference and trading boundaries.

Your turn. 8-L05 Formula-domain diagnosis: 8-B01

Take $x = (0, 0, 1)^\top$ and $y = (0, 2, 1)^\top$ with an intercept. Matrix $X^\top X = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ has determinant _____ and inverse _____. Calculate coefficients, fits and residuals, then rowwise $h_{tt} = x_t^\top (X^\top X)^{-1} x_t$ with three values _____. Why do full column rank and $n > k$ still fail to make HC3 calculable?

For a constant response, $TSS =$ _____: can R^2 be calculated? If $R = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 2 \end{smallmatrix}\right)$ repeats one restriction twice, $\operatorname{rank}(R) =$ _____: what happens to the inverse restriction matrix? How many nonredundant restrictions remain? Do valid denominators certify valid inference?

Chapter 8 solutions: 8-L01–8-L05

Solutions and derivations: 8-L01–8-L05

Definition blank

Full column rank means $\text{rank}(X) = k$.

8-L01

The four sums are $\sum x_i = 0$, $\sum y_i = 7$, $\sum x_i^2 = 2$ and $\sum x_i y_i = 3$. For $\hat{y} = 2$, fits $(2, 2, 2)$ and residuals $(-1, 0, 2)$ give $RSS = 5$; for $\hat{y} = 2 + x$, fits $(1, 2, 3)$ and residuals $(0, 0, 1)$ give $RSS = 1$. Define $Q(a, b) = \sum_i (y_i - a - bx_i)^2$, with derivatives $-2\sum_i (y_i - a - bx_i)$ and $-2\sum_i x_i(y_i - a - bx_i)$. Substitute sums to get $3a = 7, 2b = 3$ and $a = 7/3, b = 3/2$. In matrix form, $X^\top X = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$ and $X^\top y = (7, 3)^\top$ yield $\hat{\beta} = (7/3, 3/2)^\top$. Fits $(5/6, 7/3, 23/6)$ and residuals $(1/6, -1/3, 1/6)$ give $RSS = 1/6$, with $TSS = 14/3$ and $R^2 = 27/28 \approx 0.964286$. $X^\top e = 0$ means residuals are perpendicular to every design column. This hand calculation does not replace P01's code checks.

8-L02

The scaler was fitted to all of X_{train}, allowing validation folds to influence means and scales. Put it inside the estimator pipeline and pass that pipeline to cross-validation or grid search:

```
pipe = Pipeline([("scale", StandardScaler()), ("ridge", Ridge())
    ↪ ])
cv = TimeSeriesSplit(n_splits=5, test_size=30, gap=5)
alphas = [0.01, 0.1, 1.0]
search = GridSearchCV(
    pipe, {"ridge__alpha": alphas}, cv=cv,
    scoring="neg_mean_squared_error",
)
search.fit(X_train, y_train)
```

Each fold fits scaling on its training rows only. The 120-row holdout remains reserved for final scoring after parameter selection.

8-L03

Statistics t are approximately 31.116, 21.467 and 20.300. Classical covariance needs homoskedasticity and no cross-period error correlation. HC3 addresses heteroskedasticity; HAC weights cross-period score dependence subject to bandwidth, spacing and process conditions. OLS point estimates coincide, covariance estimates differ. Choose the convention from prespecified error assumptions. Dependent heteroskedastic t tails are a reference convention, not exact finite-sample inference. This t exercise does not replace P02's complete intercept/slope/standard-error/p-value report.

8-L04

Contemporaneous market return is known only after the close and cannot be that day's pre-open signal. The holdout illustration is conditional fitting, not an executable forecast. Predict subsequent labels with then-available features and explicit signal, trade and holding intervals. Training, validation, scaling and tuning use training information; set gap/purge for overlapping targets and assess covariance/HAC-bandwidth sensitivity. Define benchmarks, costs, slippage and disconfirmation; simulated fitting does not establish tradable alpha.

8-L05

Determinant 2 and inverse $\begin{pmatrix} 1/2 & -1/2 \\ -1/2 & 3/2 \end{pmatrix}$ give $X^\top y = (3,1)^\top$, coefficients $(1,0)^\top$, fits $(1,1,1)^\top$ and residuals $(-1,1,0)^\top$, with RSS 2. Leverage $(1/2, 1/2, 1)$ makes the third HC3 weight $0/0$ undefined; finite floating-point output does not validate it. Constant response gives $TSS = RSS = 0$ and undefined R^2. Duplicate restrictions have rank 1, and $R(X^\top X)^{-1}R^\top = \begin{pmatrix} 3/2 & 3 \\ 3 & 6 \end{pmatrix}$ has determinant zero. Retain one restriction with $q = 1$. Denominators check domains only; exact classical inference, HC3 and HAC each need their own error and applicability assumptions.

Chapter 9 —Optimization

> List feasible weights, then objective and constraints. Solver numbers become meaningful only when model, units and residuals agree.

Thread: two assets from a budget line to a boundary optimum

Assets 1 and 2 have annual expected total returns $(.05, .08)$ and covariance matrix

$$\Sigma = \begin{pmatrix} .04 & .01 \\ .01 & .09 \end{pmatrix}.$$

Start with budget $w_1 + w_2 = 1$, long-only bounds $w \geq 0$ and risk coefficient $\lambda = 1$. Three feasible portfolios are:

w_1	w_2	$\mu^\top w$	$w^\top \Sigma w$	$J(w)$
0	1	.08000	.09000	−.03500
.5	.5	.06500	.03750	−.04625
1	0	.05000	.04000	−.03000

These are same-year constructions. Weights are dimensionless, returns annual decimals and variance annual decimal return squared. Risk coefficient λ has inverse-return units, aligning objective terms. Three points suggest the valley ' s location without proving an interval optimum.

Trigger. Which weights do budget, leverage and long-only constraints exclude?

Definition 9.1 *(Two-asset convex quadratic objective).* Decision vector $w = (w_1, w_2)^\top$ has model

$$\min_w J(w) = \frac{1}{2}\lambda w^\top \Sigma w - \mu^\top w, \qquad \mathbf{1}^\top w = 1, \quad w \geq 0.$$

The objective scores the model, not realized returns. Expected return μ and covariance Σ must use the same holding period.

Example 9.2 *(Budget-line inputs).* Let $w = (x, 1-x)^\top$. Long-only bounds give $x \in [0, 1]$. With shorting, budget still holds but x can lie outside this interval.

Your turn. 9-L01 Mean–variance QP: 9-P01

Along the budget line, objective $J(x) = .055x^2 - .05x - .035$ gives $J'(x) =$ _____ and interior weights $w =$ _____. Calculate return, variance and objective. Remove the return reward and minimize variance only: weights become _____.

Your turn. **9-L02 Boundary KKT: 9-P03**

At $\lambda = .2$ the long-only optimum is $w = (0, 1)$. Calculate gradient $g = \lambda\Sigma w - \mu =$ _____. Write $L = J + \nu(\mathbf{1}^\top w - 1) - \ell^\top w$ and fill budget multiplier $\nu =$ _____ and bound multipliers $\ell =$ _____. Check primal/dual feasibility, stationarity and $\ell_i w_i =$ _____. Why can the full-space gradient g be nonzero at this constrained optimum?

Your turn. 9-L03 Derivation: maximum-Sharpe transformation, 9-C01

Define $\mu_i^{ex} = \mathbb{E}[R_i] - r_f$ with the same horizon/currency for returns and benchmark; declare $r_f = 0$ if assumed. Let C contain budget, long-only and optional linear caps. First check

$$D_w = \{w \in C : (\mu^{ex})^\top w > 0,\ w^\top \Sigma w > 0\} \neq \emptyset.$$

Set $y =$ _____, give positive-variance conditions and reverse map $w =$ _____, and transform $w_i \leq c_i$ to _____. With two caps $c_i = .4$, is the original domain nonempty?

Under $\Sigma \succeq 0$, why does zero optimal variance in the closed QP not yield a finite-ratio optimum? Take $\Sigma = \mathrm{diag}(0, 1)$ and $\mu^{ex} = (1, 0)^\top$. Check nonemptiness with $w = (1/2, 1/2)$, then use $y_0 = (1, 0)$ and $y_1 = (1, 1)$ to fill

$$y_\theta = (1-\theta)y_0 + \theta y_1 = _____,$$
$$q(y_\theta) = _____, \quad S(w_\theta) = _____ \quad (0 < \theta \leq 1).$$

Why cannot the same conclusion follow for an empty original domain? Is PD necessary, and can strict $q(y) > 0$ be an ordinary convex-QP constraint?

Your turn. 9-L04 Code diagnosis: solver status, 9-E07

This code normalizes and returns any nonempty variable value. Why does this certify neither success nor original feasibility?

```
problem.solve()
if weights.value is not None:
    return weights.value / weights.value.sum()
```

State necessary status, budget/long-only/cap residual and tolerance checks. When are KKT residuals diagnostics only?

Your turn. 9-L05 Objects and units

Identify decisions _____, objective _____, budget _____ and long-only bounds _____. Give units of μ _____, Σ _____ and λ _____. Gross constraint $\|w\|_1 \leq L$ restricts _____.

Your turn. 9-L06 One calculation

At $w = (.5, .5)$ and $\lambda = 1$, expected return $\mu^\top w =$ _____, variance $w^\top \Sigma w =$ _____ and objective $J(w) =$ _____. Which has squared-return units?

Your turn. 9-L07 Proof skeleton: curvature and an interior optimum

Let $w = (x, 1-x)$. Expand

$$w^\top \Sigma w = .04x^2 + 2(.01)x(1-x) + .09(1-x)^2 = _____.$$

Then

$$J(x) = \tfrac{1}{2} w^\top \Sigma w - \mu^\top w = _____,$$
$$J'(x) = _____,$$
$$J''(x) = _____.$$

Proof. Stationary point $x =$ _____ belongs to interval _____. Explain why $J'' > 0$ and convex feasibility make it the unique global minimum.[fill in: Complete the proof using convexity and endpoint checks.] ∎

Your turn. 9-L08 Complete calculation: budget, leverage and bounds

Take $\lambda = .2$. Budget-line objective $J(x) =$ _____ and derivative $J'(x) =$ _____ give budget-only weights _____. Adding $\|w\|_1 \le 1.5$ gives weights _____; adding $w \ge 0$ gives _____. State the constraint responsible at each stage.

Your turn. 9-L09 Research judgment

Do optimal weights on constructed inputs establish higher future return or live tradability? Give at least three missing checks and explain what Chapters 20 and 21 add.

Chapter 9 solutions: 9-L01–9-L09

9-L01

Budget-line derivative $J'(x) = .11x - .05$ vanishes at $x = 5/11$, giving $(5/11, 6/11)$. Return .066363636, variance .04 and objective $-.046363636$ compare with endpoint values $-.035$ and $-.030$. For variance-only objective $J_v(x) = .11x^2 - .16x + .09$, stationary point $x = 8/11$ gives $(8/11, 3/11)$ and variance .031818182.

9-L02

Gradient $g = (-.048, -.062)^\top$ with $\nu = .062$ and $\ell = (.014, 0)^\top$ satisfies $g + \nu\mathbf{1} - \ell = (0, 0)^\top$. Primal feasibility: $\mathbf{1}^\top w = 1$ and $w \geq 0$. Dual feasibility: $\ell \geq 0$. Complementarity: $\ell_1 w_1 = \ell_2 w_2 = 0$. Budget and active lower-bound normals balance a nonzero full-space gradient. Point $(.5, .5)$ strictly satisfies the long-only bounds. Convexity makes this feasible KKT point globally optimal; Slater ensures optimal-multiplier existence here.

9-L03

Let $m = \mu^{ex}$. Confirm nonempty D_w: budget feasibility and $m^\top w > 0$ alone are insufficient; also require $w^\top \Sigma w > 0$. Forward map $y = w/(m^\top w)$ gives $y \geq 0$, $m^\top y = 1$ and $q(y) = y^\top \Sigma y > 0$. Conversely retain $q(y) > 0$, set $t = \mathbf{1}^\top y > 0$ and $w = y/t$, giving $m^\top w = 1/t > 0$ and $w^\top \Sigma w = q(y)/t^2 > 0$; the forward map recovers y. Caps become $y_i \leq c_i t$. Two caps $c_i = .4$ imply $t \leq .8t$, contradicting $t > 0$, so budget and positive-variance original domains are empty.

Strict domain D_y includes $q(y) > 0$; closed domain $\overline{D}_y$ retains only nonnegativity, $m^\top y = 1$ and linear caps. Assume closed-QP attainment. With $q^\star > 0$, the reverse map gives finite maximum Sharpe $1/\sqrt{q^\star}$. With $q^\star = 0$, zero variance lies outside the strict domain and cannot be a finite-ratio optimum.

In the counterexample, $w = (1/2, 1/2)$ has excess return $1/2$ and variance $1/4$, proving nonemptiness. For $y_0 = (1, 0)$ and $y_1 = (1, 1)$, mixture $y_\theta = (1, \theta)$ gives $q(y_\theta) = \theta^2$ and reverse weights $w_\theta = (1, \theta)/(1 + \theta)$ with Sharpe $1/\theta \to +\infty$. At $x = 1/2, .9, .99, .999$ ratios are 1, 9, 99 and 999. Endpoint $w_0 = (1, 0)$ has zero variance and undefined ratio, without attaining a finite maximum.

PSD implies $q(y_0) = 0$ gives $\Sigma y_0 = 0$. With a nonempty original domain, choose feasible $q(y_1) > 0$. Convex mixing preserves linear constraints and $q((1 - \theta)y_0 + \theta y_1) = \theta^2 q(y_1) > 0$, so the supremum is infinite. An empty original domain supplies no positive-variance point: with $\Sigma = 0$ and $m = (1, 0)$, closed-QP feasibility coexists with zero variance everywhere and no finite-Sharpe sequence diverging to infinity.

$\Sigma \succ 0$ is sufficient, not necessary. Singular $\Sigma = \mathrm{diag}(0, 1)$ with $m = (0, 1)$ fixes $y_2 = 1$; closed-QP minimum variance is 1, and Sharpe is identically 1 on the positive-return domain. Strict $q(y) > 0$ generally is nonconvex: with $\Sigma = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ and $m = (1, 1)$, normalized long-only endpoints each have variance 1 but the midpoint has zero variance and is excluded. This strict constraint cannot be treated as ordinary convex QP.

9-L04

Nonempty values do not imply optimal status; optimal_inaccurate gives a low-precision candidate and infeasible has no allowed point. Normalizing alters the original solution and can violate caps. Require optimal status, finite weights, recomputed budget/long-only/cap residuals and objective, using prespecified tolerances. Even optimal is floating-point output. The helper thresholds primal feasibility at 10^{-6}; stationarity/complementarity are checked for finiteness and returned, not a complete KKT certificate.

9-L05

Decisions w, objective $\frac{1}{2}\lambda w^\top \Sigma w - \mu^\top w$, budget $\mathbf{1}^\top w = 1$ and lower bounds $w \geq 0$ apply. Weights are dimensionless, annual μ decimal expected returns, Σ annual decimal-return squared covariance and λ inverse-decimal-return risk aversion. $\|w\|_1 \leq L$ restricts gross exposure; budget restricts net weight.

9-L06

$\mu^\top w = .065$, $w^\top \Sigma w = .0375$ and $J = .5(.0375) - .065 = -.04625$. Variance has squared-return units; multiplication by λ aligns it with the expected-return term.

9-L07

$w^\top \Sigma w = .11x^2 - .16x + .09$ gives $J = .055x^2 - .05x - .035$, $J' = .11x - .05$ and $J'' = .11 > 0$. Stationary point $x = 5/11$ lies in $[0, 1]$. Strict convexity on a convex interval makes it the unique global minimum; endpoints $-.035$ and $-.030$ also exceed interior objective $-.046363636$.

9-L08

$J_{.2}(x) = .011x^2 + .014x - .071$ and $J'_{.2}(x) = .022x + .014$. Budget alone gives $x = -7/11$, weights $(-7/11, 18/11)$ and gross exposure $25/11$. Adding gross cap 1.5 gives full interval $x \in [-.25, 1.25]$: $1 - 2x \leq 1.5$ on the negative branch, $2x - 1 \leq 1.5$ above one and all $[0, 1]$ feasible. The derivative is positive throughout, so optimum $(-.25, 1.25)$. Long-only bounds narrow to $x \in [0, 1]$ with optimum $(0, 1)$. Budget restricts net weight, leverage gross exposure and lower bounds shorting.

9-L09

No. Constructed μ, Σ check consistency with the model, not estimation accuracy, future returns or trades. Needed evidence includes point-in-time data/estimation audits, out-of-sample assessment, financing/holdings/execution limits, turnover/cost ledgers and robustness. Chapter 20 develops mean–variance theory and distinguishes population from estimated moments; Chapter 21 addresses estimation error, practical constraints, costs and out-of-sample performance.

Chapter 10 —Financial Markets

> Notional measures reference exposure; margin is a performance or financing arrangement. Profit and loss depend on contract quantity, quotation units, and price or cash-flow changes. Trading, clearing, and settlement record different events.

Trigger. Why do an ETF and a futures account have different cash ledgers when the same index rises?

Reading route and coverage

10-L01, L02, and L04 retain stable numbers and fully cover 10-E03, E07, and C01. L03 adapts only the quotation conversion in 10-P02. L05 fully covers E01, E02, E04, and E05; L06 fully covers E06 and E08. L07 is an independent hand-calculation variation on P01–P03. L08 fully covers C02. L09 is a partial diagnostic variation on P04 and P05. Complete implementations and assertions for P01–P05 remain in the same-chapter body answers; hand calculations and diagnostics do not fully cover the original programming tasks. L10–L12 are additional independent exercises combining the concepts in body N01–N04.

Definition 10.1 *(Variation settlement for one MES contract).* If q is signed contract quantity, M has units USD/(contract · index point), and settlement-price change ΔF has units index point, then $VM = qM\Delta F$ has units USD. Initial margin does not enter this profit-and-loss formula.

Your turn. 10-L01 Trading, clearing, and settlement are distinct (full coverage: 10-E03)
Order trade date T, clearing, and settlement by event, and state the question answered by clearing and settlement. What scope does the SEC US-securities T+1 example support, and what does it not imply?

Your turn. 10-L02 MES variation settlement and margin (full coverage: 10-E07; supplements body Section 6)
One long MES contract has multiplier USD 5 per contract per index point and settlement prices 4,000, 3,992, and 4,005 points. Calculate each variation-margin (VM) payment and cumulative profit and loss. Assume USD 500 initial collateral; explain why it is neither futures notional nor maximum loss.

Your turn. 10-L03 Code diagnosis: FX quotation direction (partial variation: conversion in 10-P02, excluding forward programming)
The quotation is $S_{USD/EUR} = 1.08\ USD/EUR$. The code below converts EUR 25,000 into USD. Identify the incorrect operation, correct the program, and verify the result using units and an assertion.

```
# BEGIN B08_10L03_BAD
spot_usd_per_eur = 1.08
eur_amount = 25_000.0
usd_amount = eur_amount / spot_usd_per_eur
print(usd_amount)
# END B08_10L03_BAD
```

Your turn. 10-L04 Comparing an ETF and MES (full coverage: 10-C01)
A cash ETF and MES on the same index have identical initial reference exposure of USD 25,000. List at least four types of comparison data and explain how to choose a common return denominator. What economic results are omitted by comparing percentage changes in two price series alone?

Your turn. 10-L05 Rights and institutions (full coverage: 10-E01, E02, E04, E05)
Answer each part: What do brokers, dealers, and market makers do, and must they be separate institutions? How do ETF shares, net asset value (NAV), and underlying securities differ? How does securities margin lending differ from futures performance margin, and why does margin not limit loss? What other cash obligations and operational risks arise from short selling?

Your turn. 10-L06 Stock and swap cash flows (full coverage: 10-E06, E08)
Buy 100 shares at USD 50/share; terminal price is 49.80 and dividends are 0.30/share. State investment, terminal securities value, dividends, and the two return denominators. Separately, a USD 1,000,000 swap receives floating 3.6%, pays fixed 4.0%, and uses year fraction 0.25. State both interest legs, net cash, and whether notional principal changes hands.

Your turn. 10-L07 Hand calculations with units (independent variations: 10-P01–P03)

Without programming, calculate and state units: coupon for face value USD 5,000, annual coupon rate 4%, four payments per year; USD value of EUR 25,000 at USD/EUR=1.08 and a half-year reference forward with USD rate 4% and EUR rate 2%; short MES with changes (−12, +7) points and multiplier 5; long put with terminal price 85, strike 90, premium 2, and 100-share multiplier; repo financing against USD 2,000,000 collateral with a 3% haircut.

Your turn. 10-L08 Repo stress (full coverage: 10-C02)

USD 1,000,000 of bonds secures financing of USD 950,000 at a 5% haircut. Design three stresses involving collateral prices, haircut or liquidity changes, and counterparty or settlement risk. Quantify at least two, list the contract and market data needed to assess loss, and explain whether the initial haircut guarantees recovery.

Your turn. 10-L09 Event and unit diagnosis (partial variations: 10-P04, P05)
A ledger credits +50 cash on both the entitlement and payment dates of a USD 50 dividend, then adds EUR 10 to the USD balance. A contract checker permits NaN, FX quotation "1.10," a negative multiplier, and futures-change units "USD/contract." Identify each error and the fields to record or reject. This is diagnosis only, not completion of the two body programming functions.

Your turn. 10-L10 One wealth total and two paths (additional: 10-N01, N02)
One hundred shares end at 50.20, with a USD 50 dividend receivable established but unpaid. List securities, receivables, cash, and total wealth before and after payment. Starting with EUR 1, USD/EUR=1.10, annual simple USD rate 5%, EUR rate 3%, and half-year maturity, write both terminal USD values and solve for the forward and reverse quotation.

Your turn. 10-L11 Cash-flow sensitivity (additional: 10-N03)
Cash flows are USD 60 at year one and USD 1,060 at year two; the effective annual yield is 5%. From $P(y) = 60(1+y)^{-1} + 1060(1+y)^{-2}$, derive first and second derivatives, calculate modified duration and convexity, and compare first-order, second-order, and exact price changes for a 100-basis-point shock. State the unchanged cash flows and compounding conditions.

Your turn. 10-L12 Denominators and net profit (additional: 10-N04)
Cumulative MES profit is USD 50. What ratios result against USD 25,000 notional exposure, USD 1,000 hypothetical collateral, and USD 100,000 portfolio NAV? A long call has strike 100, premium 4, and a 100-share multiplier. What are payoff and profit at terminal prices 102 and 104?

Chapter 10 answers: 10-L01 through 10-L12

10-L01

Trading occurs on trade date T. Clearing then confirms transactions, calculates obligations, and applies netting and risk management under the applicable arrangements. Settlement transfers cash or securities. Clearing asks "Who owes which net obligations to whom?" Settlement asks whether and when the agreed assets were transferred. The SEC T+1 example is limited to securities transactions subject to the relevant US rules described in the FAQ, including exceptions and explicit agreements. It is not a universal cycle for every country, security, or derivative.

10-L02

From 4,000 $\rightarrow$ 3,992 the change is -8 points and VM is $-8 \times 5 = -USD$ 40; from 3,992 $\rightarrow$ 4,005 the change is $+13$ points and VM is $+13 \times 5 = +USD$ 65. Cumulative VM is $+USD$ 25, also $5 \times (4{,}005 - 4{,}000)$. USD 500 is hypothetical collateral; reference notional at 4,000 points is $4{,}000 \times 5 = USD$ 20,000. Profit and loss follow point changes and the multiplier, and subsequent adverse price changes may exceed USD 500. The nominal sum of VM excludes interest, reinvestment, and financing of cash balances after settlement.

10-L03

The error is division by the quotation. 1.08 USD/EUR means USD 1.08 per EUR, so multiply the EUR amount by the quotation: $EUR \times USD/EUR = USD$. Division does not yield USD units.

```
# BEGIN B08_10L03_FIXED
spot_usd_per_eur = 1.08
eur_amount = 25_000.0
usd_amount = eur_amount * spot_usd_per_eur
assert abs(usd_amount - 27_000.0) < 1e-10
print(usd_amount)
# END B08_10L03_FIXED
```

10-L04

Record at least the holding interval and endpoint prices; ETF distributions, fund fees, and trading costs; MES multiplier, variation settlement, margin, financing, and cash interest; contract maturity and roll costs; and execution, liquidity, and settlement times for both instruments. Compare identical initial reference exposure first, then separately report denominators based on invested capital or portfolio NAV. Do not automatically treat futures margin as all capital. Price percentages omit ETF dividends and fees, the timing of futures variation payments, rolls, financing, and collateral capital use. Alone, they cannot establish replicable net returns.

10-L05

A broker acts for or arranges client orders; a dealer trades on its own account and carries inventory; a market maker provides two-sided quotes under its arrangements. One institution may perform multiple roles; legal duties depend on jurisdiction and business. An ETF share is a traded fund-interest unit. Its intraday price reflects supply and demand, while NAV follows valuation rules, permitting a premium or discount. A secondary-market buyer does not receive underlying securities in every trade. Securities margin lending finances purchases; futures margin secures performance. Contract-based losses, margin calls, gaps, and liquidation may exceed collateral. Short sellers must buy back and return borrowed securities and may face borrow fees, dividend substitute payments, recalls, availability changes, margin calls, and forced buy-ins.

10-L06

Investment is $100\times50 = USD\,5,000$; securities value is $100\times49.80 = USD\,4,980$; dividends are USD 30; total wealth is USD 5,010. Price return is $-20/5000 = -0.4\%$ and total return $10/5000 = 0.2\%$, assuming dividends belong to the holding interval and ignoring taxes and fees. The swap receives floating $10^6 \times 0.036 \times 0.25 = USD\,9,000$, pays fixed USD 10,000, and makes a net payment of USD 1,000. Only the interest difference is exchanged here; notional principal is not transferred.

10-L07

Coupon is $5000 \times 0.04/4 = USD\ 50$. EUR 25,000 times 1.08 USD/EUR gives USD 27,000. The forward is $1.08(1 + 0.04 \times 0.5)/(1 + 0.02 \times 0.5) = 1.0906930693$ USD/EUR. Short MES cash flows are $(-1)5(-12) = USD\ 60$ and $(-1)5(7) = -USD\ 35$, totaling USD 25. Long-put payoff is $(90 - 85)100 = USD\ 500$; subtracting premium 2×100 gives profit USD 300. Repo financing is $2000000(1 - 0.03) = USD\ 1,940,000$. The USD 60,000 buffer does not guarantee full recovery.

10-L08

After a 4% price decline, collateral is USD 960,000 against debt USD 950,000, leaving USD 10,000 before interest and costs. An 8% haircut at the original market value supports USD 920,000 financing, USD 30,000 below existing debt; terms may require cash or collateral top-up. A price gap and delayed default liquidation with a 7% disposal discount yield only USD 930,000 even from USD 1,000,000, leaving a USD 20,000 shortfall. Interest, costs, or prior price declines widen it. Required data include master agreements, margining and valuation frequency, interest, insolvency and liquidation law, settlement, liquidity, and stress prices. A haircut is an initial buffer, not protection against every price or execution loss. Ten points: two quantified scenarios (4), contract and execution data (3), explanation of no guaranteed recovery (3).

10-L09

On the entitlement date, record USD 50 receivable and zero cash. On payment, reduce receivables by 50 and increase cash by 50: one cash receipt. Keep USD and EUR balances separate; conversion requires quotation direction and time. NaN, infinity, and missing amounts must fail. FX needs explicit QUOTE/BASE and a positive quotation. Multipliers must be positive and finite. Futures ΔF is in index points, not USD/contract; calculate only after matching $contract \times USD/(contract \cdot index\ point) \times index\ point = USD$. Retain date, event type, cash-transfer flag, currency, and unit fields rather than replacing invalid input with zero.

10-L10

Before dividend payment: securities USD 5,020, receivables 50, dividend cash zero, total wealth 5,070. After payment: receivables zero, dividend cash 50, wealth still 5,070. Converting an entitlement to cash creates no second return. The EUR deposit ends at EUR 1.015 and converts forward to USD $1.015F$; the USD-deposit path gives $1.10 \times 1.025 = USD\ 1.1275$. Equality gives $F = 1.1108374384$ USD/EUR, reciprocal 0.9002217295 EUR/USD. A common starting point, common maturity, positive accumulation factors, feasible borrowing/lending and covering, and frictionless conditions are essential.

10-L11

$P' = -60(1+y)^{-2} - 2120(1+y)^{-3}$; $P'' = 120(1+y)^{-3} + 6360(1+y)^{-4}$. At 5%, $P = 1018.594104$, $-P' = 1885.757478$, $P'' = 5336.0482515$; $D_{mod} = -P'/P = 1.851334$, $C = P''/P = 5.238640$. A 100-basis-point shock is 0.01: first-order change is -1.851334%; the second-order term is $C(0.01)^2/2 = 0.0261932\%$, for -1.825140% overall. Exact $P(0.06) = 1000$ gives -1.825467%. Conditions are fixed USD cash flows, effective annual compounding, one yield, positive price, and no accrued interest, fees, default, or callability. This is a local interest-rate risk estimate; changed cash flows or curve structure require revaluation.

10-L12

$50/25000 = 0.2\%$ against initial notional, $50/1000 = 5\%$ against constructed collateral, and $50/100000 = 0.05\%$ as a gross contribution to portfolio NAV. Collateral does not cap loss. At 102, payoff is USD 200 and profit −USD 200; at 104, payoff is USD 400 and profit zero. Profit subtracts the USD 400 opening premium and ignores funding and fees. Breakeven is 104.

Chapter 11 —Returns and Compounding

Define the account, endpoints, and cash flows before calculating returns. Simple returns compound over time; log returns add over time; drawdown retains the pressure of a path from historical peak to current wealth.

Trigger. Why can wealth decrease when average return is zero?

Definition 11.1 *(Wealth recursion; body Sections 4 and 6.1).* The same-currency account has no external deposits or withdrawals within an interval; valuation includes all investment cash flows. For starting $W_{t-1} > 0$ and ending $W_t \geq 0$, $R_t = W_t/W_{t-1} - 1$ and $W_T = W_0 \prod_{t=1}^{T}(1 + R_t)$. W has currency units; R is a dimensionless decimal. Raw price ratios give only price returns. Dividends, splits, taxes, fees, and reinvestment require an entitlement ledger. Do not add dividends again to a series already including distributions.

Definition 11.2 *(Means and logs; body Section 5.2).* For T equal-length intervals, arithmetic mean is $\bar{R} = T^{-1} \sum R_t$ and geometric mean is $G = [\prod(1 + R_t)]^{1/T} - 1$. If every $R_t > -1$, $g_t = \log(1 + R_t)$ and $\log(W_T/W_0) = \sum g_t$. -100% may reduce wealth to zero but has no finite logarithm. After bankruptcy, the account-return denominator is zero; a larger percentage cannot restore wealth.

Example 11.3 *(Two constructed months; body Section 8.1).* Initial wealth is 100, monthly returns are $+50\%, -50\%$, and there are no external cash flows.

Your turn. **11-L01: Growth and proof; partial adaptations of 11-E02 and 11-C01.** Wealth path _____, arithmetic monthly mean _____, cumulative return _____, geometric monthly mean _____. Which measure reconstructs constant monthly compound growth? Use log concavity to write the key step for $G \leq \bar{R}$ and the equality condition.

Endpoints, intervals, and the asset axis

Initial capital belongs in the wealth path but is not the first market zero return. Across assets, calculate simple return using beginning capital weights before taking the log of the whole portfolio.

Your turn. 11-L02: Counts and paths; partial coverage of 11-P01 and body Section 9.
Wealth is 100, 150, 75, 120. State actual interval returns, cumulative return, and maximum drawdown, explaining their two denominators. If a displayed ledger adds `ret` = 0 on the initial row, why must it be excluded from return means, volatility, Sharpe ratios, and elapsed time? Is 0% in a genuine interval retained?

Your turn. 11-L03: Portfolio code diagnosis; full coverage of 11-E03.
Invest 50 in each of two assets with returns +50%, −50% in one interval, without rebalancing. Calculate portfolio simple and log returns, weighted asset log returns, and terminal weights. What would code report if it multiplied this interval's returns by terminal weights?

Conditions for risk measures; body Sections 5.3 and 6.4

For the same interval and currency, excess return is $e_t = R_t - R_{f,t}$. The single-interval sample Sharpe ratio $\bar{e}/s(e)$ needs at least two observations and $s(e) > 0$. With identical interval variance γ_0, exact square-root-of-time scaling at specified q holds if and only if

$$\sum_{k=1}^{q-1}(q-k)\gamma_k = 0.$$

Zero covariance at every lag suffices but is not necessary for fixed q. Scaling at every horizon requires all nonzero-lag covariances to vanish under weak stationarity. The formula concerns an increment sum, not a universal annual variance identity for compounded simple return.

Frequency and research judgment

Your turn. 11-L04: RF and external cash flows; partial coverage of 11-E04 and 11-C02.
If monthly asset return is 2% and the same-month cash proxy is 0.4%, what is excess return? Why cannot annual 4% be subtracted directly? Another account starts at 100, earns 10% to 110, and receives an interval-end deposit of 90 to reach 200. Why is the unadjusted terminal-to-initial ratio not investment return?

Your turn. 11-L05: Historical chart and scope; partial coverage of 11-P03.
The current French NoDur monthly snapshot covers 120 months in 2010–2019, starts at one, and ends near 3.141. State one supported description and two unsupported strategy conclusions. Is month-end maximum drawdown equal to daily or intraday maximum drawdown? What further version, signal, or trading evidence is needed?

Your turn. 11-L06: Midyear deposits and weighted returns; independent variation, not a replacement for 11-C02.
Invest 100 at the year start, earn 20% to reach 120 in the first half, deposit another 100, then earn 10% to reach 242 in the second half. Calculate segmented time-weighted return, the unadjusted terminal-to-initial ratio, and effective annual money-weighted return. Explain the question each answers.

Your turn. 11-L07: Covariance cancellation at a fixed horizon; partial coverage of 11-N01.

Three increments have covariance matrix $\begin{pmatrix} 1 & 0.1 & -0.2 \\ 0.1 & 1 & 0.1 \\ -0.2 & 0.1 & 1 \end{pmatrix}$. Determine whether it is a valid covariance matrix, then calculate variance of the three-interval sum. Why does this not establish square-root scaling over two intervals or every horizon?

Your turn. 11-L08: Equal endpoints, different drawdown paths; independent variation, not a replacement for 11-E07.

Initial wealth is 100. The paths are $(25\%, -20\%, -10\%)$ and $(-20\%, 25\%, -10\%)$, with each interval lasting one year. Calculate both wealth paths, terminal wealth, CAGR, and maximum drawdown. Which indicators stay the same and which change?

Self-check dictionary

Price return and total return differ by cash-flow coverage; simple and log returns differ by mathematical transformation. CAGR requires a stated year span and no external cash flows. Missing intervals cannot be deleted to join paths. Initial capital enters drawdown, while the initial baseline row is excluded from return statistics; genuine 0% remains an observation. Beginning weights form current portfolio return; drifted terminal weights serve the next interval. RF must match the asset interval and currency.

Answers: check after completing the exercises

11-L01. Wealth is 100, 150, 75; arithmetic monthly mean is zero and cumulative return −25%. $G = \sqrt{0.75} - 1 \approx -13.3974596\%$ is the constant monthly compound growth rate; these months cannot automatically be called years. For positive growth factors, Jensen gives $\overline{\log(1+R)} \leq \log(1+\bar{R})$ and exponential monotonicity implies $G \leq \bar{R}$. Equality requires identical interval returns. −100% needs separate product-definition or limiting treatment and cannot be omitted in a finite-log proof.

11-L02. Four endpoints give three returns: 50%, −50%, and 60%. Terminal-to-initial return is $120/100 - 1 = 20\%$; maximum drawdown from peak 150 to 75 is −50%, which may be reported as positive loss magnitude 50%. The initial endpoint has no holding interval. Added `ret` = 0 is a display convention that inflates the return sample and time span, and must be excluded from statistics. Keep initial wealth in drawdown so first-interval losses are visible. Retain genuine 0%; do not replace unknown intervals with zero.

11-L03. Terminal asset wealth is 75 and 25, totaling 100. Portfolio simple return is zero and log return $\log 1 = 0$. Weighted asset log return is $\frac{1}{2}\log(1.5 \times 0.5) = \frac{1}{2}\log 0.75 \approx -0.143841036$, not portfolio log return. Terminal weights are 0.75 and 0.25; retrospective use gives $0.75(0.5) + 0.25(-0.5) = 25\%$ and incorrectly applies drifted weights. Use initial capital weights to aggregate simple returns first.

11-L04. Same-month arithmetic excess return is $2\% - 0.4\% = 1.6\%$. Annual 4% and monthly return cover different intervals. Assuming constant monthly compounding converts effective annual 4% into $1.04^{1/12} - 1 \approx 0.327374\%$ per month, which also differs from the supplied 0.4%. The end deposit of 90 is not investment profit: $(200 - 90)/100 - 1 = 10\%$. The unadjusted 100% includes external capital. Within-interval flows require segmentation or money-weighted measurement.

11-L05. The specified snapshot and teaching window show wealth rising from one to about 3.141. They do not establish signal predictability or executable advantage after costs. A version downloaded in 2026 does not prove 2010 availability, and a published industry portfolio is not a directly purchasable product. Month-end sampling may miss deeper intraday or between-day peaks and troughs; it is not all-frequency maximum drawdown. Trading conclusions need historical versions and security universes, publication and ingestion clocks, out-of-sample signals, executable orders, and costs. Square-root annualization is a stated convention, not evidence that next year's risk is known.

11-L06. Time-weighted return is $(1.2)(1.1) - 1 = 32\%$, removing deposits at each segment to retain investment growth. The unadjusted ratio is $242/100 - 1 = 142\%$ and includes the additional 100. Let $z = \sqrt{1+r}$; money-weighted IRR satisfies $-100 - 100/z + 242/z^2 = 0$, or $100z^2 + 100z - 242 = 0$. The positive root is $z = (-100 + \sqrt{106800})/200 \approx 1.1340135$, so $r = z^2 - 1 \approx 28.5987\%$. Time-weighted return measures the managed path; money-weighted return includes the investor's contribution amounts and timing.

11-L07. Leading principal minors are 1, $1 - 0.1^2 = 0.99$, and 0.936, all positive. The matrix is positive definite and valid as a covariance matrix. Three-interval variance is $3 + 2[2(0.1) - 0.2] = 3$, exactly $3\gamma_0$, so its standard deviation is $\sqrt{3}$. Two-interval variance is $2 + 2(0.1) = 2.2$, not $2\gamma_0$. Positive and negative covariances cancel only in the weighted sum at the fixed three-interval horizon. This establishes neither independence nor scaling at other horizons.

11-L08. The first wealth path is $100 \to 125 \to 100 \to 90$, with maximum drawdown $1 - 90/125 = 28\%$. The second is $100 \to 80 \to 100 \to 90$, with maximum drawdown 20%. Both end at 90, have total return −10% and the same arithmetic mean, and have CAGR $0.9^{1/3} - 1 \approx -3.4511\%$. Return products are unchanged by ordering, whereas drawdown depends on each endpoint relative to its preceding peak.

Return to the body

Use one ledger to check wealth recursion, interval returns, cumulative growth, matching-frequency RF, and peak drawdown in order. Historical conclusions remain bounded by cache versions, publication times, frequency, and trading information.

Project 1 —Returns, Wealth, and Risk Description

> These notes break the monthly descriptive analysis of Project 1 into steps you can recompute yourself: first identify the data object, then compound simple returns into wealth, and finally describe the same path with CAGR, conventional annualized volatility, a same-month-RF Sharpe ratio, and peak drawdown. Questions P1-01 to P1-06 correspond to Exercises 1–6 in `editions/en/projects/project01/exercises.md`, with unchanged meaning and numbers; P1-L01 is new in these notes. Prerequisite: Chapter 11, Returns and Compounding.

First identify the data object

□□□□□

> Project 1 has two data lines, and their numbers are not interchangeable. The default offline Notebook uses seed 101 to construct 180 monthly returns; NoDur, HiTec, and Enrgy are only constructed column labels, and RF is fixed at 0.001. It demonstrates methods and hand calculations. The historical report, its original figures, and the answers to Exercises 2 and 5 come from the French release snapshot downloaded on 2026-10-05, with RF joined by the same month. They are saved historical descriptions; these notes do not rerun them.

Trigger. In the constructed Notebook the energy Sharpe ratio for 2015–2019 is positive; in the historical report the same window is negative. Which one is "right"?

Definition P1.1.1 *(Data identity).* The month labels 2010–2024 of the constructed panel _____ turn it into market observations for that period. HiTec wealth 4.6270, CAGR 16.55%, and conventional annualized volatility 15.61% belong to the _____; the current constructed table cannot _____ them. Two figures with the same shape do not imply the same data or a replication.

Clock and units

`month` labels a monthly return interval: `2010-01` is the whole of January 2010, not an intraday moment. Returns are decimals, so 0.01 means 1%; RF is a decimal for the same interval. The source gives no monthly publication or ingestion time, signal time, or trade time, so this description cannot answer whether an order could be placed after the close or whether it would fill.

From simple returns to wealth

> A percentage acts on capital that has already changed, so wealth compounds multiplicatively; monthly returns are not added. Log returns turn that multiplication over time into addition, but a cross-asset portfolio still weights simple returns by beginning-of-period weights.

Definition P1.2.1 *(Wealth recursion and log returns).* With no external cash flows,

$$W_t = W_{t-1}(1 + R_t), \qquad W_0 > 0.$$

For each $R_t > -1$, define $g_t = \log(1 + R_t)$; then $\sum_{t=1}^{T} g_t =$ _____. The initial point W_0 has no preceding return; it stays in the _____ but cannot enter the _____.

Example P1.2.2 *(The Notebook hand calculation).* Start at 100, then +50% and −50%: wealth is 100, 150, _____. The arithmetic mean return is _____, the cumulative return _____, and the maximum drawdown from the peak of 150 is _____. These inputs are not the $+10\%/-10\%$ of Exercise 1.

One metric table, four different questions

Definition P1.3.1 *(Metrics used in Project 1).* The four metrics, their objects, and their conditions are as follows.

- CAGR: $\text{CAGR} = (W_T/W_0)^{12/T} - 1$, with no external cash flows over whole months; it compresses initial and final wealth with time but hides _____.
- Conventional annualized volatility: the sample standard deviation of monthly returns (ddof=1) times _____; an annual-scale convention, not an identity for compounded annual returns.
- Annualized Sharpe ratio: $e_t = R_t - RF_t$ with the _____ RF, $\widehat{SR} = \sqrt{12}\,\bar{e}/s(e)$, requiring $s(e) > 0$; it is not an alpha test.
- Drawdown: $D_t = W_t/\max_{s \le t} W_s - 1$; the maximum drawdown loss is reported as _____, with peak and trough months.

Why each segment is renormalized

Each of the three fixed slices has 60 months and restarts from wealth 1 just before its window, so segment drawdowns track peaks only within those 60 months. Segments test whether descriptive rankings depend on the period; they do not select a best window. 2020–2024 is a later historical segment observed after the data were seen, not an out-of-sample test.

Your turn. Cover the table. In one sentence each, state what CAGR, conventional annualized volatility, the Sharpe ratio, and maximum drawdown answer, and what each leaves out.

Recall: Why does volatility use $\sqrt{12}$ while CAGR uses the exponent $12/T$?

Reading figures: axes first, and returns are not prices

All four figures use the same panel: wealth index (top left), rolling volatility (top right), drawdown (bottom left), and a box plot of monthly returns (bottom right). For each one, state the month labels on the horizontal axis and the object and unit on the vertical axis before discussing shape.

Proposition P1.4.1 *(Key points for reading the figures).* The first point of the wealth chart is the initial capital 1 at the chosen December 2009 endpoint, not a _____. Rolling volatility uses the current month and the previous 11 months, so the first _____ months are blank. A drawdown of -0.5 means 50% below the _____. In the box plot, the center line is the median, the box covers the quartiles, and distant points are actual sample months that are _____ removed.

Exercises: cover the answers and recompute

Your turn. P1-01 Compounding and two return measures. Initial wealth is 100, followed by monthly simple returns of +10% and −10%. Calculate wealth at each month end, cumulative two-month simple return, arithmetic mean monthly return and mean monthly log return. Why do they differ?

Recall: What does the mean log return give after exponentiation?

Your turn. P1-02 Read the report. HiTec's mean monthly simple return exceeds NoDur's in 2010–2019. What wealth index, compound annual growth rate (CAGR) and volatility does the report give for HiTec? Do these values establish that "HiTec is a better future investment"? Explain.

Your turn. P1-03 Same-month RF. An industry returns 2.0% in a month whose RF is 0.4%. Calculate the arithmetic excess return. What unit or frequency error results from subtracting the year's annual RF of 4% from every monthly return?

Your turn. P1-04 Peak drawdown. Wealth follows `[1.00, 1.50, 0.75, 1.20]`. Calculate drawdown from the running peak at every point and its maximum loss. Calculate the full path's final cumulative simple return and explain why it differs from maximum drawdown. How does the intermediate wealth of 0.75, a −25% loss from the initial value, differ from maximum drawdown?

Your turn. P1-05 Compare segments. Use the results table to compare the three industries in 2010–2014 and 2015–2019. Which industry has the highest CAGR in each segment? How should the negative 2015–2019 Energy Sharpe ratio be interpreted? Does this test a future tradable strategy?

Your turn. P1-06 Read the figures. What questions do wealth index, rolling volatility, drawdown and boxplots answer? What does the initial point on the wealth figure's time axis represent, and why are the first 11 months of rolling volatility blank?

Your turn. **P1-L01 (New) Data identity.** A colleague puts the positive energy Sharpe ratio for 2015–2019 from the constructed Notebook next to the historical −0.033 in one table titled “Replication failed.” Identify the error, and write one conclusion that each line can support.

Project 1: Questions and Complete Answers

P1-01

Wealth rises from 100 to 110, then falls to 99. Cumulative two-month simple return is $99/100-1 = -1\%$. The arithmetic mean is $(10\% - 10\%)/2 = 0\%$. Mean log return is

$$\frac{\log(1.10) + \log(0.90)}{2} = \frac{\log(0.99)}{2} \approx -0.5025\%/\text{month}.$$

The arithmetic mean averages the two monthly returns. Wealth multiplies their growth factors, $1.10 \times 0.90 = 0.99$. Logs convert this time-series multiplication to addition; exponentiating the mean log return gives the monthly geometric return, approximately -0.5013%.

P1-02

HiTec's 2010–2019 wealth index is approximately 4.6270, starting at 1, for cumulative return of approximately 362.70%. Under the convention of 120 months equal to 10 years, CAGR is 16.55%; multiplying sample volatility by $\sqrt{12}$ gives 15.61%. These are descriptive statistics for one fixed historical window. Inputs are subsequently revised published industry portfolio returns without historical availability clocks, signals, trades or costs. They establish neither future performance nor a tradable strategy.

P1-03

Arithmetic excess return is $2.0\% - 0.4\% = 1.6\%$. Subtracting annual 4% from every monthly return treats an annual unit as a monthly unit. This is neither a same-interval comparison nor an effective-rate conversion. Use the published monthly RF when analyzing monthly returns.

P1-04

Wealth is 1.00, 1.50, 0.75, 1.20. Drawdowns from the running peak are 0%, 0%, −50%, −20%, so maximum drawdown loss is 50%. Full-path cumulative return is $1.20/1.00-1 = 20\%$. Intermediate wealth relative to initial capital gives $0.75/1.00-1 = -25\%$, whereas its drawdown from the peak of 1.50 is −50%. Cumulative return uses initial wealth as its denominator; drawdown uses a historical peak and depends on path order.

P1-05

In 2010–2014, NoDur CAGR is 17.05%, exceeding HiTec's 14.84% and Enrgy's 7.88%. In 2015–2019, HiTec's 18.30% exceeds NoDur's 7.41%; Energy is −1.84%. Growth rankings therefore vary with the fixed interval. Energy's annualized 2015–2019 Sharpe is approximately −0.033 because mean monthly excess return is slightly negative (−0.0578%) in that sample. This is not a guaranteed-loss law. Both intervals and the later segment are retrospective historical slices without real-time vintages, strategy signals, executions, costs or a reserved out-of-sample evaluation.

P1-06

Wealth index compounds monthly simple returns across time, showing how the specified path accumulates. Initial wealth 1 near December 2009 is a chosen normalization, not a market price. Rolling volatility uses the current month and previous 11 months to estimate a 12-month sample standard deviation and multiply it by $\sqrt{12}$. It displays changing local dispersion. Until 12 observations exist, the window is incomplete and the first 11 months are blank. Drawdown shows wealth below its preceding peak; a value of −0.5 means 50% below that peak. Boxplots summarize full-window monthly simple returns. The center line is the median, boxes span quartiles and individually plotted points display more distant observations. The horizontal axis identifies fixed industry portfolios; the vertical axis uses decimal monthly returns.

P1-L01

The two lines have different inputs: the constructed Notebook is generated from seed 101 and its column names are labels, while the historical report reads the French release snapshot with same-month RF. Constructed results can neither "replicate" nor "refute" the historical numbers, so "replication failed" is a category error. The constructed line supports: "The shared metric functions compute CAGR, volatility, Sharpe, and drawdown under the stated conventions on the given constructed returns." The historical line supports: "In the 2015–2019 window of that revised snapshot, the energy portfolio's mean monthly excess return was slightly negative, with a Sharpe ratio of about −0.033." Neither establishes future returns or a tradable advantage.

Fill-in key for the reading sections

Data identity: do not; original historical report; prove, update, or overturn. Wealth: $\log(W_T/W_0)$; wealth/drawdown path; return statistics. Hand calculation: 75; 0%; −25%; −50%. Metrics: path volatility and peaks/troughs; $\sqrt{12}$; same-month; $-\min_t D_t$. Figures: security price; 11; previous peak; not.

Chapter 12 —Financial Statements for Quantitative Research

Financial statements combine stocks, period flows, and different claims. Align periods, currencies, units, consolidation scope, versions, and cash-flow signs before calculating ratios. Publication and ingestion determine when a version can enter a signal.

Trigger. Why can one statement version or a single sign make the same "free cash flow" figure wrong?

Definition 12.1 *(This chapter's cash-flow proxy; main text Sections 7–8).* This chapter defines its free cash flow proxy, FCF^{proxy}, as cash flow from operations (CFO) less positive capital expenditure (CapEx). With a negative property, plant and equipment (PP&E) purchase cash-flow line, $CapEx = -PP\&E_{cash\,flow}$. Balance-sheet amounts are closing-date stocks; revenue, CFO, and CapEx are period flows. Amounts require aligned currency/unit, and consolidated operating amounts require matching capital claims.

Mapping: L01 practices entity, direction, and stock/flow distinctions; see the main text for E03's identity and E07's complete statements. L02 practices E08's four calculations and E09's version boundaries. L03 covers P01 version/sign checks and C01 mismatches, not the Apple program. L04 covers clocks/counterevidence, not P02's implementation or C02's complete model. L05 practices C01 diagnosis. All original main-text questions and full answers remain.

Your turn. 12-L01: Statements, units, and entities (partial adaptations: 12-E03, E05, E06, E07)

An AR increase usually makes CFO _____. Consolidated net income is 100, NCI profit 15, with no preferred dividends or other distributions requiring further deduction; earnings available to parent common holders are _____. An ending balance-sheet stock corresponds to _____; a CFO flow corresponds to _____. What conversion is missing when profit of 20 USD is divided directly by 500 USD millions? In one sentence, explain why the same AR change must not be deducted again from CFO to obtain this FCF proxy.

Your turn. **12-L02: Cash-flow signs and versions (partial adaptations: 12-E08, E09)** Apple FY2024 CFO is 118,254, PP&E cash flow $-9,447$, and revenue 391,035, all in USD millions. Give positive CapEx _____, FCF _____, FCF margin _____, and CFO margin _____. Spire's original inputs are $(CFO, PP\&E) = (-23,622, -30,037)$ and restated inputs $(-36,307, -17,352)$, in USD thousands. Calculate each FCF and assess whether these versions establish a uniform FCF definition.

Your turn. **12-L03: Code diagnosis—do not combine versions (partial adaptations: 12-P01, C01)**

The call below combines original and restated lines for one period. Identify the version error and add period, accession, currency, and unit checks; reject wrong signs too. The corrected program must run independently and return −53,659 USD thousands for each version.

```
# BEGIN B09_12L03_BAD
original_cfo = {"period": "FY2023", "accession": "ORIG",
                "currency": "USD", "unit": "USD thousands",
                "value": -23_622.0}
restated_ppe = {"period": "FY2023", "accession": "RESTATED",
                "currency": "USD", "unit": "USD thousands",
                "value": -17_352.0}

def fcf(cfo_row, ppe_row):
    capex = -ppe_row["value"]
    return cfo_row["value"] - capex

print(fcf(original_cfo, restated_ppe))
# END B09_12L03_BAD
```

Your turn. **12-L04: Research judgment—originals, revisions, and timing (partial adaptations: 12-E04, P02, C02)**

Spire’ s original FY2023 10-K has SEC filed date 2024-03-06. Its later 8-K, filed 2024-08-27, discloses the audit committee’ s August 23 conclusion that affected statements should no longer be relied upon. The restated 10-K/A was filed 2025-03-03. For a 2024-08-28 signal, does prior ingestion of the original alone justify use? Give conservative auditable field/clock rules and two tests exposing future-restatement leakage or late ingestion. Distinguish filed date, accepted raw time, first publication, and ingestion.

Your turn. 12-L05: Research judgment—unchanged FCF does not mean no restatement effect (partial adaptation: 12-C01)

Spire's specified CFO−CapEx proxy is −53,659 USD thousands in both versions, while CFO changes from −23,622 to −36,307 and positive CapEx from 30,037 to 17,352. Give the narrowest supported arithmetic conclusion and two unsupported conclusions. List at least four categories of restatement reconciliations, cash flow statements, or notes to examine, and two tests of whether equality merely reflects offsets between the selected lines.

Your turn. 12-L06: Recalculate statements after changing credit sales (new exercise: numerical variation of 12-E07, E01 intuition)

Use the same fictional company and quarter, with constructed unit *u*. Opening cash is 50, inventory 20, net PP&E 100, debt 40, equity 130, and AR/AP zero. There are no NCI, dividends, share issues, FX effects, or other adjustments. Quarterly sales are 100, COGS 40, cash operating expenses 25, depreciation 10, interest 5, and tax 4 (both interest and tax in CFO). Inventory purchases are 50: cash 40 and supplier credit 10. Cash equipment purchases are 8; new borrowing is 15. Change only baseline collections/credit sales of 80/20 to 70/30. Selected OWC is AR plus inventory less AP, with other selected accounts zero. Calculate gross profit, EBIT proxy, net income, Selected OWC change, CFO, FCF, ending cash, AR, assets, liabilities, and equity; verify the identity. Explain credit sales and depreciation step by step and compare baseline and variation. This varies 12-E07's chain; the complete original baseline remains in the main text.

Chapter 12 Answers: 12-L01 to 12-L06

12-L01

AR growth normally consumes cash and lowers indirect CFO. Parent net income is 100−15=85; with no additional preferred distributions, common-available earnings are also 85. The main text's parent net-margin convention uses consolidated revenue for the same group and period. No revenue is supplied, so no numerical margin or ownership-allocated "common revenue" can be invented. ROE and P/E instead pair common equity and common capitalization/earnings. Stocks refer to the closing date; CFO covers a reporting period. Convert 20 USD into the same currency/unit, for example 0.000020 USD millions, before dividing by USD millions for a dimensionless ratio. CFO already contains indirect working-capital adjustments. The proxy deducts CapEx, not the same AR change again.

12-L02

Apple positive CapEx is 9,447; FCF is $118{,}254 - 9{,}447 = 108{,}807$ USD millions. FCF margin is $108{,}807/391{,}035 \approx 27.8254\%$ and CFO margin $118{,}254/391{,}035 \approx 30.2413\%$. Spire's original CapEx is 30,037 and FCF $-23{,}622 - 30{,}037 = -53{,}659$; restated CapEx is 17,352 and FCF $-36{,}307 - 17{,}352 = -53{,}659$ USD thousands. For these two accession-specific pairs, CFO and CapEx expenditure both decrease by 12,685, offsetting in FCF. This neither establishes other companies' definitions nor generalizes to all restatements.

12-L03

The bad example combines original CFO with restated PP&E, creating a version in no accession. Each calculation requires matching period, accession, currency, and unit; PP&E must be a non-positive cash flow. The complete program below rejects version/sign errors and checks both numerical results.

```
# BEGIN B09_12L03_FIXED
from math import isfinite
scope = ("period", "accession", "currency", "unit")
def row(accession, value, period="FY2023", currency="USD",
        unit="USD thousands"):
    return dict(period=period, accession=accession, currency=
   ↪ currency,
                unit=unit, value=value)

def fcf(cfo_row, ppe_row):
    required = scope + ("value",)
    if any(key not in cfo_row or key not in ppe_row
           for key in required):
        raise ValueError("incomplete row")
    if any(cfo_row[key] != ppe_row[key] for key in scope):
        raise ValueError("mixed reporting scope")
    cfo = float(cfo_row["value"])
    ppe = float(ppe_row["value"])
    if not isfinite(cfo) or not isfinite(ppe):
        raise ValueError("non-finite amount")
    if ppe > 0:
        raise ValueError("PP&E cash flow must be nonpositive")
    return cfo + ppe

original_cfo, original_ppe = row("ORIG", -23_622), row("ORIG",
   ↪ -30_037)
restated_cfo, restated_ppe = (
    row("RESTATED", -36_307), row("RESTATED", -17_352))
original = fcf(original_cfo, original_ppe)
restated = fcf(restated_cfo, restated_ppe)
assert original == -53_659 and restated == -53_659
def rejects(cfo_row, ppe_row):
    try:
        fcf(cfo_row, ppe_row)
    except ValueError:
        return
    raise AssertionError("invalid FCF rows were accepted")

rejects(original_cfo, restated_ppe)
rejects(original_cfo, row("ORIG", -30_037, period="FY2022"))
rejects(original_cfo, row("ORIG", -30_037, currency="EUR"))
rejects(original_cfo, row("ORIG", -30_037, unit="USD millions"))
rejects(original_cfo, row("ORIG", 30_037))
print(original, restated)
```

```
# END B09_12L03_FIXED
```

12-L04

The last-ingested version alone does not establish eligibility. The later 8-K contains non-reliance information. Preserve period, form, accession, original lines, public/available times and precision, ingestion, status, currency/unit, and feature-code version. Signals read only valid versions proven to satisfy $available_at \leq signal_time$ and $ingested_at \leq signal_time$. Once invalid status is received and before a restatement is actually readable, mark the feature unavailable; never backfill future 10-K/A values. Date-level evidence cannot establish intraday instants. Tests: delaying restatement ingestion beyond a signal must leave early signals unchanged; extreme future revision values or later accessions must leave early signals and feature hashes unchanged. Without auditable publication/ingestion evidence, eligibility is unproven rather than presumed timely.

12-L05

The sole numerical conclusion is that the specified two lines from each accession give proxy −53,659 USD thousands under $CFO - CapEx$: lower CFO offsets lower positive CapEx. This does not show unchanged CFO, net income, balance sheets, operating quality, or all cash-flow definitions, nor unchanged FCF for every restatement. Check restatement reconciliation accounts/versions; complete operating/investing/financing flows and ending cash; revenue recognition, contract costs, and affected loss lines; and PP&E, capitalized software, lease, and other investment notes. Test one: if a predefined measure including software, lease principal, or broader investment differs across versions, equality belongs only to the current two-line proxy. Test two: compare classifications and cash-balance reconciliations item by item. Changes in operating cash, other investing cash, or FX/cash movements preclude calling total cash or operations neutral.

12-L06

All amounts use constructed unit u. Revenue 100 less COGS 40 still gives gross profit 60; less operating expenses 25 and depreciation 10 gives EBIT proxy 25; less interest 5 and tax 4 gives net income 16. Credit sales rise from 20 to 30, so ending AR is 30. Inventory still rises from 20 to 30 and AP by 10. Selected OWC rises from opening 20 to $30 + 30 - 10 = 50$, increasing 30. Direct CFO is $70 - 40 - 25 - 5 - 4 = -4$; indirect verification is $16 + 10 - 30 - 10 + 10 = -4$. CapEx remains 8, so $FCF = -4 - 8 = -12$. Ending cash is $50 - 4 - 8 + 15 = 53$, AR 30, inventory 30, net PP&E $100 + 8 - 10 = 98$, and total assets $53 + 30 + 30 + 98 = 211$. Liabilities remain AP 10 plus debt 55, totaling 65; equity remains 130+16=146. $65 + 146 = 211$ matches assets. Relative to 12-E07, net income, opening Selected OWC, total assets, liabilities, and equity are unchanged. Ending Selected OWC rises from 40 to 50 and AR by 10; cash, CFO, and FCF each fall by 10. AR replaces cash, leaving total assets unchanged. Credit sales 30 enter revenue and AR, while the extra uncollected 10 lowers CFO by 10. Depreciation 10 reduces EBIT, net income, and net PP&E and is added back in indirect CFO.

Transfer to the Main Text

Rerun 12-P01, 12-P02, and 12-C02, preserving raw lines and version mappings. These exercises calculate a defined proxy and version eligibility, not Apple-disclosed standardized FCF or a general claim that Spire-style restatements leave FCF unchanged.

Chapter 13 —Valuation for Quants

Enterprise value (EV) covers claims on operating capital; common value belongs only to common shareholders. Match operating scope, currency, and unit, then pair FCFF with WACC/EV and FCFE with equity cost/common value. Discount formulas do not establish data availability.

Trigger. When a DCF returns a number, how do you know whose value it estimates?

Definition 13.1 *(Match cash flow and discount rate; main text Sections 5.3, 6, 7).* Free cash flow to firm (FCFF) belongs to all capital providers before debt service and discounts at the weighted average cost of capital (WACC) to operating enterprise value. Free cash flow to equity (FCFE) belongs to common holders after interest, tax, and net-borrowing adjustments and discounts at equity cost to common value. Stable-growth terminal value (TV) uses next-period cash flow $CF_{T+1} = CF_T(1+g)$, with same-frequency r, g satisfying $r > -1$, $g > -1$, and $r > g$.

Mapping: L01 covers Lyra's bridge and E01/E02 claims, not P01's two-company ratio program. L02 practices E05/E06. L03 uses E07's three-year case; P02's five-year grid remains in the main text. L04 covers discount matching and entry-point diagnosis. L05 partially practices E08/E09 timing and C02 counterevidence, not full research design. All original questions/full solutions remain; L06/L07 are separate new exercises.

Your turn. 13-L01: EV bridge and consolidation (adaptations: 13-E01, 13-E02, 13-P01)

Lyra's constructed inputs, in USD millions, are common capitalization 480, debt 260, preferred equity 20, NCI 40, cash 30, and NOA 20. EV _____; reverse common value _____. If EBITDA includes all subsidiary profit, why include NCI in EV? For parent common P/E, should the earnings denominator use consolidated net income or common-attributable net income?

Your turn. 13-L02: FCFF, FCFE, and WACC (adaptations: 13-E05, 13-E06)

Given EBIT 80, tax 25%, D&A 12, CapEx 20, and $\Delta NWC = 5$, FCFF _____. With common net income 45, D&A 12, CapEx 20, $\Delta NWC = 5$, and net borrowing −3, FCFE _____. If $E = 700, D = 300, k_e = 9\%, k_d = 5\%, T = 20\%$, WACC _____. Match each flow to its rate and value claim.

Your turn. **13-L03: TV and year-end timing (adaptations: 13-E07, 13-P02)**
Year-end FCFF is 30, 35, and 40 USD millions; $r = 10\%$ and $g = 2\%$. Calculate year-three TV_3 _____ and today's enterprise value V_0 _____. How many periods discount TV? List perpetual-growth conditions and explain why they do not validate long-run operating forecasts.

Your turn. 13-L04: Code diagnosis—cash-flow/rate mismatch (adaptations: 13-P01, 13-P02)

Find two errors: mismatched cash-flow claims/discount rate and missing perpetual-growth validity checks. The corrected program must value year-end FCFF/WACC and reject FCFE/WACC and $g \geq r$.

```
# BEGIN B09_13L04_BAD
def value(cash_flows, wacc, growth):
    tv = cash_flows[-1] * (1 + growth) / (wacc - growth)
    pv = sum(cf / (1 + wacc) ** year
             for year, cf in enumerate(cash_flows, start=1))
    return pv + tv / (1 + wacc) ** len(cash_flows)

fcfe = [29.0, 31.0, 33.0]
print(value(fcfe, wacc=0.10, growth=0.02))
# END B09_13L04_BAD
```

Your turn. **13-L05: Research judgment—a valuation snapshot is not PIT evidence (adaptations: 13-E08, 13-E09, 13-C02)**

Constructed inputs: original FCFE 28 is public/ingested before the March 31 signal, with March 31 capitalization 600. Restated FCFE 25 is public/ingested April 2, with capitalization 620 that day; the second signal is April 3. Assume inputs are readable before their respective dates end. Calculate both FCFE/equity yields, selecting versions and capitalization dates. List at least four PIT fields per version and two counterevidence tests. Can Apple FY2024 net income and EPS alone establish actual historical P/E?

Your turn. 13-L06: Market-value WACC and consistency (main text Sections 5.1, 5.4, 7.2, 8)

Use Lyra's constructed USD-millions inputs: common capitalization 480, debt 260, preferred equity 20, NCI 40, cash 30, and NOA 20. Assume $k_e = k_{NCI} = 10\%$, $k_p = 7\%$, $k_d = 6\%$, and $T = 25\%$. Calculate EV, reverse common value, V_{cap}, and general WACC; explain V_{cap} versus EV. If rent remains operating expense, how should lease debt be treated consistently? With substantial nonoperating cash, why cannot observed whole-company k_e directly discount pure operating FCFF?

Your turn. 13-L07: Itemized CFO, FCFF, and FCFE bridges (main text Sections 5.3, 6.2, 8)

Constructed annual inputs for one company, in USD millions: EBIT 50, tax 25%, cash interest 8, D&A 10, positive CapEx 15, $\Delta NWC = 5$, and net borrowing 3. Assume interest is in CFO, cash tax is $T(EBIT - \text{cash interest})$, and the interest shield is usable. D&A and ΔNWC share the scope and are fully included in CFO, with no nonoperating gains/losses or other adjustments. Calculate common net income and CFO, then bridge CFO item by item to FCFF. Also calculate direct FCFF, the $CFO - CapEx$ proxy, and FCFE. Why must D&A not be added or ΔNWC deducted again after CFO?

Chapter 13 Answers: 13-L01 to 13-L07

13-L01

$EV = 480+260+20+40-30-20 = 750$ USD millions. Reverse bridge $750-260-20-40+30+20 = 480$ gives original common capitalization. Consolidated EBITDA includes all non-wholly-owned subsidiary operations, so EV adds corresponding NCI claims to match scope. Common P/E uses common-attributable net income, not consolidated income containing NCI. Align currency, unit, and valuation time; do not deduct/add the same cash or NOA twice.

13-L02

$FCFF = 80(1-0.25)+12-20-5 = 47$; $FCFE = 45+12-20-5-3 = 29$ USD millions. FCFF has not deducted interest from EBIT and belongs to all capital providers; discount at WACC for operating enterprise value. FCFE begins at common net income and includes net borrowing; discount at equity cost for common value. $WACC = 0.7(9\%)+0.3(5\%)(1-20\%) = 7.5\%$. These specified inputs do not estimate an actual company's capital costs.

13-L03

$TV_3 = 40(1.02)/(0.10 - 0.02) = 510$ USD millions. $V_0 = 30/1.10 + 35/1.10^2 + (40 + 510)/1.10^3 \approx 469.42$ USD millions. Year-three-end TV receives three discount periods like year-three FCFF. Gordon growth also requires same annual conventions with $r > -1$, $g > -1$, and $r > g$. These ensure series convergence, not credible FCFF, stable-period, WACC, or g assumptions. Mathematical validity does not force positive CF_3. A going-concern interpretation of TV_3 additionally requires normalized sustainable terminal cash flows.

13-L04

The bad example discounts FCFE as FCFF at WACC and returns invalid TV when $r \le g$, $r \le -1$, or $g \le -1$. The complete program checks claim/rate matching, finite values, and growth conditions. Its returned value means enterprise or common value according to the claim.

```
# BEGIN B09_13L04_FIXED
from math import isclose, isfinite
flows = [30, 35, 40]
pairs = {("FCFF", "WACC"), ("FCFE", "cost_of_equity")}
def dcf(cash_flows, cash_flow_type, discount_type, rate, growth)
   ↪ :
    cash_flows = list(cash_flows)
    if not cash_flows: raise ValueError("cash flows are required
   ↪ ")
    if (cash_flow_type, discount_type) not in pairs:
        raise ValueError("cash flow and discount rate do not
   ↪ match")
    if not all(isfinite(x) for x in [*cash_flows, rate, growth])
   ↪ :
        raise ValueError("inputs must be finite")
    if rate <= -1 or growth <= -1 or rate <= growth:
        raise ValueError("invalid terminal-growth conditions")
    terminal = cash_flows[-1] * (1 + growth) / (rate - growth)
    pv = sum(cf / (1 + rate) ** year
             for year, cf in enumerate(cash_flows, start=1))
    amount = pv + terminal / (1 + rate) ** len(cash_flows)
    if cash_flow_type == "FCFF":
        claim = "enterprise value"
    else:
        claim = "equity value"
    return claim, amount, terminal
claim, firm_value, tv = dcf(flows, "FCFF", "WACC", .10, .02)
assert claim == "enterprise value"
assert isclose(firm_value, 469.421488, rel_tol=1e-7)
assert isclose(tv, 510.0)
equity_claim, equity_value, _ = dcf(
    [29, 31, 33], "FCFE", "cost_of_equity", .10, .02)
assert equity_claim == "equity value" and equity_value > 0
def rejects(*args):
    try:
        dcf(*args)
    except ValueError:
        return
    raise AssertionError("invalid DCF inputs were accepted")
rejects(flows, "FCFE", "WACC", .10, .02)
rejects(flows, "FCFF", "WACC", .10, .10)
rejects(flows, "FCFF", "WACC", -1.0, -1.1)
rejects(flows, "FCFF", "WACC", .10, -1.0)
rejects([30, float("nan"), 40], "FCFF", "WACC", .10, .02)
print(round(firm_value, 2), round(tv, 2), round(equity_value, 2)
```

```
    ↪ )
# END B09_13L04_FIXED
```

13-L05

On March 31, $28/600 \approx 4.67\%$ uses only the then-available original and March 31 capitalization, no later than the signal. On April 3, $25/620 \approx 4.03\%$ uses the April 2 restatement and capitalization. Both use constructed USD millions. Preserve at least issuer/security, fiscal period, original line/currency/unit, form/accession/revision, `available_at`, `ingested_at`, signal time, and code version. Market snapshots also need source, as-of, class/ADR, and share rules. Delaying restatement ingestion beyond the signal must leave earlier signals unchanged; extreme future prices or revisions must not alter them either. Compare out-of-sample, industry benchmarks, costs, and negative-denominator rules. Net income/EPS alone cannot establish actual historical P/E: verify matched price/time, class/share basis, and then-public/ingested EPS version. Constructed arithmetic is not an executed factor test.

13-L06

$EV = 480+260+20+40-30-20 = 750$ USD millions; reverse bridge $750-260-20-40+30+20 = 480$ recovers specified common capitalization. The capital-weight base is $V_{cap} = 480+40+20+260 = 800$ USD millions. Its numerator is $480(10\%) + 40(10\%) + 20(7\%) + 260(6\%)(1-25\%) = 48 + 4 + 1.4 + 11.7 = 65.1$, so $WACC = 65.1/800 = 8.1375\%$. Gross financing claims of 800 exclude no cash/NOA; operating EV of 750 deducts them, serving a different purpose. With rent retained as operating expense, exclude corresponding lease liabilities from debt. Capitalizing leases requires consistent operating-profit, cash-flow, and debt-cost adjustments to avoid duplication. Substantial lower-risk cash may dilute observed common beta/required return; remove cash risk or align costs, weights, and scope before discounting pure operating FCFF. Cash is not necessarily risk-free and NOA is not cash. Costs here are constructed inputs, not calibrated actual WACC.

13-L07

Under the simplified cash-tax assumption, common net income is $(50-8)(1-25\%)=31.5$ USD millions and CFO $31.5+10-5=36.5$. From CFO, $36.5+8-8(25\%)-15=27.5$ USD millions adds interest 8, deducts its tax shield 2, and deducts positive CapEx 15 to obtain FCFF. Directly, $FCFF=50(1-25\%)+10-15-5=27.5$. The proxy $CFO-CapEx=36.5-15=21.5$ USD millions is below FCFF by after-tax interest $8(1-25\%)=6$. $FCFE=31.5+10-15-5+3=24.5$ USD millions. D&A and working-capital movements are already in CFO; adding or deducting them again duplicates the adjustments.

Transfer to the Main Text

Rerun 13-P01–13-P03 and check FCFF/EV versus FCFE/common capitalization. DCF is conditional on inputs, not observed truth. Report sensitivity when TV dominates; passing arithmetic is not strategy evidence.

Chapter 14 —Capital Asset Pricing Model and Asset Pricing (CAPM)

The stochastic discount factor (SDF) puts into prices the idea that money is worth more in some states than in others. CAPM beta is an asset's covariance with the market divided by market variance; it describes systematic risk and does not guarantee next-period returns.

Trigger. If a security pays only in low-SDF states, how does its price reflect that?

Definition 14.1 *(One-period pricing and beta).* The price is $P = \mathbb{E}[mX]$, where $m > 0$ is the SDF and X is the terminal payoff. The simple return is $R = X/P - 1$, and the gross-return factor $G = X/P = 1 + R$ satisfies $\mathbb{E}[G] - G_f = -\mathrm{Cov}(m, G)/\mathbb{E}[m]$, where $G_f = 1/\mathbb{E}[m]$. In CAPM, $\beta_i = \mathrm{Cov}(R_i, R_m)/\mathrm{Var}(R_m)$; returns and variances must use the same frequency and currency.

Your turn. 14-L01 Mapping: 14-E02

There are two equally likely states, the discount factor is $m = (1.1, 0.9)$, and a security pays $X = (0, 2)$. Find P, G_f, the risk-free simple return, the security's gross return in each state, and $\mathbb{E}[G]$ and $\mathbb{E}[R]$. Then calculate $\mathrm{Cov}(m, G)$ and explain which state gives rise to the positive expected risk premium.

Your turn. 14-L02 Mapping: 14-E03, 14-E04; review Section 8.2

An asset's covariance with the market excess return is 0, market variance is positive, and the asset's excess-return variance is 0.04. What expected excess return does CAPM require? What is its single-period standard deviation? Now let $R_f = 2\%$, $\mathbb{E}[R_m] = 8\%$, and $\sigma_m = 10\%$: what are the expected returns of an asset with beta 1.5 and an efficient market/RF portfolio with standard deviation 5%, and does each lie on the SML or CML? If both assets have a realized mean total return of 7% in a four-row constructed sample, does that contradict the SML result?

Your turn. 14-L03 Mapping: 14-P02

The call below first drops rows with missing returns and then fits CAPM:

```
valid = np.isfinite(asset) & np.isfinite(market)
fit = fit_capm_excess(asset[valid], market[valid])
```

Why does this bypass the missing-value contract of this chapter's function? Write the required failure assertion and explain what to check after restoring the original observations.

Your turn. 14-L04 Mapping: 14-C03; the research-judgment part of 14-P04

A monthly regression reports an intercept $\hat{\alpha} = 0.3\%$ and a HAC standard error of 0.1%. The reference t-statistic is 3. List statistical, data, economic-mechanism, and execution evidence needed to assess it as a potentially implementable net trading opportunity. Why cannot the current intercept be called a strategy's net return? How should the next study preserve an unsearched validation window?

From State Weights to Market Equilibrium

p_s is the probability of a state and $q_s = p_s m_s$ is its state price; no-arbitrage ensures price consistency, while CAPM also requires a common tangency portfolio and market clearing. The next three exercises complete this chain while retaining the original question numbers.

Your turn. 14-L05 New; corresponds to 14-E05

Section 8.1 uses probabilities $(0.5, 0.5)$, the SDF is $(0.9, 1.1)$, and X pays $(110, 90)$ euros. Fill in the state prices (_____, _____) and the two price contributions, then add them to get the price of X. If the only traded asset is Y, which pays 100 in both states and costs 100, write the required equation for state prices and give another pair of positive weights that satisfies it. Why cannot the expected payoff be used directly as the price?

Your turn. 14-L06 New; proof outline for 14-C01

Let $a = \mu_m - R_f > 0$ and $\sigma_m > 0$, and perturb the portfolio as $R_p = (1-\delta)R_m + \delta R_i$. Fill in $\mu_p'(0) =$ _____; the variance derivative is $2(\mathrm{Cov}(R_i, R_m) - \sigma_m^2)$, so $\sigma_p'(0) =$ _____. Substitute into the Sharpe-ratio quotient rule and set the derivative to 0; show the rearrangement leading to $\mu_i - R_f = \beta_i a$. Why must small perturbations in both directions be feasible? At what step does the common tangency portfolio become the market portfolio?

Your turn. **14-L07 New; corresponds to 14-E06**

RF=2%, the market expected return is 8%, and the market standard deviation is 10%. The market weight is -0.5 and the RF weight is 1.5. Fill in the standard deviation $\sigma_p =$ _____ σ_m, then find the expected return and beta. It has the same standard deviation as $w = 0.5$; why is its expected return different? Which branch does the positively sloped CML select?

Chapter 14 Answers: 14-L01 to 14-L07

14-L01

$P = \mathbb{E}[mX] = \frac{1}{2}(1.1 \cdot 0 + 0.9 \cdot 2) = 0.9$. $\mathbb{E}[m] = 1$, so $G_f = 1$ and the risk-free simple return is 0. The security's gross return is $(0, 2/0.9) = (0, 20/9)$, so $\mathbb{E}[G] = 10/9$ and $\mathbb{E}[R] = \mathbb{E}[G] - 1 = 1/9 = 11.11\%$. Its simple returns are $(-100\%, 122.22\%)$. $\mathrm{Cov}(m, G) = \mathbb{E}[mG] - \mathbb{E}[m]\mathbb{E}[G] = 1 - 10/9 = -1/9$, consistent with $\mathbb{E}[G] - G_f = -\mathrm{Cov}(m, G)/\mathbb{E}[m]$. The security pays only in the low-SDF state, so its payoff is negatively covarying with the state's marginal value, which gives it a positive expected premium; this does not mean it is profitable in every state.

14-L02

Beta is 0, so CAPM's conditional expected excess return is 0; however, total variance is 0.04 and standard deviation is 20%, so total risk is not zero. The SML gives $2\% + 1.5(8\% - 2\%) = 11\%$. The efficient market/RF portfolio has market weight $5\%/10\% = 0.5$, and the CML gives $2\% + 0.5(8\% - 2\%) = 5\%$. Beta is the SML's horizontal axis; standard deviation of an efficient portfolio is the CML's. The four-row sample mean is a finite realization, whereas the SML relationship concerns model expected returns. Both sample means being 7% is not a calculation contradiction and does not by itself test the model.

14-L03

Explicitly filtering out non-finite rows silently changes the sample and bypasses P02's rejection behavior; cleaning the asset and market separately could also misalign observations. The interface requires invalid inputs to raise an error rather than silently alter the sample. Keep the original equal-length arrays, construct an asset input containing NaN, and assert:

```
with pytest.raises(ValueError, match="must be finite"):
fit_capm_excess(asset_with_nan, market)
```

After restoring the original observations, verify that beta and residuals return to the accepted values in the constructed 14-P01 example; do not assert only that the function did not raise an error.

14-L04

Statistically, check the HAC bandwidth, residual dependence, structural changes, small-sample approximation, and multiple searches. For the data, verify frequency, currency, same-month alignment of RF and the market factor, input version, and when it was available. For the economics, examine proxy-market coverage, omitted risks, and mechanism stability. For execution, specify signal and trade timing, turnover, fees, financing, borrowing, taxes, and capacity, and compare against an investable benchmark. The intercept is a regression remainder for a particular sample and factor specification; it excludes these trading costs and has no signal or execution ledger, so it is not a net strategy return. Freeze hypotheses, data definitions, training and validation windows, and robustness rules in advance; test on a later interval untouched by the search and report every attempt, including failures.

14-L05

The state prices are $(0.45, 0.55)$, with contributions $0.45\times110 = 49.5$ and $0.55\times90 = 49.5$ euros, for a price of 99 euros. If only Y is traded, the constraint is $100(q_1 + q_2) = 100$; another positive pair, $(0.5, 0.5)$, also satisfies it, but assigns a price of 100 euros to X, whose payoff is not spanned by Y. No-arbitrage does not uniquely identify the price of an unspanned payoff. Probabilities give an expected payoff of 100 euros; the price must also reflect the value of the payoff across states.

14-L06

$\mu_p'(0) = \mu_i - \mu_m$; $\sigma_p'(0) = [\mathrm{Cov}(R_i, R_m) - \sigma_m^2]/\sigma_m$. The quotient rule gives

$$0 = S'(0) = \frac{(\mu_i - \mu_m)\sigma_m - a\sigma_p'(0)}{\sigma_m^2}.$$

Thus $\mu_i - \mu_m = (a/\sigma_m^2)[\mathrm{Cov}(R_i, R_m) - \sigma_m^2] = a\beta_i - a$. Adding $a = \mu_m - R_f$ to both sides gives $\mu_i - R_f = a\beta_i$. Feasibility in both directions is needed to set the derivative to 0 at an interior optimum; a one-sided constraint yields only the corresponding directional inequality. Common expectations and borrowing/lending at the same RF rate lead investors to choose a common tangency risky portfolio. Market clearing for risky assets identifies aggregate risky holdings with the market portfolio, so that tangency portfolio is M. Merely calling an index "the market" does not establish this step.

14-L07

The absolute weight is $|w| = 0.5$, so standard deviation is 5%; expected return is $2\% - 0.5 \times 6\% = -1\%$, and beta is -0.5. The positive weight 0.5 gives an expected return of 5%; the two portfolios have the same volatility but point in opposite directions. With a positive market premium, the upper efficient branch of the CML has $w \geq 0$; a negative market weight lies on the lower branch and cannot use the positively sloped formula. The SML retains the sign of beta and gives both expected returns.

Chapter 15 —Factor Models

A factor model decomposes excess returns into an intercept, common factors, and a residual; calculate total risk by forming variance from covariances first, then taking the square root.

Trigger. Is a one-month residual alpha?

Definition 15.1 *(Risk decomposition).* Excess-return variance $= \boldsymbol{\beta}^\top \Sigma_f \boldsymbol{\beta} + \sigma_\varepsilon^2$, provided the factors and residual are uncorrelated. Mkt-RF is already an excess return; SMB/HML are return differences and should not have RF subtracted again.

Your turn. 15-L01 Mapping: 15-E02

The six portfolio returns for the same month (percent) are $(SV, SN, SG; BV, BN, BG) = (4, 3, 2; 2, 1, 0)$. First construct SMB as the mean of the three small portfolios minus the mean of the three big portfolios. Then construct HML as the mean of SV, BV with high book-to-market ratios minus the mean of SG, BG with low book-to-market ratios. If only SG changes to -1%, recalculate both factors; why do both change? Can one month's spread establish a long-run relationship or factor risk premium?

Your turn. 15-L02 Mapping: 15-E03

This month $RF = 0.2\%$, the asset's total return is $R = 2.6\%$, the factors are $(Mkt - RF, SMB, HML) = (2, 1, -1)\%$, the loadings are $(0.8, 0.3, 0.5)$, and the regression intercept is $\hat{\alpha} = 0.2\%$. Find the excess return, factor contribution, and this month's residual. Why is this residual not alpha?

Your turn. **15-L03 Mapping: 15-E05**

Two factor volatilities are 4% and 2%, their loadings are 0.8 and 0.5, and residual volatility is 1%; the residual is uncorrelated with the factors. Calculate total variance and volatility when the factor correlation is 0 and when it is −0.5. Why cannot the weighted standard deviations and residual standard deviation simply be added?

Your turn. 15-L04 Mapping: 15-P01

A monthly regression test must verify that both a factor index shifted by one month and exactly collinear factors are rejected. The checker is written as:

```
try:
  fit_monthly_factor_model(y, invalid, columns)
except ValueError: pass
```

It then prints "test passed" directly. Why can this check miss a failure? Write the smallest correct rejection assertion and state which condition each of the two inputs violates.

Your turn. 15-L05 Mapping: 15-C01

A researcher tries 100 asset windows, selects the one with an FF5 intercept $p < 0.05$, and claims to have found a tradable net alpha. Give counterevidence under statistical, data, economic-mechanism, and execution headings; then design a follow-up study that preserves unsearched evidence. Why cannot the same sample be split again and the preferred split called out of sample?

From Statistical Decomposition to Conditional Interpretation

Fixed loadings, same-month returns, and the orthogonality condition each address a different issue: what is multiplied, which month is used, and whether variances can be added directly. The next exercises add calculations, conditional diagnostics, and a Project 2 reading judgment.

Your turn. 15-L06 New; corresponds to 15-E04

The true relationship is $y = \alpha + \beta x + \gamma z + u$, where u has mean zero and is uncorrelated with x, z, and $\mathrm{Var}(x) > 0$. Let $\alpha = 0, \beta = 0.8, \gamma = 0.5$, $\mathbb{E}[x] = 0.5\%, \mathbb{E}[z] = 0.3\%$, and $q = \mathrm{Cov}(x, z)/\mathrm{Var}(x) = 0.4$. Fill in the omitted-variable slope $\tilde{\beta} = \beta +$ _____ and intercept $\tilde{\alpha} = \alpha + \gamma(\mathbb{E}[z] -$ _____), then calculate their values. Why is a change in the intercept not a change in trading cash flow?

Your turn. 15-L07 New; corresponds to 15-E09

An externally specified model is $y = f + \varepsilon$, with $\varepsilon = 0.5f + u$; f,u have zero means and are uncorrelated, with variances 0.0004 and 0.0001. Find the residual variance, $\mathrm{Cov}(f, \varepsilon)$, total variance, and total volatility. Fill in the condition needed to omit the cross term: _____. Why cannot you appeal to OLS' s in-sample orthogonality here?

Your turn. 15-L08 New; corresponds to 15-E10

Two equally likely states are $(\beta, f) = (0.5, -2\%)$ and $(1.5, 4\%)$, with intercept and error equal to 0. Find the average contribution, the product of average loading and average factor return, and their difference; recalculate the covariance from centered products. If the current month's factor move determines its current-month loading, which timing contract is missing from the code?

Your turn. 15-L09 New; local variant of 15-E08 and Project 2 Questions 5, 7, and 8

Official FF3 and FF5 have different SMB values over the same window, and Mkt-RF also differs in the data. A researcher attributes the entire R^2 difference to the added RMW/CMA factors and calls same-period realized-factor explanations for 2015–2019 predictions. Identify both errors. Then state which testing family is covered by the Holm adjustment for nine asset-by-model alphas, and why it cannot correct for all earlier model searches. For a strictly nested comparison, which columns must be held fixed, and what should the model be called?

Chapter 15 Answers: 15-L01 to 15-L09

15-L01

In the original construction, the small-portfolio mean is $(4+3+2)/3 = 3\%$ and the big-portfolio mean is 1%, so $SMB_{BM} = 2\%$. The high book-to-market mean is $(4+2)/2 = 3\%$ and the low book-to-market mean is 1%, so $HML = 2\%$. After SG changes to -1%, the small-portfolio mean is $(4+3-1)/3 = 2\%$, so $SMB_{BM} = 1\%$; the growth-portfolio mean is $(-1+0)/2 = -0.5\%$, so $HML = 3.5\%$. Both factors change because SG enters both the small-stock leg and the growth leg; this does not imply that SMB and HML are orthogonal. One month supplies only one realized spread; it cannot establish a long-run statistical relationship or distinguish risk compensation, a behavioral mechanism, and chance variation.

15-L02

The asset's excess return is $2.6\% - 0.2\% = 2.4\%$. Factor contribution is $c = 0.8(2\%) + 0.3(1\%) + 0.5(-1\%) = 1.4\%$; the fitted value this month is $c + \hat{\alpha} = 1.6\%$, so the residual is $\varepsilon = 2.4\% - 1.6\% = 0.8\%$. Alpha is the estimated intercept for a given regression, sample, and factor set; a residual is one observation's deviation from its fitted value. A one-month residual is neither an intercept estimated across a sample nor a tradable remaining return.

15-L03

When factors are uncorrelated, common-factor variance is $(0.8 \cdot 0.04)^2 + (0.5 \cdot 0.02)^2 = 0.001124$. Add residual variance $0.01^2 = 0.0001$ for total variance 0.001224 and volatility 3.4986%. When $\rho = -0.5$, the cross term is $2(0.8)(0.5)(-0.5)(0.04)(0.02) = -0.00032$, so total variance is $0.001224 - 0.00032 = 0.000904$ and volatility is about 3.0067%. Covariance determines the cross term; combine the variances first and then take the square root. Standard deviations generally cannot be added directly.

15-L04

The original checker has an exception branch but no branch that fails when no exception is raised; if the function incorrectly accepts the input, the script still exits with code 0. The correct pattern is:

```
for invalid in (shifted, rank_deficient):
try: fit_monthly_factor_model(y, invalid, columns)
except ValueError: continue
else: raise AssertionError("invalid input was not
rejected")
```

A shifted-month input violates the requirement that asset and factor months match exactly and are contiguous; exact collinearity makes the design matrix rank deficient, so coefficients are not identifiable. Construct and check the two inputs separately. Continuing after the exception means only that the rejection contract held; a successful call to the function under test is not a passing test.

15-L05

Statistical counterevidence: 100 searches create multiple-testing concerns, and the result also depends on HAC bandwidth, error assumptions, model specification, and parameter stability. Data counterevidence: check financial-statement and factor versions, survivor selection, monthly alignment, source timing, and input hashes. Economic counterevidence: examine omitted risks, industry concentration, and how the strategy fares in adverse states. Execution counterevidence: account for signal availability, trades, fees, spreads, market impact, financing, borrowing, taxes, and capacity, and compare against an investable benchmark.

First freeze the data, hypotheses, sample split, cost definitions, and robustness rules; then evaluate on a future period that did not participate in the search. Preserve every attempt, failed specification, and negative result. Splitting an already searched sample remains affected by the same data selection and does not restore independence; it is not untouched out-of-sample evidence and cannot establish net alpha.

15-L06

The two blanks are γq and $q\mathbb{E}[x]$. The slope is $0.8 + 0.5 \times 0.4 = 1.0$; the intercept is $0.5(0.003 - 0.4 \times 0.005) = 0.0005$, or 0.05% per month. The omitted covariation with z enters the slope, and the part not absorbed by the mean of x enters the intercept; true alpha remains 0. Changing the explanatory benchmark changes statistical attribution, not the cash flow the asset has already realized.

15-L07

Residual variance is $0.5^2 \times 0.0004 + 0.0001 = 0.0002$; the cross-covariance is $0.5 \times 0.0004 = 0.0002$. Total variance is $0.0004+0.0002+2\times0.0002 = 0.0010$, giving volatility 3.1623%. Omitting the cross term requires $\mathrm{Cov}(f, \varepsilon) = 0$; otherwise this problem gives only 0.0006 and volatility 2.4495%. The loading 1 is specified externally, without a best linear projection; a with-intercept projection on the same population would give loading 1.5 and residual u, which would be orthogonal. The name "residual" alone does not imply orthogonality.

15-L08

The contributions are −1% and 6%, averaging 2.5%; the average loading is 1 and the average factor return is 1%, so their product is 1%. The difference is 1.5 percentage points: $[(-0.5)(-0.03) + (0.5)(0.03)]/2 = 0.015$. The missing contract runs from pre-period information availability to signal, execution, and holding-period return; the full-month factor realization is known only afterward and cannot be used to select the loading for that same month. The algebra illustrates the covariance term from time-varying loadings; it does not validate timing skill or returns after costs.

15-L09

The identically named old columns also change, so the official models are not strictly nested and the R^2 difference cannot be attributed entirely to the two added factors. Explaining an asset's return with realized same-period factors is contemporaneous attribution; a prediction must use inputs available at the forecast time to estimate a later target. Holm adjusts only the nine alpha tests for the three listed industries by three listed models; it does not cover unlisted assets, windows, lag choices, or earlier model searches. A nested isolation experiment must hold the asset, RF, months, and the same Mkt-RF/SMB/HML series fixed, then add RMW/CMA. Call it a "custom extension of a fixed FF3 baseline" and choose a test consistent with the error assumptions; do not automatically call it official FF5.

Project 2 —Factor Regressions and Evidentiary Limits

These notes practice three things: separating an asset's total return from its excess return; interpreting the conditional Betas and model intercepts of CAPM, FF3, and FF5; and judging what fit, HAC robustness, multiple testing, and a chronological split can each support. Questions P2-01 to P2-08 correspond to Exercises 1–8 in `editions/en/projects/project02/exercises.md`, with unchanged meaning and numbers. Prerequisites: Chapter 8, Regression; Chapter 11, Returns; Chapter 14, Asset Pricing; Chapter 15, Factor Models.

Two evidence lines and the dependent variable

□□□□□

The default Notebook constructs a 120-month sample with seed 102: each of five factor columns is drawn from $N(0, 0.03^2)$, RF is fixed at 0.001, and the asset's total return is built from RF, a constant 0.002, 1.1 times Mkt-RF, 0.4 times SMB, and noise with standard deviation 0.015. The historical report retains the 2010–2019 French snapshot and nine real regressions. Constructed coefficients demonstrate methods and cannot be cited as market findings.

Trigger. Only HiTec's total return is available, with no same-month RF. Can the regression still be called a CAPM excess-return regression?

Definition P2.1.1 *(Excess-return regression).* The dependent variable is $y_t = R_t - RF_t$, using the _____ RF. The right-hand Mkt-RF is already the published market excess return and _____ RF again. FF3 adds SMB and HML to CAPM; FF5 adds RMW and CMA:

$$y_t = \alpha + \beta_M \,\mathrm{MKT}_t + \beta_S \,\mathrm{SMB}_t + \beta_H \,\mathrm{HML}_t \,[+\beta_R \,\mathrm{RMW}_t + \beta_C \,\mathrm{CMA}_t] + \varepsilon_t.$$

β is conditional co-movement, not a _____; α depends on the factor benchmark and sample and is not a _____.

Whether models nest depends on the evidence line

The three constructed specifications share one set of factor columns and nest strictly. In the historical official releases, the Mkt-RF and SMB series used by FF3 and FF5 are not identical in the selected months, so the historical models are _____ nested. Do not compare fit or Alpha across the two lines and give the comparison a market meaning.

Variance accounting, HAC, and multiple testing

Proposition P2.2.1 *(OLS variance accounting in one window).* With an intercept, OLS fitted values and residuals are uncorrelated in sample, so

$$\widehat{\mathrm{Var}}(y) = \widehat{\mathrm{Var}}(\hat{y}) + \widehat{\mathrm{Var}}(\hat{\varepsilon}).$$

The fitted share is the sample R^2. This is _____, not a causal decomposition

of future risk. The constructed Notebook asserts only $y = \hat{y} + \hat{\varepsilon}$ and 120 observations; it does not compute or assert this variance identity.

What the HAC bandwidth changes

With the same asset, sample, and factor design, moving the HAC lag from 3 to 6 to 12 leaves the point estimate _____; it changes the standard error, interval, and p-value because different lagged covariance terms enter the estimate. The three calculations reuse the same 120 months and are _____.

How Holm adjusts step by step

Sort the m raw p-values as $p_{(1)} \leq \ldots \leq p_{(m)}$; the k-th adjusted value is $\max_{j \leq k} \min\{1, (m-j+1)p_{(j)}\}$. The historical report treats nine fixed asset-by-model intercepts as one family. Holm adjusts only _____, not any unrun specification search.

Your turn. Cover this section. State which quantities HAC changes and which it does not; what determines the Holm family; and why variance accounting is not a causal decomposition.

Recall: Why are three HAC bandwidths not three independent validations?

Chronological split: attribution error is not forecast error

Estimating on 2010–2014 and evaluating on 2015–2019 uses separate historical windows, but the 2015–2019 monthly factors were fully known only after each return interval was realized; an RMSE computed with them is contemporaneous attribution error. HiTec–FF5 has RMSE 1.562 and NoDur–FF5 1.932 percentage points per month. Costs, financing, and borrowing are ``not applicable'' in the historical report because there are no signals, positions, orders, or ledger; they are not computed as zero.

Exercises: cover the answers and recompute

Your turn. P2-01 Same-month RF and factor units. In a given month, HiTec's total return is 2.0%, the same month's RF is 0.3%, and the published Mkt-RF value is 1.7%. Write down the dependent variable for that month. What would be wrong with subtracting RF from Mkt-RF again? If only the asset's total return is available and RF is not, can this be called a CAPM excess-return regression?

Your turn. P2-02 Interpret a conditional Beta. HiTec's FF3 market Beta is 1.158, with a HAC 95% interval of [1.075, 1.242]. Interpret the linear conditional relationship, in percentage points, when Mkt-RF rises by 1 percentage point in the same month. Does this number prove that a market move causes HiTec's return to change?

Your turn. P2-03 Interpret model Alpha. NoDur-CAPM Alpha is 0.2603% per month, with a HAC 95% interval of [−0.1223%, 0.6429%], nominal $p = 0.1804$, and nine-test Holm $p = 1.0000$. Explain its unit, interval, and evidentiary limits.

Your turn. P2-04 Check the sample variance. The total sample variance, fitted variance, and residual variance of NoDur-CAPM excess returns are 0.001039486, 0.000582943, and 0.000456543, respectively (squared decimal returns). Check the numerical identity and convert the fitted and residual shares to percentages. Can this be called a causal decomposition of future risk?

Your turn. P2-05 Compare FF3 and FF5. HiTec's R^2 rises from 0.864 under FF3 to 0.893 under FF5. Give at least two reasons why this difference should not be interpreted directly as the "pure incremental contribution" of the newly added RMW/CMA factors.

Your turn. P2-06 HAC 3/6/12 sensitivity. Enrgy-FF3 Alpha has the same point estimate, −0.6629% per month, under HAC 3/6/12. The 95% intervals are [−1.2950%, −0.0308%], [−1.3091%, −0.0167%], and [−1.3154%, −0.0104%], respectively. Which quantities change and which stay the same? Is there a reason to treat the three bandwidths as three independent validations?

Your turn. **P2-07 Multiple testing and negative results.** Under HAC 6, the Alpha estimates for all three energy models are negative, with single-test p-values of 0.0061, 0.0444, and 0.0249. Their Holm-adjusted values are 0.0549, 0.3111, and 0.1995. How would you state a cautious conclusion? May the report retain only the CAPM nominal p-value?

Recall: What is the first Holm multiplier?

Your turn. **P2-08 Chronological split.** HiTec-FF5 Alpha estimated on 2010–2014 is −0.098% per month, and the 2015–2019 error RMSE is 1.562 percentage points per month. Why is this still not a real-time forecast or a strategy return? List at least three types of information that a future net-Alpha study would need.

Project 2: Questions and Complete Answers

P2-01

The dependent variable is HiTec's total return less RF for the same month: $2.0\% - 0.3\% = 1.7\%$. Mkt-RF is already the published spread between the market portfolio return and RF, so the right-hand side uses 1.7% directly. Subtracting 0.3% again would create a series different from the official factor and could count the risk adjustment twice. If the asset's total return is available but same-month RF is not, the excess-return response defined by this project cannot be constructed; the regression also cannot quietly be relabeled as a completed CAPM excess-return regression.

P2-02

When same-month Mkt-RF rises by 1 percentage point, the HiTec-FF3 model's fitted conditional mean excess return rises by about 1.158 percentage points. The interval [1.075, 1.242] describes coefficient uncertainty under the specified model, sample, and HAC approximation. This is conditional co-movement between the market spread and industry excess return, not a causal effect of a market move, and it does not describe future returns known before the factor occurs.

P2-03

0.2603% per month is the sample intercept for NoDur against the CAPM market benchmark. Its interval crosses zero and its nominal p-value is 0.1804; for this window, there is insufficient evidence to reject a zero model-conditional mean intercept. Holm $p = 1.0000$ is the adjustment for a family of nine fixed asset-by-model Alpha tests. Alpha depends on the factor benchmark and sample. It is not a realizable return, a demonstrated pricing error, or future Alpha.

P2-04

Fitted and residual sample variance sum to $0.000582943 + 0.000456543 = 0.001039486$, equal to total variance at the reported precision. The fitted share is $0.000582943/0.001039486 \approx 56.080\%$; the residual share is about 43.920%. This is sample variance accounting between an OLS projection and its residual over the same window. It does not mean the market caused 56.080% of returns or prove that the same proportions will explain future risk.

P2-05

First, the common Mkt-RF and SMB series in the FF3 and FF5 official releases are not identical in the selected months, so the two official models are not strictly nested. Second, factor spreads may be correlated, so the R^2 change cannot be uniquely assigned to RMW/CMA. Third, this is in-sample fit over the same historical sample; adding variables generally helps in-sample fit, but that does not establish better forecasts, a causal increment, or net returns. To isolate the contribution of the added factors, hold the same baseline factors fixed and then add RMW/CMA, labeling the result as a custom specification.

P2-06

With the same asset, sample, and factor design, the OLS intercept point estimate remains -0.6629% per month. Changing the HAC lag length to 3, 6, or 12 changes the standard error, 95% interval, and p-value because it changes which lagged covariance terms enter the covariance estimate. The three calculations reuse the same 120 months; they are neither three datasets nor independent replications. All three intervals being below zero means only that this model's single negative intercept is relatively stable across these specified bandwidths. It remains subject to the model, sample, estimation assumptions, and the nine-test Holm boundary.

P2-07

A cautious statement is: all three energy-model HAC 6 intercepts are negative, and their unadjusted p-values are below 0.05; after Holm adjustment across the nine intercepts shown in this report, none is below 0.05, and the minimum is 0.0549. Therefore, do not claim significance after adjustment, much less that future energy investing will lose money. The report must not retain only the CAPM nominal p-value: it should preserve FF3, FF5, NoDur, HiTec, and results near zero to avoid selective reporting. It should also say that Holm adjusts only the nine listed tests, not any unrun specification search. (Check: $9 \times 0.0061 = 0.0549$.)

P2-08

2010–2014 and 2015–2019 are separate historical windows, but the current data are not a month-by-month point-in-time archive, so the parameter versions may not be those available at the time. More importantly, the 2015–2019 monthly factors were fully known only after they were realized within each return interval. The RMSE therefore uses ex post, contemporaneous factors and measures historical attribution error, not forecast error known in advance or the net return of a tradable strategy.

At a minimum, a net-Alpha study would also need to define an executable signal and its availability time; the historical security universe, delisting data, and constituent versions; position and rebalancing rules; order and execution times and prices; turnover and commissions, taxes, spreads, and market impact; financing and borrow costs; cash and corporate-action treatment; and a holdings, orders, fills, and cash ledger. Without these inputs, net-of-cost returns cannot be calculated and strategy feasibility cannot be assessed.

Fill-in key for the reading sections

same-month; must not subtract; causal effect; realizable return or future Alpha; not strictly. Variance accounting: sample accounting within one window. HAC: unchanged; not three independent validations. Holm: the nine tests listed in the family.

Chapter 16 —Time Series I: Dependence and ARIMA

A weakly stationary series has a mean, variance, and autocovariance that do not change over time. An AR(1) forecast reverts toward its long-run mean, with deviations decaying as ϕ^h; forecast intervals widen with the horizon and omit parameter-estimation error.

Trigger. If this month's return is 4% and $\phi = 0.5$, why is next month's forecast not 2%?

Definition 16.1 *(Stationary AR(1)).* $X_t = c + \phi X_{t-1} + \varepsilon_t$, with causal stationarity condition $|\phi| < 1$ and long-run mean $\mu =$ _____. The centered form is $X_t = \mu + \phi(X_{t-1} - \mu) + \varepsilon_t$.

Your turn. 16-L01 AR(1) intercept and stationary moments (mapping: Section 6.1, 16-E07)

Let $c = 0.02$, $\phi = 0.6$, and $\text{sd}(\varepsilon_t) = 0.04$. Find the long-run mean, unconditional variance, $\gamma(1)$, $\rho(1)$, and $\rho(2)$. State the stability condition and innovation assumptions.

Your turn. 16-L02 Multi-step forecasts and prediction intervals (mapping: 16-E06, 16-E08)

Let $\mu = 0.005$, $\phi = 0.5$, $\sigma_\varepsilon = 0.06$, and $X_T = 0.04$. Conditional on the forecast-time information set $\mathcal{F}_T$, assume future innovations have conditional mean zero, constant conditional variance σ_ε^2 in each period, and are conditionally uncorrelated across the next two periods (independence is a stronger assumption). Find the one-step and two-step conditional means and V_1, V_2. If the next innovation is also conditionally normal and parameters are known, write the one-step 95% prediction interval. Which uncertainties does it omit?

Your turn. 16-L03 MA(1) invertibility (mapping: Section 6.3, 16-E09)
Let $X_t = \mu + \varepsilon_t + 0.5\varepsilon_{t-1}$ and $\sigma_\varepsilon = 0.06$. Find $\gamma(0)$, $\gamma(1)$, and $\rho(1)$; then, for $k \geq 2$, find $\rho(k)$ and determine whether the process is invertible under the sign convention used here.

Your turn. 16-L04 A month is not a row number: code diagnosis (adapted from Section 6.2, 16-E04)
The code below treats the previous row as the previous month. Identify the problem and change it so that it first validates the monthly index and creates a lag only if validation passes; do not impute or compress missing months.

```
# BEGIN B06R2_16_BAD
import pandas as pd
months = pd.PeriodIndex(
    ["2024-01", "2024-02", "2024-04"], freq="M"
)
returns = pd.Series([.01, .02, .04], index=months)
lag1 = returns.dropna().shift(1)
print(lag1.loc["2024-04"])
# END B06R2_16_BAD
```

Additional exercises: Process, Timing, and Forecast Reconstruction

Definition 16.2 *(Process and realization).* $\{X_t\}$ is a family of random variables indexed by time; one realization is the observed path $x_1, \dots, x_n$. A population expectation averages over possible worlds, while a sample mean averages along one path; connecting the two requires the relevant ergodic conditions.

Your turn. 16-N01 Initialization (local variant of main-text 16-N01 and Section 6.1.1)
Let $\mu = 0.005, \phi = 0.5, \sigma = 0.06$ and fix $X_0 = \mu$. Fill in the variances at $h = 0, 1, 2, 3$ and write the general formula for h. What initial variance is needed for stationary second moments from period 0? Is a stable recursion sufficient?

Your turn. 16-N02 White noise and innovations (local variant of main-text 16-N02 and Sections 6.2–6.3)
Let $e_t = \cos(2^t U)$ with $U \sim \text{Uniform}(0, 2\pi)$. The cosines at different frequencies are orthogonal, so this is second-order white noise; use the cosine double-angle identity to write e_{t+1}. If $e_t = 0.5$, what is the next-period conditional mean? Which AR conditional-forecast assumption is missing? What are the respective inputs to a price rolling mean and an MA innovation filter?

Your turn. 16-N03 Timing and reconstruction (local variant of main-text 16-N03 and Sections 6.5–6.6)
The latest price is 102 euros per share; predicted changes over two periods are 1 and 2, with forecast-error variances of 4 each and covariance 1. Find the level forecasts and the second-period variance, stating the units. If February and April are compressed into adjacent rows, which AR power is misused? Should output from an ARIMA$(p, 1, q)$ for the original level be cumulatively summed again?

Your turn. 16-N04 Research judgment from one history (local variant of Sections 3.1 and 6.2)
Each time, first draw one Z, then set $X_t = Z$ for every month. Why is the process strictly stationary but its long-path sample mean cannot estimate $\mathbb{E}Z$? Explain the difference between adding months and adding independent worlds. If the observed-return ACF is near zero, can you conclude that returns are IID or have no predictive relationship of any kind?

Chapter 16 Answers: 16-L01 to 16-L04

16-L01

$\mu = c/(1-\phi) = 0.02/0.4 = 0.05$. The innovation variance is $0.04^2 = 0.0016$, so $\gamma(0) = 0.0016/(1-0.6^2) = 0.0025$ and the unconditional standard deviation is 0.05; $\gamma(1) = 0.6(0.0025) = 0.0015$, $\rho(1) = 0.6$, and $\rho(2) = 0.36$. This is a causal AR(1) with $|\phi| < 1$; innovations must have zero mean, finite variance, and be uncorrelated with the past. Strict stationarity also requires the corresponding innovation-distribution conditions.

16-L02

$\hat{X}_{T+1|T} = 0.005 + 0.5(0.04 - 0.005) = 0.0225$; $\hat{X}_{T+2|T} = 0.005 + 0.5^2(0.035) = 0.01375$. The innovations have zero conditional means, constant conditional variance σ_ε^2, and zero conditional covariance across the two periods, so $V_1 = 0.06^2 = 0.0036$ and $V_2 = \mathrm{Var}_T(0.5\varepsilon_{T+1} + \varepsilon_{T+2}) = (0.5^2+1)0.06^2 = 0.0045$. A one-step normal prediction additionally requires $\varepsilon_{T+1}|\mathcal{F}_T \sim N(0, 0.06^2)$ and known parameters; the 95% interval is approximately $0.0225 \pm 1.96(0.06) = [-0.0951, 0.1401]$, or $[-9.51\%, 14.01\%]$. It omits parameter-estimation error and does not cover heavy tails, volatility clustering, or structural change.

16-L03

$\gamma(0) = 0.06^2(1 + 0.5^2) = 0.0045$, $\gamma(1) = 0.5(0.06^2) = 0.0018$, so $\rho(1) = 0.4$, while $\rho(k) = 0$ for every $k \geq 2$. Under the sign convention $X_t = \mu + (1 + \theta L)\varepsilon_t$, invertibility requires $|\theta| < 1$; here $\theta = 0.5$, so the process is invertible.

16-L04

The incorrect code prints 0.02, which is February's value; February and April are two months apart, so the previous row cannot be called the previous month. The answer uses a monthly-frequency validator and stops when it detects a gap. Trace the source or specify a missing-data model before taking a lag of a sequence that passes validation.

```
# BEGIN B06R2_16_FIXED
import pandas as pd
from quant_course.time_series import validate_monthly_series
months = pd.PeriodIndex(
    ["2024-01", "2024-02", "2024-04"], freq="M"
)
returns = pd.Series([.01, .02, .04], index=months)
try:
    checked = validate_monthly_series(returns)
except ValueError as exc:
    print(type(exc).__name__, str(exc))
else:
    print(checked.shift(1))
# END B06R2_16_FIXED
```

Chapter 16 Answers: Additional answers 16-N01 to 16-N04

16-N01

With a fixed starting value, the variance is 0; at horizon h the variance is $0.0036 \sum_{j=0}^{h-1} 0.25^j$. Thus, for $h = 0, 1, 2, 3$, it is 0, 0.0036, 0.0045, and 0.004725, respectively, so it changes over time and the process is not weakly stationary from the start. The initial variance must be $0.0036/(1 - 0.25) = 0.0048$, and the initial value must have mean 0.005 and be independent of future innovations to preserve stationary second moments. Strict stationarity also requires the full distribution to match; with normal innovations, initialize from the corresponding stationary normal distribution. The condition $|\phi| < 1$ only makes the effect of the starting value decay.

16-N02

$e_{t+1} = 2e_t^2 - 1$; when $e_t = 0.5$, the next value is exactly -0.5. Zero linear covariance at nonzero lags still permits nonlinear dependence, so the missing condition is that the future innovation have conditional mean zero given current information. A price rolling mean takes observed prices as input; an MA model takes current and lagged innovations, so the two inputs are not interchangeable.

16-N03

The level forecasts are $102 + 1 = 103$ and $102 + 1 + 2 = 105$ euros per share. The second-period level forecast error is the sum of the two change errors, with variance $4 + 4 + 2(1) = 10$ euros squared per share squared. Across two months, use ϕ^2 and error variance $\sigma^2(1+\phi^2)$; compressing rows incorrectly uses ϕ and σ^2. The monthly-frequency interface stops when it detects a gap. An ARIMA$(p, 1, q)$ forecast for the original level has already been reconstructed internally, so do not sum it again.

16-N04

Every finite-time vector is $(Z, \ldots, Z)$, whose joint distribution is unchanged by a time shift, so the process is strictly stationary. The path mean is always the particular draw Z; adding months does not create independent worlds and need not make it converge to $\mathbb{E}Z$, so the ergodic condition must be checked separately. A sample ACF near zero is also subject to sampling variation; even if every nonzero-lag population autocorrelation were zero, that would not imply IID or rule out nonlinear predictive relationships such as ones involving squares.

Chapter 17 —Time Series II: Unit Roots, Cointegration, and Prediction

The ADF test examines the regression of Δy_t on y_{t-1} and asks whether the latter's coefficient is zero; Engle–Granger first estimates a long-run relationship and then tests its residual; a VAR forecast recursively substitutes conditional means and omits future shocks. Cointegration, prediction, and tradable net returns are different conclusions.

Trigger. Two prices are highly correlated. Can you trade them as a pair right away?

Definition 17.1 *(ADF and Engle–Granger).* From $y_t = \phi y_{t-1} + \varepsilon_t$, we obtain $\Delta y_t = (\phi - 1)y_{t-1} + \varepsilon_t$. The ADF null is $\gamma = 0$; choose deterministic terms to match the question. The second Engle–Granger step tests the estimated residual and requires cointegration-specific critical values, not ordinary ADF critical values.

Your turn. 17-L01 ADF coefficient and deterministic terms (mapping: Section 8.1, 17-E02, 17-E06)
Let $y_t = 0.7y_{t-1} + \varepsilon_t$. Write Δy_t and find γ. If $y_{t-1} = 10$, what is this term's contribution to the change? If a linear deterministic trend is specified in advance, how should the ADF specification be chosen? Does $\gamma < 0$ by itself reject a unit root?

Your turn. 17-L02 First-stage EG OLS (mapping: Sections 8.2, 17-E04)
Let $x = (1, 2, 3, 4, 5, 6)$, $u = (1, -2, 1, 1, -2, 1)$, and $y = 2x + u$. Verify the residual mean and its inner product with centered x; find the intercept and slope, $\hat{\alpha}, \hat{\beta}$, from OLS with an intercept, and the residuals. Why cannot these six points establish cointegration? What type of critical values should the second step use?

Your turn. 17-L03 Two-step VAR conditional forecast (mapping: Section 8.3)
Let $c = (0.5, -0.2)^\top$, $A = \left(\begin{smallmatrix} 0.4 & 0.2 \\ 0.1 & 0.3 \end{smallmatrix}\right)$, and $z_T = (2, -1)^\top$ (percentage points). Calculate $\hat{z}_{T+1|T}$ and $\hat{z}_{T+2|T}$. What is substituted into the second step?

Your turn. 17-L04 A predictive relationship is not structural causality (mapping: 17-E05)
In `grangercausalitytests([y,x])`, what does rejecting "all lagged coefficients of x are zero" tell you? Which intervention claim does it not support? List two model-external factors that could affect the result.

Your turn. 17-L05 Costs on both legs and executable timing (mapping: Section 8.4)
A signal is formed at month-end t. At the next executable time, $t + 1$, use 100 dollars of notional on each leg for a long position in A and a short position in B, then close both at $t + 2$. The two legs' total returns over the holding interval are 3% and 1%, respectively. Charge 10 bp per leg at both entry and exit; the short also incurs one month of borrowing fees of 20 bp and financing fees of 5 bp, all calculated on 100 dollars of notional. Find gross P&L, total costs, and net P&L. Why cannot the full-month return for month t be used for this accounting?

Your turn. 17-L06 Code diagnosis: ordinary ADF on a residual (adapted from Sections 6.3, 8.2, and the second error in Section 12)

The code below treats the ordinary ADF p-value for first-stage OLS residuals as a cointegration test. Identify the error and give complete code that uses the cointegration-specific null distribution. The simulation is a teaching construction only.

```
# BEGIN B06R2_17_BAD
import statsmodels.api as sm
from statsmodels.tsa.stattools import adfuller
from quant_course.cointegration import (
    simulate_noncointegrated_common_trend,
)
pair = simulate_noncointegrated_common_trend(
    n=900, seed=20261017
)
design = sm.add_constant(pair["y"].to_numpy())
fit = sm.OLS(pair["x"].to_numpy(), design).fit()
residual = pair["x"].to_numpy() - design @ fit.params
ordinary = adfuller(
    residual, regression="c", autolag="AIC",
    result_object=True
)
print(f"ordinary ADF p={ordinary.pvalue:.5f}")
# WRONG: this is not the EG no-cointegration p-value.
print("reject" if ordinary.pvalue < .05 else "do not reject
   ↪ ")
# END B06R2_17_BAD
```

Additional exercises: From Accumulated Shocks to Actual Holdings

Your turn. 17-N01 High correlation and accumulated errors (local variant of main-text 17-N01 and Section 3.1)

Let $X = W + U, Y = W + V$, where the three independent random-walk innovations have variances 9, 0.25, and 0.25, and the processes start at zero. Find the population correlation for $t > 0$ and the variance of $Y - \beta X$. Which term prevents every random trend from being eliminated for any β? Does a six-point sample correlation above 0.99 change the conclusion?

Your turn. 17-N02 Which critical-value table? (local variant of main-text 17-N02 and Section 6.3)

Compare a pre-specified spread, an OLS residual estimated in the same window, and a new-window residual using frozen training coefficients; state the interpretation of each test. Why does freezing not eliminate parameter error? When a short-window ADF test fails to reject, can $\phi = 0.99$ be ruled out? At which stage should deterministic terms and lag length be selected?

Your turn. 17-N03 From log coefficient to a two-leg ledger (full calculation: main-text 17-N03 and Section 6.6)

Go long $s = \log P^A - \alpha - 1.5 \log P^B$, with $K = 100$ dollars; A/B entry prices are 20/50 and exit prices are 20.6/50.5 dollars per share. Fill in signed notionals $N_A =$ _____ and $N_B =$ _____, then find share counts, gross exposure, net notional, and gross P&L. Charge 10 bp on each leg's actual traded notional at entry and exit, with no other fees; calculate net P&L. If $\beta = -0.5$, which two legs point in which directions for a long spread? If $\beta = 1.4$ and share counts are rounded toward zero, what is the realized hedge ratio? Why may Project 5 select only $\beta > 0$ even though general cointegration has no such requirement?

Chapter 17 Answers: 17-L01 to 17-L02

17-L01

Subtracting y_{t-1} from both sides gives $\Delta y_t = (0.7 - 1)y_{t-1} + \varepsilon_t = -0.3y_{t-1} + \varepsilon_t$, so $\gamma = -0.3$; at $y_{t-1} = 10$, this term contributes -3 y-units. If the alternative includes a deterministic linear trend, choose the `ct` specification with a constant and linear trend in advance; do not try many specifications and report only the smallest p-value. A negative coefficient alone is not a test result; compare the ADF statistic with its nonstandard critical distribution under the same deterministic specification.

17-L02

$\bar{x} = 3.5$, $\sum u_i = 0$, and $\sum(x_i - \bar{x})u_i = 0$. Thus, OLS with an intercept gives $\hat{\beta} = 2$, $\hat{\alpha} = \bar{y} - 2\bar{x} = 0$, and residuals $(1, -2, 1, 1, -2, 1)$. These six points are deliberately constructed as a finite sample to check the OLS calculation; they cannot establish population integration orders or cointegration. The second step should use Engle–Granger/MacKinnon cointegration-specific critical values, rather than treating an estimated residual as a sequence specified in advance and applying ordinary ADF critical values.

Chapter 17 Answers: 17-L03 to 17-L05

17-L03

The first forecast is $c + Az_T = (0.5 + 0.4(2) + 0.2(-1), -0.2 + 0.1(2) + 0.3(-1))^\top = (1.1, -0.3)^\top$. Substitute the first-step conditional mean into the second step to obtain $c + A(1.1, -0.3)^\top = (0.88, -0.18)^\top$. Recursive forecasting takes the conditional expectation of future innovations to be zero; it does not insert unknown future realizations into the forecast.

17-L04

Given the sample, lag order, and information set, the result means that lagged x provides additional linear predictive information for y; it does not mean intervening on x would structurally change y. Omitted variables, common shocks, sampling frequency, or variable definitions can all affect the result.

17-L05

Gross P&L is $100(0.03) - 100(0.01) = \$2.00$. There are four trades across the two legs' entry and exit; each costs $100(0.001) = \$0.10$, for total trading costs of \$0.40. Borrowing costs \$0.20, financing costs \$0.05, and net P&L is $2.00 - 0.40 - 0.20 - 0.05 = \$1.35$. The accounting interval is $t + 1$ to $t + 2$; a signal available only at month-end t cannot be backfilled into the already completed return for month t. This is a fixed-notional teaching example, not an empirical cost estimate; monthly French returns cannot be decomposed into this execution interval.

Chapter 17 Answer: 17-L06

17-L06

The first-stage residual is estimated from the same level data, so the ordinary ADF unit-root null distribution does not apply to the Engle–Granger no-cointegration test. The code uses this chapter's interface, which calls a cointegration test calibrated for that null; the fixed simulation only illustrates that the two *p*-value conventions differ and does not establish a market relationship or tradability.

```
# BEGIN B06R2_17_FIXED
from quant_course.cointegration import (
    engle_granger_test,
    simulate_noncointegrated_common_trend,
)
pair = simulate_noncointegrated_common_trend(
    n=900, seed=20261017
)
test = engle_granger_test(pair["x"], pair["y"])
print(f"EG p={test.pvalue:.5f}")
print("reject" if test.pvalue < .05 else "do not reject")
# END B06R2_17_FIXED
```

Chapter 17 Answers: Additional answers 17-N01 to 17-N02

17-N01

The covariance is $9t$, each variance is $9.25t$, and the correlation is $9/9.25 \approx 0.9730$. The residual $(1-\beta)W+V-\beta U$ has variance $t[9(1-\beta)^2+0.25+0.25\beta^2]$; the independent V term of 0.25 makes the variance continue to grow with t for every coefficient. A high six-observation sample correlation does not determine population integration orders or whether a stationary combination exists; it cannot change the conclusion implied by the data-generating mechanism.

17-N02

For a combination fixed in advance and independently of this sample, an ordinary ADF reference can test that given series under the ADF conditions. An OLS residual estimated in the same window requires the Engle–Granger no-cointegration distribution. A new-window residual with frozen training coefficients diagnoses relationship stability; the coefficients were still estimated from a finite training sample, and freezing only prevents future information from being fed back. It does not remove estimation error. $\phi = 0.99 < 1$ is stable but close to a unit root; a short-window failure to reject may reflect low power and does not prove that the root is exactly 1. Specify deterministic terms, maximum lag or selection rule, and window in the training research design. Deterministic terms enter the first-stage long-run equation in EG; lagged differences enter the second-stage residual ADF regression. Do not use later windows to choose the smallest p-value.

Chapter 17 Answer: Additional answer 17-N03

17-N03

$N_A = +100, N_B = -150$ dollars, with $q_A = 5, q_B = -3$ shares. Gross exposure is 250 and initial net notional is -50; gross P&L is $5(0.6) - 3(0.5) = 1.50$ dollars. Entry notional totals 250, while exit notional is $5(20.6) + 3(50.5) = 254.5$; total traded notional is 504.5, fees are 0.5045 dollars, and net P&L is 0.9955 dollars. With $\beta = -0.5$, the notionals are $+100, +50$, so both legs are long; a general cointegrating relationship can have a negative coefficient, while Project 5 excludes negative coefficients under its preregistered rule to restrict candidates to one long and one short leg. With $\beta = 1.4$, the target is $5, -2.8$ shares; rounding toward zero gives $5, -2$, so the realized B/A notional ratio is $100/100 = 1$. Calculate actual holdings P&L from actual share counts; do not continue to book the target ratio of 1.4. The intercept is not a cash leg, and a gross-exposure P&L ratio is not automatically a capital-account NAV return.

Chapter 18 —Volatility Modeling and Forecasting

Realized variance (RV) summarizes the movements that have already occurred over an interval; EWMA, ARCH, and GARCH recursively produce the next period's conditional variance given past information. Units are squared decimal returns.

Trigger. Does a variance of 0.0004 correspond to volatility of 0.04%?

Definition 18.1 *(Recursion).* EWMA: $h_{t+1} = \lambda h_t + (1-\lambda)\epsilon_t^2$. GARCH(1,1): $h_{t+1} = \omega + \alpha\epsilon_t^2 + \beta h_t$; if $\alpha + \beta < 1$, the long-run level is $\bar{h} =$ _____, and multi-step expected variance reverts by powers of _____ toward $\bar{h}$.

Your turn. 18-L01 Interval Realized Variance and Period-by-Period Conditional Variance (adapted from: 18-E05–18-E07)

For the same values $(0.01, -0.02, 0.03)$, treat them as three subinterval log returns within one interval when calculating RV; treat them as innovations in three consecutive months when doing the recursion, with $h_1 = 0.0004$ as the first month's variance forecast. Find RV; EWMA ($\lambda = 0.9$) h_2, h_3; GARCH ($\omega = 0.00001, \alpha = 0.1, \beta = 0.8$) h_2, h_3; and ARCH(1) ($\beta = 0$) h_2, h_3. Explain what object RV represents and what object conditional variance represents.

Your turn. 18-L02 Variance Units and Multi-Step Expectations (adapted from: main text §5.1, 18-E09)

What monthly volatility corresponds to $h = 0.0004$? What is the variance in units of "percentage-return squared"? Given the next-period forecast $\hat{h}_{T+1|T} = 0.0004$, $\rho = 0.9$, and $\bar{h} = 0.0001$, list the month-by-month expected variances starting from $T+1$ for $k = 1, 2, 3$. Are these themselves the total variance over three months?

Your turn. 18-L03 Code Diagnosis: Target-Month Return Leakage (adapted from: 18-P02)

When forecasting the variance for each month, the target month's return is not known until that month ends. The code below labels the rolling-window result at the window's final month. Identify the line that leaks the target month's return, and explain why changing the data from month 3 onward changes the "forecast" for month 3.

```
# BEGIN B07R2_18_BAD
import pandas as pd
idx = pd.period_range("2000-01", periods=4, freq="M")
returns = pd.Series([.01, .03, -.01, .02], index=idx)
window = 2
forecast = returns.rolling(window).var()
changed = returns.copy()
changed.iloc[2:] = 100.0
forecast_changed = changed.rolling(window).var()
print(forecast.iloc[2], forecast_changed.iloc[2])
# END B07R2_18_BAD
```

State how the window should be realigned; which observations already known at the time should be used for the month 3 forecast?

Supplementary Conceptual Exercises

Your turn. 18-L04 Effect of Initialization (corresponds to 18-R01; new independent exercise)

Expand EWMA as in §5.3 and prove that the forecast difference between two initial values decays geometrically. If the initial values differ by 0.0005 and λ=0.9, how much do they differ after two updates?

Your turn. 18-L05 Proxy and Unit Diagnosis (corresponds to 18-R02; new independent exercise)

The true conditional mean is 0.01, the fixed center is 0, and the true conditional variance is 0.0004. What is the conditional expectation of the squared-return proxy? If decimal returns are uniformly converted to percentage returns, by how much does QLIKE increase, and does the ranking change?

Your turn. 18-L06 Research Judgment (corresponds to 18-R03; new independent exercise)

A researcher compares intraday RV with a monthly forecast and declares the proxy unbiased because its mean is close to the mean monthly squared return. What conditions and records are missing?

Chapter 18 Answers: 18-L01 through 18-L06

18-L01

RV sums the squared log returns of the three realized subintervals:

$$0.01^2 + (-0.02)^2 + 0.03^2 = 0.0001 + 0.0004 + 0.0009 = 0.0014.$$

If the three log returns are combined, the interval log return is $0.01 - 0.02 + 0.03 = 0.02$, whose square is 0.0004; this is not equal to RV. Here RV summarizes movements already realized within an interval and cannot be substituted for a single conditional variance forecast. Start the recursion from $h_1 = 0.0004$. EWMA gives, in sequence,

$$h_2 = 0.9(0.0004) + 0.1(0.01^2) = 0.00037, \qquad h_3 = 0.9(0.00037) + 0.1((-0.02)^2) = 0.000373.$$

For GARCH(1,1),

$$h_2 = 0.00001 + 0.1(0.01^2) + 0.8(0.0004) = 0.00034,$$

$$h_3 = 0.00001 + 0.1((-0.02)^2) + 0.8(0.00034) = 0.000322.$$

Setting $\beta = 0$ for ARCH(1) gives $h_2 = 0.00001 + 0.1(0.01^2) = 0.00002$ and $h_3 = 0.00001 + 0.1((-0.02)^2) = 0.00005$. Each recursive result is a conditional variance for the next period given information from the previous period, whereas RV is the sum of squared returns over an interval that has already occurred.

18-L02

Answer to the fill-in: for standard GARCH(1,1) with $\omega > 0$, $\alpha, \beta \geq 0$, and $\alpha + \beta < 1$, $\bar{h} = \omega/(1 - \alpha - \beta)$ and $\rho = \alpha + \beta$. The latter is the persistence parameter governing reversion of multi-step variance forecasts to the long-run level. Monthly volatility is the standard deviation $\sqrt{0.0004} = 0.02 = 2\%$. Converting returns from decimals to percentages multiplies returns by 100 and variance by 100^2, so the variance is $0.0004 \times 100^2 = 4$ "percentage-return squared." The mean-reversion formula is

$$\mathbb{E}_T[h_{T+k}] = \bar{h} + \rho^{k-1}(\hat{h}_{T+1|T} - \bar{h}).$$

Substituting the difference $0.0004 - 0.0001 = 0.0003$ gives, for $k = 1$, 0.0004; for $k = 2$, $0.0001 + 0.9(0.0003) = 0.00037$; and for $k = 3$, $0.0001 + 0.9^2(0.0003) = 0.000343$. These are expected conditional variances for individual months; they do not automatically equal the variance of cumulative returns over three months. Only under conditions such as zero cross-period conditional covariance and a known mean does the sum of period variances give the interval variance; otherwise covariance and uncertainty about the mean must also be included.

18-L03

The erroneous line is `returns.rolling(window).var()`: the window endpoint is the same as the target month, so the month 3 calculation includes the month 3 return. That return is not known until month-end. In the example, the original series prints approximately 0.0008; after changing month 3 onward to 100, the month 3 result becomes approximately 4997.00045. This is target-information leakage, not a forecast available at the beginning of the month. Shift the window result forward by one index period (`shift(1)`), so the forecast for target month t uses only data through the end of month $t-1$. The self-contained code below retains NaN for the first two target months, uses only January and February returns for the month 3 forecast, and leaves that forecast unchanged when the current and future suffix is modified.

```
# BEGIN B07R2_18_FIXED
import pandas as pd
idx = pd.period_range("2000-01", periods=4, freq="M")
returns = pd.Series([.01, .03, -.01, .02], index=idx)
window = 2
forecast = returns.rolling(window).var().shift(1)
changed = returns.copy()
changed.iloc[2:] = 100.0
forecast_changed = changed.rolling(window).var().shift(1)
assert abs(forecast.iloc[2] - 0.0002) < 1e-12
assert forecast.iloc[2] == forecast_changed.iloc[2]
print(forecast.iloc[2], forecast_changed.iloc[2])
# END B07R2_18_FIXED
```

Here the rolling sample variance is only the historical-variance proxy specified in 18-P02; it is not automatically a correct conditional-volatility model in every market regime. An actual study must still pre-specify the forecast window, estimation period, and missing-data treatment.

Supplementary Exercise Answers

18-L04

Subtracting the two recursions under the same shock sequence gives the difference recursion $\delta_{t+1} = \lambda\delta_t$, so after two updates the difference is $0.9^2 \times 0.0005 = 0.000405$. The initial value must be set before forecasting begins.

18-L05

The conditional expectation is $0.0004 + (0.01 - 0)^2 = 0.0005$, so the conditional bias is 0.0001. Transforming both the forecast and proxy uniformly increases the loss by log(10000), approximately 9.210340. Every model on the same sample shifts by the same constant, so the ranking is unchanged.

18-L06

First verify that intraday RV covers the full target month, and document the return definition and units, sampling noise and jumps, that the forecast was saved before the month began, and the common effective sample. Unbiasedness requires the conditional expectation to match the target; similar unconditional sample means do not establish conditional unbiasedness, and do not imply statistically significant outperformance or net returns.

Chapter 19 —State-Space Models and the Kalman Filter

The Kalman filter first uses the state equation to obtain a prior, then updates it using the innovation and gain. The gain's units depend on the state and observation; it is not a probability. The filtered state uses the current month's observation, so it cannot be backfilled into a position for that month.

Trigger. Which beta should be used to hedge at the beginning of the month?

Definition 19.1 *(Scalar Update).* Prior variance $P^- = F^2P + Q$; innovation $v = y - Hm^-$; innovation variance $S = H^2P^- + R$; gain $K = P^-H/S$; posterior mean $m^+ = m^- + Kv$; posterior variance $P^+ = (1 - KH)P^-$.

Your turn. 19-L01 Scalar Kalman Update and Gain Units (adapted from: main text §8, 19-P01)

$m = 1$, $P = 0.04$, $F = 1$, $Q = 0.01$, $H = 0.02$, $y = 0.03$, and $R = 0.0001$. Find, in order, P^-, innovation v, innovation variance S, gain K, posterior mean m^+, and posterior variance P^+. State the units of K and explain what K and Kv each represent.

Your turn. 19-L02 First-Period State Transition (adapted from: 19-E03)

The previous period's posterior is $m = 1$, $P = 0.04$; the transition is $F = 2$, with state-noise variance $Q = 0.01$; this period $H = 1$, observation $y = 2.10$, and $R = 0.03$. Find the prior, innovation, S, K, posterior mean, and posterior variance. What goes wrong if the first-period transition is skipped?

Your turn. 19-L03 When Monthly Returns Become Available (adapted from: 19-E05)

Let t denote March. Monthly market and asset returns are complete only on the last trading day of March. When can $\hat{\beta}_{t|t-1}$ and $\hat{\beta}_{t|t}$ each be used to form a position? Explain why the latter cannot be backfilled into a March position.

Your turn. 19-L04 Code Diagnosis: Same-Month Filtered Beta Look-Ahead (new; based on main text §§3, 13, related to 19-E05, 19-C02; not an adaptation of an original exercise)

In the teaching sequence below, $\hat{\beta}_{t|t-1}$ is a prior available at the start of the month, while $\hat{\beta}_{t|t}$ is the filtered state obtained only after observing that month's returns. The code uses the filtered beta to calculate the number of shares to open in the same month. Identify the timing error and give a corrected version executable at the start of the month; hold 10 asset shares each month, with the asset price at 50 and the market-proxy price at 100.

```
# BEGIN B07R2_19_BAD
import pandas as pd
months = pd.period_range("2025-01", periods=3, freq="M")
beta_predicted = pd.Series([1.0, 1.1, 1.2], index=months)
beta_filtered = pd.Series([1.1, 1.2, 1.3], index=months)
q_asset, price_asset, price_market = 10, 50.0, 100.0
q_market = (
    -beta_filtered * q_asset * price_asset / price_market
)
print(q_market.tolist())
# END B07R2_19_BAD
```

Write the corrected code's market-share positions for the three months and explain when the filtered state can affect a position.

Supplementary Conceptual Exercises

Your turn. 19-L05 Precision Derivation and Boundary Case (corresponds to 19-R01; new independent exercise)
Starting from the scalar Gaussian prior and observation densities, derive the posterior precision and mean by completing the square. Why is the weight KH? What happens when $H = 0$?

Your turn. 19-L06 Hand Calculation Across Q/R Settings (corresponds to 19-R02; new independent exercise)
Given $m = 1, P = 0.04, F = 1, H = 0.02, y = 0.03$, calculate the posterior variance for each of the three (Q, R) pairs: $(0.01, 0.0001)$, $(0.06, 0.0001)$, and $(0.01, 0.0004)$. Explain why the same innovation produces different updates (see the three-row table in main text §6).

Your turn. 19-L07 Position and Timing Diagnosis (corresponds to 19-R03; new independent exercise)
Hold 10 asset shares; the asset price is 50, the proxy price is 100, and the log-price coefficient b=1.2. How many shares are needed for a local hedge? If b is the filtered state at the end of the current month, can it determine the position for that month?

Chapter 19 Answers: 19-L01 through 19-L07

19-L01

The prior variance is

$$P^- = F^2P + Q = 1^2(0.04) + 0.01 = 0.05.$$

The predicted observation is $Hm^- = 0.02(1) = 0.02$, so $v = y - Hm^- = 0.03 - 0.02 = 0.01$. The innovation variance is

$$S = H^2P^- + R = (0.02)^2(0.05) + 0.0001 = 0.00012.$$

The Kalman gain is

$$K = \frac{P^-H}{S} = \frac{0.05(0.02)}{0.00012} = 8.333333\ldots.$$

Thus $m^+ = m^- + Kv = 1 + 8.333333\ldots(0.01) = 1.083333\ldots$, and

$$P^+ = (1 - KH)P^- = 0.0416667\ldots.$$

If the state is a dimensionless beta and the observation is a decimal return, K has units of beta/return; K is not a probability. Only after multiplying it by the return-unit innovation v does Kv become a state correction in beta units. If returns are expressed in percentages, the observation design, R, and the numerical units of the gain must also be converted consistently.

19-L02

First complete the state transition:

$$m^- = Fm = 2(1) = 2, \qquad P^- = F^2P + Q = 2^2(0.04) + 0.01 = 0.17.$$

Therefore, $v = y - Hm^- = 2.10 - 1(2) = 0.10$, $S = H^2P^- + R = 0.17 + 0.03 = 0.20$, and $K = P^-H/S = 0.17/0.20 = 0.85$. The posterior mean is $m^+ = 2 + 0.85(0.10) = 2.085$, and the posterior variance is $P^+ = (1 - 0.85)(0.17) = 0.0255$. Skipping the entire prediction step and carrying forward the previous posterior incorrectly uses $m = 1, P = 0.04$. Treating $F = 2$ as $F = 1$ but still adding $Q = 0.01$ incorrectly uses $m = 1, P = 0.05$. Both differ from the correct prior $m^- = 2, P^- = 0.17$ and lead to incorrect observation predictions and subsequent updates.

19-L03

At the start of March, only information through the end of February is available. Thus $\hat{\beta}_{t|t-1}$ can be used to form a position for March at an executable trading time, provided the relevant asset and market proxy can be traded as planned. $\hat{\beta}_{t|t}$ already incorporates March returns and can affect positions only after March ends. On entering April, use the state transition to obtain $\hat{\beta}_{t+1|t}$; do not backfill a state containing March information into March.

19-L04

The error is using $\beta_{t|t}$ to calculate the position for the same month t; it depends on that month's market and asset returns, which are not complete until month-end. Use $\beta_{t|t-1}$, which is available when the order is placed. Given the prices and position in this exercise, the market-share position is

$$q_{M,t} = -\beta_{t|t-1}\frac{q_{A,t}P_{A,t}}{P_{M,t}}.$$

The standalone code below shows both the erroneous and corrected results: the erroneous series is $[-5.5, -6.0, -6.5]$, while the tradable prior series is $[-5.0, -5.5, -6.0]$ shares. The two must not be interchanged.

```
# BEGIN B07R2_19_FIXED
import pandas as pd
months = pd.period_range("2025-01", periods=3, freq="M")
beta_predicted = pd.Series([1.0, 1.1, 1.2], index=months)
beta_filtered = pd.Series([1.1, 1.2, 1.3], index=months)
q_asset, price_asset, price_market = 10, 50.0, 100.0
q_market_bad = (
    -beta_filtered * q_asset * price_asset / price_market
)
q_market = (
    -beta_predicted * q_asset * price_asset / price_market
)
assert q_market.tolist() == [-5.0, -5.5, -6.0]
print("bad:", q_market_bad.tolist())
print("usable:", q_market.tolist())
# END B07R2_19_FIXED
```

The prior for each period must actually be available at the execution time for that month's position. The month-end filtered state can be used for a later month only after applying the state transition specified by the model. This constructed exercise demonstrates timing discipline; it is not empirical evidence about any set of betas, execution prices, or hedge returns.

Supplementary Exercise Answers

19-L05

Given a scalar Gaussian prior with $P^- > 0, R > 0$, let

$$A = \frac{1}{P^-} + \frac{H^2}{R}, \qquad B = \frac{m^-}{P^-} + \frac{Hy}{R}, \qquad C = \frac{(m^-)^2}{P^-} + \frac{y^2}{R}.$$

Multiplying the prior density by the observation density gives an exponent equal to $-1/2$ times the following quadratic:

$$\frac{(x-m^-)^2}{P^-} + \frac{(y-Hx)^2}{R} = Ax^2 - 2Bx + C = A\left(x - \frac{B}{A}\right)^2 + C - \frac{B^2}{A}.$$

The linear coefficient of the quadratic is $-2B$; multiplying by $-1/2$ makes the linear coefficient in the exponent $+B$. The last two terms do not contain x and are absorbed into the normalization constant. Comparing with the Gaussian density exponent $-(x-m^+)^2/(2P^+)$ gives

$$P^+ = \frac{1}{A} = \left(\frac{1}{P^-} + \frac{H^2}{R}\right)^{-1}, \qquad m^+ = \frac{B}{A} = P^+\left(\frac{m^-}{P^-} + \frac{Hy}{R}\right).$$

The first worked example gives $A = 24, B = 26, C = 29$, so the expanded expression is $24x^2 - 52x + 29$, the posterior variance is $1/24$, and the mean is $26/24$. $KH = H^2P^-/(H^2P^- + R)$ is a dimensionless convex weight, while K carries state/observation units. When $H = 0$, $K = 0$ and the posterior equals the prior; do not use y/H. If $P^- = 0$, the prior is a degenerate point mass, so do not use the reciprocal-precision completion of the square. For $R > 0$, the original covariance recursion instead gives $K = 0, P^+ = 0, m^+ = m^-$.

19-L06

Calculate $P^+ = P^-R/(H^2P^- + R)$. The three results, in order, are 0.041666667, 0.071428571, and 0.047619048. Increasing Q makes the prior less certain in this example, while increasing R makes the observation less precise; they respectively increase and decrease KH.

19-L07

Local hedge shares = -1.2×10×50/100 = -6 shares; writing -12 shares directly confuses notional value with share count. A local log-price sensitivity does not automatically equal a return beta. The month-end filtered state can be used for a later executable time only after the next-period state transition; it cannot be backfilled into the current month's position.

Chapter 20 —Modern Portfolio Theory

> Calculate an ordinary portfolio before optimizing it. Inputs here are constructed annualized decimals. Estimated moments give an optimum conditional on estimates. Call the cash line CAL; CML additionally needs CAPM equilibrium and the market tangency fund.

Trigger. Does a 50/50 capital allocation imply equal risk contributions?

Definition 20.1 *(Read, then calculate).* With fixed single-period weights and finite second moments, $E[R_p] = w^\top \mu$ and $\sigma_p^2 = w^\top \Sigma w$. Weights are dimensionless n-by-one; covariance is n-by-n with squared-return units; volatility has same-frequency return units. Use $\mu = (0.06, 0.10)^\top$, volatilities 0.15/0.25, correlation 0.2, and $\Sigma = \left(\begin{smallmatrix} 0.0225 & 0.0075 \\ 0.0075 & 0.0625 \end{smallmatrix}\right)$.

Your turn. 20-L01: three variance contributions

For 50/50, calculate mean, two own-variance contributions, the joint contribution, and volatility. Explain the factor 2 and why 20% is weighted-average volatility. For correlation−1, calculate equal-weight volatility and distinguish it from minimum possible volatility.

Your turn. 20-L02: generalize the calculation

Set $U_i = R_i - \mu_i$. Expand $E[(\sum_i w_i U_i)^2]$ to a double sum and a quadratic form. State the conditions for moving weights and sums outside expectation and check scalar dimensions.

20.1 From variance to optimal weights

Theorem 20.2 *(GMV and frontier conditions).* For $\Sigma \succ 0$, budget-one GMV is unique: $w_G = \Sigma^{-1}\mathbf{1}/A$, $A = \mathbf{1}^\top\Sigma^{-1}\mathbf{1}$. With unequal means, define $B = \mathbf{1}^\top\Sigma^{-1}\mu$, $C = \mu^\top\Sigma^{-1}\mu$, $D = AC - B^2 > 0$. Target-m variance is $(Am^2 - 2Bm + C)/D$; only $m \geq B/A$ is efficient. The closed forms permit shorts.

Your turn. 20-L03: GMV proof and check
For $L = \frac{1}{2}w^\top\Sigma w - \lambda(\mathbf{1}^\top w - 1)$, fill in the gradient, first-order condition, budget-determined multiplier, and strict-convexity argument. Calculate GMV weights/mean/variance and the 8% target. At what correlation does long-only GMV cease to hold both assets?

Your turn. 20-L04: the lower branch
Why does $D > 0$ require unequal means? Calculate the 5% target and equal-risk higher-mean reflection. After clipping negative GMV weights, which long-only KKT conditions must you check?

20.2 Cash, tangency, and uncertainty

Definition 20.3 *(Direction versus normalized fund).* Set $e = \mu - r_f\mathbf{1}$ and $H = e^\top\Sigma^{-1}e$. For excess target $\epsilon > 0$, minimum-risk direction is $w = (\epsilon/H)\Sigma^{-1}e$. A budget-one positive-Sharpe tangency fund additionally requires $B - r_f A > 0$, or $r_f < B/A$. Unlimited equal-rate cash borrowing/lending changes the feasible set.

Your turn. **20-L05: tangency boundaries**
At $r_f = 0.02$, calculate tangency weights/Sharpe and check with H. At 0.08, calculate the normalized fund's mean and classify its branch. At $r_f = B/A$, explain why CAL exists without a normalizable risky fund. Why can unequal borrowing/lending rates give different tangencies?

Your turn. **20-L06: code and research judgment**
Code combines monthly covariance with annual mean in a tangency function and calls clipped weights long-only GMV. Diagnose and repair both. Historical French tangency gross exposure is about 5.8/2.8 across the two windows, with NoDur+159%/−20%. For monthly volatility 4.5%,120 IID periods, calculate mean standard error. Does this establish which portfolio is more tradable?

20.3 Moments, constraints, and information clocks

Moments need an objective and a feasible set. Lag 1 changes training cutoff, not the holding month. These questions correspond to20-R01/R 02.

Your turn. 20-L07: four strategies (20-R01)
Which moments do EW, MinVar, MaxSharpe, and ERC need? Who has a 40% cap? Can slightly higher ERC net Sharpe identify a pure algorithm advantage under common constraints? At NAV 1,000,000, calculate EW amounts.

Your turn. 20-L08: two clocks (20-R02)
January 1990 holding month, $W = 60$: give lag 0/lag 1 training endpoints, virtual fill, and holding interval. Does lag 1 establish historical French availability or industry execution?

Definition 20.4 *(PSD and inversion).* A valid covariance needs PSD. Moment calculation remains defined with singular covariance. Inverse-based weights need PD. Examine means, feasible sets, zero-risk directions, and uniqueness before applying a formula.

Your turn. 20-L09: singular inputs (20-B01)
$R_1 = 0.06 + Z$, $R_2 = 0.10 + Z$, $E[Z] = 0$, $\mathrm{Var}(Z) = 0.04$. Use equal notionals against a common reference capital; directional returns mean PnL per reference unit, not division by zero net investment. Find covariance/eigenvalues and mean/variance of budget-one $(w, 1-w)$. Describe the GMV solution set and returns on $(1, -1)$ and its reverse. Set both means to 6% and reassess arbitrage. State the shorting assumptions.

Chapter 20 solutions: 20-L01 to 20-L03

20-L01

Covariance 0.0075; mean 0.08. Variance contributions are 0.005625+0.015625+0.003750=0.025; volatility 15.8114%. The square has two identical cross-products. Weighted-average 20% requires perfect positive correlation and nonnegative weights. At correlation−1, equal-weight volatility is $|0.5(0.15) - 0.5(0.25)| = 5\%$. Weights 5/8,3/8 instead achieve zero.

20-L02

$$E[(\sum_i w_i U_i)^2] = E[\sum_i \sum_j w_i w_j U_i U_j] = \sum_i \sum_j w_i w_j E[U_i U_j] = w^\top \Sigma w.$$

Expand a finite sum, use fixed weights/linearity/existing second moments, then covariance. Dimensions $(1 \times n)(n \times n)(n \times 1) = 1 \times 1$. Random dynamic weights correlated with same-period returns cannot use this unconditional fixed-weight formula.

20-L03

$\nabla_w L = \Sigma w - \lambda \mathbf{1} = 0$; $w = \lambda \Sigma^{-1} \mathbf{1}$, budget gives $\lambda A = 1$. Hessian $\Sigma \succ 0$ makes the objective strictly convex; the feasible stationary point uniquely minimizes it. Solve a linear system instead of inverting. $w = (11/14, 3/14)$, mean 6.85714%, variance 27/1400=0.0192857, volatility 13.8873%. Target 8% gives 50/50 and variance 0.025. The threshold is 0.15/0.25=0.6; at or above it, long-only minimum variance holds asset 1.

Chapter 20 solutions: 20-L04 and 20-L05

20-L04

Cauchy–Schwarz under the Σ^{-1} inner product gives $B^2 \le AC$, with equality when μ is proportional to ones. Unequal means give $D > 0$. The 5% target has weights 1.25/−0.25, variance 11/320=0.034375, and mirror mean 8.71429%. The lower target is inefficient. Long-only GMV needs $\Sigma w = \lambda \mathbf{1} + \nu$, $\nu \ge 0$, $\nu_i w_i = 0$. Positive holdings share marginal variance; zero holdings have no smaller contribution. Clipping changes hedges; resolve the QP and check these conditions.

20-L05

Direction $(1.407407, 1.111111)$ sums to 2.518518, giving 19/34,15/34. Mean 7.76471%, volatility 15.1290%, Sharpe 0.381032. $H = 98/675 = 0.145185$ gives the same ratio. At $r_f = 8\%$, the denominator is negative, weights 1.75/−0.75 have mean 3% and negative excess return on the lower branch. Cash optimization can short this fund; it solves a different feasible-set problem from a positive-Sharpe budget-one fund. At equality, the direction sums to zero and cannot normalize, while $H = D/A > 0$ still gives a CAL. Unequal borrowing/lending rates generally produce distinct tangencies and may leave an intervening risky-frontier segment. CML requires market equilibrium.

Chapter 20 solutions: 20-L06

20-L06

Match mean, covariance, and cash frequencies. Under an IID arithmetic-reporting approximation, monthly mean/covariance multiply by 12 and volatility by$\sqrt{12}$; compounding and serial dependence require separate treatment. Replace clipping with nonnegative QP and check budget, bounds, status, and KKT residuals. Mean standard error $0.045/\sqrt{120} = 0.004108$, or 0.4108 percentage points/month. This is an IID stable-moment benchmark, not a final dependent/heavy-tail interval. The inverse amplifies low-variance-direction mean errors and can produce changing weights; two windows do not identify causality or establish superiority. Freeze parameters, use available information for rolling out-of-sample evaluation, and report drifted turnover, costs, concentration, and paired differences. French revised snapshots do not supply stock-level PIT fills.

Chapter 20 solutions: 20-L07 and 20-L08

20-L07

EW estimates no moments; MinVar/ERC need covariance; MaxSharpe also needs excess means. All four are long-only and fully invested;40% caps apply only to MinVar/MaxSharpe. ERC uses positive budgets and PD covariance here, without that cap. Different objectives, inputs, and feasible sets prevent identifying a pure algorithm effect from Sharpe ranks; difference intervals also matter. EW targets 100,000 per asset at NAV 1,000,000, with drift before the next trade.

20-L08

Lag 0:1985-01–1989-12; lag 1:1984-12–1989-11. Both virtually decide/fill at December 1989 end and hold through January 1990 end. Both retain 60 months. Lag 1 advances the training cutoff without moving evaluation; it cannot supply historical reception versions, constituents, fills, or capacity logs.

Chapter 20 solutions: 20-L09

20-L09

$\Sigma = \left(\begin{smallmatrix} 0.04 & 0.04 \\ 0.04 & 0.04 \end{smallmatrix}\right)$ has eigenvalues 0.08/0. Budget-one mean is $0.10 - 0.04w$ and variance 0.04 for every real$w$. GMV is nonunique and inversion fails. The $1/-1$ direction returns-0.04 with shocks canceled; its reverse returns$+0.04$. These amounts measure PnL per equal reference-capital notional, with no division by zero net investment. Unrestricted cost-free shorts permit scaling that gain, while long-only constraints do not. Equal 6% means give zero directional return despite singular covariance. Singularity alone is insufficient for arbitrage. A diagonal addition changes the input and needs an estimation/regularization rationale.

Chapter 21 —Practical Portfolio Construction

> Portfolio construction requires separate checks of the mathematical solution, numerical residuals, and trading evidence. ERC allocates risk shares; it does not guarantee higher returns. In Project 3, ERC's Sharpe ratio is slightly higher than that of equal weighting, but the paired-difference interval crosses zero.

Trigger. Are risk shares probabilities because they sum to 1?

Definition 21.1 *(A complete reading).* For a symmetric positive semidefinite Σ and $\sigma_p = \sqrt{w^\top \Sigma w} > 0$, $\mathrm{MRC}_i = (\Sigma w)_i/\sigma_p$, $\mathrm{RC}_i = w_i \mathrm{MRC}_i$, and $p_i = w_i(\Sigma w)_i/\sigma_p^2$. The Euler identity gives $\sum_i \mathrm{RC}_i = \sigma_p$ and $\sum_i p_i = 1$, but p_i can be negative, so these are not probabilities. Under a self-financing increase in i funded by a decrease in j, the rate of change in risk is $\mathrm{MRC}_i - \mathrm{MRC}_j$.

Your turn. 21-L01: Two-asset ERC and risk amounts

Annualized $\sigma = (0.2, 0.3)$ and $\rho = 0.2$. Starting from equal contributions, derive $w = (0.6, 0.4)$, then calculate variance, volatility, and the two absolute contributions. Explain why the weights of two assets do not depend on $\rho \in (-1, 1)$ even though total volatility still changes; why can this result not simply be copied for three assets? State the sign condition for a negative risk contribution.

Your turn. 21-L02: Proving general risk budgeting

$b_i > 0$, $\sum b_i = 1$. For the objective $f(y) = \frac{1}{2} y^\top \Sigma y - \sum_i b_i \ln y_i$, $y > 0$, write the gradient, Hessian, and first-order conditions. Let $s = \sum y_i$ and $w = y/s$, *simultaneously* calculate both the numerator and portfolio variance to prove $p_i = b_i$. Is strict convexity sufficient to guarantee that a minimizer exists?

21.1 Covariance estimation and costs after drift

Definition 21.2 *(Shrinkage and the ledger).* Use the sample covariance S with divisor T, $m = \operatorname{tr}(S)/p$, and the normalized Frobenius norm $\|M\|^2 = \sum_{ij} M_{ij}^2/p$. The Ledoit–Wolf estimator is $S^* = \hat{\kappa} mI + (1-\hat{\kappa})S$; it preserves the trace and reduces some estimation noise, but does not guarantee out-of-sample returns. Pre-trade drifted weights are $w_i^+ = w_i(1+r_i)/(1+w^\top r)$; aggregate buys and sells define turnover $\mathrm{TO} = \sum_i |w_i^{\text{target}} - w_i^+|$, with no division by 2 in this textbook.

Your turn. 21-L03: Calculate shrinkage term by term

For the demeaned 4×2 data (in percentage units) $X = [[3,2],[-1,-2],[2,3],[-4,-3]]$, find $S, m, d^2 = \|S - mI\|^2$. Using row squared norms 13, 5, 13, and 25, calculate $\bar{b}^2 = [\sum_t \|x_t\|_2^4 - T\sum_{ij} s_{ij}^2]/(T^2 p)$, then calculate the intensity and the post-shrinkage correlation. If the data are changed to decimals, which quantities scale by which powers of 100?

Your turn. 21-L04: Turnover is not the change in target weights

At the start of the period, the weights are 50/50. The two assets return +10%/−10% over the month; the target for the next period remains 50/50. Find the drifted weights, turnover, and the cost as a share of pre-trade NAV at 10 bp per unit of aggregate buys and sells. How much would be omitted if the target weights were mistakenly treated as the pre-trade weights? Explain how the signal time, execution time, and return interval must be kept separate.

21.2 Views, leverage, and constraints

Definition 21.3 *(The boundaries of Black–Litterman and Kelly).* The BL prior mean $\Pi = \delta\Sigma w_{mkt}$ is an excess return at the same frequency. The prior-mean error covariance is $K = \tau\Sigma$; with view matrix P, excess-return view Q, and independent view-error covariance Ω, the posterior mean is $\mu_{BL} = \Pi + KP^\top(PKP^\top + \Omega)^{-1}(Q - P\Pi)$. The *unconstrained* mean–variance weights are $\Sigma^{-1}\mu_{BL}/\delta$ and do not automatically sum to 1. The exact log growth of a binary gamble is $G(f) = p\ln(1 + bf) + (1 - p)\ln(1 - f)$; the quadratic growth proxy used in financial Kelly is not the same formula.

Your turn. 21-L05: Updating BL step by step

Annualized $\Sigma = \left(\begin{smallmatrix} 0.04 & 0.02 \\ 0.02 & 0.09 \end{smallmatrix}\right)$, $w_{mkt} = (0.6, 0.4)$, $\delta = 2.5$, $\tau = 0.05$, $P = (1, 0)$, $Q = 0.10$, and $\Omega = \tau P\Sigma P^\top$. Find the prior, gain, posterior, and unconstrained weights. Why does the weight on asset 2 not change when its mean changes? How should the funding shortfall be handled?

Your turn. 21-L06: Exact Kelly and approximate growth

For the binary gamble $p = 0.55, b = 1$, find the optimal f, the relative growth of half-Kelly, and the sign of growth at double-Kelly. Then, in a monthly quadratic proxy with excess mean 0.008 and variance 0.0025, find full and half leverage and their growth relative to the cash benchmark. Why can the latter not be treated as exact expected log utility over a discrete period?

21.3 Research judgment and code diagnostics

Your turn. 21-L07: Neutrality and infeasible inputs

There are four assets in two sectors, $A = \{1, 2\}, B = \{3, 4\}$, with $\beta = (1, 2, 1, 2)^\top$. Construct a portfolio with zero net weight in each sector but nonzero beta. Why are a total-dollar constraint and the two sector constraints redundant? What does sector neutrality relative to a benchmark mean? Now let excess means be $(0.004, -0.004, -0.004)$, each individual upper bound be 0.4, and the budget be 1. Equal weighting is feasible, but the maximum-Sharpe code rejects the input. Find the constrained maximum excess return and explain precisely which object the code rejects.

Your turn. 21-L08: Green tests and strategy advantage

An ERC algorithm reports convergence. What numerical quantities still need checking? In Project 3, the annualized ERC-minus-equal-weight Sharpe difference is about +0.016, with a 95% paired block-bootstrap interval of $[-0.014, +0.050]$; the unadjusted interval for the annualized mean-return difference of minimum variance is about $[-3.97\%, -0.06\%]$, with 3 strategies × 2 metrics. Explain separately what can and cannot be concluded. What do the current revised French snapshot, zero-month publication lag, and cost proxy charged against pre-trade NAV restrict us from concluding about trading?

21.4 Amount ledger and research judgment

Drifted weights explain trading, and the simultaneous cost equation checks conservation of funds; descriptive volatility and an interval for a mean difference answer different questions. 21-L09/L10 correspond in full to body questions 21-R01/R02, respectively.

Definition 21.4 *(Exact model for proportional costs at a common rate).* Given pre-trade weights w^+, post-cost target weights w, and $q = c\sum_i |(1-q)w_i - w_i^+|$. Project 3 actually uses $q \approx c\sum_i |w_i - w_i^+|$; neither expression is evidence of live-market slippage.

Your turn. 21-L09: Conservation of amounts (corresponds to 21-R01)

Old weights are 55/45, NAV is 1 million yuan, and c=10 bp. For targets of 50/50 and 100/0, calculate exact costs, buy and sell amounts, and the old-NAV turnover proxy. Where does the difference between sales and purchases go?

Your turn. 21-L10: Diagnosing the object (corresponds to 21-R02)

Classify in turn: (a) the fully invested domain with upper bounds is feasible but no portfolio has positive excess return; (b) a zero-risk portfolio has positive excess return; (c) whether the BL posterior-mean covariance can directly replace return risk; (d) minimum variance has low volatility, its unadjusted single mean-difference interval is below 0, the ERC Sharpe difference crosses 0, and W24 is a post hoc exploration. For each item, state what can and cannot be concluded.

Chapter 21 Answers: 21-L01 and 21-L02

21-L01

Equal contributions, after canceling the common positive volatility denominator, give $0.04w_1^2 + 0.012w_1w_2 = 0.012w_1w_2 + 0.09w_2^2$. The cross terms cancel, and since the weights are positive, $0.2w_1 = 0.3w_2$; combined with the budget of 1, the weights are 60%/40%. $\Sigma w = (0.0288, 0.0432)$; the two variance contributions are both 0.01728, total variance is 0.03456, volatility is 0.185903, and the two absolute contributions are each 0.092952, for relative shares of 50% each. For any $\rho \in (-1, 1)$ the cross terms cancel, so the weights do not change; total variance still contains $2w_1w_2\rho\sigma_1\sigma_2$. With three or more assets, the sum of cross-contributions between an asset and the remaining assets differs across assets, so inverse-volatility weights generally do not guarantee ERC. A contribution is negative when w_i and MRC have opposite signs; when both are negative the contribution is positive, so a short position is not automatically a risk hedge.

21-L02

$$\nabla f = \Sigma y - b/y, \quad H = \Sigma + \operatorname{diag}(b_i/y_i^2), \quad y_i(\Sigma y)_i = b_i.$$

Division is componentwise. Summing the first-order conditions gives $y^\top \Sigma y = 1$. Therefore $w_i(\Sigma w)_i = b_i/s^2$ and $w^\top \Sigma w = 1/s^2$; only their ratio yields $p_i = b_i$. The two-asset example can be checked using $y = (0.6, 0.4)/\sqrt{0.03456}$. Positive semidefinite Σ plus the strictly positive diagonal term makes H positive definite, so there can be at most one minimizer; existence needs a separate argument. $\Sigma \succ 0$ is sufficient: the quadratic term controls the objective at infinity, while the log barrier prevents a coordinate from reaching zero. If there is a nonzero nonnegative zero-risk direction v, then along $y_0 + tv$ the quadratic term is unchanged, while at least one negative log term tends to negative infinity; the objective has no minimizer. Singularity does not always rule out existence: for example, the all-ones 2×2 matrix can have a unique minimizer in the positive cone. The numerical method solves $H\Delta = -\nabla f$ and uses backtracking to keep y positive and satisfy Armijo descent. The Newton decrement is not a general global error bound; after termination, also check the budget-share residual. Mathematical uniqueness, numerical convergence, and trading advantage cannot substitute for one another.

Chapter 21 Answers: 21-L03 and 21-L04

21-L03

$X^\top X/4$ gives $S = \left(\begin{smallmatrix} 7.5 & 6.5 \\ 6.5 & 6.5 \end{smallmatrix}\right)$ (percentage squared), and $m = 7$. The sum of the four squared entries of $S - 7I$ is 85, so $d^2 = 85/2 = 42.5$.

$$\sum_t \|x_t\|^4 = 13^2 + 5^2 + 13^2 + 25^2 = 988, \qquad \sum_{ij} s_{ij}^2 = 183.$$

$\bar{b}^2 = (988 - 4 \cdot 183)/(4^2 \cdot 2) = 8$; truncation does not change it, and $\hat{\kappa} = 8/42.5 = 0.188235$. $S^* = \left(\begin{smallmatrix} 7.405882 & 5.276471 \\ 5.276471 & 6.594118 \end{smallmatrix}\right)$; the correlation falls from $6.5/\sqrt{7.5 \cdot 6.5} = 0.930949$ to 0.755052.

When the data are changed to decimals, X is divided by 100; S, the shrinkage matrix S*, and m are divided by 100^2, while the fourth-order quantities $d^2, \bar{b}^2$ are divided by 100^4. The intensity and correlation are unchanged. Do not mix a covariance matrix in percentage-squared units with decimal returns. The Pythagorean expansion of theoretical loss takes $E[S] = \Sigma$ as its basis; with demeaning by the sample mean and divisor T, the finite-sample expectation is $(T - 1)\Sigma/T$. Do not present the asymptotic argument as an unconditional finite-sample identity.

21-L04

The portfolio gross return is 0, so the NAV denominator is 1 and $w^+ = (0.55, 0.45)$. Returning to 50/50 requires selling 5% and buying 5%, for total turnover of 10%; the rate 0.001 times 0.10 is 0.0001, or 1 bp of pre-trade NAV. Treating the target as the pre-trade weights would report zero turnover and omit the entire cost. First record when the information actually becomes available, then form the target, record when it can be traded, and only then calculate the holding-period return. A full month's return observed at month-end cannot be assumed to have been calculated and traded before that same close; zero-month lag is only an idealized month-boundary test. Data published later must be lagged according to receipt time, and the self-financing cost scaling must also be made explicit; this exercise uses only a proxy charged against pre-trade NAV.

Chapter 21 Answers: 21-L05 and 21-L06

21-L05

$\Sigma w_{mkt} = (0.032, 0.048)$, and multiplying by 2.5 gives $\Pi = (0.08, 0.12)$. $K = 0.05\Sigma$, $PKP^\top = 0.002 = \Omega$, and the gain is $KP^\top/0.004 = (0.5, 0.25)^\top$. The difference between the view and prior is 0.02, so the update is $(0.01, 0.005)$ and the posterior is $(0.09, 0.125)$. $\Sigma^{-1}\mu_{BL}/2.5 = (0.7, 0.4)$. The mean update for asset 2 comes from covariance, while the weight increment is $\Sigma^{-1}KP^\top$ times a scalar and therefore points along $P^\top$; only asset 1 changes.

The weights sum to 1.1, requiring a cash weight of -0.1, or 10% borrowing; the model does not impose a budget of 1. If borrowing is not allowed, solve the portfolio problem again with a budget, long-only, or upper-bound constraint; do not mechanically normalize the two weights and still call them the same optimum. Multiple views, nonorthogonal views, or constraints change this special structure.

21-L06

$G'(f) = p/(1+f) - (1-p)/(1-f) = 0$ gives $f^* = 2p - 1 = 0.10$. The exact values are $G(0.1) = 0.00500837$ and $G(0.05) = 0.00375261$, a ratio of about 74.9268%; only the quadratic proxy is exactly 75%. $G(0.2) = 0.55 \ln 1.2 + 0.45 \ln 0.8 = -0.000137741$, so the double-sized position already makes expected log growth negative; under the stated gamble repeated independently without bound, wealth tends to zero almost surely. This is not a universal claim about arbitrary financial assets.

The monthly proxy is $g(w) - r_f \approx we - \frac{1}{2}w^2\sigma^2$, where $e = 0.008$; the optimum is $w^* = e/\sigma^2 = 3.2$ times leverage and the growth proxy relative to cash is $0.0256 - 0.0128 = 0.0128$. Half leverage, at 1.6 times, gives $0.0128 - 0.0032 = 0.0096$, exactly 75%; the proxy variance is 25% of the full position's.

This omits the squared-mean term; returns can cross the bankruptcy boundary, and the unknown rf also prevents calculation of absolute growth. It cannot establish exact log utility. Real use also requires addressing estimation error, fat tails, financing, margin, and jumps; reducing the position does not guarantee a profit.

Chapter 21 Answers: 21-L07 and 21-L08

21-L07

$w = (0.5, -0.5, 0.5, -0.5)$ has zero net weight in both sectors, but $\beta^\top w = -1$. The constraint matrix

$$C = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

has its first row equal to the sum of the other two, so $CC^\top$ is singular; remove a redundant row or use independent constraints. For sector neutrality relative to a benchmark, use active weights $a = w - w_b$: net active weight within each sector is zero. A fully invested benchmark means actual weights sum to 1; it does not mean actual net dollar exposure is zero. The constrained maximum excess return assigns 0.4 to the only positive-excess asset and the remaining 0.6 to a negative-excess asset, giving $0.4(0.004) + 0.6(-0.004) = -0.0008$. The budget simplex is feasible, but no positive-excess portfolio exists; the prerequisite for the reciprocal-square QP transformation of positive Sharpe fails. The solver rejects an input to that transformation, not all portfolio weights as infeasible.

21-L08

Check finite values, positive weights, a budget sum of 1, positive volatility, and the original risk-share residual $\max_i |p_i - b_i|$; solver status cannot replace these checks, and a passing test does not establish future performance.

The ERC-difference interval crosses 0, so the current sample provides no credible evidence of a Sharpe advantage. The single unadjusted interval for minimum variance is below 0 and supports reporting evidence of a return loss for that main specification; 3 strategies × 2 metrics were not adjusted for multiple comparisons, so this cannot be generalized to statistical significance for the strategy family. Report net returns, drawdown, turnover, and economic loss alongside p-values.

The French industry series is a currently revised historical snapshot, not a point-in-time stock-level universe; zero-month lag lacks actual receipt and execution logs; the simple cost proxy does not cover financing, stock borrow, market impact, or capacity. Project 3 supports a workflow and an out-of-sample comparison for a particular specification; it cannot establish that a real fund could trade or has a reproducible advantage. Define data and cost requirements in advance and assess deployment only after filling those gaps.

Chapter 21 Answers: 21-L09 and 21-L10

21-L09

50/50: $q = 0.0001$, cost is 100 yuan, and the new amounts are 499,950 each; sell 50,050 and buy 49,950, with the difference of 100 paying the cost, for total trading of 100,000. The proxy happens to match in this example's symmetric structure.

100/0: $q = 0.001(0.90 - q) = 0.0009/1.001 = 0.0008991009$. The cost is 899.1009 yuan, and the new amount in asset 1 is 999,100.8991; buy 449,100.8991 and sell 450,000, with the difference of 899.1009 paying the cost, for total trading of 899,100.8991. The proxy gives 900,000 traded and a cost of 900, which is 0.8991 more. Conservation of funds does not establish actual slippage, market impact, or capacity.

21-L10

(a) The budget domain exists, but that does not imply the positive-Sharpe transformation applies; fall back to EW and retain the reason. (b) The finite-Sharpe denominator is invalid; do not read a QP objective of 0 as a finite ratio. (c) V_{BL} measures uncertainty about the mean; under the independent new-return-noise model, predictive covariance is $\Sigma + V_{BL}$, while the original allocation penalizes Σ. Changing the objective requires solving again.

(d) Volatility/drawdown are sample descriptions and have no corresponding difference interval; a single unadjusted interval for a mean difference cannot establish significance for a family of tests or for volatility; an ERC interval crossing 0 proves neither equivalence nor a probability of advantage. W24 was proposed after viewing the main results, is not among the 12 frozen sensitivity groups, does not replace W60, and does not provide independent validation.

Project 3 —Rolling Portfolios and Out-of-Sample Evaluation

> Project 3 links the stable estimation, long-only constraints, risk budgeting, and drift costs of Chapter 21 into one rolling evaluation: each month uses only the preceding window to estimate, holds for one month, and then computes turnover and cost from drifted weights. Questions P3-01 to P3-06 match `editions/en/projects/project03/exercises.md`, with unchanged meaning and numbers; P3-L01 is new in these notes. Prerequisites: Chapters 20 and 21.

The time contract of rolling evaluation

□□□□□

> The historical report covers French 10 industries over 1990–2026, 440 holding months, and four strategies with 1,760 ledger rows; the main specification uses a 60-month window, Ledoit–Wolf covariance, a 40% single-name cap, lag 0, and a 10 bp cost. The default Notebook runs only the constructed branch: seed 103 generates 48 months of independent normal monthly returns for four assets A/B/C/D from 2000-01 (mean 0.005, standard deviation 0.03), RF is constant at 0.001, the first 24 months form the first training window, and holdings run from 2002-01 to 2003-12.

Trigger. Equal weight has identical metrics under lag 0 and lag 1. Does that show that "when the data are available" does not matter?

Definition P3.1.1 *(One step per month).* In holding month h, the training window contains only months before h: every row must satisfy `train_end` _____ `holding_month`. Equal weight uses $1/N$; Minimum Variance and Maximum Sharpe obey the single-name cap; ERC equalizes each asset's ex ante risk contribution.

Definition P3.1.2 *(Drift, turnover, and the cost proxy).* Last month's target w drifts after this month's returns r to

$$\tilde{w}_i = \frac{w_i(1+r_i)}{1+R_p}, \qquad R_p = \sum_i w_i r_i.$$

Moving to the new target w^* at the start of the next month gives turnover $\sum_i |w_i^* - \tilde{w}_i|$ (total buys and sells, _____). Cost = c×turnover, and net return = (1−cost)(1+gross)−1. This is a cost proxy based on normalized weight changes; it _____ the self-financing transaction-amount equation. The first month starts from cash, so the turnover proxy is ____.

Lag, intervals, and post hoc exploration

Proposition P3.2.1 *(What a lag test measures).* Equal weight uses no estimate, so lag changes only training labels, not weights; Minimum Variance relies on a covariance estimate, and lag 1 shifts its window one month earlier, slightly changing weights (main-specification net Sharpe 0.596 versus 0.597). A lag test measures only the _____ of

> conclusions to the assumption that data are available at month-end; it cannot turn revised data into _____.

Reading the bootstrap intervals

The historical report uses 5,000 paired circular block-bootstrap draws with 12-month blocks and compares three strategies by two metrics without multiple-comparison correction. An interval that crosses zero _____ equivalence, and a failed screen is not a formal joint-test rejection.

Pre-specified specifications versus later attempts

The 12 pre-specified specifications were written before results were seen. A window, cap, or covariance choice proposed after the main results can be labeled only as "post hoc exploration, not pre-registered," reported separately, and never substituted for the main specification.

> **Your turn.** Cover this section. Write the drift formula; why turnover is not divided by 2; and what a lag test can and cannot show.
>
> **Recall:** If drift is forgotten, what turnover is computed?

Exercises: cover the answers and recompute

P3-01, P3-03, P3-04, and P3-06 can be answered using the inputs and historical tables provided in this book. P3-02 requires the historical `primary_ledger.csv`, and the historical calculation in P3-05 requires French inputs; neither is generated by the default constructed branch. If these materials are unavailable, read the complete historical reference solutions; there is no need to start the market-data branch.

Your turn. P3-01 (Manual ledger calculation) At the start of month 1, a two-asset portfolio has target weights (0.6, 0.4). The assets return +5% and −10% in month 1. At the start of month 2, the target weights are again (0.6, 0.4), with a cost of 25 bp per unit of turnover. The assets return +2% and +1% in month 2. Calculate the drifted weights at the end of month 1, turnover at the start of month 2, cost, and gross and net returns in month 2.

Your turn. P3-02 (Ledger identities) Read `outputs/primary_ledger.csv`. For every row, verify cost $= 0.001 \times$ turnover and net $= (1 - \text{cost})(1 + \text{gross}) - 1$. Also verify that, for each strategy, `train_end` in each of two consecutive months precedes `holding_month`.

Your turn. P3-03 (Why lag does not affect equal weight) In `sensitivity.csv`, the equal-weight metrics are identical under lag 0 and lag 1, while Minimum Variance differs slightly. Explain why, and what this implies for conclusions about the importance of the timing assumption.

Your turn. P3-04 (Interpreting intervals) ERC's annualized net Sharpe is 0.016 higher than equal weight, with a 95% block-bootstrap interval of [−0.014, 0.050], while the mean net-return difference is −0.38% per year. Can you say that ERC outperforms equal weight? Write wording for a conclusion you would accept.

Your turn. P3-05 (Unregistered exploration) Someone proposes changing the window to 24 months "to see if it gets better." Use the functions in `src/project03_backtest.py` to run Maximum Sharpe with $W = 24$ and all other main-specification parameters unchanged. Report annualized net Sharpe and explain how this result should be labeled in the report.

Your turn. P3-06 (Research design) If this project were to become a tradable study, list the data, timing, costs, and evaluation elements that must be replaced or added, and give one piece of evidence that would overturn the claim that "ERC reduces risk concentration without sacrificing returns."

Your turn. P3-L01 (New) Checking the constructed ledger. In the default constructed branch, each of the four strategies holds for 24 months. How many ledger rows exist at completion? What is the turnover proxy in the first month, which starts from cash, and why? Can the constructed $W = 24$ Maximum Sharpe result be called a replication of the historical post hoc $W = 24$ result (net Sharpe about 0.587, 25 fallbacks)?

Project 3: Questions and Complete Answers

P3-01

(Complete reference code and its historical output for P3-01, P3-02, and P3-05 are in `editions/en/projects/project03/solutions.md`; here we give the derivations and conclusions.) Month 1 gross return is $0.6(0.05) + 0.4(-0.10) = -0.01$. The drifted weights are $w_i(1+r_i)/(1+R_p)$: $0.6 \times 1.05/0.99 = 0.636364$, $0.4 \times 0.90/0.99 = 0.363636$. At the start of month 2, the weights are reset to (0.6, 0.4), so turnover is $|0.6 - 0.636364| + |0.4 - 0.363636| = 0.072727$ (total buys and sells, not divided by 2). This problem uses a cost proxy based on normalized weight changes and does not solve the post-cost self-financing transaction-amount equation. Cost is $0.0025 \times 0.072727 = 0.000182$. Month 2 gross return is $0.6(0.02) + 0.4(0.01) = 0.016$, and net return is $(1 - 0.000182)(1.016) - 1 = 0.015815$. If you forget drift and assume the weights are still (0.6, 0.4), you will calculate turnover as 0 and omit the cost.

P3-02

All three assertions pass, and the ledger contains 1,760 rows (4 strategies × 440 months). The first confirms a cost rate of 10 bp; the second confirms that cost is deducted from NAV at the trade and then applied to the holding month's growth; the third confirms that no row is trained on data from its holding month or later. (Historical run of 2026-10-06; not rerun in this round.)

P3-03

Equal weight uses no estimate from the training window, and its target weights are always 1/10, so lag changes only the `train_start`/`train_end` labels, not weights, turnover, or returns. Minimum Variance uses an estimated covariance matrix; lag 1 shifts its monthly window one month earlier, slightly changing the weights (main-specification net Sharpe 0.596 versus 0.597). This shows that the project's conclusions are insensitive to the assumption that data are available at month-end; it does not prove that the data were available then. A lag test measures sensitivity to the assumption and cannot turn revised data into historical vintages.

P3-04

You cannot say that ERC outperforms equal weight. The Sharpe-difference point estimate is positive, but its interval crosses zero; the mean net-return difference is negative, and its interval also crosses zero. The analysis compares three strategies and two metrics, without correcting for multiple comparisons. Acceptable wording: "In the idealized monthly backtest of French 10 industries over 1990–2026, this sample did not detect a net Sharpe advantage for ERC over equal weight (this is not proof of return equivalence) (difference 0.016, 95% block-bootstrap interval −0.014 to 0.050). ERC allocated the ex ante risk share evenly across industries, while equal weight had an average maximum ex ante risk share of 15.9%." This is a difference in risk structure, not an advantage in returns.

P3-05

For $W = 24$, Maximum Sharpe has annualized net Sharpe of 0.587, higher than the main specification's 0.558 at $W = 60$, but: (1) in 25 of 440 months, it fell back to equal weight because no positive-excess-return portfolio was feasible under the 40% cap or all industry excess means were negative; some of the "improvement" comes from holding equal weight; (2) average monthly turnover was 28.7%, well above $W = 60$'s 21.5%, and Sharpe fell faster as the cost tier increased; (3) this window was proposed only after the main results had been seen. The report should label it "post hoc exploration, not pre-registered," separate it from the 12 pre-specified specifications,

and not use it to replace the main specification. A formal test would require evaluating it on a new data segment or a pre-specified set of windows, with a correction for multiple attempts. (Historical run; not rerun in this round.)

P3-06

Replace or add: data—historical total returns and prices for tradable instruments (for example, industry ETFs or replicable constituent portfolios), with timestamped historical versions, recording delistings and constituent changes; timing—at each month-end, the information set contains only data published by then, using an available closing or opening price and recording execution delay; costs—bid-ask spread, commissions, market impact (as a function of trade size and liquidity), plus ETF management fees, borrowing fees if shorting, and cash drag; evaluation—a pre-specified out-of-sample period, turnover and capacity, net returns, drawdown, concentration, block bootstrap or other inference robust to dependence, and correction for multiple comparisons.

Evidence that could overturn the claim that "ERC reduces risk concentration without sacrificing returns" includes: in a pre-registered tradable sample, ERC's ex post risk shares (calculated from realized returns) remain concentrated in a few industries; or the block-bootstrap interval for its mean net return relative to equal weight lies entirely below zero (for example, an upper bound below −0.5% per year); or its after-cost Sharpe is significantly lower than equal weight.

P3-L01

Four strategies times 24 rows gives 96 rows. In the first month the account starts from cash, so all pre-trade weights are 0 and moving to the target gives total buys and sells $\sum_i |w_i^* - 0| = 1$; the turnover proxy is 1. The constructed $W = 24$ uses seed-103 constructed returns for four assets, while the historical post hoc $W = 24$ uses French 10 industries over 440 holding months. Their inputs, windows, and evidence identities differ, so the constructed result is not a replication of the historical one and cannot update or overturn it.

Fill-in key for the reading sections

precedes; not divided by 2; does not solve; 1. Lag: sensitivity; historical vintages. Bootstrap: does not prove.

Chapter 22 —Quantitative Research Methodology

Write down and freeze the hypothesis, data, signal, costs, and evaluation period before seeing the results. Log every attempt. The unit for multiple-testing correction is the number of attempts, not the number of results reported. Comparing content with a trusted saved digest supports content identity under collision-resistance assumptions; a local hash does not establish that writing preceded analysis.

Trigger. The best of five variants has $p = 0.005$. Can you call it significant?

Definition 22.1 *(Family-wise error rate and thresholds).* Under the global null (all N null hypotheses are true), if tests are independent and each has exact rejection probability α, the probability of at least one false rejection is _____. Bonferroni multiplies each p-value by _____ and compares it with α. Holm sorts the p-values in ascending order, uses $(N-k+1)p_{(k)}$, and takes the cumulative maximum. Under independence or suitable positive-dependence conditions, Benjamini–Hochberg controls the _____, not the family-wise error rate.

Your turn. 22-L01 Given four p-values $(0.01, 0.04, 0.03, 0.005)$, write the Bonferroni-adjusted values _____, Holm-adjusted values _____, and BH-adjusted values _____. At the 5% level, how many rejections does each method make?

Your turn. 22-L02 For 10 independent tests, each at $\alpha = 5\%$, the probability of at least one false rejection is _____. The two-sided Bonferroni z threshold is about _____ (the single-test threshold is 1.96). If the largest signed value, $\max Z$, is selected from 10 independent standard normal statistics, how much higher is its expected value than the single-test expectation? Is this the same object as the two-sided selection $\max |Z|$?

Your turn. 22-L03 A monthly Sharpe ratio is 0.15 over 120 months. The naive $t =$ _____ . Why can you not use this t directly after looking at 32 variants?

From hypothesis to deliverable

An economic mechanism gives the direction; point-in-time inputs provide measurable features; same-date transformations produce signals; and a weighting rule produces holdings. Predictive association and the portfolio after costs are two different evaluation objects.

Your turn. 22-L04 (Further practice) How can the economic hypothesis for high B/M be translated into a feature, a signal, a forecast, and a portfolio? Give one output for each, and say whether a percentile is already a return forecast.

Your turn. 22-L05 (Further practice) The five D01–D05 attempts in Section 5.1 have $p = (0.01, 0.04, 0.03, \text{NaN}, 0.005)$. Complete the full testing family, write the Holm sorting multipliers and adjusted values in the original row order, and identify the rejections at 5%. Can D05 retrospectively be called the prespecified primary specification?

Your turn. 22-L06 (Further practice) After reading the held-out result in H01, you change the horizon and create a new file hash. Explain what content binding, a third-party record of timing, and an exposure log can each establish. What is the status of this evaluation?

Chapter 22 answers and mapping

Answers to 22-L01–L03 and the definition $1-(1-\alpha)^N$; N; the false discovery rate. Bonferroni: $(0.04, 0.16, 0.12, 0.02)$; Holm: $(0.03, 0.06, 0.06, 0.02)$; BH: $(0.02, 0.04, 0.04, 0.02)$. At the 5% level, Bonferroni rejects 2, Holm rejects 2, and BH rejects 4. $1-0.95^{10} = 0.401$; $z \approx 2.807$. The expected value of the largest signed statistic, $\max Z$, among 10 independent standard normals is about 1.54 (exact); the expectation of one statistic is 0. Thus the increase is about 1.54. The two-sided maximum $\max |Z|$ is a different object; do not use 1.54 as its two-sided threshold. $t = 0.15\sqrt{120} = 1.64$. The Bonferroni threshold for 32 variants is about 3.16, and this statistic has not been adjusted for selection; comparing it with a single-test threshold implicitly treats it as one prespecified test (Chapter 22, Section 8).

22-L04 The input is the same-version book equity and market value that were public and ingested before the cutoff. The feature is dimensionless B/M; the signal is its same-date percentile; the primary forecast is the direction or ordering that higher signal predicts higher returns next month; and the portfolio buys the high bucket and sells the low bucket under fixed constraints. A percentile is not a return value. A numerical forecast requires an estimated and frozen mapping. IC does not replace portfolio or fee evaluation.

22-L05 Complete the family as $(0.01, 0.04, 0.03, 1, 0.005)$, with $M = 5$. The order is D05, D01, D03, D02, D04; the multipliers are 5, 4, 3, 2, 1; the products are 0.025, 0.04, 0.09, 0.08, 1; and their cumulative maxima are 0.025, 0.04, 0.09, 0.09, 1. In the original row order, the Holm-adjusted values are $(0.04, 0.09, 0.09, 1, 0.025)$, so D01 and D05 are rejected. D04 preserves the size of the attempt family; it is not a valid test. D05 is exploratory after the fact. If the discovery-window selection algorithm was defined in advance, a holdout can evaluate the selected specification, but it cannot change the specification's primary-hypothesis status.

22-L06 A comparison with a trusted saved digest supports normalized-content identity under collision-resistance assumptions; it does not rule out edits followed by restoration. A third-party timestamp supports registration chronology, and an exposure log records which results were seen. Changing a specification after H01 is a new attempt. It cannot be relabeled as an unexposed holdout. Keep the original ledger row and record the change in the research log. Even if labels were isolated, financial time dependence still needs to be checked.

Mapping: The original three unnumbered questions retain their intent as L01 (Sections 6.2–6.4; a local numerical variation, not the full E04 exercise), L02 (Sections 6.1/6.5; an independent hand calculation), and L03 (Sections 6.7/13; a local calculation). L04 follows the reader's chain in Section 3.1. L05 maps to the Holm portion of R01 as a local variation (R01 also covers Bonferroni and BH). L06 maps to Section 5.1 and local research judgments in E01/E07. The definition blanks are answered in the opening answer block.

Chapter 23 —Cross-Sectional Research

The basic unit of cross-sectional research is a cross section at a given date: rank within the same date, compute same-date IC, and form same-date quantiles. Pooling different dates into one correlation mistakes time-series differences in levels for predictive power. Keep the number of valid dates separate from the number of valid asset pairs. For overlapping models, state the conditions behind the effective date count.

Trigger. If the correlation between signals and returns pooled across all dates is 0.89, is the IC also 0.89?

Definition 23.1 *(Rank IC and average ranks).* Same-date Rank IC is the _____ correlation between signal ranks and return ranks. Ties receive _____ ranks. Compute percentiles using the full number of valid signals. For IC, first keep only the _____ valid on both sides, then rerank each side separately. The mean aggregates valid _____, not asset pairs.

Your turn. 23-L01 For signal $x = (3, 1, \text{NaN}, 3, 2, 7, 5, 5)$, write the average ranks _____ and the percentiles for $n = 7$. What ranks do each of the two 3s and the two 5s receive?

Your turn. 23-L02 Three Fama–MacBeth slopes are $0.014, 0.007, 0.013$. Their mean is _____; the naive standard error $s/\sqrt{T} =$ _____ (use the sample standard deviation for s); and $t =$ _____. Are the three date-specific slopes independent? What does this imply for t?

Your turn. 23-L03 In the zero-world, large-cross-section model in Section 6.8, use the long-run approximation at the boundary of a perfectly persistent signal ($\rho = 1$). With 120 months and a 3-month label, the effective number of dates is _____. By approximately how many times would a naive t based directly on 120 months be overstated?

From asset pairs to valid dates

First form quantiles among the available signals. For evaluation, pair observations first, rerank, and then compute one IC per date. Aggregate only dates with a defined IC. Missing months matter to the HAC interface.

Your turn. 23-L04 (Further practice) In Section 3.1, the January signal is $(1, 2, 3, 4, \text{NaN})$ and the label is $(0.01, \text{NaN}, 0.03, 0.02, 0.04)$. Give the signal count, pair count, percentiles based on all signals, two bucket assignments, the two sets of within-pair ranks, and Rank IC. If February's signal is constant and March's IC is -1, what are the number of valid dates and the time-series mean?

Your turn. 23-L05 (Further practice) For valid ICs $(0.5, -1)$, calculate the $ddof = 1$ standard deviation, the ordinary mean SE, and the Newey–West mean variance for $L = 0$. After January and March are compressed into adjacent observations, does HAC lag 1 mean one calendar month? Which dates must also be reported?

Your turn. 23-L06 (Further practice) In the constructed model in Section 6.8, what extra pair-count factor appears in the finite-T VIF relative to the long-run formula? For $T = 120, h = 3, \rho = -0.5$, calculate the long-run VIF, finite-sample VIF, and equivalent number of dates. Why is "serial correlation makes the SE too small" not generally true?

Your turn. 23-L07 (Further practice) In the two-date panel in Section 7.1, explain how the pooled correlation of 0.891, mean IC of 0, and standardized FM slopes of ± 0.02236 can all hold at once. The gross value of z/N is 0.89443. By what factor must it be scaled to reach gross 2? Which number has already been netted of costs?

Your turn. 23-L08 (Further practice) Sector 1 weights are $(0.3, -0.3)$ and sector 2 weights are $(0.2, -0.2)$. After the second name becomes untradeable and is set to zero, write the final sector exposures, net, and gross. Would scaling back to gross 1 restore neutrality? What else should a complete P4 check?

Your turn. 23-L09 (Further practice) For P102, the 2013 financial report became public on April 17 and was ingested on April 18. The April signal cutoff is May 1 at 00:00 UTC. A restatement became public on May 20 and was ingested on May 21. From when can it be used? Does the current May month-end label begin at the actual first tradable price?

Chapter 23 answers and mapping

Answers to 23-L01–L03 and the definition Pearson (equivalently, Spearman correlation of the original paired values); average; pairs; dates. The average ranks are $(3.5, 1, \text{NaN}, 3.5, 2, 7, 5.5, 5.5)$, and the percentiles are $(0.417, 0, \text{NaN}, 0.417, 0.167, 1, 0.75, 0.75)$. The mean is 0.01133, the standard error is 0.002186, and $t = 5.18$. If the net autocorrelation contribution is positive, the naive SE is too small and $|t|$ is too large; negative dependence can reverse this (use Newey–West or a block method). $120/3 = 40$. At this boundary in the model's long-run approximation, the naive $|t|$ is overstated by about $\sqrt{3} = 1.7$. The finite 120-period VIF is $3 - 8/(3 \times 120) = 2.97778$, giving 40.30 equivalent dates. Do not apply T/h directly to a general model.

23-L04 $N_{\text{signal}} = 4$ and $n_{\text{pair}} = 3$. The ranks among all signals are $(1, 2, 3, 4)$; their percentiles are $(0, 1/3, 2/3, 1)$; and their bucket assignments are $(1, 1, 2, 2)$, with the missing observation retained. For paired names A/C/D, the signal ranks are $(1, 2, 3)$ and the label ranks are $(1, 3, 2)$, giving Rank IC 0.5. Using stale ranks computed before dropping unpaired names would give about 0.655. February's constant signal has undefined IC; March's is -1. Thus $T_{\text{valid}} = 2$ and the mean is -0.25. Imputing zero would change the estimand.

23-L05 The deviations are ± 0.75, their squared sum is 1.125, and $s = \sqrt{1.125} = 1.06066$, so the ordinary mean SE is 0.75. For Newey–West with $L = 0$, dividing by $T = 2$ gives $\gamma_0 = 0.5625$ and mean variance 0.28125; the ordinary mean variance is 0.5625. A consistent $ddof$ factor cancels in a cross-sectional correlation, but the finite-sample factor does not cancel in a time-series SE. Compressed lag 1 is one step between valid observations, while January to March spans two months. Report the calendar span, the missing dates and their reasons, and calendar dates absent beyond the intersection of the input dates. Do not treat the API's missing-count field as a complete calendar audit.

23-L06 Each finite-period lag term gets an additional factor $(1 - j/T)$ because there are only $T - j$ pairs. The long-run VIF is 0.5. With finite pair counts:

$$\begin{aligned} \text{VIF}_{120} &= 1 + 2\left[\frac{119}{120}\left(-\frac{1}{2}\right)\frac{2}{3} + \frac{118}{120}\frac{1}{4}\frac{1}{3}\right] \\ &= 0.50278. \end{aligned}$$

The equivalent number of dates is 238.67. Negative correlation offsets variation in the mean, so here the naive SE is too large. For $\rho = 0.85$, the old long-run approximation is 2.615/45.9; the finite VIF is 2.59753 and the equivalent count is 46.20. These results belong to the stated constructed autocorrelation model; they are not a real finite-cross-section distribution.

23-L07 The pooled correlation includes co-movement in date means, while IC compares observations only within each date. Standardized FM slopes also incorporate the within-date dispersion of the label. The raw slopes are ± 0.02; the signal standard deviation is 1.118, giving standardized slopes of ± 0.02236. Scaling by $2/0.89443 = \sqrt{5}$ gives gross returns of ± 0.05. None of these values is net of costs. A target-weight cost proxy must separately subtract $c \sum |\Delta w|$; actual trades, financing, and benchmark risk attribution still require their own ledger.

23-L08 The final sector exposures are $(0.3, 0)$, net is 0.3, and gross is 0.7. Multiplying by $10/7$ restores gross to 1, but leaves sector 1 and net at $3/7$. Reconstruct weights on the final tradable universe and check sector, net, gross, single-name caps, cash budget, and all prespecified constraints. The A5 preparation, A6 construction ledger, and A7-B cost lesson do not amount to a complete real P4 empirical study.

23-L09 The original version is available for April: both the public-release and ingestion dates are strictly before the May 1 cutoff. The restatement becomes available starting with the May signal; it does not revise April history. The month-end label runs from the end of April to the end of May and includes a period before actual trading. An execution label must start at the actual fill price; a whole-month lag cannot fabricate intraday prices or a trading calendar.

Mapping: The original unnumbered questions retain their intent as L01 (a local example from Section 8.1), L02 (the full slope-aggregation calculation in Section 8.7), and L03 (a local boundary variation from Section 8.8, with added conditions and a finite-sample answer; this does not cover all of E08). L04 maps to the pairing and counting in R01. L05 maps to the inference interface in Section 5.1/R01. L06 covers R03. L07 maps to the portfolio portion of Section 7.1/R02. L08 maps to the constraint portion of Section 13.1/R02. L09 maps to the local timing diagnosis in Section 5.2/E01. The definition blanks are answered in the opening answer block.

Chapter 24 —Value Factors

> "Cheap" must pass three checks: the ratio's direction and capital claim must match, nonpositive denominators need an explicit rule, and statement versions must be available at the signal time. Every step can change the ranking and create a value signal that only appears effective.

Trigger. A ratio looks high: does it really mean the company is cheap, or has its denominator lost its economic meaning?

Check direction first, then ask whether the comparison is valid

B/M, E/P, and FCF Yield are yield measures; EV/EBITDA is a multiple. FCFE belongs to shareholders and is divided by market capitalization; FCFF belongs to all capital providers and is divided by enterprise value (EV). EV bridge items must use the same currency and scale and be known at the same signal cutoff; record timestamps separately for market and statement figures. All amounts in the examples below are constructed for teaching.

Same-direction news does not guarantee higher correlation: normalization also changes. Section 6.5 gives a counterexample. The calendar-IC axis is signal month, with next-month returns as labels.

Your turn. 24-L01 (old question: direction and matching) Fill in the blanks: B/M and E/P are _____-form measures (larger means cheaper); EV/EBITDA is a _____-form measure (smaller means cheaper). Under this chapter's rule, nonpositive book equity makes B/M _____. Cash flow to equity holders is divided by _____; cash flow for the whole firm is divided by _____.

Your turn. 24-L02 (old question: Orion and Lyra) Orion: market cap 600, EV 650, book equity 180, net income 30, EBITDA 75; Lyra: market cap 480, EV 750, book equity 120, net income 20, EBITDA 75. Calculate B/M, E/P, and EV/EBITDA for each company. Which one is cheaper under each of the three measures?

Your turn. 24-L03 (old question: reporting period and filing date) For this question, assume each version is ingested on its publication date. Version table: FY2021 book equity 80 (published 2022-03-15); FY2022 book equity 100 (published 2023-03-15); FY2022 restated to 90 (published 2023-11-20); FY2023 book equity 120 (published 2024-03-12). Which value is used at the month-ends 2023-02, 2023-03, 2023-10, 2023-11, 2024-02, and 2024-03? If `period_end` were treated as the availability date, which value would be used incorrectly at the end of 2023-01?

Your turn. 24-L04 (new: EV bridge and cash-flow matching) Equity market capitalization $M = 600$, debt 200, cash 100, noncontrolling interests 20, preferred equity 10, and nonoperating assets 30. Calculate EV using the complete bridge. If FCFF is 56, after-tax interest is 12, and net borrowing is 4, with no separate preferred-equity or noncontrolling-interest cash-payment adjustment in this simplified bridge, calculate FCFE, FCFF/EV, and FCFE/M. Now consider a net-cash company: $M = 100$, debt 20, cash 80, all other bridge items zero, and FCFF 8. Calculate EV and FCFF/EV. Must EV always be higher than market capitalization?

Your turn. 24-L05 (new: negative values and missing-value diagnosis) Assume three companies all have positive market capitalization. Company A has book equity −20, net income 3, EV 40, and EBITDA 2; company B has book equity 10, net income −3, EV 40, and EBITDA −2; company C has book equity 10, net income 3, EV −5, and EBITDA 2. For each company, should B/M, E/P, and EV/EBITDA retain a value or be NaN? Why should a negative P/E not be called "cheapest"? What should you report when excluding NaN values?

Your turn. **24-L06 (new: separate available and ingested cutoffs)** Version table (book-equity units are consistent): FY2021, 80, externally available 2022-03-15 and ingested 2022-03-15; first FY2022 version, 100, available 2023-03-15 and ingested 2023-03-16; FY2022 restatement, 90, available 2023-11-20 and ingested 2023-12-02; FY2023, 120, available 2024-03-12 and ingested 2024-04-03. Select the value used in 2023-02, 03, 10, 11, and 12 and in 2024-02, 03, and 04, requiring both `available_at` and `ingested_at` to be no later than month-end. If you check only the external availability date, in which two months would you adopt a version too early?

Your turn. **24-L07 (new: risk compensation or mispricing?)** Two companies have the same high B/M. One is deteriorating and has rising debt; the other's fundamentals are stable. Explain how risk compensation and mispricing would each explain the low price and subsequent return. Can B/M or one period of positive returns distinguish the explanations? Give one research-design feature that could help distinguish them.

Chapter 24 Answers

24-L01 Yield form; multiple form; NaN; market capitalization; EV. Under this chapter's rule, valid B/M observations require positive book equity. E/P may retain the sign of negative earnings when market capitalization is positive. FCFE/M and FCFF/EV match cash flow to shareholders and to all capital providers, respectively.

24-L02 Orion: B/M $= 180/600 = 0.30$, E/P $= 30/600 = 0.050$, EV/EBITDA $= 650/75 \approx 8.67$. Lyra: $120/480 = 0.25$, $20/480 \approx 0.0417$, $750/75 = 10$. All three measures say Orion is cheaper: its two yields are higher and its EV/EBITDA multiple is lower.

24-L03 The six month-ends use 80, 100, 100, 90, 90, 120, respectively. The FY2022 version is not published until 2023-03-15, so February month-end still uses FY2021. The restatement is not published until 2023-11-20, so October month-end still uses 100. If reporting-period end were treated as the availability date, the end of 2023-01 would use FY2022's 100 too early.

24-L04 The complete EV bridge is $600+200-100+20+10-30 = 700$. FCFE $= 56-12+4 = 48$; FCFF/EV $= 56/700 = 8\%$; FCFE/M $= 48/600 = 8\%$. The two matched yields happen to be equal in this example. The net-cash company's EV is $100 + 20 - 80 = 40$, below its market cap of 100; FCFF/EV $= 8/40 = 20\%$. EV can be lower than market capitalization. If FCFF in the first example were incorrectly divided by market cap, the result would be $56/600 \approx 9.33\%$, mixing capital-structure differences into the signal.

24-L05 A: B/M is nonpositive, so set it to NaN; E/P $= 3/M$ can retain a positive value; EV/EBITDA $= 40/2 = 20$. B: B/M $= 10/M$ can be retained; E/P $= -3/M$ retains its negative sign; EV/EBITDA is NaN because EBITDA is nonpositive. C: B/M and E/P can be retained; EV/EBITDA is NaN because EV is nonpositive. A negative P/E arises from negative earnings. The reciprocal is strictly decreasing on each side of zero, but the positive-value ranking cannot be carried across zero; ranking a negative P/E as an ordinary multiple would incorrectly call it cheap. NaN is not zero: report the count and share excluded for each measure.

24-L06 The values, in order, are 80, 100, 100, 100, 90, 90, 90, 120. Although the restatement was externally published in November, it was not ingested until December 2, so November still uses 100 and December uses 90. FY2023 was published on 2024-03-12 but not ingested until April 3, so March still uses 90 and April uses 120. If you check only `available_at`, you would adopt 90 too early in 2023-11 and 120 too early in 2024-03.

24-L07 Under risk compensation, the low price reflects worse or more uncertain fundamentals, and holders require higher expected returns for bearing describable systematic risk. Under mispricing, the market is too pessimistic about fundamentals, so the price may later recover toward them. B/M and one period of positive returns cannot identify the explanation. Preregister a prediction that can separate them, such as grouping by independently measured risk exposure and testing returns after risk controls, or testing for a price correction in the direction of fundamental news. Use point-in-time data and an out-of-sample test; do not label a result as risk or mispricing by itself.

Chapter 25 —Momentum

> A momentum signal is defined jointly by its time window, return convention, and holding rule. 12–1 uses the 11 complete months before the signal month and skips the most recent month; shift an endpoint, omit dividends, or stitch futures prices across contracts, and the result is a different quantity.

Trigger. The signal month is December 2025: which intervals are the formation period, skipped month, and next-month label?

Fix the clock first

At month-end s, the 12–1 formation period is $s-11, \dots, s-1$, 11 months in all; month s is skipped, and month $s+1$ is the holding-return month. Returns must correspond to the assets actually held. The numerical examples below are constructed for teaching.

Compounded futures price ratios assume a teaching index with full collateral at each segment's starting price, wealth-adjusted notional, and no fees or collateral interest. A fixed contract requires separate P&L and a capital denominator.

Your turn. 25-L01 (old question: window and overlapping holdings) Fill in the formation months for 12–1: _____. The vintage share replaced each month in a K-month overlapping-holdings portfolio is _____; if $B_s = \frac{1}{K}\sum_{j=0}^{K-1} w^{(s-j)}$, write the change $B_s - B_{s-1}$ between adjacent months.

Your turn. 25-L02 (old question: compounding and a shifted window) Fourteen monthly returns (percentage points) are $3, -2, 5, 1, -4, 6, 2, -1, 4, 3, -2, 5, -6, 2$, and the signal month is month 13. Calculate the compounded and arithmetic-sum returns for 12–1 (months 2–12), then calculate the compounded return for the window shifted one month later (months 3–13). Why is the latter not standard 12–1?

Your turn. 25-L03 (old question: TSMOM volatility scaling) Three assets have 12–0 signals $+, -, +$, ex ante annualized volatilities $10\%, 20\%, 40\%$, and target volatility 40%. Calculate the three weights by taking each position as $s_i(40\%/\sigma_i)$ and then dividing by $S = 3$; find gross and net exposure. If next-month returns are $1\%, -2\%, 3\%$, find the portfolio return.

Your turn. 25-L04 (old question: stitched futures-roll return) Contract A is priced at $100, 102$; after the roll, contract B is priced at $106, 107$. What are the period-by-period returns on the stitched sequence $100, 102, 106, 107$? What are the returns from actually holding A through the roll and then holding B? Identify the part that cannot be attributed to a holding.

Your turn. **25-L05 (new: check 12–1 against the calendar)** Signal month s = 2025-12; the following month is the holding/diagnostic-label month. The monthly returns in the table are percentage points of simple total return. Use the calendar to identify the 11 formation months for 12–1, and calculate the compounded return and arithmetic sum. Calculate the compounded return if the window is shifted one month later and mistakenly includes December. Which interval contains the −6% return in January 2026?

Month	Return (%)	Role
2025-01	3	Candidate formation month
2025-02	−2	Candidate formation month
2025-03	5	Candidate formation month
2025-04	1	Candidate formation month
2025-05	−4	Candidate formation month
2025-06	6	Candidate formation month
2025-07	2	Candidate formation month
2025-08	−1	Candidate formation month
2025-09	4	Candidate formation month
2025-10	3	Candidate formation month
2025-11	−2	Candidate formation month
2025-12	5	Signal month; skipped
2026-01	−6	Next-month label, not an input

Your turn. **25-L06 (new: three momentum questions)** In one sentence each, explain what XSMOM, TSMOM, and short-term reversal compare and how they set direction. Which is usually one long and one short? Which can be net long? Why should TSMOM 12–0 and XSMOM 12–1 not be described as the same signal?

Your turn. 25-L07 (new: total return and futures-roll information) A stock is priced at 100 before a 2-for-1 split; afterward, each share is priced at 51 and pays a cash dividend of 1. Calculate the unadjusted price return and total return including the split and dividend. If a futures backtest has only a continuous series made by joining prices, what holding or roll information is still needed before it can report an actual holding return?

Your turn. 25-L08 (new: a P4 label is not an execution) Project 4 uses monthly returns constructed from a known generating mechanism and labels the next month. Diagnose this call: does `formation_return(r,12,1)` implement 12–1, and if not, how should it be changed? Which signal-code properties can it validate? List at least three kinds of evidence that still require real data or execution records to support "tradable performance."

Chapter 25 solutions

25-L01 The formation months are $s-11,\dots,s-1$, 11 months in all. Each month replaces a vintage share of $1/K$; actual trading volume depends on the difference between old and new weights and is not guaranteed to fall by exactly $1/K$. Subtracting the two sums gives $B_s - B_{s-1} = (w^{(s)} - w^{(s-K)})/K$.

25-L02 The compounded return for months 2–12 is $\prod_{k=2}^{12}(1+R_k)-1 = 17.716\% \approx 17.72\%$; the arithmetic sum is 17.0%. The later window, months 3–13, compounds to $12.911\% \approx 12.91\%$. It includes the −6% in signal month 13 and omits month 2, so it is not the standard window from $s-11$ through $s-1$.

25-L03 Unnormalized positions are $(4,-2,1)$; after dividing by 3, weights are $(4/3,-2/3,1/3)$. Gross exposure is $|4/3|+|-2/3|+|1/3| = 7/3$, and net exposure is $4/3-2/3+1/3=1$. Portfolio return is $(4/3)(1\%)+(-2/3)(-2\%)+(1/3)(3\%) = 3.67\%$.

25-L04 Returns on the stitched series are $102/100-1 = 2.00\%$, $106/102-1 \approx 3.92\%$, and $107/106-1 \approx 0.94\%$. Actual returns from first holding A and then B are 2.00% and about 0.94%; $106/102-1$ crosses two different contracts. It is a price gap and does not correspond to holding either contract.

25-L05 The formation period is January through November 2025, exactly 11 months; December 2025 is skipped, and January 2026 is the next-month label. The correct compound return is $\prod_{m=\text{Jan}}^{\text{Nov}}(1+r_m)-1 \approx 15.47\%$, and the arithmetic sum is 15.00%. Shifting the window one month later uses February 2025 through December 2025 and compounds to about 17.72%; it omits January's 3% and includes the skipped December return of 5%. January 2026's −6% is a holding/diagnostic label, not a formation input.

25-L06 XSMOM ranks assets by their formation returns on the same date, goes long winners and short losers, and has zero nominal net weight when both legs are normalized to 1 (this does not guarantee zero market beta). TSMOM compares each asset with its own historical return and uses the sign to set long or short direction; the portfolio can be net long or short when most signals have the same sign. Short-term reversal negates the most recent month's return and bets on a short-term giveback. In this course, TSMOM uses 12–0, includes the latest month, and scales by ex ante volatility; standard XSMOM uses 12–1, skips that month, and ranks assets cross-sectionally. They answer different questions.

25-L07 Unadjusted price return is $51/100-1 = -49\%$. After the split, the holder owns two shares and receives a dividend of 1 per share, so total return is $(2\times 51+2\times 1)/100-1 = 4\%$. A futures strategy needs the actual contract held in each period, settlement or execution prices, roll rules and timing, and the contract-multiplier or funding convention. A price gap in a continuous series cannot replace those position records. A net-of-costs result also needs roll trading costs and margin/financing assumptions.

25-L08 That call takes 12 observations and skips one month, so it is not 12–1, which takes 11 observations; change it to `formation_return(r,11,1)`. Constructed data can check window endpoints, signal direction, compounding, weight mapping, and label alignment against the declared generating mechanism. It cannot establish real security returns, actual information availability or execution prices, or profitability after fees. At least these inputs are still needed: real total returns adjusted for dividends and splits, historical investable constituents and delisting returns, order and price records at the next executable time after the signal, and trading-cost and borrow/financing assumptions. A strategy conclusion also needs out-of-sample evaluation. A month-end to next-month return label is a time-label approximation, not execution evidence.

Chapter 26 —Quality

> Quality is not a single number. Leverage can raise ROE, accruals separate earnings from cash, and the numerator and balance must come from the same known financial statement. Each measure should state what it captures, how its denominator is calculated, and how much coverage is lost as a result.

Trigger. When ROE rises, what traces distinguish operating improvement, a change in leverage, and near-zero equity?

Separate operations, financing, and cash

ROE decomposes into net margin, asset turnover, and the equity multiplier. GP/A uses gross profit before interest, but the course interface uses average assets. A negative accrual ratio $(N - F)/\bar{A}$ means operating cash flow exceeds net income; Project 4's `neg_accrual` has the opposite sign. All monetary examples below are constructed for teaching.

Your turn. 26-L01 (old question: DuPont and same-version rule) Complete the DuPont identity: ROE $= \frac{N}{S} \cdot \frac{S}{A} \cdot$ _____. Before averaging equity, first require both year-end balances to be _____; a restated numerator cannot be divided by a denominator from _____. Briefly explain why.

Your turn. 26-L02 (old question: leverage and ROE) Let $\rho = 8\%$, after-tax debt cost $i = 3\%$, and $D/E = 0, 0.5, 1, 2$. Calculate ROE $= \rho + (\rho - i)D/E$ for each value. Then let $\rho = 3\%$, $i = 4\%$, and $D/E = 1.5$; find ROE and explain the direction of the leverage effect.

Your turn. **26-L03 (old question: average and near-zero equity)** Net income is 40, beginning equity is 100, and ending equity is −30. What are unguarded average equity and ROE? What is the result if both balances must be positive before averaging? If ending equity were instead 2, what would unfiltered ending ROE be, and why should this ratio not be interpreted as exceptionally strong operating quality?

Your turn. **26-L04 (old question: mixing restated values)** Originally reported net income is 40 and equity is 400; after restatement, net income is 28 and equity is 350. Find both same-version ROEs, the mixed value using the new numerator with the old denominator, and the apparent and true changes.

Your turn. **26-L05 (new: same-version flows and average balances)** One FY2024 filing was released on 2025-03-20 and ingested on 2025-03-21. Revenue 1000, cost of goods sold 600, SG&A 200, interest 20, net income 90, and operating cash flow 120; beginning/ending assets are 800/1000, beginning/ending equity 300/400, and debt is 300. Calculate course GP/A, average-equity ROE, accrual ratio and `neg_accrual`; then calculate OP under this chapter's definition using ending equity and debt/assets. The signal month-end is 2025-03.

Your turn. 26-L06 (new: course average denominator and gross profitability) Continue with 26-L05. What is gross profit? Calculate the ratio using the course interface's average-assets denominator and again using ending assets of 1000. Explain why the two values differ; then state which parts of the term "gross profits-to-assets" can be accurately attributed to the source record and which should not be attributed to the original without checking. Explain whether a financing change mechanically changes GP/A when operations and assets are held fixed.

Your turn. 26-L07 (new: OP, P4, and RMW) Project 4 has fields `gp_a`, `neg_accrual`, and `roe_avg`, but no OP; French RMW is a published portfolio spread. Can RMW fill in company-level OP, or can we say P4 has calculated OP? State the numerator or denominator differences between OP and GP/A.

Your turn. 26-L08 (new: ROE coverage threshold) Project 4's `roe_avg` requires both year-end equity balances to be positive and average equity to be strictly greater than 5% of average assets. Apply the threshold to the company in 26-L05, then consider another company with average assets of 900 and beginning/ending equity of 30/50. If 4 of 20 otherwise complete companies are excluded by this filter, calculate the coverage loss. Why is 5% an interface rule rather than a universal theorem?

Chapter 26 solutions

26-L01 A/E; positive; the denominator from the pre-restatement version (the same original version). The DuPont identity is ROE $= (N/S)(S/A)(A/E)$. Require both year-end balances to be positive before averaging to avoid a positive average hiding negative year-end equity. After a restatement, select the numerator and denominator from the same financial-statement version rather than mixing versions.

26-L02 The four values are $8\%, 10.5\%, 13\%, 18\%$. When $\rho = 3\% < i = 4\%$, ROE $= 3\% + (3\% - 4\%) \times 1.5 = 1.5\%$, below the operating return on assets. With a negative spread, more leverage reduces ROE; leverage also amplifies the effect of operating-return volatility on ROE.

26-L03 Unguarded average equity is $(100-30)/2 = 35$, giving ROE $= 40/35 \approx 114.3\%$. Because year-end equity is nonpositive, requiring both balances to be positive makes the result NaN. If ending equity alone is 2, ROE is $40/2 = 2000\%$; a near-zero denominator magnifies small accounting differences into extreme values, and the economic meaning fails for negative or near-zero equity.

26-L04 Original version: $40/400 = 10\%$; restated version: $28/350 = 8\%$; mixed version: $28/400 = 7\%$. Mixing shows a 3-percentage-point decline from 10%, while the true same-version change is 2 percentage points, so the false change adds 1 percentage point.

26-L05 Gross profit is $1000 - 600 = 400$. Average assets are $(800 + 1000)/2 = 900$, so course GP/A is $400/900 \approx 44.44\%$. Average equity is $(300 + 400)/2 = 350$, and ROE is $90/350 \approx 25.71\%$. Accrual ratio is $(90-120)/900 = -30/900 \approx -3.33\%$; the opposite-sign `neg_accrual` is $+3.33\%$, meaning operating cash flow exceeds net income. OP is $(1000 - 600 - 200 - 20)/400 = 45\%$; debt/assets is $300/1000 = 30\%$. By the end of March 2025, the filing has both been released and ingested, so it is available. A signal on March 20 or earlier cannot use a version that has not yet been ingested.

26-L06 Gross profit is 400. With average assets, the ratio is $400/900 \approx 44.44\%$; using ending assets gives $400/1000 = 40\%$. The difference comes from the denominator, not a change in gross profit. The source record supports describing the original measure as gross profits-to-assets; the course and P4 explicitly use average assets. Unless checked, do not claim that "average assets" is the paper's exact denominator. If operations and assets are fixed, changing debt/equity financing alone does not mechanically change GP/A: gross profit does not deduct interest and total assets are unchanged. If financing subsequently changes operations or assets, actual GP/A may still change.

26-L07 No. OP's numerator is revenue minus cost of goods sold, SG&A, and interest, and its denominator is book equity. GP/A's numerator is revenue minus cost of goods sold, and its denominator is average assets. OP deducts expenses and interest and uses equity; GP/A uses gross profit and assets. RMW is an aggregate return spread from portfolios formed on company profitability and size. It provides no company-level OP values and cannot establish that P4 has calculated OP.

26-L08 For 26-L05, 5% of average assets of 900 is 45. Average equity of 350 is strictly above 45, and both beginning and ending equity are positive, so it passes. For the other company, average equity is $(30+50)/2 = 40$. Both balances are positive, but $40 \leq 45$, so it is excluded. The coverage loss is $4/20 = 20\%$; report both coverage counts and the exclusion rate before and after filtering. The 5% value is a near-zero-equity threshold fixed in advance for P4's interface, not a universal theorem about every ROE definition. Coverage loss may also change the composition of the remaining sample.

Chapter 27 —Carry, Defensive Investing, and Low Volatility

First identify the cash flows and capital denominator, then calculate carry under unchanged conditions, and finally consider actual price changes and financing. BAB's estimated-beta neutrality, net dollar exposure, and true beta are three different quantities. The exercises below involve hand calculations and diagnostics; they do not provide real individual-stock or FX strategy evidence.

Trigger. If capital is halved for the same contract, how do PnL and return change, respectively?

Read first: cash flows, conditions, and timing

The excess return on a contract expiring in one period is $(S_{t+1} - F_t)/X_t$; total return adds the collateral rate. Carry counterfactually sets future spot equal to today's spot: $C = (S_t - F_t)/X_t$. Bond holding return under an unchanged curve also includes roll-down; on a roll, record PnL using the same contract at both endpoints. BAB ranks first and then scales by positive leg betas; true exposure must be reported separately.

Definition 27.1 *(FX quotation and denominator).* With a local/foreign quote, $F = S(1 + r)/(1 + r^*)$. Under full collateralization, $X = F$, long-foreign-currency carry is _____; applying the same formula to the reciprocal quote gives $F/S - 1$, and the sign is _____; whether the absolute magnitudes match depends on the denominator.

Your turn. 27-L01 (existing: FX and break-even) $S = 100, r = 2\%, r^* = 6\%$, one-year simple rates. Calculate F, carry, and the exchange rate giving zero excess return relative to the local deposit. If expiry spot is 96, calculate the forward return and verify it with a deposit comparison.

Your turn. **27-L02 (existing: roll-down on an unchanged curve)** Zero-coupon annual-compounded yields are 2% at 1 year, 3% at 2 years, and 3.5% at 3 years. Buy a 3-year bond and hold for one year; calculate the yield component, roll-down, and total return. Explain why the two components cannot simply be added, then subtract one-year financing.

Your turn. **27-L03 (existing: BAB construction and true exposure)** Estimated betas are $0.6, 0.9, 1.1, 1.4$; assign rank weights. Next month's excess returns are $1.2\%, 1.5\%, 1.0\%, 2.0\%$. Find the two leg weights, betas, leverage, and BAB return. If true leg betas are 0.7 and 1.3, while scaling uses 0.82 and 1.18, what is true exposure?

Read again: capital, rolls, and financing cannot be omitted

Contract PnL is a price difference; a return must state its capital denominator. BAB's risky-asset net exposure is $L - H$, and gross exposure is $L + H$. Adding the financing leg $-(L - H)$ gives self-financing PnL, but does not make the risky assets dollar-neutral. The course's leg-by-leg borrowing cost is an explicit sensitivity convention.

Your turn. **27-L04 (new: capital ledger)** $S = 102, F = 100, r^f = 1\%$, and actual expiry spot is 99. For each of $X = 50, 100$, record carry, price change, excess return, collateral interest, ending equity, and total return. Which amounts stay the same, and which ratios change?

Your turn. 27-L05 (new: same contract and rolling) Contract A is priced at 102, 101, and 100; at the end of period 2, roll into B, which is priced at 103 then and 102 next period. Ignore collateral interest and fees. Fully collateralize at each period's starting contract price and adjust notional with wealth (fractional units allowed). For the roll period, calculate the spliced-price return minus the tradable same-contract rebalanced-index return, and calculate both cumulative returns. Explain why buying a more expensive new contract does not earn the price spread that day.

Your turn. 27-L06 (new: proof outline and two kinds of neutrality) For the legs in 27-L03, calculate risky-asset net exposure, gross exposure, and the risk-free financing leg. Let true betas be $x_L = 0.7 < x_H = 1.3$, with shrinkage $0.6x + 0.4$. Differentiate $g(x) = x/(0.6x + 0.4)$ and complete the proof: on the same _____ denominator interval, $g(x_L) < g(x_H)$, so true net beta is _____. If $x_L = -1, x_H = 2$, can the same conclusion still be used?

Your turn. 27-L07 (new: research judgment and P4 gaps) Current P4 uses seven stock-value, momentum, and quality fields. List three categories of required inputs each for FX carry and BAB. Which inputs can French industry returns replace, and which can they not replace? Does "positive portfolio carry" prove positive expected realized return?

Chapter 27 answers: 27-L01 through 27-L03

27-L01 Fill in $(r^* - r)/(1 + r)$, and "reversed" . $F = 100(1.02)/1.06 = 96.226415$, carry $= 100/F - 1 = 3.921569\%$. Break-even relative to the local deposit is $S_T = F$; tolerable depreciation is $1 - F/100 = 3.773585\%$. At spot 96, return is $96/F - 1 = -0.235294\%$. The foreign-currency deposit converts back to local currency at $1.06(0.96) = 1.0176$, versus 1.02 in the local deposit; relative to the local deposit, $(1.0176 - 1.02)/1.02 = -0.235294\%$. Carry under the reciprocal quote is $F/100 - 1 = -3.773585\%$: the sign reverses and the magnitude differs.

27-L02 Current price is $1.035^{-3} = 0.901943$; after one year on an unchanged curve, price is $1.03^{-2} = 0.942596$. Yield component is 3.5%, roll-down is $(1.035/1.03)^2 - 1 = 0.973230\%$, and total return is $1.035^3/1.03^2 - 1 = 4.507293\%$. The multiplicative decomposition $(1.035)(1.00973230) - 1$ includes a cross-term, so 3.5 and 0.9732 cannot simply be added. Subtracting the 2% one-year risk-free rate gives 2.507293% excess holding return. Dividing by the financing terminal value instead gives $1.04507293/1.02 - 1$, a different denominator convention that should not be conflated with it.

27-L03 Ranks are 1 through 4, mean rank is 2.5, deviations are $(-1.5, -0.5, 0.5, 1.5)$, their absolute sum is 4, and $k = 0.5$.

Low leg $(0.75, 0.25, 0, 0)$; high leg $(0, 0, 0.25, 0.75)$. Betas are 0.675 and 1.325; leverage is 1.481481 and 0.754717. Leg returns are 1.275% and 1.75%, so BAB return is $1.275\%/0.675 - 1.75\%/1.325 = 0.568134\%$. True exposure is $0.7/0.82 - 1.3/1.18 = -0.248036$; the scaled betas make both legs equal to 1 under estimated beta, but not under true beta.

Chapter 27 answers: 27-L04 through 27-L07

27-L04 Contract carry PnL is 2, actual PnL is -1, and spot change is -3. At $X = 50$: carry 4%, price change -6%, excess return -2%, collateral interest 0.5, ending equity 49.5, and total return -1%. At $X = 100$: carry 2%, price change -3%, excess return -1%, interest 1, ending equity 100, and total return 0. Contract PnL is the same; capital, interest amount, and normalized returns differ.

27-L05 Under the stated full-collateralization and notional-rebalancing assumptions, successive same-contract returns are $101/102 - 1$, $100/101 - 1$, and $102/103 - 1$. Cumulative return is $(100/102)(102/103) - 1 = 100/103 - 1 = -2.912621\%$. The spliced sequence 102, 101, 103, 102 has cumulative return 0. The roll-period difference is $103/101 - 100/101 = 3/101 = 2.970297\%$; old-contract return is -0.990099%, and spliced return is $+1.980198\%$. Rolling into B means buying another expiry contract; 103 is its quote, not a gain on a held contract that rose from 100 to 103.

27-L06 Risky-asset net exposure is $1.481481 - 0.754717 = 0.726765$; gross exposure is 2.236198; the risk-free financing leg is -0.726765. Estimated risk-asset beta is $1 - 1 = 0$, while net dollar exposure is not zero. $g'(x) = 0.4/(0.6x + 0.4)^2 > 0$, so fill in "positive" and "negative." The two positive scaling denominators lie in the same interval to the right of the pole at $-2/3$, so $g(0.7) - g(1.3) = -0.248036$. In the counterexample $x_L = -1$, the denominator is -0.2 and $g(-1) - g(2) = 5 - 1.25 = 3.75 > 0$; the values cross the pole, so the original proof does not apply, and the BAB interface rejects nonpositive leg beta. For the noisy case, see the full conditions for optimal shrinkage and conditional expectation in Section 6.9; do not extend the noiseless proof to that case.

27-L07 FX requires same-time, same-tenor spot and forward quotes, quote direction and settlement identity, bid/ask, and collateral/financing terms. BAB requires point-in-time security returns and a market benchmark, estimation window and shrinkage rule, and financing, borrow, and margin inputs. French industry returns can support aggregate industry risk estimates and diagnostics, but cannot supply point-in-time individual-stock data, FX quotes, or actual execution and borrow records. A carry portfolio constructed by sorting has nonnegative carry; realized returns also include price changes, which can offset or exceed carry. Positive true expected return requires a separate test. Current P4 does not add this chapter's factors automatically.

Chapter 28 —Multi-Factor Models: Preprocessing, Combination, and Constraints

> A row of raw values first becomes residuals and z-scores, then a composite score, and finally weights on a support set. Neutralizing a signal is not the same as neutralizing a quantile portfolio; a neutral target is not the same as neutral executed holdings. State the parameter and support-set clocks explicitly.

Trigger. If we project first and then zero out an unlisted name, does industry neutrality survive?

Read first: three distinct objects

Preprocessing uses same-date fields, controls, and industry labels in the versions known at signal time; combination follows the predeclared weights and coverage rules; portfolio constraints apply to the final weights. Homogeneous industry constraints permit proportional gross restoration; nonzero targets do not. The small examples below use separate hand-calculation conventions and do not replace the production minimum of 60 names.

Definition 28.1 *(Same-date residuals and combination).* In a same-date regression on all industry indicators, the residual equals _____. The equal-weight composite IC is $(\rho_1 + \rho_2)/\sqrt{2(1 + r_{12})}$, where r_{12} is the _____ correlation. Full-sample standardization used _____.

Your turn. **28-L01 (old: neutralization and IC)** Industries A and B contain the first three and last three names, respectively. Signal $(1, -1, 0, 11, 10, 9)$; next-month returns $(5, 3, 4, -3, -4, -5)\%$. Find the residuals, raw and neutralized ICs, and explain the sign reversal.

Your turn. 28-L02 (old: combination and order) $\rho_1 = 0.04, \rho_2 = 0.02, r_{12} = 0.5$. Find the equal-weight IC, the standardized-residual composite IC for order 1 then 2 and order 2 then 1, and the optimal $R^{-1}\rho$ IC. Explain why the optimum uses only the first factor.

Your turn. 28-L03 (old: timing and diversification) For $(1, 2, 3, 10)$, find the expanding and full-sample z-scores of the third value. For three factors with Sharpe ratio 0.3 and average correlation 0.2, find the equal-risk average Sharpe ratio; explain why the formula is conditional.

Read next: listed set and valid-alpha set

$\mathcal{S}_t$ is the signal-time listed support set; nonmissing alpha determines only the initial quantile ranking. Optional industry projection allows listed names without scores to carry hedge weights; names outside the support set must have weight zero. A singleton industry has only the zero direction; if the projection is numerically zero, hold cash rather than amplify noise.

Your turn. 28-L04 (new: complete one-row calculation) Using the residuals from 28-L01, find the sum of squares, sample standard deviation, and z-scores. Let $z_2 = (1, 0, -1, 0, -1, 1)/s$; find the six-name equal-weight composite. If f is missing z_2, what are its values under reweighting over available factors and zero fill? If the two factors are different styles and at least two are required, what is the result?

Your turn. 28-L05 (new: support-set projection) Industry A contains a,b and B contains c,d; weights $(0.4, -0.4, 0.3, -0.3)$; a is unlisted. After zeroing a, find industry net, total net, and gross. Project only on b,c,d: write A, $AA^\top$, and Au, find the nearest point, then restore gross to 2. Explain why the singleton industry b also goes to zero.

Your turn. 28-L06 (new: missing alpha and thresholds) All four names are listed; b and d have no alpha; initial quantile weights are $(1, 0, -1, 0)$. Find the weights after industry projection. Is this still a pure quantile portfolio? Then classify production dates with 59 or 60 listed names and a field with 29 values: what are the universe eligibility, field winsorization, and portfolio target?

Your turn. 28-L07 (new: proof and execution diagnosis) Given $Av = 0$ and gross $g > 0$, let $k = 2/g$. Complete $A(kv) =$ _____ and prove that industry and net-zero constraints remain. Use $A = (1, 1), b = 1, v = (0.5, 0.5)$ as a nonzero-target counterexample. The target is $(0, 0, 1, -1)$; executed holdings are $(0, 0, 0.8, -0.6)$. Under the same NAV convention, what are industry net, total net, and gross? Can the holdings be written back into the signal?

Chapter 28 answers: 28-L01 to 28-L03

28-L01 The definition blanks are within-group demeaning, same-date cross-section, and future data. Industry means are 0 and 10; residuals $(1, -1, 0, 1, 0, -1)$. The raw signal mean is 5, the squared deviations sum to 154, the label mean is 0, the label sum of squares is 0.01, and the cross-product sum is -1.16. Thus IC $= -1.16/\sqrt{1.54} = -0.934755$. Residual sum of squares is 4, cross-product sum is 0.04, and IC $= 0.04/\sqrt{0.04} = 0.2$. Industry B's signal level is higher overall while its returns are lower, masking the positive within-industry relationship. This is the stated example; it does not show that real industry shifts are necessarily unrelated noise.

28-L02 Equal-weight IC is $0.06/\sqrt{3} = 0.034641$. Order 1 then 2: $(0.04+0)/\sqrt{2} = 0.028284$. Order 2 then 1: $[0.02 + 0.03/\sqrt{0.75}]/\sqrt{2} = 0.038637$. $R^{-1}\rho = (0.04, 0)$ and optimal IC is 0.04. The second factor's correlation, 0.02, equals 0.5(0.04); its predictive power comes from correlation with the first factor, with no additional label covariance. The optimal formula still requires a positive-definite correlation matrix, nonzero rho, and jointly feasible true values; do not estimate it on the evaluation period and call it optimal in advance.

28-L03 The first three values have mean 2 and sample standard deviation 1, so the third value has expanding z-score 1. The full-sample mean is 4, variance 50/3, standard deviation 4.082483, and z-score -0.244949. The future value 10 changes the sign. The diversified Sharpe is $0.3\sqrt{3}/\sqrt{1.4} = 0.439155$. This assumes equal risk, equal component Sharpe, and the stated correlation structure; changes in correlation alter the denominator, so future Sharpe is not guaranteed.

Chapter 28 answers: 28-L04 to 28-L07

28-L04 Residual sum of squares is 4, with mean 0, so $s = \sqrt{4/5} = 0.894427$. The z-scores are $(1, -1, 0, 1, 0, -1)/s$. The composite is $(1, -0.5, -0.5, 0.5, -0.5, 0)/s$, or $(1.118034, -0.559017, -0.559017, 0.559017, -0.559017, 0)$. If f lacks the second factor, reweighting gives $-1/s = -1.118034$ and zero fill gives $-0.5/s = -0.559017$. If the contract requires at least two factors across styles, f is missing and has no alpha ranking eligibility; reweighting cannot bypass the minimum.

28-L05 After zeroing a, weights are $(0, -0.4, 0.3, -0.3)$: industry A net -0.4, B net 0, total net -0.4, gross 1. On coordinates (b, c, d), $A = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{smallmatrix}\right)$, $AA^\top = \mathrm{diag}(1, 2)$, and $Au = (-0.4, 0)$. Subtracting industry means gives $v = (0, 0.3, -0.3)$, gross 0.6; multiply by $10/3$ and embed to get $(0, 0, 1, -1)$. Outside-support weight is zero, industry and total net are zero, and gross is 2. b is the only coordinate in industry A, so the zero-net constraint requires $b = 0$; the two independent rows are not rank deficient. Only v is the nearest point; after restoring the budget, final weights are generally no longer nearest.

28-L06 Initial industry net positions are 1 and -1, with means 0.5 and -0.5. Projection gives $(0.5, -0.5, -0.5, 0.5)$, net 0, gross 2. b and d hedge without alpha, so this is no longer a pure quantile portfolio. With 59 names the entire date is ineligible, preprocessing is missing, and the target is cash. Sixty names pass only the universe threshold. A 29-value field fails the 30-value winsorization minimum and should be missing. Other fields still need the 10-name z-score minimum, one field within a style, and two styles across the composite; the target depends on these separate requirements. If every industry has only one name, there is no nonzero long-short direction, so hold cash.

28-L07 Fill in $kAv = 0$; the sum of industry rows is the net-position row, so total net is also zero. Positive rescaling makes gross $kg = 2$; do not scale a numerically zero vector. In the nonzero example $Av = 1$ and gross is 1; after multiplying by 2, $A(2v) = 2 \neq 1$. The executed B net and total net are both 0.2, and gross is 1.4, showing that partial fills and price drift after execution need not preserve target neutrality in realized holdings. A real ledger reports closing holdings divided by current NAV and records filtering and fees. Those data may be used only for execution and diagnosis; they cannot change the signal retroactively. Current listing status is a constructed input, not evidence of actual borrow or halt eligibility.

Chapter 29 —Building a Backtesting Engine

A target is an instruction; cash and shares are the account state. Execute an earlier close's signal at a later price-index observation, solve for post-fee NAV, then mark at the close. Between trades, weights follow relative returns. A zero reconciliation residual checks internal accounting, not price quality or point-in-time validity. Monthly lag 1 is one monthly observation; default $E = P$ is a closing-mark proxy, not an opening-price observation.

Trigger. After deciding on 50% A, 30% B, and 20% cash, what shares and cash actually remain after a 20-bps fee?

Definition 29.1 *(Post-fee NAV and self-financing).* With no external flows, $V' = V_{\text{pre}} - c\sum_i |w_i V' - x_i|$. Old holding values x_i use the _____ price. If $c\sum_i |w_i| < 1$, the real root is _____; it is positive exactly when $c\sum_i |x_i| <$ _____. Given consistent signs s_i, $V' = (V_{\text{pre}} + c\sum_i s_i x_i)/(1 + c\sum_i s_i w_i)$. Without trading, $w_{i,t+1} = w_{i,t}(1 + r_i)/$ _____.

Your turn. 29-L01 Cash is 1000, $c = 0.002$, targets are A 0.5 and B 0.3, and $E = (10, 20)$. Find V' _____, A shares _____, B shares _____, fees _____, and cash _____. At closing prices $(11, 19)$ find NAV _____ and A's actual weight _____.

Your turn. 29-L02 At the February 27 close $(12.1, 19)$, NAV is _____. At the March 2 execution prices $(12, 20)$, $V_{\text{pre}} =$ _____. Sell A and buy B: use the closed form to find V' _____, A shares sold _____, B shares bought _____, and fees _____. At the close $(12, 21)$, NAV is _____.

Your turn. 29-L03 Cash is zero, long value 1000, short value −990, and $c = 0.01$. Why is the rebalance infeasible within the monotone target-gross domain? Distinguish nonpositive execution equity from positive equity with unaffordable fees.

Your turn. 29-L04 Begin with 50% in each of A and B. A returns +100% on day 1 and −50% on day 2; B is unchanged. Do not trade between them. Ledger cumulative return is _____; naive fixed-target `shift(1)` gives _____. State four sufficient conditions for equality, in addition to aligned initial state, execution dates, and policy. Why are they not necessary?

Follow the cash and shares through one day

Settle interest on old cash. Apply splits before dividends on new shares. Value at execution, solve the allocation base, trade, and only then mark at the close. An order's dollar value divided by its execution price gives shares. Old and new holdings belong to different price segments.

Your turn. **29-L05** Previous cash is 200, shares 8, and price 100. Cash rate is 0.001. A 2:1 split precedes a dividend of 1 per new share. Let $E = 49, P = 50, w = 0.8, c = 0.002$.
Interest _____, split shares _____, dividend cash _____; pre-trade cash _____, x _____, V_{pre} _____.
Assume a buy, write the scalar equation, and find V', ΔM, Δq, fees, cash, and closing NAV. Verify the buy sign. Attribute the previous split-adjusted price to E using old shares, and E to P using new shares.

Your turn. **29-L06** Diagnose: default $E = P$ means an opening fill; monthly lag 1 means tomorrow; equity -1 at execution can continue if the close recovers to $+2$; negative cash pays no interest. Where does each of the three NAV boundaries stop or reject the account?

Chapter 29 —Answers and checks

Definition blanks: execution; unique; V_{pre}; $1 + r_p$.

29-L01 $V' = 1000/1.0016 = 998.4026$. A shares $= 0.5V'/10 = 49.92013$; B shares $= 0.3V'/20 = 14.97604$. Fees $= 0.002(0.8V') = 1.5974$. Cash $= 1000 - 0.8V' -$ Fees $= 199.6805 = 0.2V'$. Closing NAV $= 199.6805 + 49.92013(11) + 14.97604(19) = 1033.3466$, using unrounded intermediate values. A weight is 0.5314.

29-L02 February 27 NAV is $199.680511 + 49.920128(12.1) + 14.976038(19) = 1088.2588$. At execution, $x_A = 599.0415$, $x_B = 299.5208$, and $V_{\text{pre}} = 1098.2428$. Thus

$$V' = \frac{1098.2428 - 0.002(599.0415 - 299.5208)}{0.9996} = 1098.0830.$$

The target-dollar changes are -50.0000 and 29.9041, giving a sale of 4.1667 A shares and purchase of 1.4952 B shares, confirming the signs. Fees are 0.1598; cash is 219.6166 and new shares $45.75346, 16.47125$. Closing NAV at $(12, 21)$ is 1114.5542. All calculations retain full precision until display.

29-L03 Equity is $1000 - 990 = 10$ while liquidation costs are $0.01(1000 + 990) = 19.9$. Within $c\sum|w_i| < 1$, $f(0) = 9.9 > 0$ and the unique real root is nonpositive; the engine rejects this rebalance. If equity at E is itself nonpositive, it stops without trading or solving. These are distinct policies.

29-L04 After day 1, values are 100 and 50, NAV 150, and weights $2/3, 1/3$. Day 2 returns A's value to 50, so NAV returns to 100 and cumulative return is 0%. A fixed 50/50 calculation gives $1.5(0.75) - 1 = 12.5\%$. With aligned initial states, already lagged execution dates, and policy, sufficient conditions are zero fees, zero cash rate, $E = P$, and exact drift with no unmodeled corporate actions. Use old weights for the period return, then reset at the trading close. Positive configured fees need not break equality in a no-trade interval because actual fees are zero.

29-L05 Interest is 0.2; split shares 16; dividend cash 16; pre-trade cash 216.2; $x = 16(49) = 784$; $V_{\text{pre}} = 1000.2$. The buy-segment solution is

$$V' = \frac{1000.2 + 0.002(784)}{1 + 0.002(0.8)} = 1000.1677.$$

Then $\Delta M = 0.8V' - 784 = 16.1342 > 0$, $\Delta q = \Delta M/49 = 0.3293$, fees 0.0323, cash 200.0335, new shares 16.3293, and closing NAV 1016.4970. Old-share PnL $16(49 - 50) = -16$ offsets dividend cash 16. New shares earn $16.3293(50 - 49)$. Add interest and subtract fees to get the NAV gain 16.4970. Round only the displayed final values.

29-L06 All four statements are false. Default $E = P$ is a closing-mark proxy; monthly lag 1 is the next monthly observation; execution-time ruin excludes a later recovery; negative cash pays the same simple rate. The three boundaries are: $V_{\text{pre}} \le 0$ stops at E without solving; positive equity with $c\sum|x_i| \ge V_{\text{pre}}$ in the monotone domain rejects the rebalance; a valid execution followed by closing NAV ≤ 0 stops at P. Later NAV rows are not generated. The simple cash convention excludes borrow, rebates, and margin calls.

Chapter 30 —Backtesting Biases

Yuxiao Ma

> For each bias, check in the same order: what information did the erroneous method use, was it available at decision time; what is the numerical difference after the fix on the same data and ledger; and what does the fix still fail to address? The erroneous version is not guaranteed to look better. Report the sign of the difference and its distribution across multiple seeds.

Trigger. After-hours announcements, a universe containing only survivors, and the top three selected on the full sample: at what step did each method use future information?

Definition 30.1 *(Availability time and survivor truncation).* Information available at signal time s, whose price-index position is k, satisfies $\max(t_{\text{avail}}, t_{\text{ingest}}) \le$ _____; execution-index position $\ge k+$ _____. If cumulative return $X \sim N(\mu_X, \sigma_X^2)$ and survival requires $X \ge d$, then $\mathbb{E}[X \mid X \ge d] = \mu_X + \sigma_X \lambda(\alpha)$, where $\alpha = (d - \mu_X)/\sigma_X$ and $\lambda(\alpha) =$ _____. Bias increases _____ with delisting probability, but not in direct proportion.

Your turn. 30-L01 $\mu_X = 0$, $\sigma_X = 0.3$, $d = -0.3$: $\alpha =$ _____, delisting probability _____, $\lambda =$ _____, and the conditional mean when observing survivors only _____.

Your turn. 30-L02 Take the maximum z across 50 independent zero-effect trials: exact expectation _____ (numerical integration), Bailey–López de Prado approximation _____, and first-order Gumbel asymptotic _____. Why must the number of trials (the Chapter 22 ledger) be reported with the maximum?

Your turn. 30-L03 Hold a stock through a 2:1 split without recording the split in the ledger: the stock's market value is recorded as _____ its prior value; this chapter's portfolio return booked on the split date is _____ (correct: −1.29%). What is wrong with treating a "total-return level" as the execution price and also passing dividends through the dividend table?

Your turn. 30-L04 French 10 industries: choose the top three by Sharpe over the full sample (1926–2026), then their Sharpe over 1976–2026 is _____; choose the top three using only 1926–1975, then their 1976–2026 Sharpe is _____; the equal-weighted 10-industry Sharpe is _____. Which is selection bias, and which selects using the earlier period then evaluates on the later one? Why are these data not evidence of tradability?

Numerical direction of errors and scope of the fixes

Your turn. 30-L05 In the saved results, Sharpe for look-ahead wrong/fixed is 0.3730 and −0.7760; for the survivor universe it is −0.2950 and −0.5565. First calculate wrong minus fixed from the displayed values, then compare with the unrounded JSON differences and explain the rounding discrepancy. Does the negative wrong Sharpe for the survivor universe mean there is no bias? Why can adding lag not fix survivorship bias, a complete trial ledger not fix duplicate dividends, and recording a split not replace a time holdout? Can changing display rounding alone be called new strategy evidence?

Chapter 30 —Answers and Checks

s; 1; $\phi(\alpha)/(1-\Phi(\alpha))$; monotonically increasing.

30-L01 $\alpha = -1$; $\Phi(-1) = 0.1587$; $\lambda = 0.2876$; conditional mean 0.0863 (while the true mean is 0).

30-L02 2.2491, 2.2763, 2.3072; the distribution of the maximum depends on the number of trials. Without that count, one cannot tell whether a value around 2.25 for z is just the result of selection.

30-L03 One-half; −26.20%; a total-return level already includes dividends, so adding them again double-counts them. It cannot be passed off as an execution price in a true share ledger. A declared synthetic-proxy-unit teaching model that does not book dividends again is a different, valid teaching model.

30-L04 Full-sample selection (NoDur, Hlth, Shops): 0.6519—it uses second-period results and is selection bias. Selection on the first period (Hlth, Telcm, HiTec): 0.5658—this follows the time order by selecting first and evaluating later. Equal-weighted: 0.6147. French industry returns are aggregate, revised, non-point-in-time portfolio returns, not executable prices.

30-L05 Calculate the differences from the unrounded JSON first, then display four decimal places: 1.1490 and 0.2616, respectively. Subtracting the displayed four-decimal survivor Sharpe values directly gives 0.2615, showing that calculations should use original precision before rounding. The wrong result being negative does not remove the erroneous selection mechanism. Preserve the boundary on direction across seeds; a single difference does not generalize to all markets. Lag fixes execution timing, and a historical asset universe fixes end-of-period survivor selection. A trial ledger preserves the search count and all results; dividend conservation separately requires unadjusted execution prices and corporate actions. A split fixes the account ledger, while a time holdout fixes evaluation selection. Each of the eight fixes has its own target. Display updates and correction of the same-day dividend teaching formula use saved outputs; they are not a new strategy run or evidence from real trading.

Chapter 31 —Financial Model Validation

Yuxiao Ma

For sample i, features are observed through the close at t_i, and its label uses returns after t_i through e_i, so its support is the half-open interval $(t_i, e_i]$. Purge any training sample whose interval overlaps any test sample. Embargo training samples after the test block when their feature lookback reaches into test labels. Gap is measured in rows; it corresponds to the boundary purge count only for a single-asset regular calendar, fixed-horizon, forward split. Generate candidates for the evaluation purpose: deployment uses only historical rows whose labels are available; diagnostic K-fold validation may include post-test blocks.

Trigger. With a five-day return label and a train/test split on one day, do the last few training labels already contain returns from the test period?

Definition 31.1 *(Interval overlap, purge, and embargo).* Intervals $(t_i, e_i]$ and $(t_j, e_j]$ overlap if and only if $\max(t_i, t_j) <$ _____. If $e_j = t_k$ (the intervals touch at an endpoint), they _____ overlap: the first uses daily returns $r_{t_j+1}, \dots, r_{e_j}$, and the second begins at r_{t_k+1}. Embargo removes training samples located _____ the test block, with observation time $t \in [e_{\max}, e_{\max} + m)$. m must be at least the feature _____. For a pure-history walk-forward split that is first filtered by availability and still checks label intervals, under this chapter's post-test definition it _____ a post-test embargo.

Your turn. 31-L01 One asset, $t = 0, \dots, 14$, $h = 5$, test block $\{6, 7, 8\}$. Purge removes the pre-test rows _____ and post-test rows _____. Why is $t = 1$ retained? With $m = 3$, $e_{\max} =$ ____, embargo removes _____, and the final training set is _____.

Your turn. 31-L02 Change to 3 assets ordered by (date, asset), with testing starting at date 6. sklearn's `gap`$= h - 1 = 4$ deletes _____ rows; the true overlaps number _____ rows, leaving _____ rows. Give the general formula $N_{\text{leak}} =$ _____.

Your turn. **31-L03** Chapter's constructed experiment (linear Ridge, 5 seeds): shuffled K-fold R^2 _____, time split without purge _____, purged walk-forward _____. For 1-nearest-neighbor, Rank IC without purge _____, after purge _____. What does each comparison show?

Delete first, then retain; a deleted branch never returns

Definition 31.2 *(Two deletion sets and two retained sets).* Let C be candidates and Q the test set. R_{purge} contains candidates whose labels overlap any test label; $S_{\text{after_purge}} = C \setminus$ _____.
Only among survivors, delete samples whose observation times fall in the post-test buffer $[e_{\max}, e_{\max}+m)$, denoting them R_{embargo}; $S_{\text{final}} = S_{\text{after_purge}} \setminus$ _____. Thus the final set must be a _____ of the purge survivors.

Your turn. 31-L04 Return to the example with 15 observations and write $C, Q, R_{\text{purge}}, S_{\text{after_purge}}, R_{\text{embargo}}, S_{\text{final}}$. Why do the merely touching samples 1 and 13 have different final fates? Someone connects purge-deleted sample 2 to the "historical sample exempt from embargo" branch, then retains it again. What is wrong?

Your turn. 31-L05 Deployment predicts at 6; sample 1 has $e_1 = 6, a_1 = 7$. At which gate is it first removed? If 13 and 14 are kept for training, is this deployment walk-forward or offline diagnosis that allows future blocks? Can the closed-form gap leakage count for $h = 5, N = 3$ be applied directly to a missing or irregular panel?

Chapter 31 —Answers and Checks

$\min(e_i, e_j)$; do not; after; lookback length; does not require.

31-L01 Purge removes $\{2, 3, 4, 5\}$ and $\{9, 10, 11, 12\}$. The interval $(1, 6]$ for $t = 1$ uses $r_2..r_6$, while test interval $(6, 11]$ uses $r_7..r_{11}$; they only touch at the endpoint. $e_{\max} = 13$; embargo removes $\{13, 14\}$; the final set is $\{0, 1\}$.

31-L02 The gap deletes 4 rows; 12 rows overlap; 8 remain (all rows on dates 2 and 3, and two rows on date 4). $N_{\text{leak}} = (h-1)(N-1)$, restricted to a date/asset-sorted, complete balanced panel with fixed horizons, a test split at a full-date boundary, and enough prior history.

31-L03 0.5327, 0.4160, and 0.4143: the first two protocols differ by 0.1167, while TSS and the final protocol differ by about 0.0017. But training dates/sample sizes, full-date blocking, purge, and scaling all differ at once; this cannot be decomposed into causal shares for shuffling or purge. For 1-NN the values are 0.1194 and 0.0108: in this constructed data and 1-NN setup, overlapping labels clearly change the result. This does not prove that model capacity and bias have a generally monotone relationship.

31-L04 The blanks in the definition are, in order, R_{purge}, R_{embargo}, and subset.

$$
\begin{aligned}
Q &= \{6, 7, 8\}, \\
C &= \{0, 1, 2, 3, 4, 5, 9, 10, 11, 12, 13, 14\}, \\
R_{\text{purge}} &= \{2, 3, 4, 5, 9, 10, 11, 12\}, \\
S_{\text{after_purge}} &= \{0, 1, 13, 14\}, \\
R_{\text{embargo}} &= \{13, 14\}, \quad S_{\text{final}} = \{0, 1\}.
\end{aligned}
$$

Sample 1 ends with r_6; test sample 6 starts with r_7. Sample 13 starts with r_{14}; test sample 8 ends with r_{13}. Neither interval overlaps. Sample 13 is in the post-test buffer $[13, 16)$ and is removed by embargo, while 1 is a historical candidate and remains. Sample 2 already terminated on the purge-deletion branch. "Exempt from embargo" can retain only a sample that still exists; it cannot resurrect a deleted one.

31-L05 Since $a_1 = 7 > 6$, the availability gate removes sample 1 before candidate generation; a settled end time $e_1 = 6$ cannot substitute for its readable time. Keeping the full-axis candidates 13 and 14 is a diagnostic example, not a deployment training set at time 6. For the complete, balanced three-asset panel, the leakage count at $h = 5$ is $(5 - 1)(3 - 1) = 8$ rows. Missing securities, irregular dates, or variable-length labels break the row-count mapping, so filter each actual interval. Clean inner and outer folds separately, fit the scaler only on each training fold, and return global row indices.

Chapter 32 —Performance Statistics & Inference

Yuxiao Ma

First distinguish description from inference. CAGR, drawdown, and Sharpe describe realized returns. PSR and DSR depend on assumptions about the return distribution, sample length, and search count. CSCV measures the relative out-of-sample performance of candidates selected in sample. No single statistic proves that future alpha is tradable. All Sharpe inference begins with excess returns measured at one common frequency.

Trigger. You run 100 trials and report only the best-looking return curve. Why should its Sharpe not be interpreted in the same way as the result of a single prespecified trial?

Definition 32.1 *(PSR, DSR, and search correction).* Let $\widehat{SR}$ be the sample Sharpe of monthly excess returns, SR^* the monthly threshold, n the sample size, γ_3 skewness, and γ_4 Pearson raw kurtosis. Under i.i.d. returns, a finite fourth moment, and an asymptotic approximation,

$$z = \frac{(\widehat{SR} - SR^*)\sqrt{n-1}}{\sqrt{1 - \gamma_3 \widehat{SR} + \frac{\gamma_4 - 1}{4}\widehat{SR}^2}}, \qquad PSR = \Phi(z).$$

The one-sided normal-approximation p-value is _____. The raw kurtosis of a normal distribution is _____. Skewness 0 and kurtosis 3 _____ prove normality. DSR replaces the threshold with the expected maximum noise Sharpe based on the effective number of independent trials, rather than continuing to use 0.

Before substituting values, align frequencies: returns and the risk-free series, observed Sharpe, threshold, and standard error must use the same frequency. n counts observations at that frequency. Annualization multiplies Sharpe by $\sqrt{A}$ and variance by A. Scaling a sum of returns depends on independence or covariance assumptions; compounded wealth still follows a product.

Your turn. **32-L01** There are 24 monthly observations, $\widehat{SR} = 0.458$, and $SR^* = 0$. First calculate z and PSR under normal moments. Then use skewness -2.448 and raw kurtosis 10.164 and compare the denominator and PSR. Why can a larger sample not fix the absence of a fourth moment?

Your turn. 32-L02 There are 1,250 daily returns and an annualized Sharpe of 2.5, using 250 trading days per year. What is the per-period Sharpe? If the effective number of independent trials is 100 and the daily Sharpe variance is 0.002, the estimated daily noise threshold is 0.113172. What is wrong with comparing this number directly with annualized 2.5? Can the effective trial count be set equal to the number of saved candidate columns without explanation?

Your turn. 32-L03 CSCV has 8 periods and 3 candidates divided into 4 blocks. In each partition, use 2 blocks to select the best candidate and the other 2 to evaluate it. How many partitions are there? If the selected candidate's out-of-sample rank is 1, with rank 3 best, calculate $\omega = \text{rank}/(N+1)$ and $\lambda = \log(\omega/(1-\omega))$. Does this partition count toward the $\lambda \leq 0$ PBO? Does this procedure simulate deployment in real time?

Your turn. **32-L04** Monthly returns follow a zero-mean AR(1) with $\phi = 0.5$. Under the long-run variance approximation, what fraction of the original sample size is the effective sample size? Why does using only $1 + 2\rho_1$ understate long-run variance? Give two data limitations that must be disclosed for the French industry-return diagnostic.

Your turn. **32-L05** The observed daily Sharpe is 0.10 and the daily threshold is 0.15. If a change in moments increases the variance coefficient D from 1 to 4, in which direction do z and PSR move? What if the threshold is 0? State the condition missing from the claim that "heavy tails always lower PSR."

Chapter 32 —Answers and Checks

Definition blanks: $1 - PSR$; 3; does not.

32-L01 Under normal moments, the denominator is $\sqrt{1 + 0.458^2/2} = 1.0511$, and the numerator is $0.458\sqrt{23} = 2.1965$, so $z \approx 2.0897$ and PSR ≈ 0.9817. With non-normal moments, the denominator is $\sqrt{1 + 2.448(0.458) + (10.164 - 1)0.458^2/4} \approx 1.6132$, so $z \approx 1.3616$ and PSR ≈ 0.9134. Here negative skew and high kurtosis lower the approximate confidence measure. If the fourth moment does not exist, the asymptotic variance formula lacks a required condition; no chosen finite sample length can make the population fourth moment finite. A $t(4)$ distribution is a counterexample. Even for a distribution with $\nu > 4$, dependence and estimation error still need to be checked.

32-L02 The per-period Sharpe is $2.5/\sqrt{250} = 0.158114$. The threshold 0.113172 is daily, so compare it with the daily 0.158114; their difference is 0.044942. The corresponding annualized threshold is about 1.7894. The trial count affects selection bias: correlated trials, repeated configurations, and unrecorded failed experiments all matter. Candidate-column count cannot be treated mechanically as the number of independent trials. The course calculation uses explicitly supplied distributional parameters; it does not supply the researcher ' s unknown true effective trial count.

32-L03 There are $\binom{4}{2} = 6$ partitions; $\omega = 1/4$ and $\lambda = \log(1/3) \approx -1.0986$, so the partition is included under this version of PBO. CSCV may use later blocks for in-sample selection, so it is a symmetric diagnostic of the candidate matrix, not a time-ordered deployment result. Actual deployment requires a separate Walk-Forward analysis. The strict definition uses $r < N/2$, while CSCV uses $r \leq (N + 1)/2$; the thresholds should both be disclosed because they differ for even N too.

32-L04 For AR(1), $\sum_{k\geq 1} \rho_k = \sum_{k\geq 1} \phi^k = \phi/(1 - \phi)$, so the long-run variance factor is $(1 + \phi)/(1 - \phi) = 3$ and effective sample size is approximately $n/3$. The expression $1 + 2\rho_1 = 2$ keeps only the first lag and omits later autocorrelations. This is an approximation for inference about the mean, not a general PSR correction. The French data are a revised historical series rather than point-in-time data, and industry portfolio returns have no actual execution, borrowing, or capacity records. The diagnostic therefore does not establish tradable net alpha.

32-L05 Let $k = \sqrt{n - 1} > 0$. The original z is $-0.05k$; after D increases from 1 to 4, it is $-0.025k$. The negative z rises toward zero and PSR moves up toward 0.5. With a zero threshold, z moves from $0.10k$ to $0.05k$, so PSR moves down toward 0.5. The direction of the decrease requires the observed Sharpe to be above the threshold; equality leaves PSR at 0.5. This comparison holds the stated inputs fixed and does not claim that the moments of an actual distribution can be changed independently.

Chapter 33 —Risk Management

Yuxiao Ma

Define loss as $L = -(V_{t+h} - V_t)$, so positive values are losses. VaR is a quantile; ES is the average loss in the worst tail. Report both with their horizon, units, and estimation conditions. A stress scenario does not have an invented probability, and liquidity or margin constraints cannot be inferred from daily-return standard deviation. Feed the risk limit into target weights, then send them to the Chapter 29 ledger for execution.

Trigger. When VaR contains a large atom of probability mass, why is the mean loss "above VaR" not ES?

Definition 33.1 *(Quantile and tail mass).* $VaR_\alpha = \inf\{\ell : F_L(\ell) \geq \alpha\}$,

$$ES_\alpha = \frac{\mathbb{E}[L\mathbf{1}_{L>v}] + v(F_L(v) - \alpha)}{1 - \alpha}, \quad v = VaR_\alpha.$$

The probability mass at VaR that must be allocated to the tail is _____. If there is no atom at that point and the tail is integrable, ES can be written as the conditional mean over the corresponding continuous tail.

Start with units: monetary L and return loss $-R$ are separate; VaR and ES inherit their respective units and horizons. Shares × currency/share × return gives PnL; NAV-normalized weights must not be divided by NAV again. Actual positions on the signal date, the next order, and exposures after execution are three distinct objects.

Your turn. 33-L01 Losses in dollars are $-10, 0, 5, 20$, with probabilities $0.50, 0.30, 0.15, 0.05$. The confidence level is 90%. Find VaR; how much probability at 20 and 5 belongs to the worst 10% tail; and what is ES? Calculate the conditional mean strictly above VaR and the conditional mean at or above VaR, and explain why neither is correct.

Your turn. **33-L02** NAV is USD 1,000; A holds 10 shares at USD 50, B holds −5 shares at USD 40, and cash is USD 700. Under a stress scenario, A falls 20% and B rises 10%; ignore fees and financing. Revalue NAV and loss, then calculate net exposure, gross exposure, and leverage. Why can this scenario not be called a 99% loss?

Your turn. 33-L03 You need to sell 12,000 shares; ADV is 10,000 shares per day and the participation rate is 10%. How many days are required to exit? If the limit requires exit within 5 days, how many shares are allowed at most? If portfolio weights are all multiplied by 0.4, must NAV immediately reflect the same risk reduction?

Your turn. 33-L04 A 99% one-day VaR has no exceedances in 250 forecasts made in advance. What is the binomial probability of zero exceedances? What is the Kupiec unconditional coverage statistic? Can non-rejection prove that the risk model is accurate? What happens if the threshold is fitted on those same 250 days?

Your turn. **33-L05** Continue 33-L02. Before the stress, risky-asset net exposure is USD 300 and NAV is USD 1,000. State the units of the monetary net amount, net weight, and net leverage. If the signal-date target is scaled to 0.4 but has not yet traded, is the loss on the current shares USD 120 or USD 48? When can USD 48 apply?

Chapter 33 —Answers and Checks

Definition blank $F_L(v) - \alpha$.

33-L01 The cumulative probabilities at losses $-10, 0, 5, 20$ are $0.50, 0.80, 0.95, 1.00$, so $VaR_{0.9} = 5$. The 10% tail takes all 5% of the probability at 20 and 5% of the probability at 5; $ES = (0.05 \times 20 + 0.05 \times 5)/0.10 = 12.5$. The conditional mean strictly above VaR is 20; the mean at or above VaR is $(0.15 \times 5 + 0.05 \times 20)/0.20 = 8.75$. The first has less than 10% tail mass; the second includes 20%. ES takes exactly the worst 10%, not an arbitrary one-sided conditional event.

33-L02 Initial NAV is $700 + 500 - 200 = 1000$; stressed NAV is $700 + 400 - 220 = 880$, so the loss is USD 120. Initial risky-asset net market-value exposure is $500 - 200 = 300$ USD, or 0.3 of NAV; gross exposure is $500 + 200 = 700$ USD, or 0.7 of NAV, so gross leverage is 0.7. After stress, net and gross exposure are $180/880 \approx 0.2045$ and $620/880 \approx 0.7045$. These ratios are normalized by current NAV. The specified shocks are scenario assumptions, not estimated probabilities, and do not automatically correspond to 99% VaR.

33-L03 Daily capacity is $0.1 \times 10000 = 1000$ shares, so exit takes 12 days; the 5-day limit allows 5,000 shares. This linear calculation ignores market impact, changes in ADV, and queueing, so it is a teaching approximation before actual execution constraints. Scaling target weights does not mean immediate execution: the target is formed after the signal and must be filled at a later permitted time under the lag. Risk from actual positions remains until the trade. Scaling can also change cash, financing, and costs, so they must be entered in the ledger again.

33-L04 Under independent daily exceedances with probability 0.01, the probability of zero is $0.99^{250} \approx 0.08106$. At the boundary $x = 0$, use $0 \log 0 = 0$; $LR_{uc} = -2(250 \log 0.99) \approx 5.0252$, with asymptotic χ^2_1 p-value about 0.025. The 8.1% is the point probability of exactly zero exceedances; it is not the same as an asymptotic p-value. A two-sided exact test must first specify a rejection region. This is a 99% daily hand calculation, not the body's constructed 95% monthly backtest with 1,500 training plus 1,500 test observations. For small samples, examine exact binomial probabilities and test power; do not treat an asymptotic p-value as exact. Unconditional coverage does not test clustering. Fitting the threshold on the same 250 days reuses fitting information; instead, estimate tomorrow's threshold from a past training window and then collect the subsequent realized loss.

33-L05 The monetary net amount is USD 300; net weight $300/1000 = 0.3$ is dimensionless, and net leverage is 0.3. Once normalized, do not divide by NAV again. Before execution, the portfolio still holds 10 shares and -5 shares, so the stress loss remains USD 120. The loss falls to USD 48 only after the actual share quantities are scaled to 0.4, assuming the marks and shocks stay the same and costs are ignored. After execution, recalculate using the new shares, prices, and NAV.

Chapter 34 —Market Microstructure

A quote is a venue state, and the midpoint is a mark; an order is a conditional instruction, and a fill changes cash and holdings. A buy order can sweep several ask levels, and a limit-price touch does not guarantee a fill. Queues and priority rules depend on the venue, order class, and rule version. A snapshot or OHLCV cannot recover the complete historical matching process. All numbers in this chapter are constructed for teaching.

Trigger. The midpoint lies between the two quotes. Why does it not promise that even one share can trade at the midpoint?

Definition 34.1 *(Two-Sided Quotes).*

$$b_t = \text{best Bid}, \quad a_t = \text{best Ask}, \quad m_t = \frac{a_t + b_t}{2}, \quad S_t = a_t - b_t.$$

Here b_t, a_t, m_t, S_t are all in USD per share; use this definition only for valid two-sided quotes with $a_t > b_t$. The relative spread $s_t = S_t/m_t$ is dimensionless. The midpoint is a reference mark, not a fill record.

Your turn. 34-L01 Mapping: 34-E01. Explain Bid, Ask, Mid, and Quoted Spread separately. Can a screenshot of the best bid and ask prove that 1,000 shares filled at the Mid? What data are still missing?

Your turn. 34-L02 Mapping: 34-E02. Snapshot: Bid 100.00 with 300 shares, 99.98 with 400 shares; Ask 100.02 with 200 shares, 100.04 with 300 shares. Buy 350 shares, then sell 350 back against the same frozen book. List the fills at each level, both VWAPs, cash flows, midpoint-mark PnL, and hypothetical round-trip PnL.

Your turn. **34-L03** Mapping: 34-E03. Buyer-initiated fill, $P = 100.02, m_0 = 100.01, m_1 = 100.03$. Set $d = +1$ and calculate Effective Spread, Realized Spread, and Signed Price Impact; verify the identity. Which conclusions are markouts only, and which require causal identification?

Your turn. **34-L04** Mapping: 34-P01, 34-P02. Suppose the actual result is requested=350, filled=100, remaining=250, vwap=100.02, notional=10,002. Diagnose the settlement code below and explain its cash change, position change, and what happens with no fill. Then write corrected code and map it to the validation evidence for P01/P02.

```
cash_delta = -result.requested * result.vwap
position_delta = result.requested
```

Your turn. **34-L05** Mapping: 34-E04, 34-C01. An aggregate snapshot shows 8,000 shares at one price. List the events, rules, and clock fields needed to reconstruct a historical limit-order queue. Give at least two failure conditions that require rejecting a definite matching conclusion.

Your turn. **34-L06** Mapping: 34-C02. The same OHLCV bar is O=100, H=101, L=99, C=100, V=400 shares. Write two opposite paths with four prints of 100 shares each, event times 1, 2, 3, and 4, with each record available at `event_time+0.1`. The same 100-share DAY buy limit order at 99 is decided/sent at $t = 2.5$, with zero assumed latency and arrival at that time. Assume no ask at or below 99 is executable on arrival. Add the aggressor-side and toy-priority assumptions outside OHLCV: the order remains valid through the close, there is zero queue at 99 on arrival, and a later 100-share seller-initiated print at 99 matches this sole resting order first; other prints do not consume it. Explain the opposite fill outcomes and the research boundary.

Connection Exercise (Keep Its Own Number)

Your turn. **34-T01** Partial coverage: body 34 §3.1. A static sell request has size 100 shares and direction $d = -1$; the calculator returns a nonnegative fill quantity of 60 shares. Convert these to signed Q, F, R and identify which value updates holdings. Does a quote snapshot prove that 60 future shares will actually fill?

Chapter 34 —Answers and Recomputations

Definition blank The Mid does not guarantee a fill; it is only a reference mark. Depth, order arrival and routing, protection conditions, venue rules, and time order can all affect an actual fill.

34-L01 The Bid is the buyer's displayed price, and the Ask is the seller's displayed price; the Mid is their arithmetic average; Quoted Spread is Ask minus Bid. A screenshot cannot establish that there was a counterparty at the Mid. We also need executable contra-side depth when the order arrives, venue and routing, order parameters, time sequence, and rule state; historical matching also requires order-level events.

34-L02 Buy: 200 shares at 100.02 and 150 shares at 100.04; notional USD 20,004 + 15,006 = 35,010, VWAP 100.028571 ... USD/share. Sell back: 300 shares at 100.00 and 50 shares at 99.98; proceeds USD 30,000 + 4,999 = 34,999, VWAP 99.997142 ... USD/share. Initial cash is USD 35,010; after the buy, cash is 0 and holdings are 350 shares. Marked at the arrival Mid of USD 100.01/share, equity is USD 35,003.50 and midpoint-mark PnL is -USD 6.50. Hypothetical round-trip PnL after the sale is -USD 11. The Mid mark does not execute the sale, and the frozen book is only a constructed assumption.

34-L03 Effective Spread $= 2(100.02 - 100.01) = 0.02$; Realized Spread $= 2(100.02 - 100.03) = -0.02$; Signed Price Impact $= 2(100.03 - 100.01) = 0.04$ USD/share, so $0.02 = -0.02 + 0.04$. The factor of 2 gives a full-spread-equivalent measure; for a one-sided nonnegative fill size $Q_f = |F|$, arrival-midpoint cost is $Q_f \times EffectiveSpread/2$. The identity is an algebraic decomposition. Ex post midpoint changes can reflect news, other orders, and common market movements; causal impact or private information requires a prespecified counterfactual or identification design.

34-L04 The incorrect code treats all 350 requested shares as filled: it records a cash outflow of USD 35,007 and a position of 350 shares, overstates the cash deduction by USD 25,005, and overstates the position by 250 shares compared with the actual 100 shares filled at 100.02 for USD 10,002. With no fill, `result.vwap=None`; multiplying an integer by it raises `TypeError`. The runnable code below uses `filled` and `notional` for posting and asserts both partial-fill and no-fill states.

```
from decimal import Decimal
from quant_course.microstructure import Level, OrderBook, sweep_displayed_book

thin = OrderBook(bids=(), asks=(Level(Decimal("100.02"), 100),))
partial = sweep_displayed_book(thin, "buy", 350)
assert (partial.requested, partial.filled, partial.remaining) == (350, 100,
    ↪ 250)
assert partial.notional == Decimal("10002.00")
assert partial.vwap == Decimal("100.02")
cash_delta = -partial.notional
position_delta = partial.filled
assert (cash_delta, position_delta) == (Decimal("-10002.00"), 100)

empty = OrderBook(bids=(), asks=())
no_fill = sweep_displayed_book(empty, "buy", 350)
assert (no_fill.filled, no_fill.remaining, no_fill.notional, no_fill.vwap) ==
    ↪ (
    0, 350, Decimal("0"), None
)
assert (-no_fill.notional, no_fill.filled) == (Decimal("0"), 0)
```

The P02 input-validation mapping is to the following named test nodes:

```
test_rejects_crossed_unsorted_or_off_tick_book
test_empty_contra_side_returns_no_fill
```

A crossed book has a best bid at or above the best ask, violating the valid uncrossed-book condition. Bid levels must be ordered from highest to lowest and ask levels from lowest to highest; the input is rejected rather than silently sorted. An off-tick price violates the permitted price increment and is rejected rather than rounded. The parameterized node covers crossed, unsorted, and off-tick books; the other covers an empty contra side. From the project root, run the complete command in body 34-P02. The API uses `filled` for executed quantity and `notional` for actual cash amount; VWAP is defined only when `filled>0`.

34-L05 At minimum, provide order ID, side, price, and display class; order add, execution, quantity reduction, cancel, delete, and reprice events; event sequence and `event_time`; `available_at`/`ingested_at`; venue-receive, acknowledgement, and fill times; session state; feed-sequence continuity; and the applicable rule version. Same-timestamp order follows the feed or venue definition. A message-sequence gap, unknown rule version, clocks that cannot order events, or a missing required order event each requires refusing to report an exact queue position or fill quantity.

34-L06 For each record below, the pair is price (USD/share), shares. Time is in constructed teaching units; event times t_1, t_2, t_3, t_4 are 1, 2, 3, and 4, and each record becomes available 0.1 after its event time.

Path A: $(100, 100), (101, 100), (99, 100), (100, 100)$.

Path B: $(100, 100), (99, 100), (101, 100), (100, 100)$. Both have $O = 100, H = 101, L = 99, C = 100, V = 400$.

In both paths, the same DAY buy limit order for 100 shares at 99 is decided and sent at $t = 2.5$, with zero assumed latency and arrival at that time. Assume the order remains active through the close, there is no ask at or below 99 on arrival, and the queue at 99 is empty. In path A, a 100-share seller-initiated print occurs at 99 at $t_3 = 3$; under the toy-priority rule, it matches this sole resting buy order first, so the order fills. Other prints do not consume it. In path B, the 99 print occurred before arrival and there is no later seller-initiated print at 99; the order remains unfilled at the close. Continued order validity, aggressor side, empty queue, and toy priority are assumptions outside OHLCV; print order alone does not prove a fill.

A defensible statement is: "Daily-bar proxy results do not prove a real limit-order fill; identical OHLCV cannot recover order sequence, contra-side quantity, or queue."

Connection Exercise Answer

34-T01 $Q = -100, F = -60, R = -40$, so $Q = F + R$. Only F changes holdings, reducing the position by 60 shares; the unfilled 40 shares do not change the ledger. The calculator's output is a conditional sweep under the assumption that visible static depth is executable; it does not prove a future fill. Queue, cancellations, hidden quantity, and latency have not been reconstructed.

Chapter 35 —Transaction Costs

> Paper target weights must pass through explicit fees, bid-ask spreads, market impact, financing, and borrow frictions before they become filled shares, cash balances, and portfolio NAV. First accounting rule: the fill price already embeds half-spread and temporary impact, so the cash ledger must not deduct them again. A fill VWAP exists only for a nonzero actual fill; fee, cash, and fill-price terms in IS are summed only over the nonzero-fill set, with an empty sum equal to zero. This model calls a change in the post-trade reference price permanent impact, but a finite observation window cannot prove that real-world impact is permanent. Keep one-sided stock turnover distinct from total traded notional; self-financing buy quantity is constrained by a quadratic equation. The data, capacity estimates, and ablation examples in this chapter are constructed for teaching and do not represent real-market Alpha or universal empirical laws.

Trigger. The optimizer recommends 30% turnover, and the buy fill price already includes half-spread and impact. Why must the spread never be deducted again in the cash ledger? If the account has only USD 100,000 in cash, how many shares can it buy after frictions?

Unit bridge: prices, proportions, and slopes

Absolute spread S is USD/share; relative spread $s = S/P^{\text{mid}}$ is dimensionless. A price can only be adjusted by $S/2 = P^{\text{mid}}s/2$. A linear-impact slope λ_{impact} is USD/share^2; turnover τ_{turnover} is dimensionless. In the abstract volatility model, first convert daily return volatility into $\sigma_{P,1\text{d}} = P^{\text{mid}}\sigma_{r,1\text{d}}$ to obtain a per-share price impact; the daily-return standard deviation uses a return window frozen before the decision. The current engine uses a dimensionless scenario coefficient multiplied by `current_mid` and a power of participation; it has no calibrated daily-volatility input. The original 24-month ablation chart and the rebound described in 35-L06 retain their historical R2R6 identity; see Section 35§10.3 for the current refresh, and do not mix the two versions.

Definition 35.1 *(Trading frictions and cash-settlement contract).* Define $\mathcal{J}_F = \{i : F_i \neq 0\}$. A fill VWAP P_i^{fill} is defined only for $i \in \mathcal{J}_F$; at zero fill it is undefined, and the empty sum is zero. For a buy ($F_i > 0$), the fill price is $P_i^{\text{fill}} = P_i^{\text{mid}} + _____ + g_i(|F_i|)$; for a sell ($F_i < 0$), it is $P_i^{\text{fill}} = P_i^{\text{mid}} - _____ - g_i(|F_i|)$. The cash change for fills satisfies

$$\text{cash}_{\text{after}} = \text{cash}_{\text{before}} - \sum_{i \in \mathcal{J}_F} \left(F_i P_i^{\text{fill}} + _____ \right) + \text{Interest}_{\text{net}} - \text{Borrow}.$$

Spread and temporary impact are embedded in P^{fill}; you _____ deduct them from cash a second time. In this model, permanent impact changes the subsequent _____ path; a finite observation window cannot prove that real-world impact is permanent. One-sided turnover for tradable risky securities is $\tau_{\text{risky, one-sided}} = \frac{1}{2}\sum_{i \in \mathcal{S}_{\text{risky}}} |\Delta w_i|$. If pre- and post-trade weights use the same NAV and a common mark, and all orders fill at that mark, $V_{\text{trade}} = 2\tau_{\text{risky, one-sided}}\,\text{NAV} =$ _____ is actual two-sided stock traded notional. In general it is only a requested-notional proxy inferred from weights; with partial fills, calculate actual traded notional from filled shares and fill prices. Cash weights are used for settlement only and are excluded from stock trading statistics.

Your turn. 35-L01 Full coverage: 35-E02. Portfolio NAV is NAV = \$5,000,000. Assume the pre- and post-trade weights use the same NAV and a common mark and all orders fill at that mark. Stock weights move from $[0.40, 0.30, 0.20]$ to $[0.20, 0.50, 0.10]$; cash weight also moves from 0.10 to 0.20. For $\mathcal{S}_{\text{risky}} = \{1, 2, 3\}$, calculate one-sided risky-security turnover $\tau_{\text{risky, one-sided}}$ _____ and actual two-sided stock traded notional V_{trade} _____. At a cost rate of 12 bps of actual stock traded notional, calculate friction cost _____ and basis points of NAV _____. What separate statistic can be reported for all-asset weight changes? Why can it not replace stock turnover?

Your turn. 35-L02 Full coverage: 35-E01; partial coverage: the core anti-double-charge assertion in 35-P01. Diagnose the code below, write the correct cash recursion, and calculate the numerical difference:

```
cash_balance -= Q * fill_price
cash_balance -= Q * (0.5 * S)          # Researcher note:
   ↪ deduct spread loss.
cash_balance -= Q * (impact_coef * Q)  # Researcher note:
   ↪ deduct impact loss.
cash_balance -= commission
```

Treat the order input as a confirmed actual fill, so the quantity in the code should be F, not the parent-order request Q. Identify the serious accounting error. When $Q = 1{,}000$ shares, $P^{\text{mid}} = \$50.00$, $S = \$0.04$, $\lambda_{\text{impact}} Q = \$0.08$, and commission is USD 5.00, how much extra cash does the incorrect logic deduct? What serious additional defects arise if there is no fill ($F = 0$) or only a partial fill ($F < Q$)?

Your turn. 35-L03 Full coverage: 35-E03. A buy order has decision price 80.00, arrival price 80.10, and a fill of 8,000 shares at 80.25; the remaining 2,000 shares are canceled at a cutoff price of 80.50. Commission on actual fills is USD 0.005 per share. Fill in delay, execution, opportunity, and fee terms using the signed IS convention _____, then recompute the total using the direct-dollar identity. Which items are cash flows and which are attribution relative to an execution benchmark?

Your turn. 35-L04 Full coverage: 35-E04; partial coverage: the single-recall risk condition in 35-C02. Over 30 days on ACT/360, five short positions have absolute market values [USD 4.9M, 0.1M, 0, 0, 0] and annual borrow rates [0%, 12%, 0, 0, 0] (rates and positions remain constant within the interval). Negative cash is USD 2M, with a 6% financing rate. Calculate each borrow fee _____, financing interest _____, and total _____. Explain why an equal-weight average rate cannot be multiplied by total short market value. If the recalled security jumps by u, short exposure/NAV is L_s, and cover friction is c_{liq}, write a sufficient condition for one event to exhaust NAV.

Your turn. 35-L05 Partial coverage: the self-financing root in 35-P02 and the linear-cost subproblem in 35-C01. Let $C = \$100{,}000$, $P^{\text{mid}} = \$50.00$, $S/2 = \$0.02$, $\lambda_{\text{impact}} = 0.0001$ USD/share^2, and $c_{\text{share}} = \$0.005$/share. Find the nonnegative integer share quantity satisfying $F(P^{\text{mid}} + S/2 + \lambda_{\text{impact}}F) + c_{\text{share}}F \le C$. For 35-C01 with fixed total duration T, assume liquidity resets in each slot and g does not vary with T/N; βN is an exogenous per-slot penalty. For $J(N) = \eta X^{1+\alpha} N^{-\alpha} + \beta N$, derive the continuous optimizer when $\alpha = 1$. State the feasible domain $N \in \{1, 2, \ldots\}$; if $N^* < 1$, evaluate from $N = 1$, and compare costs at feasible adjacent integers.

Your turn. 35-L06 Partial coverage: 35-P03 and the monthly capacity arithmetic in Project 4 Lab Section 8; related to the scope of cost-ablation conclusions. All inputs are monthly: $\alpha_{\text{gross}} = 1.20\%$, one-sided monthly turnover of tradable risky securities $\tau_{\text{turnover}} = 1.0$, $D = 21$ trading days, $c_0 = 8.25$ bps, $\gamma = 0.05$, and average daily ADV of USD 100M. Using this month's $V_{\text{trade}} = 2\tau_{\text{turnover}} A$, calculate break-even AUM and the AUM maximizing dollar return. Annualized turnover may be reported separately as monthly turnover times 12, outside the formula. Finally, explain in one sentence why a 24-month layered ablation is not an orthogonal causal decomposition.

Your turn. **35-L07** Related to the event clock and accrual convention in Section 5.5. Use ACT/360, a 4% annual cash rate, and a 1.5% annual borrow rate. On day 0, short 100 shares of A at 100 with initial cash of USD 100,000. On day 10, first settle the old balances, then buy 50 shares of B at 100, and observe a new mark of 105 for A. Settle at month-end on day 30. Fill in cash after the day-0 short sale _____, day 0–10 interest and borrow fee _____, cash after buying B _____, and day 10–30 interest and borrow fee _____. Can the new mark be backfilled into the first 10 days? Even a slot with no fills produces a mark at slot end: from which interval does the new mark become the borrow-fee basis?

Connection Exercise (Keep Its Own Number)

Your turn. **35-T01** Full coverage: main-text Section 35§4.3. Initial cash is 2,000 and shares are 0. Buy 100 shares, but fill only 60 at 10.05 with a 0.30 fee. Marks are first 10.02 and then 10.10. Calculate cash, holdings, and NAV at both marks. If IS of 7.30 is deducted again, what is the erroneous NAV? Now use price 50 and $s = 0.0008$ to calculate the half-spread using both the absolute and relative spread.

Chapter 35 —Answers and Reconciliation

Definition blanks $S/2$ (half-spread); $S/2$; commission$_i$; must not; midpoint (mark); AUM $\cdot$ $\sum_{i=1}^{3} |\Delta w_i|$.

35-L01 The sum of absolute stock-weight changes is $0.20 + 0.20 + 0.10 = 0.50$, so one-sided turnover for the tradable risky securities is $\tau_{\text{risky, one-sided}} = 0.50/2 = 0.25$. If the cash-weight change of 0.10 is also included in an all-asset statistic, one-sided all-asset weight reallocation is $(0.50+0.10)/2 = 0.30$, but that figure does not correspond to stock orders. Two-sided stock traded notional is $V_{\text{trade}} = 2\times0.25\times5{,}000{,}000 = \$2{,}500{,}000$. The combined friction rate is 12 bps $= 0.0012$. Total friction cost is Cost $= \$2{,}500{,}000\times0.0012 = \$3{,}000.00$. In basis points of total portfolio AUM, $\$3{,}000/\$5{,}000{,}000 = 0.0006 = 6.0$ bps. Cash is a passive settlement pool: changes in cash are the monetary counterpart of stock purchases and sales, not orders matched on the exchange limit order book, so they do not create stock spread, impact, or exchange fees. Including cash in the friction denominator would overstate turnover cost.

35-L02 Error diagnosis: by definition, the buy fill price $P^{\text{fill}} = P^{\text{mid}} + 0.5S + \lambda_{\text{impact}}Q$ already includes half-spread and temporary impact, so the purchase payment is already $Q \cdot P^{\text{fill}}$. The second and third lines deduct those spread and impact costs again, double charging them. The extra amount is the sum of the second and third lines:

$$\Delta\text{Cash}_{\text{err}} = Q(0.5S) + Q(\lambda_{\text{impact}}Q) = 1{,}000 \times 0.02 + 1{,}000 \times 0.08 = \$20.00 + \$80.00 = \$100.00.$$

The incorrect logic removes an extra USD 100.00 from the account. Zero- and partial-fill defects: if $F < Q$, the code uses the parent-order request Q to calculate payment and commission, posting payment for unfilled shares and overcharging commission. If there is no fill ($F = 0$), the fill price is undefined (for example, `None`); evaluating `Q*fill_price` raises `TypeError`, and the code also incorrectly charges commission on a zero fill. A self-contained correct ledger entry and anti-double-charge unit test (mapped to `tests/test_execution_costs.py`, test `test_no_duplicate_fees_in_cash_ledger`):

```
import datetime as dt
import math
from quant_course.execution_costs import ExecutionConfig,
    ↪ ExecutionPortfolioLedger, Fill

t0 = dt.datetime(2026, 10, 9, 9, 30, tzinfo=dt.timezone.utc)
fill_time = t0 + dt.timedelta(minutes=1)
cfg = ExecutionConfig(commission_rate=0.001, half_spread_bps=5.0,
    ↪ temp_impact_coef=0.10)
ledger = ExecutionPortfolioLedger(100_000.0, cfg, fill_time)
# Fill price embeds half-spread and temporary impact; post only signed_qty *
    ↪ fill_price + commission.
fill = Fill(
    fill_id="fill-1", order_id="order-1", parent_id="order-1", slot_index=0,
    fill_time=fill_time, signed_quantity=100.0,
    mid_price=100.0, fill_price=100.15, notional=10_015.0,
    half_spread_usd=0.05, temp_impact_usd=0.10,
    perm_impact_usd=0.0, commission=10.015,
)
cash_before = ledger.cash
ledger.post_fill("XYZ", fill, {"XYZ": 100.0})
assert math.isclose(cash_before - ledger.cash, 100.0 * 100.15 + 10.015,
    ↪ abs_tol=1e-10)
```

35-L03 Inputs: $Q = +10{,}000$, $F = +8{,}000$, $R = +2{,}000$; $P^{\text{dec}} = 80.00$, $P^{\text{arr}} = 80.10$, $P^{\text{fill}} = 80.25$, $P^{\text{term}} = 80.50$; and $c_{\text{share}} = 0.005$. Calculate each of the four terms: - Delay cost: Delay $= Q(P^{\text{arr}} - P^{\text{dec}}) = 10{,}000 \times (80.10 - 80.00) = 10{,}000 \times 0.10 = +\$1{,}000.00$. - Execution friction: Execution $= F(P^{\text{fill}} - P^{\text{arr}}) = 8{,}000 \times (80.25 - 80.10) = 8{,}000 \times 0.15 = +\$1{,}200.00$. - Opportunity cost: Opportunity $= R(P^{\text{term}} - P^{\text{arr}}) = 2{,}000 \times (80.50 - 80.10) = 2{,}000 \times 0.40 = +\$800.00$. - Explicit fee: Explicit $= |F|c_{\text{share}} = 8{,}000 \times 0.005 = +\$40.00$. Total implementation shortfall is $\text{IS}_{\text{total}} = 1{,}000.00 + 1{,}200.00 + 800.00 + 40.00 = +\$3{,}040.00$. Direct-dollar cross-check:

$$\begin{aligned}\text{IS}_{\text{direct}} &= F(P^{\text{fill}} - P^{\text{dec}}) + R(P^{\text{term}} - P^{\text{dec}}) + \Xi \\ &= 8{,}000 \times 0.25 + 2{,}000 \times 0.50 + 40.00 \\ &= 2{,}000.00 + 1{,}000.00 + 40.00 = +\$3{,}040.00.\end{aligned}$$

The two methods agree exactly (error $< 10^{-10}$). Accounting boundary: unfilled opportunity cost is a hypothetical attribution relative to the paper-portfolio target benchmark. It measures the theoretical gain missed when shares not purchased continue to rise in the market; the account never owned those 2,000 shares, and no cash flow for them occurred. Deducting opportunity cost from cash again would break reconciliation with the custodian's cash ledger.

35-L04 Calculate borrow fees security by security: $4{,}900{,}000 \times 0\% \times 30/360 + 100{,}000 \times 12\% \times 30/360 = \$1{,}000$. Financing interest on negative cash is $2{,}000{,}000 \times 6\% \times 30/360 = \$10{,}000$, for a total of \$11,000. Applying the equal-weight average rate of 2.4% to total short market value gives a borrow fee of \$10,000, ten times the correct amount because the positions are not equally weighted. If the loss at one recall is $L_s u + c_{\text{liq}} \geq 1$ as a fraction of NAV, that event can exhaust NAV. A positive recall probability implies a positive-probability tail event, but it is not enough to infer net-return survival from gross Sharpe.

35-L05 1. Closed-form self-financing purchase quantity. The equation is $\lambda_{\text{impact}}F^2 + (P^{\text{mid}} + S/2 + c_{\text{share}})F - C = 0$. Let $A = \lambda_{\text{impact}}$ and $B = P^{\text{mid}} + S/2 + c_{\text{share}}$. If $A = 0$ (no impact), $F^* = \lfloor C/B \rfloor$. If $A > 0$, take the positive root:

$$F_{\text{exact}} = \frac{2C}{B + \sqrt{B^2 + 4AC}}, \qquad F^* = \lfloor F_{\text{exact}} \rfloor.$$

Actual cash used is $C_{\text{used}} = F^*(P^{\text{mid}} + S/2 + \lambda_{\text{impact}}F^*) + c_{\text{share}}F^*$, and residual cash is $R_{\text{cash}} = C - C_{\text{used}} \geq 0$. Substituting the given values gives a positive root of about 1,991.0757 shares. Thus $F^* = 1{,}991$ shares can be purchased; outlay is about USD 99,996.1831 and residual cash is about USD 3.8169. 2. Optimal number of slots. Under this problem's fixed T, per-slot liquidity reset, and exogenous βN penalty, the continuous relaxation is $J(N) = \eta X^2 N^{-1} + \beta N$; this is not a standard delay-risk term implied automatically by fixed duration. Feasible execution counts are $N \in \{1, 2, \dots\}$. The first-order condition is $\frac{dJ}{dN} = -\eta X^2 N^{-2} + \beta = 0 \implies N^* = X\sqrt{\frac{\eta}{\beta}}$. The second derivative is $\frac{d^2J}{dN^2} = 2\eta X^2 N^{-3} > 0$, so the function is strictly convex and has a unique global minimum. For an engineering choice, compare $J(\max(1, \lfloor N^* \rfloor))$ and $J(\max(1, \lceil N^* \rceil))$; if the continuous solution is below 1, choose $N = 1$. For general $\alpha > 0$, the continuous solution is $N^* = (\alpha\eta X^{1+\alpha}/\beta)^{1/(1+\alpha)}$; this exercise calculates only the $\alpha = 1$ case.

35-L06 1. Monthly capacity calculation: Effective fixed-cost-adjusted Alpha is $\alpha_{\text{eff}} = \alpha_{\text{gross}} - 2\tau_{\text{turnover}}c_0 = 0.012 - 2(1.0)(0.000825) = 0.01035$. The nonlinear-impact coefficient is

$$k = \frac{\gamma(2\tau_{\text{turnover}})^{1.5}}{\sqrt{D \cdot \text{ADV}}} = \frac{0.05 \times 2^{1.5}}{\sqrt{21 \times 10^8}} = \frac{0.14142136}{45{,}825.757} \approx 3.086067 \times 10^{-6}.$$

Break-even scale is

$$A_{\text{break}} = \left(\frac{\alpha_{\text{eff}}}{k}\right)^2 = \left(\frac{0.01035}{3.086067 \times 10^{-6}}\right)^2 \approx (3{,}353.78)^2 \approx \$11{,}247{,}862.5$$

(about USD 11.25M). The scale maximizing net dollar profit is

$$A_{\text{opt}} = \frac{4}{9}A_{\text{break}} \approx \$4{,}999{,}050.0$$

(about USD 5.00M). 2. Frequency boundary, ablation scope, and evidence limits. The stated $\tau_{\text{turnover}} = 1.0$ is one-sided monthly turnover, not annualized turnover. If annualized statistics are reported, separately calculate $12 \times 1.0 = 12.0$ times per year; do not insert this value into the monthly impact formula with $D = 21$. The cost ablation is a sequential demonstration on a fixed sample path: each later stage inherits prior holdings and cash changes, so it is not an orthogonal causal effect for each cost component. Project 4's ablation adds spread, temporary impact, commission, and financing/borrow in sequence; it is a path-dependent chronology. At stage five, net return rebounds from −4.99% to −2.66% because the constructed path holds net cash and earns positive interest, which offsets borrow cost. This depends strongly on cash holdings on that particular simulated path; it does not mean "borrow fees improve strategy performance," and it is not an orthogonal causal decomposition of cost components. The experiment does not use real exchange-level order-book data (L3), broker fill records, or timestamped historical borrow rates and availability. All capacity thresholds and cost curves therefore validate only the self-financing accounting workflow; they must not be interpreted as live capacity or evidence of tradability for a real strategy.

35-L07 Cash after the day-0 short sale is $100{,}000 + 100 \times 100 = \$110{,}000$. For the first 10 days, interest is $110{,}000 \times 0.04 \times 10/360 = \$122.2222$ and the borrow fee is $100 \times 100 \times 0.015 \times 10/360 = \$4.1667$. Post net accrual of 118.0556, then buy B for USD 5,000; cash becomes \$105,118.0556. Over the next 20 days, interest is $105{,}118.0556 \times 0.04 \times 20/360 = \$233.5957$. A's new mark is

effective only from day 10, so the borrow fee is $100 \times 105 \times 0.015 \times 20/360 = \$8.75$. Total interest over both intervals is \$355.8179 and total borrow fee is \$12.9167. Accrual uses simple ACT/360 within each interval; after each settlement, interest is included in cash and becomes part of the next interval's cash base. A mark first received on day 10 cannot be backfilled into the earlier fee base. Even a slot with no fills records a new available mark at slot end: accrue through slot end using the old mark, and use the new mark only for subsequent intervals.

Connection Exercise Answer

35-T01 Cash is $C = 2{,}000 - 60(10.05) - 0.30 = 1{,}396.70$ and holdings are $H = 60$. NAV at the first mark is $1{,}396.70 + 60(10.02) = 1{,}997.90$; NAV at the cutoff mark is $1{,}396.70 + 60(10.10) = 2{,}002.70$. The new mark does not change cash. If IS of 7.30 is deducted again, erroneous NAV is 1,995.40. For price 50 and $s = 0.0008$, $S = 50(0.0008) = 0.04$ USD/share and $S/2 = 0.02 = P^{\text{mid}} s/2$. Do not add $s/2 = 0.0004$ directly to the price.

Chapter 36 —Algorithmic Execution and Trade Scheduling

Algorithmic execution decomposes the optimizer's target into discrete child orders with causal timestamps. The central task is to balance market impact, which grows with trading speed, against price risk, as delay moves the benchmark. TWAP allocates evenly over time; forecast-volume VWAP uses a curve frozen in advance; POV uses only lagged external volume that is fully observed, never future volume. With no fills, fill VWAP is mathematically undefined. The four-part implementation-shortfall decomposition equals the direct-amount method, and the Almgren–Chriss model reduces to TWAP in the risk-neutral limit. All schedules, algorithm comparisons, and routing models in this chapter are constructed for instruction and do not guarantee optimal execution.

Trigger. A large order must be completed over 13 intervals. Why must the VWAP child schedule never use the day's realized market volume after the close? Why must POV send zero shares in slot 0?

Local Notation and Evaluation Target

Here Δt is the slot width, σ is a price-diffusion coefficient in USD/share/$\sqrt{\text{time}}$, and λ_{risk} is in USD^{-1}. Do not confuse them with Chapter 35's dimensionless turnover or impact slope in USD/share^2. Almgren–Chriss requires $\tilde{\eta} = \eta/\Delta t - \gamma/2 > 0$. Remaining inventory decreases along a convex curve; when both risk aversion and diffusion are positive, more is sold early. If either is zero, take the equal-quantity TWAP limit. This analytic model and the scenario-matching model based on ExecutionSlot do not use the same engineering inputs.

Definition 36.1 *(Execution Clock and Causal Constraints).* Record signal-freeze time t_{signal}, decision time t_{dec}, parent-order receipt time t_{parent}, and for every child order its send, venue-arrival, fill, POV-observation-availability, and forecast-freeze times. For each child order, require $t_{\text{signal}} \le t_{\text{dec}} \le t_{\text{parent received}} \le t_{\text{arrival benchmark}} \le t_{\text{send},k} = t_{\text{venue},k} \le t_{\text{fill},k} \le t_{\text{term}}$; also require $t_{\text{obs},k} \le t_{\text{send},k}$ and $t_{\text{forecast},k} \le t_{\text{send},k}$. Slot windows may overlap. At each send, count only fills strictly earlier than the send and reserve sent-but-unconfirmed quantities; a send at the same instant as a fill confirmation comes first. Arrival is only the IS benchmark. POV observations must still be available before that child send, and the forecast curve must be frozen before the first send. Forecast-volume VWAP uses only an ex-ante frozen _____ curve; ex-post market VWAP is a _____ benchmark, never an ex-ante schedule input. POV child quantity depends only on external volume from the _____; in the first slot, with no prior volume, the request is ____. A subsequent child order may be sent before the previous slot ends, but the schedule must reserve the shares still in flight; it must not count an unknown future fill as filled quantity. For a single-direction parent, all fill-price sums run over the nonzero-fill set $\mathcal{F} = \{k : F_k \neq 0\}$, and an empty sum is zero. Do not replace a missing average price by zero or compute $0 \times$ undefined. Parent-order conservation is $Q =$ _____ $+ R$. With no fills, P_{fillVWAP} is _____, and explicit commission is ____.

Your turn. **36-L01** Full coverage: signed decomposition in 36-E01. A 20,000-share sell parent order fills 15,000 shares, leaving 5,000. Prices are 50.00 at decision, 49.80 at arrival, 49.50 average fill, and 49.00 at the deadline; commission is USD 0.005 per filled share. Compute all four components and the direct amount, then explain why falling prices create positive cost for a sell order.

Your turn. **36-L02** Full coverage: 36-E02. Allocate a 6,000-share buy over three equal time intervals, with prices 20.10, 20.30, and 20.50 and frozen forecast-volume proportions 1:2:3. Compare the quantities and fill VWAP under uniform and forecast-volume schedules. Which is an ex-ante input and which is an ex-post benchmark? What can this monotone-price example establish?

Your turn. 36-L03 Full coverage: 36-E03. The target participation rate is 20%, and previous fully observed external volume is 40,000 shares. Compare child quantities under exclusive and inclusive denominators; derive the limit as the inclusive rate approaches 100%, and explain why both participation and parent-remainder caps must remain in force.

Your turn. 36-L04 Full coverage: 36-E04; partial coverage: the no-fill boundary in 36-P02. A 5,000-share buy order has decision price 30.00, arrival price 30.10, terminal price 31.50, and no fills. Explain why fill VWAP is undefined; use the empty-fill-set convention for execution cost and fees, then verify total IS. Can IS be deducted from cash?

Your turn. 36-L05 Partial coverage: 36-P01 and Section 6 of INTERFACE_CONTRACT on causal POV. A trader writes the following child-order logic:

```
# Incorrect POV child-order logic (diagnose it)
def compute_pov_child(parent_order, current_slot,
   ↪ target_rate):
    # Bug: reads external volume realized before the current
   ↪  slot has ended
    market_vol = current_slot.realized_external_volume
    child_qty = target_rate * market_vol
    return min(child_qty, parent_order.remaining_qty)
```

Identify the serious look-ahead leak. Explain the noncausal behavior this code creates in slot 0. Give corrected logic based on lagged observed volume and an assertion checking `observation_time<=child_send_at`. Then use a volume perturbation to explain why the schedule input should remain unchanged. If a later child order is sent before the previous slot ends and its fill is unknown, explain how to treat the already-sent quantity; state the ordering when send and fill confirmation share a timestamp. Finally, give an example of cumulative fills exceeding the parent quantity or reversing its direction, and identify the assertion that the shared entrypoint should use to reject it.

Your turn. 36-L06 Full coverage: 36-E05; partial coverage: 36-C01; connection: 36-C02. First derive the interior difference equation for discrete Almgren–Chriss and its TWAP limit as $\lambda_{\text{risk}} \to 0$. Then take two buy paths with common endpoints and let $\Delta x(t) = x_{\text{fast}}(t) - x_{\text{slow}}(t)$ be the nonnegative excess early inventory. Derive the derivative of signal reward B with respect to decay rate ρ under fixed peak drift and under fixed finite-horizon signal area. Must the derivative signs always be opposite? Finally distinguish dark-pool per-share difference $M(z)$ from the marginal cost of changing route share; give the derivative when p, Δ depend on z, and explain whether a single point with $M(z) > 0$ implies $z^* = 0$.

Your turn. **36-L07** Partial coverage: the performance boundary test in 36-P01; connection: the maximum-drawdown definition in Section 12.6. Starting with NAV 1, compute wealth paths and maximum drawdown for monthly returns $[-10\%, -10\%]$ and $[-10\%]$, then determine drawdown for a monotonically rising path. Explain why there are still N returns but $N + 1$ wealth points for drawdown.

Connection Exercise (Keep Its Own Number)

Your turn. **36-T01** Full coverage: the connection example in body Section 4. A 100-share buy fills only 60 shares; decision price is 10.00, arrival 10.02, fill 10.05, terminal 10.10, and fees are 0.30. Calculate all four IS components and the direct amount. Repeat with $Q = -100, F = -60, R = -40$ and prices 10.00/9.98/9.95/9.90. Why must neither positive shortfall be deducted from cash a second time?

Chapter 36 —Answers and Recomputations

Definition blanks Ex-ante frozen forecast volume; ex-post evaluation; previous fully observed external volume; 0 shares; F; undefined (None/NaN); USD 0.00.

36-L01 Inputs: $Q = -20{,}000$ shares (sell), $P_{\text{dec}} = 50.00$, $P_{\text{arr}} = 49.80$, $F = -15{,}000$, $P_{\text{fillVWAP}} = 49.50$, $R = -5{,}000$, $P_{\text{term}} = 49.00$, and $c_{\text{comm}} = 0.005$. Compute each component: - Delay: $\text{IS}_{\text{delay}} = Q(P_{\text{arr}} - P_{\text{dec}}) = (-20{,}000)(49.80 - 50.00) = (-20{,}000)(-0.20) = +\text{USD } 4{,}000.00$. - Execution: $\text{IS}_{\text{exec}} = F(P_{\text{fillVWAP}} - P_{\text{arr}}) = (-15{,}000)(49.50 - 49.80) = (-15{,}000)(-0.30) = +\text{USD } 4{,}500.00$. - Opportunity: $\text{IS}_{\text{opp}} = R(P_{\text{term}} - P_{\text{arr}}) = (-5{,}000)(49.00 - 49.80) = (-5{,}000)(-0.80) = +\text{USD } 4{,}000.00$. - Explicit commission: $C_{\text{fees}} = |F|c_{\text{comm}} = 15{,}000(0.005) = +\text{USD } 75.00$. Total IS is $4{,}000.00 + 4{,}500.00 + 4{,}000.00 + 75.00 = +\text{USD } 12{,}575.00$. Direct amount:

$$\begin{aligned} \text{IS}_{\text{direct}} &= F(P_{\text{fillVWAP}} - P_{\text{dec}}) + R(P_{\text{term}} - P_{\text{dec}}) + C_{\text{fees}} \\ &= (-15{,}000)(-0.50) + (-5{,}000)(-1.00) + 75 \\ &= 7{,}500 + 5{,}000 + 75 = +\text{USD } 12{,}575.00. \end{aligned}$$

Economic meaning: the ideal sell target is to sell at a high price. Prices keep falling from decision through execution to the close ($P_{\text{term}} < P_{\text{fillVWAP}} < P_{\text{arr}} < P_{\text{dec}}$). All price changes are negative, as are signed sell quantities Q, F, R, so each product is positive. Positive shortfall measures the shortfall in proceeds relative to the paper target of selling all shares at decision price. A sell can reduce a long holding; it is not short-strategy PnL, and an existing short may benefit from the decline.

36-L02 1. TWAP schedule: $N = 3$, so $q_1 = q_2 = q_3 = 6{,}000/3 = 2{,}000$ shares. Fill VWAP: $P^{\text{TWAP}}_{\text{fillVWAP}} = 2{,}000(20.10 + 20.30 + 20.50)/6{,}000 = (20.10 + 20.30 + 20.50)/3 = \text{USD } 20.3000/\text{share}$. 2. Forecast-VWAP schedule: total forecast volume is 60,000 shares and the weights are $1/6, 2/6, 3/6$. Child quantities are $q_1 = 1{,}000, q_2 = 2{,}000, q_3 = 3{,}000$ shares. Fill VWAP: $P^{\text{VWAP}}_{\text{fillVWAP}} = [1{,}000(20.10) + 2{,}000(20.30) + 3{,}000(20.50)]/6{,}000 = (20{,}100 + 40{,}600 + 61{,}500)/6{,}000 = 122{,}200/6{,}000 \approx \text{USD } 20.3667/\text{share}$. 3. Economic logic and research boundary: for these constructed monotonically rising prices, buying earlier lowers average price. Forecast VWAP assigns 50% of the order to slot 3, shifting more buying into the higher-price period, so its average is worse than uniform TWAP. Ex-post market VWAP is a passive benchmark calculated after the close and is not known in advance; using realized future volume as a scheduling input would be look-ahead bias. This result applies only to this constructed rising path and does not rank VWAP for other rising, falling, or oscillating paths; it is not general empirical evidence.

36-L03 1. Child quantities under each definition: - Exclusive denominator (Definition A): $q = \rho V_{k-1}^{\text{obs}} = 0.20(40{,}000) = 8{,}000$ shares. - Inclusive denominator (Definition B): $q = \frac{\rho}{1-\rho} V_{k-1}^{\text{obs}} = \frac{0.20}{1-0.20}(40{,}000) = 0.25(40{,}000) = 10{,}000$ shares. 2. Singularity and economic meaning: the inclusive-denominator equation is $\rho = q/(V^{\text{obs}} + q)$, so $q = \rho V^{\text{obs}}/(1-\rho)$. As $\rho \to 1$, $1-\rho \to 0$ and $q \to +\infty$. The inclusive denominator counts own child quantity in total volume. When external volume is positive, any finite child quantity is strictly less than 100% of external plus own volume; mathematically, reaching 100% requires an infinite order. Engineering limits: participation thresholds are configurations to calibrate; the formula does not imply a safe range of 1–25%. Before sending, subtract confirmed fills and in-flight reservations from parent capacity. Use current-slot realized external volume for the actual matching cap, not stale pre-send observations as if they revealed current capacity. The uncapped inverse diverges, but the actual request remains finite under a finite remainder cap.

36-L04 1. Fill VWAP status: because actual fill quantity $F = 0$, the weighted-average denominator is zero and $P_{\text{fillVWAP}} = 0/0$ is undefined. The system must return `None` or `math.nan`. Assigning 0.00 would falsely imply that shares were bought at zero dollars and could produce nonsensical profits or break benchmark comparisons. 2. Decomposition and check: $R = Q - F = 5{,}000 - 0 = 5{,}000$ shares. - Delay: $\text{IS}_{\text{delay}} = 5{,}000(30.10 - 30.00) = +\text{USD } 500.00$. - Execution: the nonzero-fill set is empty, so $\sum_{i \in \mathcal{F}} f_i(P_{i,\text{fill}} - P_{\text{arr}}) = \sum_{i \in \emptyset}(\cdot) = 0$. Do not read undefined P_{fillVWAP} or construct $0 \times$ undefined. - Opportunity: $\text{IS}_{\text{opp}} = R(P_{\text{term}} - P_{\text{arr}}) = 5{,}000(31.50 - 30.10) = 5{,}000(1.40) = +\text{USD } 7{,}000.00$. - Explicit commission: $0(0.002) = \text{USD } 0.00$. Total IS is $500.00 + 0.00 + 7{,}000.00 + 0.00 = +\text{USD } 7{,}500.00$. Direct amount is $R(P_{\text{term}} - P_{\text{dec}}) = 5{,}000(31.50 - 30.00) = 5{,}000(1.50) = +\text{USD } 7{,}500.00$. The results agree exactly. 3. Ledger meaning: with no fills, the entire difference relative to decision price is attributed to delay and unfilled opportunity cost. These are evaluation benchmarks; actual cash change in the ledger is zero, and IS must not be deducted from cash.

36-L05 Diagnosis: `current_slot.realized_external_volume` is the volume that accumulates during the slot after the child order has been sent. At decision/send time t_{send} it is unknown future data. Reading that field creates look-ahead leakage. In slot 0 the incorrect code directly sizes from that slot's future volume, violating the causal rule that the initial request is zero when no prior volume exists. A repair that maps to `quant_course.execution_costs.size_pov_slot` and `test_execute_order_handles_overlap_without_using_unavailable_fills` is:

```
from quant_course.execution_costs import size_pov_slot

def compute_pov_child(
    parent_order,
    current_slot,
    cumulative_filled,
    reserved,
    target_rate,
):
    assert (
        current_slot.observation_time is None
        or current_slot.observation_time <= current_slot.child_send_at
    )
    # Reserve unconfirmed child quantity; current realized volume is for
    ↪ matching capacity only.
    return size_pov_slot(
        parent_order,
        current_slot,
        cumulative_filled,
        target_rate,
        reserved_quantity=reserved,
    )
```

Adversarial tests should mutate `current_slot.realized_external_volume` arbitrarily, for example multiply it by 100 or set it to zero; child-request quantity must remain unchanged because this field is used only for matching capacity. If the previous child order's fill time is not strictly earlier than the current send, POV sizing uses the quantity already sent as `reserved_quantity`; it must not count that fill as completed or release the same capacity twice. A send at the same instant is processed before fill confirmation. The `size_pov_slot` entrypoint first checks cumulative fills and reservations for finiteness, direction, and total quantity. For example, $Q = 100, F = 120$, $F = -1$, or $F = 90$ with a 20-share reservation should raise `ValueError`, not reverse direction or over-send.

36-L06 1. Discrete Euler–Lagrange equation and degenerate limit: the constant term $\gamma X^2/2$ is path-invariant only after an algebraic regrouping of the discrete cost; it is not the original discrete permanent-impact contribution. For example, with $\gamma = 0.01, \eta/\Delta t = 0.02, X = 1{,}000$, the original permanent term is 0 for $[1{,}000, 0]$ and 2,500 for $[500, 500]$, while the regrouped constant is 5,000 in both cases. For a sell path expressed using cumulative sales y_t, the instantaneous permanent-impact cost rate is $\gamma y_t \dot{y}_t$, and its cumulative cost is $\int_0^T \gamma y_t \dot{y}_t \, dt = \gamma X^2/2$. With remaining inventory $x_t = X - y_t$, the same integral under fixed endpoints can be written $-\int_0^T \gamma x_t \dot{x}_t \, dt = \gamma X^2/2$. The integrals agree, but their integrands are not equal at every instant. The objective is $U(x) = \frac{1}{2}\gamma X^2 + \tilde{\eta} \sum_{k=1}^{N} (x_{k-1} - x_k)^2 + \lambda_{\text{risk}} \sigma^2 \Delta t \sum_{k=1}^{N-1} x_k^2$. Differentiating at an interior node and setting the derivative to zero gives

$$\frac{\partial U}{\partial x_k} = 2\tilde{\eta}(x_k - x_{k-1}) - 2\tilde{\eta}(x_{k+1} - x_k) + 2\lambda_{\text{risk}} \sigma^2 \Delta t \, x_k = 0,$$

and therefore

$$x_{k+1} - \left(2 + \frac{\lambda_{\text{risk}} \sigma^2 \Delta t}{\tilde{\eta}}\right) x_k + x_{k-1} = 0.$$

As $\lambda_{\text{risk}} \to 0$, the coefficient $\lambda_{\text{risk}}\sigma^2\Delta t/\tilde{\eta} \to 0$ and the equation reduces to $(x_{k+1}-x_k)-(x_k-x_{k-1}) = 0$. Thus the first difference, and shares liquidated per slot $n_k = x_{k-1} - x_k$, is constant. With endpoints $x_0 = X, x_N = 0$, $n_k = X/N$ and $x_k = (1 - k/N)X$, proving the risk-neutral limit is constant-speed TWAP. 2. Decaying signal and dark-pool boundary: Let $\Delta x(t) = x_{\text{fast}}(t) - x_{\text{slow}}(t) \geq 0$ denote the extra early inventory on two buy paths with the same endpoint. Their signal-reward difference is $\Delta B(\rho) = \int_0^T \alpha_\rho(t)\Delta x(t)\,dt$. Under fixed peak, $\alpha_\rho(t) = \alpha_0 e^{-\rho t}$, so $\frac{d\Delta B}{d\rho} = -\alpha_0 \int_0^T te^{-\rho t}\Delta x(t)\,dt \leq 0$. Faster decay reduces the reward captured by the same early-inventory difference. For fixed finite-horizon signal area S_0, set $\alpha_\rho(t) = S_0\rho e^{-\rho t}/(1 - e^{-\rho T})$ and $m_\rho = 1/\rho - T/(e^{\rho T} - 1)$. Then $\frac{d\Delta B}{d\rho} = \int_0^T \alpha_\rho(t)(m_\rho - t)\Delta x(t)\,dt$. Its sign depends on the distribution of excess inventory $\Delta x(t)$ around m_ρ; with greater weight early, the derivative may be positive. Fixed area alone does not imply a global reversal of the optimal speed. The global schedule also depends on temporary impact, risk, capacity, and speed constraints. The dark-pool per-share difference is $M(z) = (1 - p(z))\Delta(z) - p(z)(c_A - c_B)$ and total cost satisfies $C(z)/Q = c_A + zM(z)$. $M(z) > 0$ means that, at this route share, the average cost per share assigned to dark routing is above the all-lit alternative. If p, Δ vary with z, the marginal derivative is $M(z) + zM'(z)$. Therefore one point with $M(z) > 0$ does not imply positive marginal cost from increasing dark routing or $z^* = 0$. If $M(z) \geq 0$ throughout the feasible interval, the all-lit corner is a global minimum; capacity or nonlinear impact can change that condition and yield an interior solution.

36-L07 For two losses, wealth is $W_0 = 1, W_1 = 0.90, W_2 = 0.81$. The historical peak is 1, so maximum drawdown is $1 - 0.81/1 = 19\%$. A single period with return -10% has 10% drawdown; a monotonically increasing path has zero drawdown. Returns still have N entries for volatility and Sharpe, but the initial value W_0 must be added, so drawdown uses $N + 1$ wealth points.

Connection Exercise Answer

36-T01 Buy order: delay $100(0.02) = 2.00$; execution $60(0.03) = 1.80$; opportunity $40(0.08) = 3.20$; commission 0.30; total 7.30. Direct amount is $60(0.05) + 40(0.10) + 0.30 = 7.30$. For the sell, price differences and signed quantities are both negative, so the four components remain 2.00, 1.80, 3.20, and 0.30; direct amount is $(-60)(-0.05) + (-40)(-0.10) + 0.30 = 7.30$. In the buy case terminal actual NAV is USD 2,002.70, below the paper USD 2,010. In the sell case the initial account holds 100 shares and USD 1,000 cash; after the sale it has USD 1,596.70 cash and 40 shares, with terminal actual NAV USD 1,992.70 versus ideal USD 2,000. IS compares two accounts; it is not a second cash outflow.

Project 4 —From Fields to Signals to Execution

> Project 4 has three layers. The signal layer asks which fields are known at the decision and how they are combined and constrained into a target. The ledger and cost layer asks how a target becomes fills, fees, cash, and holdings. The validation layer asks which rows can train, how a parameter is selected, and what being selected means. These notes keep two sets of original identifiers: cost Exercises 1–6 of the cost lab (questions and complete solutions in `editions/en/projects/project04/cost_execution_lab/solutions.md`) and the new integrative exercises P4-T01–T03 (`editions/en/projects/project04/exercises.md`). All numbers come from the stated inputs or labeled constructed scenarios; none is real-market research. Prerequisites: Chapters 24–31 and 35–36.

Signal layer: what is known at the decision

□□□□□

> Value, momentum, and quality fields differ in units and coverage. First decide whether a field is known at signal time, then preprocess and combine under same-day rules, and finally specify the support set where holdings are allowed. The default Notebook's constructed world uses seed 20261210, 240 companies (40 of them banks), and 96 months, with no real security-level data.

Trigger. Two versions for the same reporting period both happen to give ROE of 10%. May v2's profit be combined with v1's equity?

Definition P4.1.1 *(Version availability time).* A filing version becomes available at $a = \max(\text{publication}, \text{ingestion})$. A decision at time s may use only versions with $a \leq s$, and must take _____; numerators and denominators cannot be spliced across versions.

Proposition P4.1.2 *(Order of checks in the signal method).* Check in this order, using only information known at the decision.

1. Choose the support set from listing status known at the time, with at least _____ names; filings must meet both publication and ingestion cutoffs.
2. The seven fields keep their economic direction; nonpositive denominators or bank inapplicability are missing values and _____.
3. Same-day 1%/99% winsorization needs at least 30 values; industry/size OLS uses one complete paired set with at least 2 residual degrees of freedom; sample z-scores need at least 10 values.
4. Within a style, at least one available field with nonzero weight; across styles, at least two; available weights are renormalized, and below-threshold cases stay missing.
5. The alpha-valid set only determines ranks; the optional industry projection uses the full listed support set; outside the support set, weights are exactly zero.
6. When a homogeneous industry/net-zero projection is nonzero, one positive scale

restores gross 2; with no long–short direction or a degenerate case, hold cash explicitly.

7. Report industry/net/gross, support-set residuals, coverage, and missingness separately; post-execution information is used only for the ledger and diagnostics and _____.

Three objects

The signal target, the execution-filtered target, and realized holdings are three objects. The execution copy sets names delisted by then to zero under eligibility at that time, without writing back to the original signal or rescaling other names; a neutral target does not guarantee neutral holdings after fills. The base `run_backtest` _____ for a nonzero target on a security already delisted on the execution date.

Costs and execution: where the money goes

Definition P4.2.1 *(Implementation shortfall for one parent order).* A parent order $Q = F + R$, with filled quantity $F = \sum_i f_i$ and remainder R. The direct form is

$$\mathrm{IS}_{\mathrm{direct}} = \sum_i f_i(P_{\mathrm{fill},i} - P_{\mathrm{dec}}) + R(P_{\mathrm{term}} - P_{\mathrm{dec}}) + \mathrm{Fees}.$$

The decomposition consists of delay cost $Q(P_{\mathrm{arr}} - P_{\mathrm{dec}})$, filled execution cost $\sum_i f_i(P_{\mathrm{fill},i} - P_{\mathrm{arr}})$, unfilled opportunity cost $R(P_{\mathrm{term}} - P_{\mathrm{arr}})$, and Fees. Equality of the two forms proves only _____; it does not prove that the fee convention is correct.

What is already inside the fill price

For a buy, $P_{\mathrm{fill}} = S + \mathrm{sgn}(f)$(half-spread + temporary impact). Spread and temporary impact are already paid through the price, so cash is reduced additionally only by _____; permanent impact changes later _____ and enters NAV through valuation of holdings.

Two denominators for participation

The exclusive rate is $\rho_{\mathrm{excl}} = f/V$ and the inclusive rate $\rho_{\mathrm{incl}} = f/(V + f)$, with $\rho_{\mathrm{incl}} = \rho_{\mathrm{excl}}/(1 + \rho_{\mathrm{excl}})$. This project's configuration and `simulate_slot_fill` use the _____ rate.

Schedulers must not peek

VWAP requests follow a forecast curve frozen before sending; POV requests follow lagged observed volume. Realized volume can only cap actual fill capacity after the fill event. Past fills already confirmed may change later requests; that is legitimate feedback, not leakage.

Your turn. Cover this section. Write the names and formulas of the four IS components, and explain why "direct form equals decomposition" cannot detect a double-charged fee.

Recall: Why can the fill VWAP not be written as 0 when $F = 0$?

Validation layer: which rows can train

A label covers $(i, e_i]$ and becomes available at $a_i \geq e_i$. A fit at time f may use only rows with $a_i \leq f$, and retained labels must not intersect test intervals. Inner validation selects parameters; the outer test is evaluated once. If a model is changed after seeing the holdout, the holdout has become discovery data.

Exercises I: cost Exercises 1–6

These are the original cost-lab questions with unchanged identifiers and numbers. Exercise 6 retains the original data and negative results of the old R2R6 signal layer; it is not a new market result from the 2026-10-14 current-signal summary.

Your turn. **Cost Exercise 1 Algebraic proof and verification of the implementation shortfall direct identity.** Given a signed parent order Q, filled child quantities f_i summing to $F = \sum_i f_i$, unexecuted remainder $R = Q - F$, decision price P_{dec}, arrival price P_{arr}, terminal mark price P_{term}, and explicit fees $\text{Fees} = \sum_i c_i$: (1) prove algebraically that the four-part implementation shortfall decomposition is identically equal to the direct amount method; (2) what is the value of realized fill VWAP when $F = 0$, and how does the decomposition evaluate under zero fill?

Your turn. **Cost Exercise 2 Participation-rate boundary and denominator conversion.** A trader configures a participation limit of $\rho = 20\%$. (1) If ρ is the exclusive rate (own fill / external volume), what fraction of the consolidated tape does the trader execute? (2) If the exchange reports an inclusive rate $\rho_{\text{incl}} = 20\%$, convert it to the equivalent exclusive rate. (3) State the boundary conversions at $\rho = 0$ and $\rho = 1$.

Your turn. **Cost Exercise 3 Cash-ledger conservation and duplicate-fee prevention.** All amounts are USD. A long order of 1,000 shares executes at mid price $S = 100.00$, with half-spread 0.03 per share, temporary impact 0.02 per share, permanent impact 0.01 per share, and a 5 bp commission rate. (1) Compute the fill price and the cash deducted at the fill. (2) Explain why deducting the spread or temporary impact a second time from cash is an accounting error. (3) How does permanent impact enter portfolio NAV?

Your turn. **Cost Exercise 4 Causal volume-leakage defense via adversarial perturbation.** Describe an adversarial unit test that checks a VWAP or POV execution algorithm for future-volume leakage.

Your turn. **Cost Exercise 5 Trade-offs among execution schedulers.** Why did VWAP have lower total implementation shortfall than TWAP, while POV was markedly higher, in the fixed-parent-order experiment?

Your turn. **Cost Exercise 6 Turnover, fees, and net results in the constructed path.** In the old-signal R2R6 constructed walk-forward run, filled one-sided annual turnover is about 5.2383× and filled total-notional turnover about 10.4767× NAV. Commission is 5 bp and half-spread 2.5 bp per traded dollar. Estimate the mechanical fee-and-spread drag, compare it with the GROSS/TWAP CAGR difference, and explain why the values need not match.

Exercises II: integrative exercises P4-T01–T03

The three shared price-observation dates are D0=2026-01-30, D1=2026-02-02, and D2=2026-02-27. The decision is made at the D0 close; lag=1 executes at the next price-index observation, D1. The 60 constructed securities are S01..S60: the first 30 are in industry G1 and the remaining 30 in G2. In each industry, the first 15 securities have a processed score of +1 and the remaining 15 have −1; the common size exposure is set to 0. The scores are supplied, already-processed teaching inputs: do not re-estimate the seven-field alpha or present them as the original 240-company × 96-month DGP. All securities are listed at D0 and have complete scores, so the support threshold is at least 60. Industry demeaning and gross-2 rescaling are the portfolio steps in this exercise; other real-research gates must be checked separately. For the version-timing example, S01 has two versions for the same reporting period. Version v1 is published at D0 14:00 and ingested at 14:05, with profit 10/equity 100. Version v2 is published at D0 16:30 and ingested at D1 09:45, with profit 5/equity 50. Both ROE values happen to be 10%; versions must not be spliced. S60 delists on D1 with liquidation at 2 per share; S02 delists on D2 with liquidation at 2 per share. All other securities survive throughout. Initial NAV/cash is 1000 USD with no holdings. All execution prices are 10 USD; at the D1 close S01 is 11 and the other surviving securities are 10; closing prices remain unchanged on D2. S60 has no valid quote from D1 onward, and S02 has no valid quote from D2 onward; these are represented as NaN and accompanied by explicit delisting events. Assume no interest, dividends, splits, impact, borrowing fees, lot-size constraints, or margin calls. The fee rate is 0.001 on total traded notional. The base engine must be called with allow_short=True and max_gross=2; this does not establish that real borrowing is available.

Your turn. **P4-T01 Versions, Support, and Signals** (1) Which version is available at the D0 close? Can v2 be used at the D1 close? What if the decision is made before 09:30 on D1? Explain why 5/100 is invalid. (2) Use the generation rule to calculate the mean score in each industry, the 60-name support, each security's weight after rescaling to gross 2, the portfolio net weight, and each industry's net weight. (3) At D1, only 59 securities remain. Can a new signal that passes the ≥60-name gate be generated? How should S60's weight in the original D0 signal be retained? Distinguish the signal available at that time from execution eligibility.

Your turn. **P4-T02 Execution Copy, Post-Fee NAV, and Realized Holdings** (1) At execution on D1, set S60 to 0 without rescaling the other 59 weights. Calculate the execution copy's gross and net exposure and cash target. Explain what happens if the base run_backtest receives the original nonzero target for a security already delisted on the execution date. (2) Solve for post-fee NAV V′, the fee, shares in each surviving security, and cash. Give D1 closing NAV and S01's realized closing weight. Why is it not equal to 1/30? (3) On D2, how are the existing S02 shares liquidated? Find the liquidation cash, price loss, and final NAV. Explain why S60's and S02's delistings cannot both be handled by simply "dropping the column and ignoring the loss." Do not overwrite the D0 signal with the execution copy or restore gross/neutrality. (4) Call the base ledger on this small input to check shares, fees, cash, NAV, and reconciliation; do not run the full P4/A6 CLI, Notebook, or historical research. The answer gives an independently checkable table and entry-point code.

Recall: Why does the fee denominator contain V' itself?

Your turn. **P4-T03 Label Availability, Forward Selection, and Result Identity** This is a separate, small label timeline. Do not describe its few samples as passing the 60-name portfolio gate. Sample start positions are i=0..14, with label intervals (i,i+2] and a_i=i+2, except a_4=7. Fitting and prediction take place at 6, with test start positions 6..8. At a shared endpoint, settle and ingest first, then fit and predict. (1) Which of the candidate historical rows 0..5 can train at the fit time? Do any overlap test intervals? Is there an embargo under the chapter's post-test definition? (2) Inner validation A: at 2, train on the available history and validate on sample 2. Inner validation B: train at 3 and validate on sample 3. List the training rows for each. Given validation labels (1,−1), candidate predictions A=(0.5,−0.5) and B=(1,0), calculate the MSE in both folds and select a parameter. Which rows are used to fit scaling? (3) Run the outer test only for selected candidate A. The labels are (0.2,−0.2,0), and predictions are (0.1,0.1,0.1); find the MSE. Given the complete family's valid nominal p-values (0.04,0.20), apply Bonferroni/Holm at the 5% level. Does selecting A establish tradable alpha? (4) Compare this new exercise with the old R2R6 results, the 2026-10-14 current-signal cost summary, the I3 default Notebook's fresh computation/replay, and real research, which is still planned. If holdout performance is negative or one algorithm performs worse, what must be preserved? Which teaching, consumer, assembly, and data-rights gates remain?

Project 4: Questions and Complete Answers

Cost Exercise 1

(1) Rewrite each price difference in the direct form relative to the arrival price: $P_{\text{fill},i} - P_{\text{dec}} = (P_{\text{fill},i} - P_{\text{arr}}) + (P_{\text{arr}} - P_{\text{dec}})$ and $P_{\text{term}} - P_{\text{dec}} = (P_{\text{term}} - P_{\text{arr}}) + (P_{\text{arr}} - P_{\text{dec}})$. Substituting and collecting terms,

$$\text{IS}_{\text{direct}} = \sum_i f_i(P_{\text{fill},i} - P_{\text{arr}}) + R(P_{\text{term}} - P_{\text{arr}}) + \Big(\sum_i f_i + R\Big)(P_{\text{arr}} - P_{\text{dec}}) + \text{Fees}.$$

Since $\sum_i f_i + R = F + R = Q$, this is the decomposition. The formula applies to one parent order whose child fills share its direction and conserve shares; fill tables netted across parent orders cannot be spliced into it.

(2) With $F = 0$, the fill VWAP has denominator 0 and is mathematically undefined; record None or NaN, not 0.0. Filled execution cost is 0, $R = Q$, Fees = 0 because nothing traded, and total IS $= Q(P_{\text{arr}} - P_{\text{dec}}) + R(P_{\text{term}} - P_{\text{arr}}) = Q(P_{\text{term}} - P_{\text{dec}})$; the identity still holds. Equal values prove only arithmetic consistency: a wrong duplicate fee placed in Fees on both sides still leaves a zero difference.

Cost Exercise 2

(1) $f = 0.20V$, so $\rho_{\text{incl}} = 0.20V/(1.20V) = 1/6 \approx 16.67\%$. (2) $f = 0.20(V + f) \Rightarrow 0.80f = 0.20V \Rightarrow \rho_{\text{excl}} = 0.25 = 25.0\%$. (3) At $\rho = 0$ both rates are 0 and capacity is 0 shares; at $\rho_{\text{excl}} = 1$, $\rho_{\text{incl}} = 1/2 = 50\%$; at $\rho_{\text{incl}} = 1$, $\rho_{\text{excl}} = \rho_{\text{incl}}/(1 - \rho_{\text{incl}})$ diverges, and the capacity helper represents it as unbounded. A mathematical bound is not a fill guarantee. This project's configuration and `simulate_slot_fill` use the exclusive rate; do not treat an inclusive 20% as this configuration's 20%.

Cost Exercise 3

(1) $P_{\text{fill}} = 100.00 + 0.03 + 0.02 = 100.05$ USD per share; notional $1{,}000 \times 100.05 = 100{,}050.00$; commission $100{,}050.00 \times 0.0005 = 50.025$; cash deducted 100,100.025, i.e. $C_{\text{after}} = C_{\text{before}} - 100{,}100.025$. (2) The fill price already contains 0.03 half-spread and 0.02 temporary impact per share, so 1,000 shares have already paid 50.00 through the price. Deducting 50.00 again from cash, while the explicit fee field still holds only the 50.025 commission, makes cash inconsistent with $C_{\text{after}} = C_{\text{before}} - \sum_i f_i P_{\text{fill},i} - \sum_i \text{commission}_i$ and leaves a 50.00 residual. If the wrong 50.00 were also added as a second explicit fee to both IS calculations, they could still agree; embedded spread/impact, the commission field, and cash entries must be checked independently. (3) Permanent impact moves the later mid to 100.01; NAV is then measured from holdings at the new mark, e.g. 1,000 shares at 100.01. It enters NAV through later marks, not as a separate cash fee.

Cost Exercise 4

Build time-ordered execution slots, each with the forecast volume frozen before sending, the lagged observation available before sending, the volume realized after sending, and fill occurrence/confirmation times. Run the schedule and save each sending time's child requests; then change only the current or future slots' `realized_external_volume` (for example, multiply by 100 or set to 0) while keeping frozen forecasts, available observations, confirmed fills, and pending reservations before that sending time identical, and compare requests at each sending time. Assertion: with the same sending-time information set, perturbing realized volume not yet visible cannot change that child request. VWAP requests follow the frozen forecast curve; POV requests follow lagged observed volume; realized volume only caps actual capacity after

the fill event. If an earlier slot's fill changes and is confirmed before a later send, later requests may change with known fills and the remainder; that is legitimate historical feedback. The existing tests `test_vwap_schedule_adversarial_perturbation_invariance` and `test_pov_schedule_lagged_causality` cover the frozen and lagged cases, but they do not guarantee path-by-path identical later requests when past fills and remainders differ.

Cost Exercise 5

External volume on this constructed path is U-shaped. TWAP requests $Q/13$ per slot and so participates more heavily in low-volume midday slots; VWAP allocates along the constructed volume curve frozen before the first send. In the 10,000-share buy example, total IS is 541.70 bp for VWAP and 552.74 bp for TWAP; in the sell example, 248.95 and 257.01 bp. These numbers describe only this constructed path, not a general ranking. POV has no observed volume in the first slot, so it sends 0 shares; later child sizes equal lagged external volume times the participation cap, bounded by the remainder. Fixed POV does not automatically catch up on unfilled quantity; with insufficient observed volume or lower actual capacity, the remainder is left at the deadline and incurs opportunity cost.

Cost Exercise 6

Commission 10.4767×5 bp $= 52.4$ bp and half-spread 10.4767×2.5 bp $= 26.2$ bp, about 78.6 bp in total before impact and financing. Filled total notional is about $2\tau_{\text{one}}$; filled dollar notional uses actual fill prices while requested notional uses decision prices, and this small difference does not mean more shares filled than the parent order. On this corrected constructed path, GROSS CAGR is -0.26% and TWAP CAGR -0.73%, a difference of about -46.9 bp. They need not match: CAGR compounds the whole monthly path, whereas the simple estimate omits temporary/permanent impact, cash interest, borrow fees, and changes in the order of monthly marks. GROSS is already negative, so this example cannot show that costs erased a positive alpha. The 2026-10-14 signal refresh has a different saved-summary identity and cannot replace these original R2R6 values.

P4-T01

Version clock: at the D0 close only v1 is usable, because both its publication and ingestion are complete; v2 has $a = \max(\text{publication}, \text{ingestion}) =$ D1 09:45, so it is usable at the D1 close but not for an order before D1 09:30. 5/100 splices v2 profit with v1 equity and wrongly gives 5%; the same-version 5/50 is still 10%. A coincidental equal value does not show that the version join is correct.

Support and target: each industry has fifteen $+1$ and fifteen -1, so the industry mean score is 0 and demeaning does not change scores. The sum of 60 absolute scores is 60, so the gross-2 scale factor is $2/60 = 1/30$. Each target is $\pm 1/30$; portfolio net $= 0$, gross $= 2$, and each industry's net weight is 0. This checks only the portfolio step on processed scores; it does not claim market validation of the real seven fields, ingestion quality, or all gates.

At D1 the support is $59 < 60$, so no new signal passing this gate can be declared. The D0 signal is retained unchanged; S60's $-1/30$ was a valid target at that time and is set to 0 only at execution under eligibility then. The post-execution ledger need not stay neutral or at gross 2, and an execution adjustment must not be disguised as a historical alpha correction.

P4-T02

(1) Removing S60's short target leaves 30 longs and 29 shorts: gross = 59/30, net = 1/30, cash target 1 − net = 29/30. The base `run_backtest` raises an error for a nonzero target on a security delisted by the execution date; the wrapper/execution layer must create an eligibility copy explicitly and preserve the original signal, without rescaling or re-neutralizing.

(2) Valuing the unpositioned account at execution prices gives $V_{\text{pre}} = 1000$. New traded notional is $(59/30)V'$, so

$$V' = 1000 - 0.001\,(59/30)\,V' = \frac{30{,}000{,}000}{30{,}059} = 998.0371935194.$$

The fee is $59000/30059 = 1.9628064806$ USD; each surviving security holds $\pm V'/300 = \pm 3.3267906451$ shares; cash is $(29/30)V' = 964.7692870688$ USD. Crediting full short proceeds to cash is this base engine's simplification, not a realistic model with borrowing and segregated collateral. At the D1 close S01 rises from 10 to 11, gaining $V'/300$, so NAV $= 301V'/300 = 1001.3639841645$. S01's realized closing weight is $(11V'/300)/(301V'/300) = 11/301 = 3.65448505\%$, while the execution target is $1/30 = 3.33333333\%$; the difference comes from the post-execution price move and the closing denominator. The target is hit relative to post-fee execution NAV; closing weights are not required to stay undrifted.

(3) On D2 the long S02 position is closed out, recovering $2V'/300 = 6.6535812901$ with a price loss of $8V'/300 = 26.6143251605$; final NAV $= 293V'/300 = 974.7496590040$. S60 delisted before any position was opened, so there is no liquidation P&L and it cannot be shorted; S02 was held and must be liquidated at the explicit 2 per share, without deleting the historical loss.

(4) Call `run_backtest` with `BacktestConfig(initial_cash=1000,fee_rate=0.001,lag=1,allow_short=True,max_gross=2)`, the execution copy, and `delistings={S60:(D1,2), S02:(D2,2)}`; assert D1 fees $= 1000 - V'$, cash $= 29V'/30$, S01 shares $= V'/300$, D2 S02 shares 0, D2 NAV $= 293V'/300$, and an absolute `reconcile` residual below 10^{-9}. Full code is in `editions/en/projects/project04/solutions.md`; 10^{-9} is an absolute currency threshold for this 1000-dollar ledger, and reconciliation checks only internal accounting.

P4-T03

(1) At fit = 6, candidates with $a \le 6$ are 0, 1, 2, 3; row 4 has $e_4 = 6$ but $a_4 = 7$, and row 5 has $e_5 = a_5 = 7$, so neither is usable. Retained labels end by 5 and the test starts at $(6, 8]$, so there is no intersection. Purely historical candidates have no post-test embargo; this does not deny other historical dependence gaps.

(2) Fitting at 2, row 0 has $e = a = 2$ and is available while row 1 has $a = 3$ and is not, so train $\{0\}$ and validate $\{2\}$; fitting at 3, use $\{0, 1\}$ and validate $\{3\}$. Each fold fits scaling only on its own training rows; after selection, preprocessing and the model are refit on the outer available rows $\{0, 1, 2, 3\}$. Two-fold MSE: A is $(0.25 + 0.25)/2 = 0.25$ and B is $(0 + 1)/2 = 0.50$, so the inner loop selects A.

(3) Outer MSE $= [(0.2 - 0.1)^2 + (-0.2 - 0.1)^2 + (0 - 0.1)^2]/3 = 11/300 \approx 0.03666667$; the outer result does not switch the choice to B. Bonferroni gives 0.08 and 0.40; Holm's first step has $0.04 > 0.05/2$, so it stops, with adjusted values 0.08 and 0.20; neither rejects at the 5% family level. A slightly better MSE, being selected, and tradable alpha are three different things; p-values must themselves be valid, cannot be produced automatically from this MSE table, and the family size is not reduced because the models are correlated.

(4) The 60-name, 3-date example and the small label example are new teaching, not new research in the original 240×96-month signal world or the cost world. Old R2R6 full-period GROSS CAGR −0.26% and TWAP −0.73% are kept separate from the 2026-10-14 refresh's full-period GROSS −0.2316% and TWAP −0.4592%; negative and zero results for algorithms or scenarios are not deleted, and an already-negative gross return is not described as "positive alpha erased by fees." The I3 cost Notebook recomputes the fixed-parent and partial/zero-fill examples and replays saved summaries otherwise; none of this reruns historical results. Real P4 market research remains planned: PIT data and usage rights, real constituents, tradable prices, financing and borrowing, capacity, and trading calendars are not resolved by this exercise, and G05/A10-B is not authorized. Any negative holdout result is reported as is; a model revised after seeing the holdout has turned it into discovery data.

(The following records the gate status as written in the accepted answer; it is retained verbatim.) The root may register complete P4 teaching only after accepting B14 signal, A ledger/cost, and B validation integration. BIL17 consumer review and BIL23 chapter-level English Notebooks, full-package assembly, cold start, and global references remain pending. Real P4 market research remains planned; this exercise does not resolve PIT data, data rights, real membership, tradable prices, financing/borrowing, capacity, or calendar requirements. G05/A10-B has not been authorized. Report any negative holdout result. If the model is changed after looking at holdout data, that holdout has become discovery data and cannot again be called untouched test data.

Fill-in key for the reading sections

the whole same version; 60; are not filled with zero; never written back to the signal; raises an error. IS: arithmetic consistency. Fill price: commission; mid/mark prices. Participation: exclusive.

Project 5 —Pairs Trading

> A pairs trade must pass five gates. Screening asks whether the spread may revert; hedging translates β into dollars and shares; signals use only information then available; the ledger charges each leg correctly; evaluation reports losing as well as winning results. These notes follow the five gates and use roughly 70% reading and 30% fill-in work. Empirical tests on real prices are blocked pending data permission. All numbers come from hand calculations in the exercises or explicitly labelled constructed teaching panels; they demonstrate methods and are not market evidence.

The spine: five gates

Gate	Question	Takeaway (fill in the blanks)
Screening	Will the spread revert?	_____ measures comovement; cointegration asks whether the _____ is stationary; test it against _____ critical values.
Hedging	How much to buy and sell?	Log β is an _____; the dollar ratio is β, and the absolute share ratio is _____.
Signal	When do we trade?	Estimate and freeze parameters on the _____; a closing signal can first trade at the _____.
Ledger	Where does the cash go?	Spread and impact are in the _____; separately deduct _____. Charge interest and borrow fees using the balance and mark known at the interval _____; _____.
Evaluation	What can we claim?	Compare with _____, report _____, and conclude under the preregistered rules.

Exercise mapping: P5-L01–L04 correspond to P5-01–P5-04; P5-L05 to P5-07; P5-L06 to P5-08; P5-L07 to P5-09; P5-L08 to P5-10; P5-L09 to P5-05; and P5-L10 to P5-06. Prerequisites: Chapters 16, 17, 29, 31, 35, and 36.

Your turn. Cover the text and write one sentence for each gate: what goes in, and what comes out for the next gate.

Gate 1: Screening —Will the spread revert?

□□□□□

> Two prices moving together show synchrony; the trade is on their spread, so the real question is whether that spread has a mean to return to. Cointegration addresses this; correlation does not. Test the residual estimated on the training window, using critical values for cointegration residuals.

Trigger. Two ETFs have a level correlation of 0.97 over the past two years. A colleague says, "This pair is stable; trade it." What do you check first?

Definition P5.1.1 *(Cointegration, Chapter 17).* Two $I(1)$ series, $\log TR_y$ and $\log TR_x$, are cointegrated if there is a β such that the linear combination $\log TR_y - \beta \log TR_x$ is a _____ series. Correlation measures the degree to which two series _____. It can be high even when both share a random trend and each also has its own random walk.

Proposition P5.1.2 *(Critical values for the EG residual test).* When a unit-root test is run on formation-period OLS residuals $\hat{e}_t$, the null distribution is shifted further to the left than the ordinary ADF distribution. At 5% with a constant and about 500 observations, the critical value is about -3.35; the ordinary ADF value is about -2.87.

Reasoning clue: RSS is not proof of mean reversion

The first stage of EG estimates the cointegrating-regression coefficient, so under the no-cointegration null the residual test has a nonstandard distribution; its critical value must match the deterministic terms and dimension of the regression (Chapter 17). OLS minimizes RSS, but that does not prove faster mean reversion. Let e_t be one random walk and multiply it by a positive constant $c = 0.1$. RSS becomes _____ times its original value; with the same ADF specification, scaling both the dependent variable and its lagged term leaves the unit-root coefficient's t statistic _____. The counterexample retains a unit root while lowering RSS. The MacKinnon critical value returned by `coint` addresses the test distribution for estimated residuals; it does not use RSS to prove stationarity.

Example P5.1.3 *(Constructed panel TA/TB, eight 504-day formation windows).* TA and TB share one random trend and each has an independent random walk added (constructed, not market data). Level correlation ranges from 0.54 to 0.98, daily-return correlation is about 0.78 to 0.81, and EG p-values range from 0.35 to 0.95. None of the eight windows has $p < 0.05$.

Your turn. **P5-L09** Explain the example: why is return correlation almost unchanged while the cointegration test fails throughout? If pairs were selected only when the level correlation exceeded 0.9, in which windows would TA/TB be selected? What could happen to the spread afterward?

Your turn. **P5-L05** The formation-period residual EG statistic is -3.06, the EG 5% critical value is -3.35, and the ordinary ADF 5% critical value is -2.87. The conclusion is _____. Using the ADF critical value by mistake would _____. Does failure to reject establish that there is no cointegration? CA/CB are constructed to be cointegrated ($\phi = 0.97$); why do they pass in only three of eight windows?

Recall: Why is the EG critical value more negative than the ADF value?

The screening gate admits pairs whose spread looks mean-reverting in the training window. It can miss a true relationship because power is limited, and it can admit a spurious one because chance selection can occur across eight pairs and multiple windows. The fifth gate addresses the latter risk.

Gate 2: Hedging —Translate the coefficient into dollars and shares

□□□□□

> In a log-price regression, β describes percentages: if x rises 1%, y rises β% on average. For an order, we need the dollar amounts and share counts in both legs. Mixing these three units leaves directional exposure where we thought we had a hedge.

Trigger. The training-window estimate is $\beta = 1.2$, so someone says, "Buy 1 share of y and sell 1.2 shares of x." What is wrong with that statement?

Definition P5.2.1 *(Dollar weights for two legs, preregistered clarification A1).* For a long spread ($s = +1$) or short spread ($s = -1$), $w_y = s/(1 + |\beta|)$ and $w_x =$ _____; gross exposure $|w_y| + |w_x| =$ ____. Signal share count is $n_t = \text{trunc}(wW_t/C_t)$: first truncate toward zero using signal-close prices, then convert the share unit for the next session by its split ratio. Dividends do not change the target share count; the IS reference price is $C_t/r_{t+1} - D_{t+1}$.

Proposition P5.2.2 *(Dollar ratio and share ratio).* For a log-price spread, the dollars in x needed to hedge are β times the dollars in y (with sign: $n_x P_x = -\beta\, n_y P_y$). Before integer rounding, the absolute share ratio is $|n_x|/|n_y| = |\beta| P_y/P_x$ and the signed ratio is $n_x/n_y = -\beta P_y/P_x$; after truncation toward zero, both hold only approximately. For the level spread $P_y = a + bP_x + e$, the signed hedge ratio is $-b$ and its absolute value is $|b|$.

Proof. Assume fitted coefficients are frozen, the residual is held constant, and price changes are small; this is a first-order approximation. Corporate actions first put shares and prices in consistent units. Coefficient drift, residual changes, and rounding risk remain. The dollar PnL in y is approximately $n_y P_y\, d\log P_y$, and in x it is approximately $n_x P_x\, d\log P_x$. Substitute $d\log P_y = \beta\, d\log P_x$ and set the two amounts to offset: [fill in: the relation between $n_x P_x$ and $n_y P_y$]. Divide by P_x to obtain the share ratio. ∎

Example P5.2.3 *(What happens when $\beta < 0$?).* When $\beta < 0$, w_y and w_x have the same sign: a long spread buys both legs. Same-direction positions can hedge negative correlation to first order. With $\beta = -1.2$, the two dollar amounts are USD 454,545 and USD 545,455 in the same direction; if $d\log y = -1.2\, d\log x$, total first-order PnL is zero. In a constructed panel, independent random walks IA/IB pass EG in one window with $p = 0.047$ and $\hat{\beta} = -1.907$. This is a false positive (there is no cointegration by construction), not a problem with the sign of β. Preregistered v2 requires $\beta > 0$ to limit the project to one-long, one-short relative-value pairs.

Further thought: If β drifts during the holding period, fixed share counts move away from the hedge. This project does not rebalance during a holding period; rebalancing would incur costs on both legs.

Your turn. **P5-L01** $\beta = 1.2$, $P_y = 80$ and $P_x = 50$. For this simplified entry, assume no split or dividend between signal close and entry, enter at 80/50, and do not separately model overnight price changes or execution costs. Take a long spread with gross exposure USD 1,000,000. Dollar exposure in y _____, in x _____; shares of y _____, shares of x _____; absolute share ratio _____, signed share ratio _____; net dollar exposure _____. Why is nonzero net dollar exposure not an error? After rounding, is the share ratio still exactly $|\beta|P_y/P_x$?

Recall: What does log-price β measure?

Gate 3: Signals —Use only what was known at the time

At the end of the formation period, freeze α, β, the spread mean, and its standard deviation. During the trading period, insert new prices into those frozen formulas. A signal calculated at the close can first trade at the next session's open. Any peek into the future brings information into backtest returns that could not have been traded.

Trigger. The chart looks better if the z-score uses the spread mean from the full sample. Is that allowed?

Definition P5.3.1 *(Frozen spread and z-score).* $s_t = \log TR_{y,t} - \hat{\alpha} - \hat{\beta} \log TR_{x,t}$, $z_t = (s_t - \bar{s})/\hat{\sigma}_s$, where $\hat{\alpha}, \hat{\beta}, \bar{s}, \hat{\sigma}_s$ are estimated using only _____ data. Enter when $|z| \geq 2$; exit when $|z| \leq 0.5$ or when it crosses zero; stop out when $|z| \geq$ ____ and do not re-enter in the same window. At the last close in the window, issue a _____; completion remains subject to capacity and the execution domain.

Time fields

Field	Project value
`signal_time`	Close at t (16:00 New York time)
`trade_time`	_____
Decision price	$C_t/r_{t+1} - D_{t+1}$: the close converted to the next open's _____ share unit
ADV and volatility	20-day window through the signal close; open-fill budget = 5% × ADV_{20} (shares adjusted for the execution-day split). Full-day volume is known only after the close and is used only for _____.
Interest rate	Use the rate _____ at the previous close for the overnight interval.

Example P5.3.2 *(Future-perturbation check).* Multiply all prices after the first formation-window end by 1.5 and re-estimate that window: $\hat{\beta}$ is 1.191788 both before and after, and Formation is identical. The project tests also verify that perturbing prices after close t leaves orders signalled no later than t and NAV before t unchanged.

Your turn. P5-L10 The formation-period spread mean is 0 (the OLS includes an intercept); the full-sample mean is 0.011635. What changes if the full-sample mean is used for standardization? Run `leakage_checks()` in `solutions_code.py`. Write the three channels it checks and explain why passing all three does not mean "there is no leakage."

Recall: When can a close signal first trade?

Gate 4: The ledger —Each leg pays its own costs

□□□□□

> Spread and impact are already in the fill price; cash separately deducts only commission. Charging the same amount twice falsifies NAV. Interest and borrow fees use the balance and prices available at the start of each interval. Splits change share count; dividends change cash. An unfilled exit remainder cannot be cancelled as if it had closed the position: daily replacement orders remain subject to finite sample length, capacity, and integer-share support.

Trigger. An exit order fills only 60% on the first day. What happens to the rest, and what must the ledger record?

Definition P5.4.1 *(Fill price and cash equation, Chapter 35).* $P^{fill} = P^{open}\big(1 + \mathrm{sgn}(F)(h + \eta\sigma_{20}\sqrt{|F|/ADV_{20}})\big)$. Here F is a confirmed, nonzero, integer-share fill; the quote or requested quantity Q does not enter cash. With no fill, do not read an undefined fill price. Cash is $C^+ = C^- - FP^{fill} - $ _____. Interest over $[t_{j-1}, t_j]$ is $C_{j-1}\, r\, \Delta d/360$; borrow fee is |short shares|× _____ $\times b\, \Delta d/360$. Each leg can fill at most $5\% \times ADV_{20}$ at 09:30, using ADV known at the signal close. This is an uncalibrated opening-fill scenario, not a guarantee of actual participation.

Proposition P5.4.2 *(Two equivalent IS calculations, Chapter 36).* Let $\mathcal{F}$ be the set of nonzero actual fills, with an empty sum equal to zero. Fills in one parent order have the same direction, and $Q = \sum_{k\in\mathcal{F}} F_k + R$. Each parent order has one initial arrival price P^{arr}. Then

$$\begin{aligned} IS &= Q(P^{arr} - P^{dec}) + \sum_{k\in\mathcal{F}} F_k(P_k^{fill} - P^{arr}) \\ &\quad + R(P^{term} - P^{arr}) + \text{fees} \\ &= \sum_{k\in\mathcal{F}} F_k(P_k^{fill} - P^{dec}) \\ &\quad + R(P^{term} - P^{dec}) + \text{fees}. \end{aligned}$$

Proof. Use $Q = \sum_{k\in\mathcal{F}} F_k + R$ to split the first term on the left; [fill in: what remains after combining it with the second and third terms?] If the full initial delay on Q is retained, re-anchoring Day 2 execution at that day's open and adding $F_2(a_2 - d)$ leaves an extra [fill in: which part of the original remainder's delay?]. Re-anchoring is still correct when the initial delay is assigned only to F_1. ∎

Example P5.4.3 *(Partial-fill policies).* `matched`: fill both legs in the same proportion and cancel the entry remainder when it arrives. `independent`: fill each leg up to its own capacity. In a constructed stress case ($W_0 = 10^9$, budget is 1% of ADV_{20}), the actual share ratio on the first fill under the latter policy differs from the order ratio by about 64%.

Integer-share support comes before ledger posting

Decimal equity conversion does not guarantee fractional-share execution. 0.3333333333333333 is not exactly 1/3. A flat-account entry target of 7.5 shares can fill 7 and cancel the entry remainder; if a split would turn an existing holding into 7.5 shares, the whole account fails integer-share support before overnight accrual or corporate-action posting. At the sample end, a remainder is assigned opportunity cost and an open episode; failed windows cannot be dropped or described as unconditionally flattened.

Your turn. P5-L02 An independent corporate-action scenario starts with pre-event holdings of 5,681 shares of y and $-10{,}909$ of x, and preceding closes of 80/50; it does not carry forward the same entry opening in L01. The next day, y has a 2:1 split and x goes ex-dividend by 0.30. New y share count _____, cash dividend on x _____; TR step for y with prior close 80 and post-split close 40.6 _____; TR step for x with prior close 50 and ex-date close 49.9 _____. What happens if adjusted prices are used for fills and the ledger also records the corporate-action cash?

Your turn. P5-L03 Return to the original P5-L01 account; do not carry forward P5-L02's post-action state. Initial cash is USD 1,000,000, there are no initial holdings, and entry fees are zero. In this constructed model, all short-sale proceeds earn interest. Cash after entry _____. From Friday 16:00 to Monday 09:30, over a weekend in UTC or without a daylight-saving-time change, the interval is 65.5 hours, or _____ days. Assume DFF of 5.33% was known at the interval start (not historically verified for this exercise); credit rate is DFF minus 25 bps, borrow fee is 25 bps, the mark is Friday's close of 50, and the convention is ACT/360. Interest _____, borrow fee _____, ending cash _____. Can the new Monday opening price be used for this interval?

Your turn. **P5-L04** Exit order: sell $Q = -10{,}000$ shares; decision price 41.00. Day 1 open is 40.80; 6,000 shares fill (half-spread 2 bps, impact 5 bps). Day 2 open is 40.50; the remaining 4,000 fill (half-spread 2 bps, impact 4 bps). Commission is 0.005 per share. p_1 _____, p_2 _____; delay _____, execution _____, fees _____, IS _____. If the full Q first-delay term is retained, Day 2 execution is re-anchored at its open and $F_2(a_2 - d)$ is added, the result is too high by _____.

Your turn. **P5-L07** When both legs can only be partially filled, how do `matched` and `independent` handle them? What can cancelling an entry remainder mean for PnL? Why does cancelling an exit remainder not mean that the position has closed? What intent do daily retries serve under this protocol, and what constrains them? If a new entry signal arrives before the previous exit finishes, how is IS recorded on the remainder?

Recall: Where are spread and impact charged?

Gate 5: Evaluation —Report the results that did not make money

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> The more tests we run, the more often one passes by chance. Report every pair and every window, then compare with an ungated strategy that trades every window. If screening helps, screened performance should reliably beat ungated performance and retain that advantage across the cost grid. v1 was frozen before downloading real prices. v2 was frozen after seeing the constructed results but before real-data analysis, and that design change is disclosed. Conclusions must follow the relevant version's rules; the entire v2 process was not frozen before all data were seen.

Trigger. The v1 backtest shows screened portfolio Sharpe of 0.39 and ungated Sharpe of −0.47; rerunning under v2 (which adds $\beta > 0$) gives −0.15. Which result supports screening?

Definition P5.5.1 *(Preregistered decision rule* `claim_rule`*).* A positive result under the primary specification must hold under both the _____ and the _____ to count as support. Otherwise write that evidence does not support this strategy and retain all zero and negative results. Threshold grids and the 252-day formation window are only _____.

Example P5.5.2 *(Constructed-panel results, not market evidence).* Across 3 pairs and 8 windows there are 24 tests; 4 pass the EG test. If all nulls are true, about 1.2 passes are expected. Under v1 all four EG-passing windows are selected; most screened excess performance comes from one false-positive selection of IA/IB. Under v2 that window ($\beta < 0$) is excluded: 3 windows are selected, screened excess Sharpe is −0.15, with 4 closed trades and a 25% win rate. The truly cointegrated CA/CB pair trades less often under screening, with screened Sharpe −0.14 versus ungated 0.48. All 27 points on the cost grid are negative; with a 252-day formation window, screened Sharpe is −0.02. v2 was frozen after viewing constructed results to repair scope and timing, so this change cannot be read as evidence that v2 is better.

When each null p-value is correctly calibrated, linearity of expectation gives $E[\sum I_j] = \sum E[I_j] = 24(0.05) = 1.2$; independence is not needed. Overlapping 504-day windows and shared series are not 24 independent trials. BH q-values within windows are descriptive only; they do not establish FDR control across all correlated windows.

> Open items: FRED DFF has not been downloaded; borrow fees are scenarios, not historical data; field coverage and institutional permission for real prices remain unconfirmed. v2 is frozen before access to real prices; no design changes are allowed during the real-data stage.

Your turn. **P5-L06** There are _____ tests, with _____ expected $p < 0.05$ under the null; v2 selects _____, and the screened portfolio's excess Sharpe is _____. State the conclusion under `claim_rule`; then explain why IA/IB's profitable selection is not evidence of cointegration.

Your turn. **P5-L08** Real prices are blocked pending permission. Write three sentences: what can you claim, what can you not claim, and from which step should you rerun the work when data become available?

Recall: What does the ungated comparison reveal?

Project 5: Answers and Recalculations

Spine table on the front page

Correlation; residual stationarity (cointegration); EG (cointegration-residual) critical values; elasticity; $|n_x|/|n_y| \approx |\beta|P_y/P_x$; formation (training) period; next session's open; fill price; commission; start; a split changes share count and a dividend changes cash; ungated; every pair and window.

Definition blanks

Gate 1: stationary ($I(0)$); move together. Gate 2: $-s\beta/(1+|\beta|)$; 1. Gate 3: formation (training) period; 4; forced exit request (not guaranteed execution); next day's 09:30 open; after the event (split or ex-dividend); after-close diagnostic (it does not change the 09:30 fill); available. Gate 4: commission; P_{j-1}^{avail} (the mark available at the start of the interval: prior close overnight, or that day's 09:30 open intraday). Gate 5: cost grid; the screened-versus-ungated selection-effect comparison; descriptive.

Proof blanks

EG: RSS shrinks to 0.01 times its original value, while the ADF t statistic is unchanged; scaling does not remove the unit root. First-stage estimation makes the residual's null distribution nonstandard, so use EG critical values matching the deterministic terms and dimension; a lower RSS does not imply faster mean reversion. Hedging: $n_x P_x = -\beta\, n_y P_y$; before rounding the signed share ratio is $-\beta P_y/P_x$ and the absolute ratio is $|\beta|P_y/P_x$ ($\beta < 0$ gives same-sign legs). IS: $Q(P^{arr} - P^{dec}) = \sum F_k(P^{arr} - P^{dec}) + R(P^{arr} - P^{dec})$; combine with the execution and opportunity terms to obtain the right-hand side. Under the specified bad algorithm, which retains the full first-delay term and re-anchors Day 2 execution while adding a second delay term, the extra amount is $R_1(P_1^{arr} - P^{dec})$, or 800 in P5-L04 ($-4{,}000 \times (40.80 - 41.00)$).

Exercises

P5-L01 Use the simplified entry: no split or dividend between signal close and entry, entry prices 80/50, and no separate overnight-price or execution-cost model. +454,545.45, −545,454.55; 5,681, −10,909; absolute ratio 1.920261, signed ratio −1.920261 (before rounding, ±1.92); net dollars −90,970. Log-price β hedges percentage risk, so the dollar ratio is β and nonzero net dollars are by design. This is only a first-order hedge for a common move with frozen coefficients and a constant residual; it does not remove all risk. After rounding, the share ratio is only approximately $|\beta|P_y/P_x$.

P5-L01 recalculation note This is a separate protocol-alignment constructed example, not a replacement for the original exercise or its answer. NAV is USD 1,000,000, single-leg weight is +0.5, signal-close price is USD 100, the next-day dividend is USD 1, and there is no split. Target: 5,000 shares; IS reference price: USD 99. If only the next-day split ratio changes to 2:1, the target is 10,000 shares and the IS reference price is USD 49. For the truncation-order example, retain weight +0.5, signal-close price USD 100, and next-day 2:1 split; set NAV to USD 1,199 and next-day dividend to zero (as in the rounding counterexample). Truncate 5.995 first to get 5, then multiply by 2 to get 10. Converting first and truncating 11.99 would give 11, which violates the required order. Negative weights are also truncated toward zero before conversion.

P5-L02 In this independent scenario, 5,681/−10,909 are existing pre-event holdings and 80/50 are preceding closes; the same opening event in L01 is not converted twice. 11,362 shares (cash unchanged); −3,272.70 (dividend compensation paid on the short); 1.015; 1.004. Adjusted prices already include the split and dividend; posting the events again double counts them.

P5-L03 Use the original P5-L01 account, with initial cash USD 1,000,000, no holdings, and no entry fees; short-sale proceeds earn interest. $C_0 = 1{,}000{,}000 - 5{,}681(80) + 10{,}909(50) = 1{,}090{,}970$; marked position value is $-90{,}970$, so NAV remains 1,000,000. Over the weekend from Friday 16:00 to Monday 09:30 in UTC or without a daylight-saving-time switch, the interval is 65.5 hours and $\Delta d = 131/48 \approx 2.729167$ days. Assume DFF was known at the interval start:

$$I = C_0(0.0533 - 0.0025)\frac{131/48}{360} \approx 420.1497,$$

$$B = 10{,}909(50)(0.0025)\frac{131/48}{360} \approx 10.3377,$$

$$C_1 = C_0 + I - B \approx 1{,}091{,}379.8121.$$

The new Monday open price cannot be used for this interval; it is available only for later intervals. This exercise does not carry forward the post-split and post-dividend P5-L02 state.

P5-L04 $p_1 = 40.80(1 - 0.0007) = 40.77144$, $p_2 = 40.50(1 - 0.0006) = 40.47570$. Let $d = 41$, $a_1 = 40.8$, $a_2 = 40.5$, $F_1 = -6{,}000$, $F_2 = -4{,}000$, and $Q = F_1 + F_2$. Under the first-arrival anchor,

$$D = Q(a_1 - d) = 2{,}000, \quad E_1 = F_1(p_1 - a_1) = 171.36,$$

$$E_2 = F_2(p_2 - a_1) = 1{,}297.20, \quad C = 50,$$

$$IS = D + E_1 + E_2 + C = 3{,}518.56.$$

The direct-dollar calculation is $F_1(p_1 - d) + F_2(p_2 - d) + C = 1{,}371.36 + 2{,}097.20 + 50 = 3{,}518.56$. The specified bad algorithm keeps $D = 2{,}000$, re-anchors Day 2 execution at a_2 so $E_2' = F_2(p_2 - a_2) = 97.20$, and adds $D_2 = F_2(a_2 - d) = 2{,}000$. Thus $IS_{bad} = 2{,}000 + 171.36 + 97.20 + 2{,}000 + 50 = 4{,}318.56$. The excess 800 is $F_2(a_1 - d)$. If the first-day delay is assigned only to F_1, then $D_1' = 1{,}200$ and $D_1' + E_1 + D_2 + E_2' + C = 3{,}518.56$; re-anchoring remains correct. Adding only the Day 2 delay to the original execution term would overstate by 1,200 if measured from first arrival, or by 2,000 if measured from the decision price. These are different errors from the specified 800 algorithm.

P5-L05 Do not reject no cointegration; at the 5% level, using the ordinary ADF critical value would falsely reject and declare cointegration. Non-rejection does not establish no cointegration: with 504 observations and $\phi = 0.97$, power is limited, and CA/CB pass in only 3 windows.

P5-L06 24 tests; 1.2 expected (calibrated null p-values and linearity of expectation; independence is not required); 3 selected; screened excess Sharpe -0.15. The v2 primary result is negative and all points on the cost grid are negative, so there is no positive result to support the strategy. The conclusion is that evidence does not support this strategy, with all negative results retained. IA/IB are known independent random walks; their selection is a multiple-testing false positive, with no cointegration basis. This follows from their constructed process and cannot be inferred from the sign of β alone; same-sign positions can hedge negative correlation to first order.

P5-L07 `matched` fills both legs in the same proportion and cancels the entry remainder; `independent` fills each leg to its own capacity and can leave a mismatch. Cancelling an entry remainder changes exposure but has no predetermined PnL sign: the reduced position may gain less or lose less. Integer rounding under `matched` still leaves a small hedge error; different leg proportions under `independent` create a mismatch. Cancelling an exit remainder does not remove the position: depending on actual fills, it may remain a matched pair, a mismatched pair, or a single leg. From Y100 and X−200, a matched exit of Y50 and X−100 leaves Y50 and X−100: ratio 2 remains, but the account is not flat. With $\beta = 2$, equal leg prices, and a constant residual, its common-factor exposure is still zero to first order for a small move. Daily replacement orders serve the intended exit under this protocol; cancellation is not a completed close. Completion remains subject to sample length, capacity, and integer-share support, with no unconditional

promise to reach zero. If a new signal replaces an unfinished exit, record opportunity cost on the remainder as remaining shares times (new decision price minus first arrival price), with prices converted to the same post-event share unit.

P5-L08 We can claim that implementation, ledger, IS, leakage checks, and selection effects were validated on constructed data. We cannot claim that any real pair is cointegrated or profitable, or establish real costs and capacity. Once the data are frozen, rerun from the data step: TR data, formation tests, screened and ungated ledgers, cost grid, figures, notebook, then report.

P5-L09 The common trend drives most daily returns, keeping return correlation near 0.8, but the independent random walks in each series do not cancel in any linear combination. The spread is nonstationary, so EG keeps failing. A level-correlation rule above 0.9 selects windows 3 and 4 (0.964 and 0.976); afterward the spread has no dependable center and may drift.

P5-L10 The full-sample mean brings trading-period prices into standardization, shifting threshold-crossing entry dates and leaking future information into backtest returns. From the project root, run `solutions_code.py` (the line after P5-06 prints `leakage checks`), or use the cases whose names include `leakage` and `fill_day_volume` in `tests/test_p05_pairs.py` (with `pytest -k`). The three channels are: (a) multiply prices after formation ends by 1.5; Formation remains identical; (b) multiply prices after close t by 1.35; orders signalled at or before t and NAV before t remain unchanged; and (c) shrink only execution-day full-day volume; the 09:30 fill remains unchanged. All three are necessary checks, but they exclude only these channels and do not prove the entire implementation is free of leakage.

Chapter 37: Accuracy, Decisions, and Strength of Evidence

A prediction model produces a direction or a value; a trading system turns it into a position; the account records $p_t r_t - \kappa\tau_t$. Accuracy weights every day equally, while the account records each day in proportion to the size of its return. Equal magnitudes guarantee that the two agree; when magnitudes differ, they generally cannot substitute for one another, though they can happen to coincide. This chapter adds three more points: the positive overlap in this chapter's labels reduces the effective sample size for the mean; multiple attempts make the best result look better; and prediction relationships can flip with the regime. All numerical examples are constructed for teaching.

Trigger. Test-set directional accuracy rises from 52% to 55%. Which three tables do you still need before approving deployment?

Definition 37.1 *(Decision ledger, ±1 positions).* The prediction is made after the close on Day $t-1$, the position is established at the open on Day t, and r_t is the simple return from that open to the next open. Turnover is $\tau_t = |p_t - p_{t-1}|$, with $p_0 = 0$; when reversing from long to short, $\tau_t =$ ____. Net return is $n_t = p_t r_t - \kappa\tau_t$, and wealth evolves as $W_t = W_{t-1}(1+n_t)$; T daily returns correspond to _____ wealth points.

Proposition 37.2 *(Identity between accuracy and gross returns).* If $p_t \in \{-1,+1\}$ and $r_t \neq 0$, define $c_t = \mathbf{1}\{\operatorname{sign} p_t = \operatorname{sign} r_t\}$. Then

$$\sum_t p_t r_t = \Big(\sum_t |r_t|\Big)\big(2\,\mathrm{Acc}_{|r|} - 1\big), \qquad \mathrm{Acc}_{|r|} = \frac{\sum_t |r_t| c_t}{\sum_t |r_t|}.$$

Proof. When $c_t = 1$, $p_t r_t =$ _____; when $c_t = 0$, $p_t r_t =$ _____. Combine the two cases to [fill in: express $p_t r_t$ using $|r_t|$ and c_t]. Sum over t and substitute the definition of $\mathrm{Acc}_{|r|}$. ∎

Your turn. 37-L01 Full coverage: 37-E01. Daily returns (%) are $r = (+1.0, -0.2, +0.3, -4.0, +0.2)$; Model C positions are $(+1,-1,+1,+1,+1)$ and Model D positions are $(-1,-1,-1,-1,+1)$; $\kappa = 0.1\%$, $p_0 = 0$.

	Acc	$\mathrm{Acc}_{\lvert r\rvert}$	Gross return sum	Turnover	Cost	Additive approximate net return
C	____	____	_____	___	____	_____
D	____	____	_____	___	____	_____

Which model would you choose? Check D's gross return using the identity.

Recall: What weight does accuracy give to each day?

Your turn. **37-L02** Full coverage: 37-E02. With ±1 positions, the direction reverses independently each day with probability q. Write the expected turnover over T days: _____. For $q = 0.5$, $T = 252$, and $\kappa = 0.1\%$, annual cost on an additive basis is about _____. Also explain: if direction is symmetric between signs ($E[p_t] = 0$ each period) and independent of future returns, why is expected net return negative with positive fees and positive expected turnover? Must a finite sample lose money?

Your turn. 37-L03 Full coverage: 37-E03. Let $y = b_t x + \varepsilon$, with independent $x, \varepsilon \sim N(0,1)$; the slope is $+b$ in Regime 1 and $-b$ in Regime 2. Use Regime 1's $\hat{b}$ to predict Regime 2, with zero prediction as baseline. First complete the derivation:

$$E[(y-\hat{b}x)^2] = \text{[fill in: expand and take expectations]} = (b+\hat{b})^2+1, \qquad R^2_{\text{OS}} \approx 1-\frac{(b+\hat{b})^2+1}{b^2+1}.$$

When $b = \hat{b} = 0.2$, $R^2_{\text{OS}} =$ _____; if the true slope in Regime 2 is 0, $R^2_{\text{OS}} =$ _____. Can in-sample R^2 warn you?

Your turn. 37-L04 Full coverage: 37-E04 and 37-E05. (a) For $y = bx+\varepsilon$, $\sigma_x = 1$ and $\sigma_\varepsilon = 2\%$, to obtain $\rho = 0.05$, $b =$ _____, $R^2 =$ _____. (b) If daily returns are independent, the autocorrelation of the h-day overlapping label $Y_t = \sum_{j=1}^{h} u_{t+j}$ is $\rho_k =$ _____ $(0 \le k < h)$. Complete the variance-inflation factor for the sample mean:

$$1 + 2\sum_{k=1}^{h-1}\frac{h-k}{h} = 1 + \frac{2}{h}\cdot \text{[fill in: } \sum_{k=1}^{h-1}(h-k)\text{]} = ___.$$

For $h = 10$ and 2,520 daily labels, the effective sample size is about _____.

Recall: By what factor is the t statistic inflated when $h = 5$?

Your turn. 37-L05 Full coverage: 37-E06; partial coverage: the selection-bias diagnosis in 37-P03.

```
import numpy as np
rng = np.random.default_rng(0)
r = rng.normal(0, 0.01, (504, 100))          # 100 zero-mean strategies
sr = np.sqrt(252) * r.mean(0) / r.std(0, ddof=1)
best = int(np.argmax(sr))
print("discovered Sharpe:", sr[best])        # reported as evidence
```

(a) Why is the number reported by this code not evidence? Under the zero null its expected value is about $\sqrt{252/T}\,E[\max Z_N] =$ _____ ($E[\max Z_{100}] \approx 2.51$). (b) Rewrite the last two lines: how can you obtain a result that can serve as evidence? (c) For $T = 252$, $N = 50$, and $E[\max Z_{50}] \approx 2.25$, someone reports "the best strategy has an annualized Sharpe of 2.0." How would you respond?

Your turn. 37-L06 Full coverage: 37-E07; partial coverage: the evaluation protocol in 37-C02. (a) Assign each item to one of the five leakage types and give a fix: full-sample median imputation; feature selection using test-period IC; use of a post hoc revised analyst database; random 10-fold evaluation of 20-day labels. (b) For "20 closing-price features predict 5-day cross-sectional returns," list the five checks to conduct together (one concrete metric for each), three baselines, and two falsification conditions.

Your turn. **37-T01** New core-line diagnostic. A researcher reports 55% accuracy without specifying information-ingestion time, execution time, or costs. Give one required field for each of the information, prediction, and ledger tables, and explain whether the next-open example in Chapter 37 and the next-close example in Chapters 38–39 can use the same label directly.

Chapter 37: Answers and Checks

Definition and proof blanks

When reversing from long to short, $\tau_t = 2$; there are $T + 1$ wealth points. Identity: $|r_t|$; $-|r_t|$; $p_t r_t = |r_t|(2c_t - 1)$.

37-L01

$\sum |r| = 5.7$. C: wrong on Day 4, Acc = 0.80, $\text{Acc}_{|r|} = 1.7/5.7 = 0.298$, gross return $1.7-4.0 = -2.3\%$; turnover $1 + 2 + 2 = 5$, cost 0.5%, approximate net return -2.8%. D: wrong on Days 1 and 3, Acc = 0.60, $\text{Acc}_{|r|} = 4.4/5.7 = 0.772$, gross return $4.4 - 1.3 = +3.1\%$; turnover $1 + 2 = 3$, cost 0.3%, approximate net return +2.8%. Choose D. Identity check: $5.7 \times (2 \times 0.772 - 1) = 5.7 \times 0.544 = 3.10$.

37-L02

$1 + 2q(T - 1)$. With $q = 0.5$ and $T = 252$: 252 units, and annual cost is about 25.2%. If direction is independent of future returns and symmetric, expected gross return is $E[p_t r_t] = E[p_t]E[r_t] = 0$; positive fees and positive expected turnover make expected net return negative. A finite sample can still be profitable; it is not "almost certain" to lose. Here position-difference turnover is a proxy, not the exact self-financing turnover in Chapter 29 that includes drift and post-cost rebalancing.

37-L03

$E[(y - \hat{b}x)^2] = E[((b_2 - \hat{b})x + \varepsilon)^2] = (b_2 - \hat{b})^2 + 1$, and with $b_2 = -b$ this is $(b + \hat{b})^2 + 1$. For $b = \hat{b} = 0.2$: $1 - 1.16/1.04 = -11.5\%$. When the Regime 2 slope is 0: $1 - 1.04/1 = -4\%$. In-sample R^2 uses only Regime 1 and cannot see the reversal.

37-L04

(a) $b^2(1-0.0025) = 0.0025 \times 0.0004$, so $b = 0.001001$; $R^2 = 0.0025$. (b) $\rho_k = (h-k)/h$; $\sum_{k=1}^{h-1}(h-k) = h(h-1)/2$, so the variance-inflation factor is $1 + (h-1) = h$. With $h = 10$, effective sample size is about 252. For $h = 5$, the naive t statistic is inflated by about $\sqrt{5} \approx 2.24$ relative to the correction using the known true long-run variance. This sum has no kernel tapering; fixed-bandwidth Bartlett NW with $q = h - 1$ uses weights $1 - k/h$, giving limiting factor $(2h^2 + 1)/(3h)$, which is 3.4 at $h = 5$, not 5. Covering the MA order does not guarantee an exact correction with a fixed bandwidth.

37-L05

(a) Selecting the maximum from 100 zero-return strategies makes the reported value selection-driven; its expected value is about $\sqrt{252/504} \times 2.51 = 1.77$. (b) After freezing the selection, calculate the same strategy's Sharpe in a holdout period not used before, report the number of attempts ($N = 100$), and evaluate with Chapter 22's multiple-testing correction and Chapter 32's DSR:

```
holdout = rng.normal(0, 0.01, (504, 100))
print("holdout Sharpe:", np.sqrt(252) * holdout[:, best].mean() / holdout[:,
    ↪ best].std(ddof=1))
```

(c) $\sqrt{252/252} \times 2.25 = 2.25 > 2.0$. The reported value is below the expected best value under the zero null and cannot be treated as a discovery; require a holdout result and DSR.

37-L06

(a) Median imputation: preprocessing leakage; fit on training rows or use the same-day cross-sectional median. Feature selection by test-period IC: selection leakage; move it to inner validation. A post hoc revised database: look-ahead feature; use a point-in-time version. Random 10-fold: overlap leakage; use date-based walk-forward and exclude training rows according to label intervals. (b) See the 37-C02 rubric in Section 18 of Chapter 37. Examples: out-of-sample (IC Newey–West t statistic, lag 4; R^2_{OS} relative to training mean); stability (annual IC sign); costs (net returns and turnover at two cost levels); tails (maximum drawdown and worst 1% of days); economic logic (who bears the cost of this return). Baselines: zero prediction, training mean, and a simple rule. Falsification examples: "after-cost net return ≤ 0" and "the result appears in only one subperiod."

37-T01 Full Answer

Information table: one of feature available_at, label_available_at, fit_time, or the signal cut-off, with a check that it was available then. Prediction table: signal date, asset, prediction, or evaluation-date set. Ledger: trade_time, position, turnover, or cost. A 55% figure does not include these three contracts and cannot imply net returns. Chapter 37 ' s return starts at the next open, while Chapters 38–39 use the next close; the return intervals in the labels differ, so they cannot be used interchangeably.

Chapter 38: Time Contracts, Splitting, and Training-Only Fitting

Record six times for every row: feature availability, signal, execution, label start, label end, and label availability. A signal date before fit time does not mean the label has finished accruing; split by date, not by row; fit every transform with parameters estimated across dates inside the training fold. All numbers in this chapter are teaching constructions. Question map: 38-L01 ↔ 38-E01; 38-L02 ↔ 38-E02, 38-P01; 38-L03 ↔ 38-E03 and Section 6.1; 38-L04 ↔ 38-E05, 38-P02; 38-L05 ↔ 38-E06, 38-E07, and Section 12.1; 38-L06 ↔ 38-E08, 38-C02.

Trigger. You build a model after Monday's close, and the label is "buy at tomorrow's close and hold for two days." Can the sample from last Thursday enter training?

Definition 38.1 *(Available Training Set).* Execution lag ℓ, holding period h; the label interval for signal date D_k is _____, and the label is known at the close on _____. At fit time D_s close, row (i,k) is available if and only if $k < s$ and _____, with nonmissing features and label.

Your turn. 38-L01 Full coverage: 38-E01. The default cutoff τ_k is 16:10 after the closing price for D_k is finalized. Give `available_at` and decide whether each item can be used for the D_k signal: (a) D_k closing price _____; (b) full-day trading volume on D_k _____; (c) quarterly earnings released at 16:30 on D_k _____; (d) an index change announced after hours on D_k, effective on D_{k+5} _____. Which item is most easily mistaken for the "reporting-period end date"? If the cutoff moves to 20:00 that evening (with data ingested beforehand), which conclusions change?

Your turn. 38-L02 Full coverage: 38-E02; partial coverage: 38-P01. Closing prices ($k = 1..12$): B $= (50, 50.5, 50, 49.5, 49, 49.5, 50, 51, 51.5, 52, 51, 52.5)$, C $= (20, 20, 20.2, 20.4, 20.2, 20, 20.4, 20.6, 20.6, 20.8, 21, 20.8)$; $\ell = 1, h = 2$. (a) For B at $k = 5$, execution price _____, endpoint price _____, $y =$ _____, interval _____. For C at $k = 6$, $y =$ _____. (b) If fit time changes to D_8 and test dates are $k = 8, 9, 10$, available signal dates _____, excluded rows ____ (3 assets).

Proposition 38.2 *(Excluded Dates Versus Overlap Purging).* With no missing data and equally spaced date indices, the availability rule excludes the last $\ell + h - 1$ signal dates before the test; overlap-only purging excludes $h - 1$.

Proof. Available rows satisfy $k + \ell + h \le s$. Candidate dates are $k = 1..s - 1$, so the number excluded is $= (s - 1) - (s - \ell - h) =$ _____. Overlap: the training interval $(D_{k+\ell}, D_{k+\ell+h}]$ intersects the first test interval $(D_{s+\ell}, D_{s+\ell+h}]$ if and only if [fill in: write the condition in terms of k, s, h], that is, $k > s - h$, for a total of _____. The additional ℓ dates have label intervals that end before the test starts but _____.

Your turn. 38-L03 Full coverage: 38-E03. Complete the table and answer the question for $h = 1$.

(ℓ, h)	$(1, 5)$	$(2, 3)$	$(1, 1)$
Excluded by availability rule	_____	_____	_____
Excluded by overlap purging alone	_____	_____	_____

When $h = 1$, overlap purging excludes no rows. Why does the availability rule still exclude rows? Is embargo needed with expanding-window walk-forward?

Recall: What determines the length of an embargo?

Your turn. 38-L04 Full coverage: 38-E05; partial coverage: 38-P02. (a) Complete the proof: with-intercept OLS fitted values are invariant to affine standardization $\tilde{X} = (X - \mathbf{1}\mu^\top)S^{-1}$ because $[\mathbf{1}, X]$ and $[\mathbf{1}, \tilde{X}]$ have the same _____, and fitted values are the _____ of y onto that space. What condition is also needed for new-point predictions to be unchanged? _____. Rank-deficient example: the columns of X are $(1, 2, 3)$ and $(2, 4, 6)$, $y = (1, 2, 3)$, new point $(1, 5)$: raw `LinearRegression` predicts _____, standardized regression predicts _____. Why do they differ? (b) Diagnose:

```
keep = SelectKBest(f_regression, k=10).fit(X_all, y_all).get_support()
for tr, te in date_folds:
    model = Ridge(alpha=1.0).fit(X_all[tr][:, keep], y_all[tr])
```

What leaks in this code? In the chapter's construction with 200 pure-noise features, its R^2_{OS} is _____, its IC is _____; after moving selection inside a Pipeline, it is _____. Rewrite it with training-only selection.

Your turn. 38-L05 Full coverage: 38-E06, 38-E07; also see Section 12.1. (a) AR(1) features with $\rho = 0.98$: $E(x_k - x_{k+j})^2$ at lags of one and three days is _____, _____; five-day label correlations are _____, _____. Explain why kNN gets about +57% under shuffled K-fold. (b) Outer fit time D_{600}, $\ell = 1, h = 5$, inner validation blocks of 63 days, at least 252 training days. The final two inner validation blocks are _____ and _____, with fit times _____; inner training ends at $k =$ _____. (c) Why can't `TimeSeriesSplit(gap=5)` replace date-based rules in a panel with four rows per day?

Your turn. 38-L06 Full coverage: 38-E08; partial coverage: 38-C02. (a) Derive $\sum(y - \hat{y})^2 = \sum_d N_d(\bar{y}_d - \bar{\hat{y}}_d)^2 + \sum(y^\circ - \hat{y}^\circ)^2$: the cross term is zero because _____. (b) Date 1: $y = (0.02, 0.00)$, $\hat{y} = (0.03, 0.01)$; date 2: $y = (-0.01, 0.01)$, $\hat{y} = (0.00, -0.02)$: total error _____, level term _____, cross-sectional term _____. Is the model useful for a dollar-neutral portfolio? (c) Cross-sectional calibration slope $b^\circ = 0.6$ (estimated on the inner fold), one-way cost 0.10%, and demeaned predictions of +0.25% and −0.05% for two assets sampled from a larger cross-section: calibrated values _____, _____. Under "trade only when the forecast exceeds round-trip cost," do you trade?

Your turn. **38-T01** Full coverage: body question 38-T01. Close at 16:00; same-day features ingested at 16:08; snapshot locked and fitting starts at 16:09, completes at 16:09:40; signal at 16:10; historical label B ended at 16:00 but was ingested at 16:20; earnings ingested at 16:30. Historical label A was ingested at 15:55, and both historical feature sets were readable. Training-eligible label _____, same-day data available for the prediction _____; which row is wrongly included if "label ended" is mistaken for "label mature"? _____. Write the three exact-time inequalities. If fitting finishes at 16:12, can the result still be recorded as a 16:10 signal?

Chapter 38: Answers and Recomputations

Definition and proof blanks

$(D_{k+\ell}, D_{k+\ell+h}]$; $D_{k+\ell+h}$; $k+\ell+h \le s$. The availability rule excludes $\ell+h-1$ dates; overlap occurs when $s+\ell < k+\ell+h$, for $h-1$ dates. Their labels end after fit time D_s, so they are not yet known.

38-L01

Default cutoff: (a) available after the close is finalized and before the cutoff. (b) Available if delivered before 16:10; otherwise use the actual delivery time, and it is unavailable. (c) 16:30 is after the cutoff, so it cannot be used for D_k; it can first be used for the D_{k+1} signal. (d) The announcement time is after the cutoff, so it cannot be used for D_k; D_{k+5} is the effective date. Item (c) is easiest to mistake for the reporting-period end. With a 20:00 cutoff, (c) and (d) can be used for the D_k signal; execution remains on D_{k+1}. State the cutoff and time zone in the data dictionary.

38-L02

(a) D_6 is 49.5 and D_8 is 51, so $y = 51/49.5 - 1 = 0.030303$, interval $(D_6, D_8]$. For C: $20.6/20.4 - 1 = 0.009804$. (b) $k+3 \le 8$, so $k = 1..5$ are available; exclude $k = 6, 7$, a total of 6 rows.

38-L03

$(1, 5)$: 5 and 4; $(2, 3)$: 4 and 2; $(1, 1)$: 1 and 0. When $h = 1$, the label for $k = s-1$ ends at $D_{s+\ell}$, after D_s, so it is unknown. Expanding-window walk-forward has only training rows before the test, so no embargo is needed. First purge true label overlap, then compare feature and test intervals using the full intersection condition. For a single test-label interval $(D_a, D_e]$ and feature support $(D_{k-L}, D_k]$, overlap requires both $k > a$ and $k - L < e$. If rows are explicitly restricted to $k > e$, the latter condition alone is enough. Deleting through $e + L - 1$ can be a conservative envelope, not a generally exact minimum. For a noncontiguous test-label union, check each interval; EMA has no exact finite lookback.

38-L04

(a) Column space; orthogonal projection (training fitted values are unchanged, with no full-rank requirement). New-point predictions are unchanged if $[\mathbf{1}, X]$ has full column rank, so the OLS solution is unique; then $\tilde{a} + \tilde{b}^\top S^{-1}(x-\mu) = a + b^\top x$. The rank-deficient example gives 2.20 and 1.75. Both satisfy $b_1 + 2b_2 = 1$; minimizing $\|b\|$ in original coordinates gives $(0.2, 0.4)$, while minimizing $\|Sb\|$ in standardized coordinates gives $(0.5, 0.25)$. The new point does not satisfy $x_2 = 2x_1$, so its prediction depends on the unidentified component in the null-space direction $(2, -1)$. (b) Feature selection used all samples, including test-period labels, causing selection leakage. The construction gives +0.617%, IC 0.0476; inside Pipeline it gives −1.214%. Rewrite:

```
for tr, te in date_folds:
    model = Pipeline([("s", StandardScaler()), ("k", SelectKBest(f_regression,
    ↪  k=10)),
                      ("m", Ridge(alpha=1.0))]).fit(X_all[tr], y_all[tr])
    pred = model.predict(X_all[te])
```

38-L05

(a) 0.04, 0.1176; 0.8, 0.4. After shuffling, nearest neighbors of a test row are almost all adjacent dates for the same asset (feature distance much smaller than the distance of 2 between assets); their labels are highly correlated with the test label, and the prediction is a "lookup." The chapter construction gives +56.67% (walk-forward: −40.73%). (b) Outer-fold data are available through $k = 594$; inner blocks are 253..315, ⋯, 442..504, 505..567 (568..594 is too short); the final two are 442..504 and 505..567, with fit times D_{442} and D_{505}; inner training ends at $k = 436$ and 499. (c) `gap` counts rows: five rows with four rows per day are only about one date, and rows from the same date can be split across folds. The contract requires excluding $\ell + h - 1$ complete signal dates; the date-index difference must be at least $\ell + h$.

38-L06

(a) Demeaned residuals sum to zero within each date. (b) 0.0012; 0.0004; 0.0008. The ranking is correct on date 1 and reversed on date 2, so the two-date mean cross-sectional IC is 0. The mean gross advantage of symmetric equal-weight positions is 0; two observations do not establish that the population has no information. (c) 0.15%, −0.03%; neither exceeds 0.20%, so neither is traded. Since $0 < b^\circ = 0.6 < 1$, the forecast amplitude is too large relative to the linear calibrated conditional mean; the scale adjustment brings both expected excess returns below costs. A zero slope would indicate no linear information, and a negative slope would indicate reversal, not just a dispersion problem.

38-T01 Complete Answer

Use A for training: 15:55 is no later than 16:09. Do not use B: 16:20 is after `fit_time`. Use the 16:08 same-day feature for prediction; do not use 16:30 earnings. The same-day feature is for prediction only; do not add a row without a mature label to training. Three gates: training data readable by `fit_time`; prediction feature `available_at` $\leq \tau_s$; fit completion time $\leq \tau_s$. A fit completed at 16:12 cannot generate a 16:10 signal; under a predeclared protocol, delay the signal and adjust the execution convention, or use an already completed older model. Execution is at the next close, unlike Chapter 37's next-open example.

Chapter 39: Shrinkage, Thresholding, and Baselines

Under fixed-design and conditional zero-mean error assumptions, OLS is unbiased but can have high variance; in the orthogonal-design teaching example, Ridge shrinks proportionally, Lasso subtracts a fixed amount and stops at zero, and Elastic Net subtracts first and then shrinks. In scikit-learn 1.9.1, Ridge does not divide the residual sum of squares by $2n$, while Lasso and Elastic Net do, so α values across estimators are not comparable. All numbers are constructed teaching examples. Question mapping: 39-L01 ↔ 39-E01; 39-L02 ↔ 39-E02, 39-E03; 39-L03 ↔ 39-E04, 39-E05; 39-L04 ↔ 39-E06 and Section 12.1; 39-L05 ↔ 39-E07, 39-E08; 39-L06 ↔ 39-P02, 39-C02.

Trigger. Two models both use $\alpha = 1$. Are the Ridge and Lasso penalties equally strong?

Definition 39.1 *(Three objective functions (scikit-learn 1.9.1)).* Ridge: $\|y - Xw\|_2^2 + \alpha\|w\|_2^2$. Lasso: _____. Elastic Net: $\frac{1}{2n}\|y - Xw\|_2^2 +$ _____. The intercept is _____ penalized; standardization parameters are estimated only on _____.

Your turn. 39-L01 Full coverage: 39-E01. The columns of X are $(1, 1, -1, -1)$ and $(1, -1, 1, -1)$, with $X^\top X = 4I$; $y = (2, 0, 2, -4)$. $X^\top y =$ _____, $z = X^\top y/n =$ _____, OLS = _____, intercept _____. Why does OLS equal z under orthogonality?

Proposition 39.2 *(Bias–variance for a single Ridge coefficient).* $x^\top x = n$, $y = wx + \varepsilon$, $\operatorname{Var}\varepsilon_i = \sigma^2$, independent zero-mean errors, fixed design, $w \neq 0, \sigma^2 > 0$: $\hat{w} = \frac{x^\top y}{n+\alpha}$, $\mathrm{MSE}(\alpha) = \frac{\alpha^2 w^2 + n\sigma^2}{(n+\alpha)^2}$, and the minimum is at $\alpha^* = \sigma^2/w^2$. When $w = 0$ with nonzero noise, MSE tends to 0 as α increases; this ratio is not a finite optimum.

Proof. $E\hat{w} =$ _____, bias = _____; $\operatorname{Var}\hat{w} =$ _____. Set the derivative numerator to 0 and divide by $2(n + \alpha)$: [fill in: simplify to $n\alpha w^2 = n\sigma^2$]. ∎

Your turn. 39-L02 Full coverage: 39-E02, 39-E03. (a) Continuing from 39-L01, Ridge at $\alpha = 2$: _____; at $\alpha = 12$: _____. Another package writes the objective as $\frac{1}{2n}\|y - Xw\|^2 + \frac{\lambda}{2}\|w\|^2$. The equivalent λ for $\alpha = 12$ is _____. (b) For $n = 250, w = 0.05, \sigma = 1$: α^* = _____, bias _____, variance _____, MSE _____, OLS _____. What is the cost of shrinkage?

Recall: Why does α^* not depend on n?

Proposition 39.3 *(Lasso and Elastic Net under an orthogonal design).* $X^\top X = nI$: $\hat{w}_j^{\mathrm{L}} = S(z_j, \alpha)$, $\hat{w}_j^{\mathrm{EN}} = S(z_j, \alpha\rho)/(1 + \alpha(1-\rho))$, $S(z,t) = \mathrm{sign}(z)(|z| - t)_+$.

Proof. Expand $\frac{1}{2n}\|y - Xw\|^2 =$ constant $-z^\top w + \frac{1}{2}\|w\|^2$; the objective separates by coordinate into $\frac{1}{2}w_j^2 - z_j w_j + \alpha|w_j|$. The subgradient condition is $0 \in$ _____. When $w_j > 0$, $w_j =$ _____ (requiring $z_j > \alpha$); $w_j = 0$ if and only if _____. The Elastic Net quadratic coefficient becomes _____.

Your turn. 39-L03 Full coverage: 39-E04, 39-E05. $z = (1, 2)$. Lasso: at $\alpha = 0.5$ _____, at $\alpha = 1$ _____, at $\alpha = 2$ _____; threshold for all coefficients to be 0: _____. Elastic Net at $\alpha = 1$: $\rho = 0.8$ _____, $\rho = 0.2$ _____. Geometrically, why does the L1 constraint often push the solution onto a coordinate axis?

Your turn. 39-L04 Full coverage: 39-E06; linked to Section 12.1.

```
X_pct = np.column_stack([x1, x2 * 100])     # x2 recorded in percent
coef = Lasso(alpha=0.3).fit(X_pct, y).coef_  # printed [0.2408 0.0054]
```

Converted back to decimal units, the x_2 coefficient is _____, and x_1 is shrunk from about 0.5 to _____. Why does the same information receive a different penalty when its recording units change? Which of OLS, Ridge, and Lasso is unaffected, and under what conditions? How should it be fixed (specify the rows used to estimate the mean and standard deviation)?

Your turn. 39-L05 Full coverage: 39-E07, 39-E08. (a) For $r = 0.95$, $n = 100$, and $\sigma = 1$: VIF _____, OLS single-coefficient standard deviation _____; for Ridge at $\alpha = 10$, shrinkage in the "sum" direction _____ and "difference" direction _____. In the body simulation ($r = 0.99$), the fraction of OLS estimates with a negative sign is _____, and the fraction of samples where Lasso retains only one feature is _____. (b) Ridge estimates a standardized "book-to-market ratio" coefficient of 0.002 (5-day return), with training standard deviation 0.4: the per-unit coefficient is _____. Give two reasons why it is not a causal effect.

Your turn. 39-L06 Partial coverage: 39-P02, 39-C02. Section 8.4 (constructed panel, one-way costs of 10 basis points):

Model	R^2_{OS} vs. zero	Own-date mean IC (valid days)	After-cost Sharpe	Annual turnover
OLS	−0.699%	0.0316 (750 days)	1.23	24.2
Ridge	−0.001%	0.0349 (750 days)	1.33	24.2
Lasso	−0.037%	0.0380 (441 days)	1.08	13.9
Elastic Net	+0.025%	0.0499 (624 days)	1.75	19.0

(a) Why is OLS R^2 the worst while the portfolio still makes money? (b) What does Ridge's frequent choice of $\alpha = 10^7$ show? (Hint: as $\alpha \to \infty$, consider the level and cross-sectional direction of predictions.) (c) Lasso, Elastic Net, and Ridge have 441, 624, and 750 valid dates, respectively. Without Chapter 39 row-level predictions, can you calculate the mean difference on common dates? Can P06 rows substitute? (d) State a falsification condition you would test on real data.

Your turn. 39-T01 Full coverage of body 39-T01. Original-unit b and standardized coefficient w satisfy $w =$ _____; the standardized L2 penalty in original units is _____. If x_2 is multiplied by 100 and its slope divided by 100, the unstandardized L1 and L2 penalties become _____ and _____ of their original values. If y is multiplied by 100: Ridge α is _____; Lasso α is _____. For Elastic Net with original $\alpha = 0.01$, $\rho = 0.5$, equivalent α', $\rho' =$ _____.

Your turn. 39-T02 Full coverage of body 39-T02. Ridge has three daily ICs (0.1, 0.9, 0.2); m has (0.2, 0.3) only on dates 1 and 3. Write their own-date sets and common set, calculate the two own-date means _____ and the mean difference on common dates _____. Why does the direction of the conclusion reverse? Can the 750/441/624 summary counts or P06 rows supply the Chapter 39 historical paired difference?

Chapter 39: Answers and Recomputations

Definition and proof blanks

Lasso: $\frac{1}{2n}\|y - Xw\|_2^2 + \alpha\|w\|_1$; Elastic Net: $\alpha\rho\|w\|_1 + \frac{\alpha(1-\rho)}{2}\|w\|_2^2$; the intercept is not penalized; training rows. Bias–variance: $\frac{n}{n+\alpha}w$; $-\frac{\alpha}{n+\alpha}w$; $\frac{n\sigma^2}{(n+\alpha)^2}$; $\alpha w^2(n+\alpha) - \alpha^2 w^2 - n\sigma^2 = 0 \Rightarrow n\alpha w^2 = n\sigma^2$. Soft thresholding: $w_j - z_j + \alpha\,\partial|w_j|$; $z_j - \alpha$; $|z_j| \le \alpha$; $1 + \alpha(1-\rho)$.

39-L01

$(4, 8)$; $(1, 2)$; $(1, 2)$; 0. Under orthogonality, $(X^\top X)^{-1}X^\top y = \frac{1}{n}X^\top y = z$.

39-L02

(a) $\frac{4}{4+\alpha}z$: $(0.6667, 1.3333)$; $(0.25, 0.5)$; $\lambda = \alpha/n = 3$. (b) 400; −0.03077; 0.000592; 0.001538; 0.004. Cost: the estimate is only $250/650 = 38\%$ of the true value on average. α^* does not depend on n because the Ridge objective is not divided by n: n appears on both sides of the optimality condition and cancels.

39-L03

$(0.5, 1.5)$; $(0, 1)$; $(0, 0)$; $\alpha \ge 2$. For $\rho = 0.8$: $(0.2, 1.2)/1.2 = (0.1667, 1)$; for $\rho = 0.2$: $(0.8, 1.8)/1.8 = (0.4444, 1)$. The L1 constraint set is a diamond with corners. Expanding the residual contours outward from the OLS center, the first point of contact is often a corner (on a coordinate axis); the L2 circle has no corners.

39-L04

0.54; 0.24. The penalty acts on coefficients; a feature with units scaled up by 100 needs only 1/100 of the coefficient, so its penalty is almost 0. With full column rank, OLS is unaffected (training fit and new-point predictions are invariant to the affine transformation); with exact collinearity, OLS predictions for new points can change (Chapter 38, Section 6.3); Ridge and Lasso are affected. Fix: estimate the mean and standard deviation on training rows, then standardize (Pipeline); this example gives $(0.2243, 0.2528)$ after standardization.

39-L05

(a) VIF $= 1/(1 - 0.9025) = 10.26$; $\sqrt{10.26/100} = 0.320$; $195/205 = 0.951$ and $5/15 = 0.333$; 41.8%; 44.8%. (b) $0.002/0.4 = 0.005$. Reasons: the coefficient is a conditional linear prediction in the training distribution, not an intervention effect; the feature may proxy for an omitted variable or be collinear with other features; Ridge shrinkage makes its size depend on α.

39-L06

(a) R^2 includes level error, and OLS's large coefficients on 46 features make the level prediction noisy; dollar neutrality removes the common level, while actual continuous weights still use relative demeaned predictions, so IC remains positive. (b) Inner MSE penalizes magnitude and calibration errors; at low signal-to-noise ratios, heavier shrinkage may reduce estimation noise and brings the level closer to the training mean, without implying that level error dominates; the cross-sectional direction approaches $X^\top y$ and is stable in the nondegenerate limit when α is sufficiently large relative to the eigenvalues. Selection of the largest grid value alone does not prove this is the cause, and does not justify changing the tuning objective based on outer results. (c) Lasso has five folds with all coefficients equal to 0 and constant predictions, so IC is undefined on those dates; the summaries confirm only 441 own-date valid days for Lasso, 624 for Elastic

Net, and 750 for Ridge. Common dates, paired means, and the reason for each daily omission could not be recomputed; P06 cannot substitute. The old NW calculation dropped missing dates and then compressed the observation index, so it cannot be called four-trading-day HAC; do not present the old t-statistic as new inference. (d) Examples: after-cost Sharpe is not positive; a paired test on dates frozen in advance does not support an advantage (the historical paired quantity could not be evaluated); or results hold in only a few folds. The constructed ranking is not a real-world conclusion.

39-T01 Complete Answer

$w = Sb$; $\sum_j s_j^2 b_j^2$. Unstandardized L1 retains 1/100 and L2 retains 1/10,000 of its original value; if the training standard deviation also increases by 100, $w_2 = s_2 b_2$ is unchanged. Multiplying y by 100 leaves Ridge α unchanged and multiplies Lasso α by 100. For Elastic Net, $\alpha' = 0.505, \rho' = 100/101$; $\alpha'\rho' = 0.5$ and $\alpha'(1-\rho') = 0.005$, preserving the L1 term's 100-fold scaling and the unchanged L2 term, respectively. The intercept is never penalized.

39-T02 Complete Answer

$D_R = \{1, 2, 3\}$, $D_m = D_{mR} = \{1, 3\}$. Own-date means: 0.4 and 0.25; common-date means: 0.15 and 0.25, with $\Delta_{mR} = 0.10$. Date 2 enters only Ridge's own-date mean, so the questions differ. Historical summaries contain neither date membership nor daily values, so the paired quantity could not be recomputed; P06 is a different experiment and cannot substitute.

Chapter 40: Splitting, Averaging, and Stepwise Correction

Trees balance the difference in mean returns across two sides against sample balance; forests average away the part on which trees differ; and, under squared loss, boosting fits residuals round by round. Depth, leaf size, number of rounds, and learning rate must be selected using date-based validation windows inside the training rows; importance is a diagnostic for one model on one window. All numbers are constructed for teaching. Exercise mapping: 40-L01 ↔ 40-E01, 40-E02, and Section 6.1; 40-L02 ↔ 40-E03 and Section 6.2; 40-L03 ↔ 40-E04, 40-E05; 40-L04 ↔ 40-E08 and Sections 12.1–12.2; 40-L05 ↔ 40-E06, 40-E07; 40-L06 ↔ Sections 8.5–8.6 and 40-C01.

Trigger. Does strict monotonicity preserve training order yet necessarily preserve predictions for new points? Can preprocessing use the full sample?

Definition 40.1 *(Regression-tree split).* At node S, choose feature j and threshold s to minimize _____; the leaf value is _____. `min_samples_leaf` requires _____.

Section 6.1: split gain. Within-group and between-group decomposition: $\mathrm{SSE}(S) = \mathrm{SSE}(S_L) + \mathrm{SSE}(S_R) + n_L(\bar{y}_L - \bar{y})^2 + n_R(\bar{y}_R - \bar{y})^2$. Since $\bar{y} = (n_L\bar{y}_L + n_R\bar{y}_R)/n$, [fill in: write $\bar{y}_L - \bar{y}$ and $\bar{y}_R - \bar{y}$ as multiples of $\bar{y}_L - \bar{y}_R$ and substitute], giving $\Delta =$ _____.

Your turn. 40-L01 Fully covers 40-E01 and 40-E02. (a) For $x = (1, \dots, 6)$, $y = (1, 1, 2, 4, 5, 5)$%: gains at the five thresholds _____; best threshold _____, leaf values _____. (b) For the Section 3 data ($y = (-2, -1, -1, 0, 2, 1, 3, 2)$), with `min_samples_leaf=3`, feasible thresholds _____, best _____. Change y_8 to 9: unconstrained best _____ (gain _____), constrained best _____. What does the constraint prevent?

Proposition 40.2 *(Section 6.2: variance of an average).* For B identically distributed trees with variance σ^2 and pairwise correlation $\rho \geq 0$: $\mathrm{Var}\left(\frac{1}{B}\sum T_b\right) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$.

Proof. Expand the variance of the sum: $\frac{1}{B^2}[$ _____ variances + _____ covariances$]$; each covariance is _____. [fill in: Simplify to the result.]

Your turn. **40-L02** Fully covers 40-E03. For $\sigma^2 = 4, \rho = 0.25$, the values at $B = 1, 4, \infty$ are _____. For $\sigma^2 = 5, \rho = 0.1$, the value at $B = 4$ is _____ and at $B = \infty$ is _____. Which setting is better for many trees? How does this relate to random-forest `max_features`?

Recall: Why does adding trees beyond 100 do so little?

Your turn. 40-L03 Fully covers 40-E04 and 40-E05. (a) For the Section 3 data and learning rate 0.25, do one round of stump boosting: $F_0 =$ ____, selected threshold ____, leaf values _____, $F_1 =$ _____, training MSE _____. Why is the negative gradient the residual under squared loss? (b) An HGB leaf has 4 samples, residual sum 6, and $\lambda = 4$: leaf value at learning rate 1 _____, at learning rate 0.1 _____. Compare it with the Ridge shrinkage factor in Chapter 39: _____.

Your turn. 40-L04 Fully covers 40-E08 and Sections 12.1–12.2.

```
gbm = HistGradientBoostingRegressor(max_iter=1000, early_stopping=True,
                                    validation_fraction=0.1,
    ↪ n_iter_no_change=20)
gbm.fit(X_panel_train, y_panel_train)     # rows = (date, stock), 5-day
    ↪ labels
# 12 outer folds: n_iter_ = 1000 every time
#   holdout MSE 22.06e-4, test MSE 26.01e-4
# rounds from inner date folds: median 8 rounds
#   test MSE 24.72e-4 (zero prediction: 24.76e-4)
```

(a) What two things do randomly held-out rows share with training rows? (b) Why did early stopping never trigger? (c) Another error: choosing the best number of rounds on the test window gives $R^2_{\text{OS}} = +0.26\%$, while inner-fold selection gives +0.16%. Why can the +0.26% not be used? (d) Give the two key steps in the fix.

Your turn. 40-L05 Fully covers 40-E06 and 40-E07. (a) Four stocks on one day, $x_2 = x_1 = (-1.5, -0.5, 0.5, 1.5)$, $y = (-0.9, -0.4, 0.2, 1.1)$, model $f = 0.6x_1$, baseline MSE 0.015. Under permutation $(3, 4, 1, 2)$, importance from permuting only x_1 is _____, only x_2 _____, and both together _____. Does x_2 contain no information? (b) For $f = x_1 + 2x_1x_2$, $x = (1, 1)$, and background $(0, 0)$, $v(\varnothing), v(\{1\}), v(\{2\}), v(\{1, 2\}) =$ _____; $\phi_1 =$ _____, $\phi_2 =$ _____. How is the interaction contribution allocated? Is this a discovery or a rule?

Your turn. 40-L06 Partially covers Sections 8.5–8.6 and 40-C01. Same fold, same accounting, one-sided cost of 10 bp:

Panel	Model	Mean IC	After-cost Sharpe	Annual turnover
Interaction (registered 4e-4)	Ridge	−0.0098	−1.25	24.4
Interaction (registered 4e-4)	Gradient boosting	0.0069	−0.63	36.7
Interaction (fivefold, post hoc)	Ridge	0.0571	2.54	23.9
Interaction (fivefold, post hoc)	Gradient boosting	0.1190	5.55	34.9
Linear (fivefold, post hoc)	Ridge	0.2149	10.27	22.2
Linear (fivefold, post hoc)	Gradient boosting	0.2136	10.25	25.0

(a) Why did tree models fail to find the interaction at the registered signal strength? What do the inner-fold round counts reveal? (b) Why is tree-model IC about twice Ridge's on the fivefold interaction panel? (c) Why do Sharpes on the fivefold panel have no real-world meaning? Why must these results be labeled post hoc? (d) State a falsification condition for adopting a tree model on real data.

For training data $x = (0, 2), y = (0, 1)$, the midpoint threshold is 1. The strictly increasing square transform on the nonnegative domain changes the training threshold to 2; at new point 1.2, the original tree predicts 1, but after squaring, $1.44 \leq 2$ and it predicts 0. Strict

> monotonicity preserves separable training order, but nonlinear midpoint interpolation may change predictions for new points; a positive affine rescaling preserves the corresponding threshold. Winsorization and binning are not one-to-one transformations; wait until the complete available cross-section is ready before computing cross-sectional ranks.

Chapter 40: Answers and Recalculation

Definition and proof blanks

$\mathrm{SSE}(S_L) + \mathrm{SSE}(S_R)$; the mean of y within the leaf; each leaf must contain at least that many training rows (otherwise the split is infeasible). $\bar{y}_L - \bar{y} = \frac{n_R}{n}(\bar{y}_L - \bar{y}_R)$, $\bar{y}_R - \bar{y} = -\frac{n_L}{n}(\bar{y}_L - \bar{y}_R)$, between-group term $= \frac{n_L n_R^2 + n_R n_L^2}{n^2}(\bar{y}_L - \bar{y}_R)^2$, $\Delta = \frac{n_L n_R}{n}(\bar{y}_L - \bar{y}_R)^2$. Variance expansion: B variances and $B(B-1)$ covariances, each $\rho\sigma^2$; $\frac{1}{B^2}[B\sigma^2 + B(B-1)\rho\sigma^2] = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$.

40-L01

(a) Gains: 4.8, 12.0, 16.667, 12.0, 4.8; best threshold 3.5, leaf values 1.333 and 4.667 (closed form $\frac{9}{6}(10/3)^2 = 16.667$). (b) Feasible thresholds: 3.5, 4.5, 5.5; best 4.5 (gain 18). With $y_8 = 9$, the unconstrained best threshold is 7.5, with gain $\frac{7}{8}(0.286 - 9)^2 = 66.45$, isolating stock 8 in its own leaf; the constrained best is 4.5 (45.1). The constraint prevents a leaf containing only one extreme stock. Lower training error does not prove better or worse next-period performance; use validation data from later dates.

40-L02

4, 1.75, 1; then 1.625, 0.5. The second setting is better with many trees: only $\rho\sigma^2$ remains in the limit, and $0.5 < 1$. When `max_features` is less than 1, each split sees only some of the features; this may lower correlation ρ, but it may also weaken an individual tree and change its σ^2 and bias. For large B, compare the overall error, including $\rho\sigma^2$ and bias; a smaller feature fraction is not guaranteed to be better. Beyond 100 trees, the second term $(1-\rho)\sigma^2/B$ is already small, while $\rho\sigma^2$ no longer changes with B.

40-L03

(a) $F_0 = 0.5$; threshold 4.5; leaf values ∓ 1.5; F_1 is 0.125 for the first four stocks and 0.875 for the last four; training MSE 1.7656. Since $L = \frac{1}{2}(y - F)^2$, $-\partial L/\partial F = y - F$, which is the residual. (b) $6/(4+4) = 0.75$; at learning rate 0.1 it is 0.075. The leaf value is $\nu\frac{n_R}{n_R+\lambda}\bar{r}$, which has the same form as Ridge's $\frac{n}{n+\alpha}$; smaller leaves receive more shrinkage.

40-L04

The excessive number of selected rounds is observed in this panel; it is not a theorem that random holdout always has this effect. (a) The same-day market shock (other stocks from that day remain in training), and overlapping five-day labels (adjacent signal dates share four days of returns). (b) Validation labels have almost been "seen" on the training side. As the model fits the training data more closely, validation error keeps falling, so the condition of 20 rounds without improvement is never met; all 12 folds run to 1,000 rounds. Random-holdout MSE of 22.06 is about 15% below the actual test MSE of 26.01. (c) Round selection uses test-window labels, which are unavailable when orders are placed; +0.26% is the best value selected using test data, not achievable out-of-sample performance. (d) Set early_stopping=False; select the number of rounds at the minimum of the mean validation curve using staged_predict on inner date-based folds, then refit on all outer-training rows with that number of rounds (the gbm factory in tree_models rejects early-stopping-related parameters).

40-L05

(a) 1.560, 0, 1.560. x_2 contains the same information as x_1; importance of 0 means the model did not use x_2, not that x_2 contains no information. If the model used x_2 instead, the ranking would reverse; group permutation measures the information in the pair. (b) $0, 1, 0, 3$; $\phi_1 = \frac{1}{2}(1) + \frac{1}{2}(3) = 2$, $\phi_2 = \frac{1}{2}(0) + \frac{1}{2}(2) = 1$, and their sum is 3. The main effect 1 goes to feature 1; the interaction contribution 2 is split equally. This allocation follows the Shapley symmetry axiom; it is a rule, not a discovery about the mechanism.

40-L06

(a) The daily conditional-mean standard deviation of the interaction signal is about 3.4×10^{-4}, idiosyncratic volatility is 2%, and assuming a nearly constant five-day signal and independent idiosyncratic shocks, ignoring compounding and the heterogeneous-beta common term, the cumulative signal/noise variance ratio is about $5(3.4 \times 10^{-4}/0.02)^2 = 0.001445$ (0.145%, on the order of 0.1%). This is a scale approximation, not a measured panel statistic; this run did not detect the interaction in about 750 test days. Inner validation selects only 1–14 rounds for boosting in 11 of 12 folds and often selects depth 2 for the single tree: validation error is lowest before much fitting, indicating that the training window contains no structure that appears to persist. (b) Linear models can use only the average slope on q_1 (half the conditional slope) and the linear projection of $|q_3|$ (zero); trees can split on q_2, then on q_1, and split each tail of q_3. (c) Fivefold loadings make the constructed signal far stronger than reality; Sharpes of 5–10 reflect only that constructed signal strength. The run was added after viewing the main results, so it is a post hoc sensitivity, cannot replace registered results, and cannot turn "trees are better" into a registered conclusion. (d) Example: under a frozen protocol, do not adopt tree models if gradient boosting's paired daily IC difference relative to Ridge has NW $t < 2$, if Sharpe after 20 bp is no higher than Ridge's, or if the advantage appears in only some folds.

Chapter 41: Variance, Distance, and When Information Is Available

PCA ranks variance; k-means ranks distance. Neither uses labels. To use them in research, fit scaling, directions, and centers within the training window, then only project or assign later dates. Noise inflates in-sample explained shares; smoothed probabilities and centered windows see the future. All numbers are constructed for teaching. ID map: 41-L01 ↔ 41-E01 and Section 6.1; 41-L02 ↔ 41-E02 and 41-E03; 41-L03 ↔ Sections 8.2 and 12.1; 41-L04 ↔ 41-E05 and 41-E08; 41-L05 ↔ 41-E06, 41-E07, and Section 6.6; 41-L06 ↔ Section 8.4 and 41-C02.

Trigger. A direction explains 99% of the variance. Must it be useful for prediction?

Definition 41.1 *(Principal components and explained share).* The first loading is $v_1 = \arg\max_{\|v\|=1}$ _____; the solution is an _____ of the covariance matrix. Explained share = $\lambda_j /$ _____. Estimate means, standard deviations, and loadings using the _____ only. For later dates, the score is $z_t = V_k^\top \tilde{x}_t$, where $\tilde{x}_t$ is centered using the _____.

Section 6.1: Why the solution is an eigenvector. The Lagrangian is $v^\top \Sigma v - \lambda(v^\top v - 1)$. Differentiate with respect to v and set it to zero: [fill in: Write the first-order condition and explain why the objective equals λ]. Therefore, choose the eigenvector corresponding to the _____.

Your turn. 41-L01 Fully covers 41-E01. Let $\Sigma = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$. Characteristic equation: _____; $\lambda =$ _____. Loadings (use "positive sum; if the sum is zero, choose the first element positive"): _____ and _____. The first principal component explains _____.

Your turn. 41-L02 Fully covers 41-E02 and 41-E03. (a) Two uncorrelated columns with variances 9 and 1: first covariance-PCA loading _____, explained share _____; after standardization the eigenvalues are _____. Is the principal component unique? _____. (b) Noise correlation matrix with $N = 100, T = 400$: $\lambda_\pm =$ _____; the first in-sample component appears to explain about _____, while its true share is _____. (c) When does the "first principal component" depend on the measurement units of the data?

Recall: Why do sample eigenvalues of noise spread out?

Your turn. 41-L03 Fully covers Sections 8.2 and 12.1. Noise panel (50 independent columns, daily volatility 1.5%):

```
full  = ut.pca_fit(noise, 3, standardize=True)          # all 1000 dates
train = ut.pca_fit(noise.iloc[:250], 3, standardize=True)
test  = noise.iloc[250:271]
# printed: 0.115 0.077 0.112   (full-on-test, train-on-test, train in-
    ↪ sample)
```

(a) The true share for the three directions is _____. Why is the training-window in-sample share of 0.112 inflated? (b) Why does full-sample fitting yield 0.115 on the test window, higher than the training-only fit at 0.077? (c) In the factor panel in Section 8.2 with $T = 60$, the minimum principal-angle cosine between adjacent three-dimensional subspaces, averaged across windows, is 0.736. What does this show? Does it mean "the third principal component rotated by arccos 0.736 degrees each month" ?

Your turn. 41-L04 Fully covers 41-E05 and 41-E08. Data $\{0.10, 0.12, 0.15, 0.30, 0.35, 0.40\}$, $K = 2$, initial centers 0.10 and 0.15.

Step	Assignment (Cluster 0 | Cluster 1)	Objective	Updated centers
0	_____	_____	_____
1	_____	_____	_____
2	_____	_____	Converged

Partitions $[0, 0, 1, 1, 2, 2]$ and $[1, 1, 0, 0, 2, 2]$: ARI _____, elementwise agreement _____. With $[0, 1, 0, 1, 2, 2]$: ARI _____.

Proposition 41.2 *(Section 6.6: Filtering recursion).* Prediction: $\Pr(s_t = j \mid x_{1:t-1}) = \sum_i \alpha_{t-1}(i) P_{ij}$; update: $\alpha_t(j) \propto \phi(x_t; \mu_j, \sigma_j) \Pr(s_t = j \mid x_{1:t-1})$. With training-fitted parameters held fixed, α_t depends only on $x_{1:t}$.

Proof. The first equation uses _____ and _____. The second uses Bayes' rule and the fact that, "conditional on the entire state path, observations are _____ across dates." Smoothed probabilities also multiply by the backward vector β_t, which depends on[fill in: Write the observations on which β_t depends].

Your turn. 41-L05 Fully covers 41-E06 and 41-E07. Let $P = \begin{pmatrix} 0.98 & 0.02 \\ 0.05 & 0.95 \end{pmatrix}$; calm returns follow $N(0.05\%, 0.7\%^2)$ and volatile returns follow $N(-0.1\%, 2\%^2)$. (a) Yesterday's $\alpha_{t-1} = (0.6, 0.4)$; today's return is $x_t = -1.0\%$. Convert all values to decimal returns for the density calculations: $-1\% = -0.010$, $0.7\% = 0.007$ (convert the means too). Density units are per unit decimal return. Find the prediction _____, densities _____, and $\alpha_t =$ _____. Why is the posterior almost the same as the prior? (b) Which are available at close t for execution at $t + 1$: past 21-day volatility; volatility calculated with `center=True`; filtered probability with training-window parameters; filtered probability with full-sample parameters; full-sample smoothed probability; nearest-center assignment of past features using training-window k-means?

Your turn. 41-L06 Partially covers Section 8.4 and 41-C02. D11 post hoc diagnostic: medians over 50 seeds (4200–4249; not a registered primary result or confirmatory evidence), 2,000 test days, and one-way cost of 10 basis points. Valid timing: signal at close t, execution at close $t+1$, and return held through close $t+2$. Let u_d be the signal formed at close. Index returns by their ending date, $r_d = C_d/C_{d-1} - 1$, and use weight u_{d-2}, shifting the signal by two periods.

Rule	Median Sharpe	Median annual turnover
Smoothed probability, same-day alignment (leaks)	0.76	32.3
Smoothed probability, execution at $t+1$ (leaks)	0.05	32.3
Centered volatility, lagged (leaks)	0.63	8.8
Filtered probability, training-window parameters	−0.03	38.3
Filtered probability, true parameters (unavailable)	0.38	12.2
Online k-means state	0.29	6.1
Past volatility	0.35	8.8

Check adjacent signal dates: a signal at the October 12 close executes at the October 13 close and is paired with the return ending October 14; a signal at the October 13 close executes at the October 14 close and is paired with the return ending October 15.
(a) The same-day Sharpe exceeds the $t+1$ execution Sharpe by 0.7. Which day's data does it use? (b) Why does centered volatility still lead even though execution is lagged? (c) Why does the training-window-parameter filter perform worst? Give a testable explanation (hint: estimated transition-matrix diagonal entries are 0.874 and 0.371). (d) State one falsification condition you would test on a real index.

With the training center, positive scales, and orthogonal loadings fixed, orthogonal decomposition guarantees reconstruction $R^2 \in [0, 1]$ when total squared norm is positive; predictive R^2 can be negative. If the denominator is zero, the API returns 1 by convention, while the mathematical value $0/0$ is undefined. Sign orientation cannot fix rotations within a repeated-eigenvalue subspace. Dollar neutrality does not guarantee beta neutrality. The state results use the fixed-weight proxy $w_d r_d - c|w_d - w_{d-1}|$, with $w_d = u_{d-2}$ and costs charged on the return-ending row. This differs from the Chapter 29 event-based ledger. Training a Gaussian mixture, hard-classifying, and counting transitions is not Baum–Welch maximum likelihood for an HMM.

Chapter 41: Answers and Recalculation

Definition and proof blanks

$v^\top \Sigma v$; eigenvector (largest eigenvalue); the sum of all eigenvalues, $\operatorname{tr}\Sigma$; training window; the training-window mean (and standard deviation). First-order condition $2\Sigma v - 2\lambda v = 0 \Rightarrow \Sigma v = \lambda v$; objective $v^\top \Sigma v = \lambda v^\top v = \lambda$; largest eigenvalue. Filtering: law of total probability and Markov property; conditional independence; β_t contains $x_{t+1}, \ldots, x_T$.

41-L01

$(3-\lambda)^2 - 1 = 0$; $\lambda = 4, 2$. For $\lambda = 4$: $(0.7071, 0.7071)$. For $\lambda = 2$: the sum is zero, so choose the first element positive, $(0.7071, -0.7071)$. Explained share $4/6 = 66.7\%$.

41-L02

(a) Covariance diag(9, 1): loading (1, 0) and share 90%. After standardization, the identity matrix has eigenvalues 1 and 1; every direction is an eigenvector, so the principal component is not unique and each direction explains 50%. (b) $q = 0.25$, $\lambda_\pm = (1 \pm 0.5)^2 = 2.25,\ 0.25$; in-sample share about 2.25%, true share 1%. The MP edge is only an IID, large-sample, fixed-ratio reference; the finite-sample maximum can exceed it. (c) When columns are weakly correlated and variance differences mainly come from units, the first covariance-PCA direction is the highest-variance column; changing units changes the result. For feature reduction, standardize on training rows.

41-L03

(a) $3/50 = 6\%$. In-sample eigenvalues include noise-driven spread (Section 6.3), systematically inflating the largest three; the effect is stronger for smaller T (18.6% at $T = 60$). (b) The full-sample PCA's mean, scale, and directions use the test window itself, so it naturally fits the test window better; this is leakage, not a better model. (c) One direction in the leading three-dimensional subspaces of adjacent windows aligns poorly, showing that the subspace itself changes over time. Risk factors built from these components need a cross-window stability check. It is incorrect to say "the third principal component rotated by a certain number of degrees": principal angles describe subspaces, not a particular axis (swapping PC1 and PC3 within the same subspace leaves all three principal-angle cosines at 1, although the axis cosines are 0, 1, and 0). The value 0.736 is also an average cosine across windows, and arccos is nonlinear, so arccos of the average cosine is not the average angle.

41-L04

Step	Assignment	Objective	Centers
0	$\{0.10, 0.12\} \mid \{0.15, 0.30, 0.35, 0.40\}$	0.1254	0.11, 0.30
1	$\{0.10, 0.12, 0.15\} \mid \{0.30, 0.35, 0.40\}$	0.0143	0.1233, 0.35
2	Unchanged	0.006267	Converged

The objective is calculated using "this step's assignments and the previous centers,"and decreases monotonically. ARI is 1 (swapping labels has no effect); elementwise agreement is 1/3. The ARI with $[0, 1, 0, 1, 2, 2]$ is $1/6 \approx 0.167$.

41-L05

(a) Substitute decimal returns into $\phi(-0.010; 0.0005, 0.007)$ and $\phi(-0.010; -0.001, 0.020)$. Prediction $(0.608, 0.392)$; densities 18.50 and 18.03 (per unit decimal return); after normalization the state probability is dimensionless: $\alpha_t = (0.614, 0.386)$. A -1% return is 1.5 standard deviations from the calm mean and 0.45 standard deviations from the volatile mean. The two densities are nearly equal, so the likelihood ratio is close to 1 and the posterior is determined by the prior. (b) Available: past 21-day volatility; filtered probability with training-window parameters; nearest-center assignment from training-window k-means. Unavailable: centered window (uses the next 10 days); filtered probability with full-sample parameters (parameters contain test-period information); full-sample smoothed probability (uses $x_{t+1:T}$).

41-L06

Check the timing first: r_{14} uses u_{12}, and r_{15} uses u_{13}. Shifting by two periods pairs the signal with a return earned after execution. The D11 cross-seed medians are post hoc diagnostics only and do not replace the registered single-path result.

(a) Under same-day alignment, the position on day t is determined by that day's smoothed probability, which uses the return on day t itself (and later returns): after a large decline, that day is classified as volatile, so the rule is "conveniently" out of the market that day. (b) At t, the centered window uses volatility from $t+1, \ldots, t+10$. The signal executes at close $t+1$ and earns the return on $t+2$; data from $t+3, \ldots, t+10$ still arrive after the return being earned. (c) Daily hard classification labels many ordinary calm-period fluctuations as volatile. The estimated calm-state persistence, 0.874, is far below the true value 0.98; filtered probabilities frequently cross 0.5, producing 38 turnovers per year and costs that consume returns. With true parameters, the same filter has median Sharpe 0.38 and turnover 12, suggesting the main problem is parameter estimation rather than the filtering recursion. Testable check: change only the training-window parameter-estimation method to EM, then compare turnover and Sharpe; do not assume in advance that the estimates will move toward the true-parameter version. (d) Example: if the filter with training-window EM parameters has out-of-sample after-cost Sharpe no higher than the past 21-day volatility rule, or its advantage appears in only one subperiod, do not conclude that the state model adds value.

Chapter 42: Deep Learning Foundations and Quantitative Applications

> The essence of a network architecture is a structural inductive bias on function space. In low-signal-to-noise financial time series, adding parameters does not automatically create a trading advantage. Strict boundary purging and an execution ledger with explicit trading costs are central to evaluating complex models.

Trigger. Can deeper layers and more advanced attention masks rescue low signal-to-noise data or look-ahead in features?

The triangular mask here serves real-time prediction at each historical position. If an entire window is already known at one cutoff and only its final representation predicts a later target, bidirectional encoding inside that window is not automatically leakage; full availability and fitting boundaries still apply.

Implementation path for this chapter

The MLP is the complete training path: matrix backpropagation, mini-batch SGD, validation-set early stopping, and restoration of the best parameters. The CNN only demonstrates one-dimensional valid cross-correlation with a fixed kernel; the RNN is a scalar linear recursion with identity activation; the LSTM only calculates one cell step; the Transformer only demonstrates positional encoding, masked softmax, and Q/K/V causal attention. The local hand calculations below do not mean that all of these architectures were trained or compared.

In this chapter, the global-mean baseline is fitted using only the 2,136 rows from 178 mature training dates; validation and test data are not used.

Definition 42.1 *(MLP forward pass and dimension contract).* Let the batch size be B and the input feature dimension be D_{in}. The first-layer weight is $\mathbf{W}_1 \in \mathbb{R}^{D_{\text{in}} \times D_{\text{hidden}}}$ and its bias is $\mathbf{b}_1 \in \mathbb{R}^{1 \times D_{\text{hidden}}}$. The hidden-layer pre-activation is $\mathbf{z}_1 = \mathbf{X}\mathbf{W}_1 + \mathbf{b}_1$ and the activation function is the hyperbolic tangent, $\mathbf{a}_1 = \tanh(\mathbf{z}_1)$. The second-layer weight is $\mathbf{W}_2 \in \mathbb{R}^{D_{\text{hidden}} \times D_{\text{out}}}$ and its bias is $\mathbf{b}_2 \in \mathbb{R}^{1 \times D_{\text{out}}}$; the output is $\hat{\mathbf{Y}} = \mathbf{a}_1\mathbf{W}_2 + \mathbf{b}_2$.

Your turn. 42-L01: Hand calculation of the 2→2→1 MLP forward pass and matrix dimensions (maps to: 42-E01 / 42-P01)

Take a single-sample input $x = [0.5, -0.2]$, weights $\mathbf{W}_1 = \begin{pmatrix} 0.2 & -0.1 \\ 0.4 & 0.3 \end{pmatrix}$, $\mathbf{b}_1 = [0.1, 0.0]$, $\mathbf{W}_2 = \begin{pmatrix} 0.5 \\ -0.2 \end{pmatrix}$, and $\mathbf{b}_2 = [0.05]$. Calculate each step: (1) $\mathbf{z}_1$; (2) $\mathbf{a}_1 = \tanh(\mathbf{z}_1)$ (to 4 decimal places); and (3) scalar output $\hat{y}$. State the exact dimension of the tensor calculated at each step.

Your turn. 42-L02: Full-parameter chain-rule gradient outline and numerical differences (maps to: 42-E02 / 42-P01)

Let the batch loss be $\mathscr{L} = \frac{1}{B \cdot D_{\text{out}}} \|\hat{\mathbf{Y}} - \mathbf{Y}\|_F^2 + \lambda(\|\mathbf{W}_1\|_F^2 + \|\mathbf{W}_2\|_F^2)$. Set $\boldsymbol{\delta}_2 = \frac{\partial \mathscr{L}}{\partial \hat{\mathbf{Y}}} = \frac{2}{B \cdot D_{\text{out}}}(\hat{\mathbf{Y}} - \mathbf{Y})$. Complete the backpropagation-gradient outline: (1) $\frac{\partial \mathscr{L}}{\partial \mathbf{W}_2} =$ _____ + _____; (2) $\frac{\partial \mathscr{L}}{\partial \mathbf{b}_2} =$ _____; (3) hidden-layer local error $\boldsymbol{\delta}_1 = \frac{\partial \mathscr{L}}{\partial \mathbf{z}_1} = (\boldsymbol{\delta}_2 \mathbf{W}_2^\top) \odot$ _____; (4) $\frac{\partial \mathscr{L}}{\partial \mathbf{W}_1} =$ _____ + _____; (5) Explain why this chapter does not add an L_2 penalty to bias terms $\mathbf{b}_1, \mathbf{b}_2$. Distinguish the algebraic equivalence for ordinary SGD from the condition for positive contraction ($0 < 2\eta\lambda < 1$), and state how to choose a step size for central-difference gradient checks.

42.1 Loss Function, Optimizer, and Model Selection

Definition 42.2 *(Training-only preprocessing and early-stopping protocol).* In quantitative time-series modeling, feature means $\boldsymbol{\mu}_X$, standard deviations $\boldsymbol{\sigma}_X$, and label mean μ_y and standard deviation σ_y must be estimated on training samples only and then frozen. Mixing validation- or test-set statistics into training causes global look-ahead leakage. Early stopping must monitor loss on an independent validation set that was not used in gradient updates. When patience is exhausted, it must restore the parameters from the epoch with the lowest historical validation loss (Best Epoch), not directly use weights from the stopping epoch.

Your turn. **42-L03: Diagnose train-only standardization and mini-batch SGD updates** (maps to: 42-E02 / 42-P02)

A quantitative researcher writes the following training-code fragment:

```
scaler = StandardScaler().fit(X_all) # X_all contains the full sample
X_scaled = scaler.transform(X_all)
for epoch in range(100):
    W -= lr * grad_W # learning rate lr=0.5, no weight decay
```

Identify the definite leakage caused by full-sample preprocessing and list what other information is missing before optimization stability can be judged. Write the standard train-only standardization repair. If label variance is σ_y^2, explain how to invert the model prediction $\hat{y}_{\text{scaled}}$ back to the original return scale.

Your turn. 42-L04: Analyze validation-set early stopping and test-set contamination (maps to: 42-E03 / 42-C01)

In the frozen Chapter 42 experiment, the 4→8→1 MLP is trained for up to 200 epochs. Validation loss reaches its historical minimum of 0.9812 at epoch 5; early stopping is then triggered at epoch 20 with patience=15. (1) Explain why the Chapter 42 protocol selects the checkpoint with the lowest validation error instead of using the epoch-20 weights, and why this does not guarantee lower error on the true future; (2) If a researcher mistakenly monitors the test set for early stopping, what methodological error (Data Snooping / Information Leakage) does this create? Explain how it can deceptively affect reported test MSE.

42.2 Structural Inductive Bias: CNN, RNN, and Transformer

Definition 42.3 *(One-dimensional cross-correlation and causal attention).* (1) A one-dimensional convolutional layer in deep learning actually computes cross-correlation (it does not flip the kernel): $(x \star w)[i] = \sum_{j=0}^{K-1} x[i+j]w[j]$; (2) Transformer scaled dot-product attention is $\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} + \mathbf{M}\right)\mathbf{V}$. The upper-triangular elements ($j > i$) of causal mask $\mathbf{M}$ must be set strictly to $-\infty$, making normalized attention weights $\text{softmax}(\cdot)_{i,j} \equiv 0$.

Your turn. 42-L05: CNN one-dimensional valid cross-correlation, parameter sharing, and translation equivariance (maps to: 42-E04)

Take input sequence $x = [1, 2, 3, 4]$ and one-dimensional convolution kernel $w = [1, -1]$ (stride 1, valid padding). (1) Calculate each element of the output vector by hand and confirm the result is $[-1, -1, -1]$; (2) Compare the sign of the result with convolution using a flipped kernel; (3) Explain how CNN local connectivity and parameter sharing substantially reduce parameter count compared with a fully connected layer, and why convolution gives translation equivariance rather than translation invariance.

Your turn. 42-L06: RNN gradient-product attenuation and the modern LSTM constant error carousel (maps to: 42-E05)

(1) The scalar Simple RNN state equation is $h_t = 0.5h_{t-1} + x_t$. For input $x = [1, 2, 0]$ and initial state $h_{-1} = 0$, calculate h_0, h_1, h_2 by hand; calculate state derivative $\frac{\partial h_t}{\partial h_{t-k}}$ across k steps and describe the gradient reduction at $k = 10$; (2) Write the four modern LSTM gate/candidate-state equations (forget gate f_t, input gate i_t, candidate cell $\tilde{c}_t$, and output gate o_t); (3) With scalar state and fixed gates and candidate, prove that the direct-path cell-state partial derivative is $\frac{\partial c_t}{\partial c_{t-1}} = f_t$. Explain how the constant error carousel (CEC) mitigates vanishing gradients and why it cannot be claimed that "LSTM completely solves all gradient problems."

42.3 Attention and the Financial-Research Boundary

Theorem 42.4 *(Future-perturbation invariance of causal attention and independence from feature leakage).* In causal-mask attention, arbitrarily perturbing input features $(\mathbf{q}_t, \mathbf{k}_t, \mathbf{v}_t)$ at any future time $t > \tau$ leaves the output vector $\mathbf{o}_\tau$ at historical time τ strictly unchanged. Corollary: a causal mask can only prevent look-ahead inside the attention layer. If the input feature itself already contains future information from its construction (such as forward returns or labels overlapping an unpurged boundary), the causal mask cannot repair any underlying research leakage.

Your turn. **42-L07: Calculate Transformer causal masking and test future perturbation** (maps to: 42-E06 / 42-P03)

Given sequence length $T = 3$, $d_k = 2$, and input matrices: $\mathbf{Q} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$, $\mathbf{K} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$, $\mathbf{V} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$. (1) Calculate scaled dot-product score matrix $\mathbf{S} = \frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{2}}$ and masked matrix $\tilde{\mathbf{S}}$; (2) Calculate normalized attention-weight matrix $\mathbf{A} = \text{softmax}(\tilde{\mathbf{S}})$ by hand (to 4 decimal places); (3) Calculate attention output $\mathbf{O} = \mathbf{A}\mathbf{V}$ step by step (A is displayed rounded; calculate O with unrounded weights); (4) Why must an engineering implementation use strict $-\infty$ together with stable numerical subtraction, instead of treating -10^9 as an unconditional equivalent? (5) Design a code unit test proving that future-token perturbations change historical-token outputs by strictly 0.0.

Your turn. **42-L08: Interpret low-SNR financial results and trading-cost drag** (maps to: 42-C02)

In the frozen constructed Chapter 42 experiment, test-set performance is as follows for the 4→8→1 MLP (49 parameters), Ridge regression (5 parameters), and global mean (1 parameter):

- Test-set MSE: mean baseline 0.00010135 < Ridge 0.00010144 < MLP 0.00010186;
- Test-set R^2: Ridge is −0.000892, and MLP is −0.004977;
- Portfolio ledger: MLP gross Sharpe is 0.501 (annualized turnover 360.2); with a one-way cost of 10 bps, net Sharpe falls to −4.688.

(1) Why does the MLP, which has the most parameters, have worse MSE on low-SNR financial data than the one-parameter global training-mean baseline? (2) What central tension in high-frequency/daily quantitative signals is revealed by positive gross Sharpe and deeply negative net Sharpe? Explain why an event-by-event ledger is needed to evaluate actual net returns instead of an approximate linear subtraction. (3) In a synthetic-data and non-independent-sample comparison, why must an unpurged daily-compressed Newey-West t statistic not be mislabeled as a "trading-day HAC t" ?

Chapter 42 Answers: 42-L01 to 42-L04

42-L01

(1) Calculate the pre-activation:

$$\begin{aligned}\mathbf{z}_1 &= x\mathbf{W}_1 + \mathbf{b}_1 \\ &= [0.5, -0.2]\begin{pmatrix} 0.2 & -0.1 \\ 0.4 & 0.3 \end{pmatrix} + [0.1, 0] \\ &= [0.5(0.2) - 0.2(0.4) + 0.1,\ 0.5(-0.1) - 0.2(0.3)] \\ &= [0.12, -0.11].\end{aligned}$$

Dimensions: $(1 \times 2) \times (2 \times 2) + (1 \times 2) = 1 \times 2$. (2) Calculate activations:

$$\mathbf{a}_1 = [\tanh(0.12), \tanh(-0.11)] \approx [0.1194, -0.1096].$$

Dimensions: 1×2. (3) Calculate the output layer:

$$\begin{aligned}\hat{y} &= \mathbf{a}_1\mathbf{W}_2 + \mathbf{b}_2 \\ &= [0.1194, -0.1096]\begin{pmatrix} 0.5 \\ -0.2 \end{pmatrix} + [0.05] \\ &= 0.1194(0.5) + (-0.1096)(-0.2) + 0.05 \\ &= 0.0597 + 0.0219 + 0.05 = 0.1316.\end{aligned}$$

Dimensions: $(1 \times 2) \times (2 \times 1) + (1 \times 1) = 1 \times 1$ (a scalar).

42-L02

(1) $\frac{\partial \mathscr{L}}{\partial \mathbf{W}_2} = \mathbf{a}_1^\top \boldsymbol{\delta}_2 + 2\lambda \mathbf{W}_2$; (2) $\frac{\partial \mathscr{L}}{\partial \mathbf{b}_2} = \sum_{i=1}^{B} \boldsymbol{\delta}_{2,i}$ (sum over rows, preserving the $1 \times D_{\text{out}}$ dimension); (3) Hidden-layer local error $\boldsymbol{\delta}_1 = (\boldsymbol{\delta}_2 \mathbf{W}_2^\top) \odot (1 - \mathbf{a}_1^2)$, since $\frac{d\tanh(z)}{dz} = 1 - \tanh^2(z) = 1 - a^2$; (4) $\frac{\partial \mathscr{L}}{\partial \mathbf{W}_1} = \mathbf{X}^\top \boldsymbol{\delta}_1 + 2\lambda \mathbf{W}_1$; (5) Why the biases are not penalized: biases determine activation shifts (the mean of the baseline output). Penalizing them with L_2 would pull the baseline toward zero and reduce the model's ability to adapt to labels that are not centered at zero. This is the choice made in this chapter; it does not mean that biases have no effect on local derivatives or curvature. Under ordinary SGD, the update is $\mathbf{W}^{(t+1)} = \mathbf{W}^{(t)} - \eta(\nabla_{\mathbf{W}} \mathscr{L}_{\text{MSE}} + 2\lambda \mathbf{W}^{(t)}) = (1 - 2\eta\lambda)\mathbf{W}^{(t)} - \eta \nabla_{\mathbf{W}} \mathscr{L}_{\text{MSE}}$, equivalent to weight decay with contraction factor $(1 - 2\eta\lambda)$. The effective contraction condition is $0 < 2\eta\lambda < 1$. Choose central-difference step size $\epsilon \sim 10^{-6}$: if ϵ is too large, truncation error dominates; if too small (for example, 10^{-15}), double-precision rounding and cancellation error dominate.

42-L03

Issue 1: Fitting the scaler on the full sample causes look-ahead leakage; future and test-set variances and means flow into the training input. Other risks: The fragment does not show how gradients are recalculated as parameters and batches change, whether validation is monitored, or whether a checkpoint is restored. The learning rate $lr = 0.5$ and absence of decay alone are not enough to determine whether training diverges; stability also depends on loss curvature and feature scale. Repair:

```
scaler = StandardScaler().fit(X_train) # training set only
X_train_s = scaler.transform(X_train)
X_val_s = scaler.transform(X_val)
X_test_s = scaler.transform(X_test)
```

Inverse transform: $\hat{y} = \hat{y}_{\text{scaled}} \cdot \sigma_y + \mu_y$. This restores the units exactly; it does not automatically correct prediction bias. With positive standard deviation it preserves rank ordering.

42-L04

(1) After index 5, validation loss fluctuates and does not reach a new low for 15 consecutive epochs. This is consistent with overfitting risk, but the curve alone cannot identify noise fitting as the mechanism. Epoch-20 validation error is above epoch 5, so restore epoch 5 under the frozen rule. Validation error is a noisy selection criterion and cannot prove that epoch 20 is worse on every new sample. Epoch numbering starts at 0; 21 epochs were actually run. (2) If the test set is used for early stopping, it is effectively treated as the validation set (model-selection set). The test set loses its methodological and live-trading credibility as an "unbiased test of true out-of-sample ability," creating data snooping. Test MSE then participated in model selection and cannot also serve as a final evaluation that was not used for selection. The contaminated Chapter 42-C01 checkpoint was epoch 17, with MSE 0.0001015535; no numerical improvement, however large, removes the contamination.

Chapter 42 Answers (continued): 42-L05 to 42-L08

42-L05

(1) Sliding one-dimensional cross-correlation:

$$\begin{aligned}\text{out}[0] &= x[0]w[0] + x[1]w[1] = 1(1) + 2(-1) = -1,\\ \text{out}[1] &= x[1]w[0] + x[2]w[1] = 2(1) + 3(-1) = -1,\\ \text{out}[2] &= x[2]w[0] + x[3]w[1] = 3(1) + 4(-1) = -1.\end{aligned}$$

The output vector is $[-1,-1,-1]$. (2) Mathematical convolution flips the kernel left-to-right to $[-1,1]$ and returns $[2(1)-1(1), 3(1)-2(1), 4(1)-3(1)] = [1,1,1]$. Deep-learning libraries such as PyTorch implement cross-correlation internally in 'Conv1d'; since weights are learned end to end by gradient descent, flipping the kernel only mirrors the sign convention. (3) The number of fully connected connections grows linearly with input length, while a CNN kernel has fixed size (for example, $K = 2$) and shares the same weights at each time step, sharply reducing effective degrees of freedom. Convolution is translation equivariant: if the input sequence is shifted by Δt, the output feature sequence shifts by the same amount ($f(g(x)) = g(f(x))$). This matches the prior that patterns in financial indicators, such as momentum or filter differences, are consistent across different historical time steps. Compare finite valid boundaries only at common valid positions. Pooling can be invariant when translation only reorders the same feature set; cropping, stride, or padding can change the set and must still be checked.

42-L06

(1) Recursion: $h_0 = 0.5(0) + 1 = 1.0$; $h_1 = 0.5(1.0) + 2 = 2.5$; $h_2 = 0.5(2.5) + 0 = 1.25$. The gradient product is $\frac{\partial h_t}{\partial h_{t-k}} = (0.5)^k$. At $k = 10$, $(0.5)^{10} \approx 0.000976$ (attenuation by more than a factor of 1,000), leaving the gradient of early sequence information on the current loss very small (vanishing gradient). (2) Modern LSTM equations: $f_t = \sigma(\mathbf{W}_f x_t + \mathbf{U}_f h_{t-1} + b_f)$; $i_t = \sigma(\mathbf{W}_i x_t + \mathbf{U}_i h_{t-1} + b_i)$; $\tilde{c}_t = \tanh(\mathbf{W}_c x_t + \mathbf{U}_c h_{t-1} + b_c)$; $c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$; $o_t = \sigma(\mathbf{W}_o x_t + \mathbf{U}_o h_{t-1} + b_o)$; $h_t = o_t \odot \tanh(c_t)$. (3) Direct-path partial derivative: with scalar state and fixed gates and candidate (the vector Jacobian matrix is $\text{diag}(\mathbf{f}_t)$), $\frac{\partial c_t}{\partial c_{t-1}} = \frac{\partial (f_t \odot c_{t-1} + i_t \odot \tilde{c}_t)}{\partial c_{t-1}} = f_t$. When the forget gate is saturated at $f_t \approx 1$, error can flow through the additive linear path with a one-step gain close to 1; across multiple steps, forget gates are still multiplied, e.g. $0.99^{1000} \approx 4.32 \times 10^{-5}$ (the CEC mechanism). The total derivative also includes nonlinear gate-dependent paths through h_{t-1} (such as $\frac{\partial f_t}{\partial h_{t-1}}$, $\frac{\partial i_t}{\partial h_{t-1}}$, etc.). With greater depth or an unsaturated forget gate, gradients can still decay or oscillate, so one cannot claim that all gradient problems are solved.

42-L07

(1) Dot product and mask:

$$\mathbf{QK}^\top = \begin{pmatrix}1 & 0\\ 0 & 1\\ 1 & 1\end{pmatrix}\begin{pmatrix}1 & 1 & 0\\ 1 & 0 & 1\end{pmatrix} = \begin{pmatrix}1 & 1 & 0\\ 1 & 0 & 1\\ 2 & 1 & 1\end{pmatrix}.$$

Divide by $\sqrt{2} \approx 1.4142$:

$$\mathbf{S} = \begin{pmatrix}0.7071 & 0.7071 & 0.0000\\ 0.7071 & 0.0000 & 0.7071\\ 1.4142 & 0.7071 & 0.7071\end{pmatrix}.$$

Apply the causal mask (upper triangle is $-\infty$):

$$\tilde{\mathbf{S}} = \begin{pmatrix}0.7071 & -\infty & -\infty\\ 0.7071 & 0.0000 & -\infty\\ 1.4142 & 0.7071 & 0.7071\end{pmatrix}.$$

(2) Softmax-normalized weights $\mathbf{A}$: Row 0: only the first entry is valid, $\mathbf{A}[0] = [1.0000, 0.0000, 0.0000]$. Row 1: valid entries are $[0.7071, 0.0000]$; $\exp(0.7071 - 0.7071) = 1.0$, $\exp(0 - 0.7071) = 0.4931$, and the sum is 1.4931. $\mathbf{A}[1] = [1/1.4931, 0.4931/1.4931, 0.0] \approx [0.6698, 0.3302, 0.0000]$. Row 2: valid entries are $[1.4142, 0.7071, 0.7071]$; $\exp(0) = 1.0$, $\exp(-0.7071) = 0.4931$, $\exp(-0.7071) = 0.4931$, and the sum is 1.9862.

$$\begin{aligned}\mathbf{A}[2] &\approx [1/1.9862,\ 0.4931/1.9862,\ 0.4931/1.9862] \\ &\approx [0.5035, 0.2483, 0.2483].\end{aligned}$$

(3) Output matrix $\mathbf{O} = \mathbf{AV}$: use unrounded A below; the earlier four-decimal entries are display values only. $\mathbf{O}[0] = 1.0 \times [1, 2] = [1.0000, 2.0000]$. $\mathbf{O}[1] = A_{10}[1, 2] + A_{11}[3, 4] \approx [1.6605, 2.6605]$.

$$\begin{aligned}\mathbf{O}[2] &= A_{20}[1, 2] + A_{21}[3, 4] + A_{22}[5, 6] \\ &\approx [2.4895, 3.4895].\end{aligned}$$

(4) Using strict $-\infty$ with the explicit check 'np.where(isneginf, 0.0, ...)' guarantees a future-position weight exactly equal to 0.0 at floating-point precision. A finite large negative number can still have a positive exponential at ordinary scales or fail to mask after a sufficiently large positive score is added; it is not unconditionally equivalent to negative infinity. (5) Unit-test outline:

```
_, out = causal_attention(Q, K, V, mask=True)
Q_pert, K_pert, V_pert = Q.copy(), K.copy(), V.copy()
Q_pert[2] += 10.0 # future token 2
K_pert[2] -= 5.0
V_pert[2] += 8.0
_, out_pert = causal_attention(Q_pert, K_pert, V_pert, mask=True)
change = np.max(np.abs(out[:2] - out_pert[:2]))
assert change == 0.0 # historical outputs are unchanged exactly
_, u = causal_attention(Q, K, V, mask=False)
_, up = causal_attention(Q_pert, K_pert, V_pert, mask=False)
assert np.max(np.abs(u[:2] - up[:2])) > 1e-3
```

Perturbing only future Q is a weak diagnostic: historical rows use only their own Q, even without a mask. Future K/V must also be perturbed, with an unmasked negative control. The operator's domain here is finite input and at least one unmasked entry per row.

42-L08

(1) The MLP's higher test MSE shows that, with this fixed training setup and split, the extra nonlinearity did not improve error. Capacity may increase sensitivity to noise, but a single result cannot prove that variance grows monotonically with parameter count. The 178 training dates / 2,136 rows are not the same as the effective number of independent samples; dependence, regularization, and nonstationarity also matter. (2) Daily prediction volatility causes frequent position changes (annualized turnover as high as 360.2 times). Annualized turnover multiplied by the one-way fee indicates the scale of trading losses, but actual net returns must be evaluated by continuously simulating capital in an execution ledger, rather than subtracting an approximate linear cost. High turnover erodes weak gross excess returns, turning net Sharpe from positive to −4.688. (3) Newey–West/HAC can address heteroskedasticity and weak serial dependence under suitable conditions; independent observations are not required. The problem here is calculating lags by array position after missing dates are dropped, which treats gaps between compressed observations as if they were the original trading-day intervals. With missing data, report the effective dates, common paired dates, and lag-index rule. Without calculations based on the original calendar, the result cannot be called trading-day HAC. This chapter reports no HAC significance; the direction of bias cannot be inferred from missingness alone.

Chapter 43: Bayesian Inference, Causal Identification, and Alternative Text Data

Probability updating represents uncertainty; causal identification estimates counterfactual outcomes under interventions; and text and alternative data require feature extraction under strict contracts for physical timing, delayed ingestion, and corpus cutoffs. Confusing conditional independence given a parameter with posterior predictive dependence, treating a confounded association as trading alpha, or treating an LLM pretraining cutoff as historical PIT can mislead statistical judgment and backtests.

Trigger. If a strategy has 8 winning trades out of its first 10, is the probability of two future wins $(8/10)^2$? If companies holding supplemental earnings calls reach their target the next day 64% of the time, compared with 26% for companies that did not hold a call, can we conclude that the call has a positive causal effect?

Definition 43.1 *(Conjugate Posterior, Posterior Prediction, Positivity, and PIT Eligibility).* For $y \sim \text{Bin}(n, p)$ with prior $p \sim \text{Beta}(\alpha, \beta)$, the posterior is Beta(_____, _____). Two future trials are conditionally _____ given p, and after integrating over p they are marginally _____; their covariance is exactly _____. Causal positivity requires $0 < P(T = 1 \mid Z = z) <$ ____ in target-population strata with positive weight. It means a nonzero treatment-assignment probability; it does not mean that a variable follows a _____ distribution. Under the valid fields, time ordering, version semantics, and single-cutoff conditions in the main text, the text-eligibility rule is max(available_at, ingested_at) $\leq$ _____.

Your turn. 43-L01 (Mapping: 43-E01 / 43-P01: Beta-Binomial Conjugate Updating and Predictive Covariance) The prior is $p \sim \text{Beta}(2, 2)$; 8 of 10 observed trials are successes. The posterior parameters are Beta(____, ____); posterior mean $E[p \mid y] =$ _____; posterior variance $\text{Var}(p \mid y) =$ _____. The predicted probability of success on one future trial is $P(\tilde{y} = 1 \mid y) =$ _____; the joint success probability for two future trials is $E[p^2 \mid y] =$ _____. Calculate the difference $E[p^2 \mid y] - (E[p \mid y])^2 =$ _____ and state how it relates to the posterior variance _____. Why are the two future trials positively correlated marginally?

Your turn. **43-L02** (Mapping: 43-E02: Normal Hierarchical Mean and Partial-Pooling Shrinkage with Known Hyperparameters) Group priors are independent given the known hyperparameters; observations are conditionally independent homoskedastic normals given the group mean, and new noise is independent of the mean and old observations. Dependence or heteroskedasticity would invalidate direct use of these weights. The hyperparameters are known: $\mu = 0.0, \tau = 0.02, \sigma = 0.04$. Returns are expressed as decimals ($0.02 = 2\%$); means and standard deviations have the same units, while variances have squared decimal-return units. The variance ratio is $\sigma^2/\tau^2 =$ ____. Group 1 ($n = 4, \bar{y}_1 = 0.03$): shrinkage weight $w_1 =$ ____, posterior mean $\hat{\theta}_1 =$ ____, posterior variance $V_1 =$ ____, and posterior predictive variance for a future observation $\mathrm{Var}_{\mathrm{pred},1} =$ ____. Group 2 ($n = 36, \bar{y}_2 = 0.01$): shrinkage weight $w_2 =$ ____, posterior mean $\hat{\theta}_2 =$ ____, posterior variance $V_2 =$ ____, and posterior predictive variance for a future observation $\mathrm{Var}_{\mathrm{pred},2} =$ ____. What do the two groups show about how sample size affects the degree of trust assigned under partial pooling?

Your turn. 43-L03 (Mapping: 43-E03 / 43-P02: Simpson Confounding Example and Causal Standardization) The sample size is $N = 200$, confounder $Z \in \{\text{high}, \text{low}\}$, and target distribution $P(Z = \text{high}) = 0.5$. High-attention group $Z = \text{high}$: for $T = 1$, $63/90 = 0.70$; for $T = 0$, $8/10 = 0.80$; within-stratum observed difference _____; Low-attention group $Z = \text{low}$: for $T = 1$, $1/10 = 0.10$; for $T = 0$, $18/90 = 0.20$; within-stratum observed difference _____; Unadjusted overall averages: success rate for $T = 1$ is $64/100 = 0.64$, and for $T = 0$ is $26/100 = 0.26$; unadjusted crude difference _____. The standardized difference (interpretable as an ATE only under consistency, conditional exchangeability, positivity in the target support, and SUTVA) is _____. Explain mathematically why the sign reverses from $+0.38$ to -0.10.

Your turn. 43-L04 (Mapping: 43-E04 / 43-C01: Positivity-Violation Guard and Algebra of Collider Bias) 1. Suppose z^* has positive weight in the target population. If every unit in the stratum receives treatment ($P(T = 1 \mid Z = z^*) = 1$), explain why the nonparametric causal ATE is mathematically unidentifiable. How should quantitative code respond? _____. 2. Let the true DAG be $T \to C \leftarrow Y$, where independent $T, Y \sim \text{Bernoulli}(0.5)$ and the unconditional covariance is $\text{Cov}(T, Y) = 0$. If a factor backtest incorrectly selects on the collider condition $C = \mathbb{I}(T = 1 \vee Y = 1) = 1$ (for example, selecting only among large-cap index constituents), derive the exact theoretical conditional covariance $\text{Cov}(T, Y \mid C = 1)$ and conditional correlation _____, and explain the warning for quantitative research (a DAG alone does not determine the sign of an association).

Your turn. 43-L05 (Mapping: 43-E05 / 43-P03: After-Hours Text Timing Contract and Version Exclusion) A company holds an earnings call after the close at 16:05 Eastern Time on trading day D_k. The vendor transmission delay is 20 minutes (available_at = _____); internal ingestion takes another 5 minutes (ingested_at = _____). The next-day decision signal is at signal_time = D_{k+1} 09:00; execution occurs at trade_time = D_{k+1} 09:30. If the company releases a revised transcript (Version 2) at 09:15 on D_{k+1}, which version should the strategy use at its 09:00 decision time, and why? _____.

Your turn. 43-L06 (Mapping: 43-E05 / 43-P03: Daylight-Saving-Time Shift and Absolute Comparison of Full Timestamps) Eastern Time switches to daylight saving time (EST → EDT) on March 13, 2022. The UTC timestamp for 16:05 EDT on March 14, 2022 is _____. If ingestion finishes at 20:30:00 UTC and the pre-opening signal is at 13:00:00 UTC the next day (09:00 EDT), explain why comparison must use full absolute timestamps including the date rather than only within-day time strings. _____.

Your turn. 43-L07 (Mapping: 43-E06 / 43-P04: Training-Only TF-IDF and Negation-Aware Sentiment Reversal) 1. Why must the TF-IDF vocabulary and IDF be fit on the training set only, never on the full sample? How are unseen new terms (OOV) in the test set transformed? _____. Diagnose the following text preprocessing: which step leaks information, and how should it be repaired?

```
vectorizer.fit(train_docs + test_docs)
X_train = vectorizer.transform(train_docs)
X_test = vectorizer.transform(test_docs)
```

2. The fixed lexicon assigns +1 to each of improve/growth and 0 sentiment weight to all other words; not/no activate the stated polarity-reversal rule for the next three words. Sentence: "earnings performance did not improve showing no growth despite aggressive restructuring." Naive unigram bag-of-words (BoW) score: _____; negation-aware sliding-window score: _____. Explain why the two methods reverse the polarity.

Your turn. 43-L08 (Mapping: 43-C02 / 43-C03: LLM Ledger, Template Generalization, and Pretraining-Cutoff Safeguards) The researcher evaluates an offline LLM assistant for historical earnings calls. The 120 events repeat four fixed templates, and every time block contains all four. Can the researcher claim generalization to new text based on the 30 test events or the unchanged-vocabulary OOV test? Why? _____. 1. The model declares a pretraining-corpus cutoff in December 2023. Can the researcher claim that the study "strictly satisfies historical point-in-time requirements with no leakage"? Why? _____. 2. List the core elements required in this chapter's offline LLM teaching ledger (why must the digest include the actual input and use an independent trusted reference? What happens if both the input and digest are changed?) _____.

Chapter 43: Answers and Recalculation

$\alpha + y$; $\beta + n - y$; independent; correlated; posterior variance $\mathrm{Var}(p \mid y)$; 1; normal; signal_time.

43-L01 Beta$(10, 4)$; $10/14 = 5/7 \approx 0.7143$; $40/(196 \times 15) = 2/147 \approx 0.0136$; $10/14 = 5/7$; $110/210 = 11/21 \approx 0.5238$; $2/147 \approx 0.0136$; exactly equal. Explanation: The two trials are independent conditional on p. Integrating over their shared uncertain parameter gives predictive covariance $\mathrm{Var}(p \mid y) > 0$, so they are positively correlated under the marginal predictive distribution. If a new first-trial success is subsequently observed, the posterior for p is then updated, raising the second trial's conditional prediction given that new outcome.

43-L02 V_j describes uncertainty about the group mean; the predictive variance of a future individual return also includes observation noise σ^2. $\sigma^2/\tau^2 = 0.0016/0.0004 = 4$. Group 1: $w_1 = 4/(4+4) = 0.5$; $\hat{\theta}_1 = 0.015$; $V_1 = 1/(2500 + 2500) = 0.0002$; $\mathrm{Var}_{\mathrm{pred},1} = 0.0002 + 0.0016 = 0.0018$. Group 2: $w_2 = 36/(36+4) = 0.9$; $\hat{\theta}_2 = 0.009$; $V_2 = 1/(2500 + 22500) = 0.00004$; $\mathrm{Var}_{\mathrm{pred},2} = 0.00004 + 0.0016 = 0.00164$. As sample size n grows, sampling variance σ^2/n falls, sample precision receives more weight, and shrinkage weight w approaches 1, so the model puts more trust in the group sample. A small-sample group is pulled more strongly toward the whole-market prior center $\mu = 0$.

43-L03 -0.10; -0.10; $+0.38$; -0.10. Mathematical reason: 90% of the treatment group ($T = 1$) is concentrated in the originally high-success stratum $Z = \text{high}$, while 90% of the control group ($T = 0$) is concentrated in the low-success stratum $Z = \text{low}$. The standardized difference is interpretable as an ATE only if the identification assumptions hold. Differences between the observed stratum means (0.71 versus 0.19) strongly mask the negative observed treatment difference within each stratum (-0.10).

43-L04 1. Because z^* has positive target-population weight, if the stratum contains only treated units, the counterfactual control outcome $E[Y(0) \mid Z = z^*]$ has no corresponding observed facts in the sample; the sample-mean denominator is zero, so this standardized estimate cannot be calculated directly. If the population treatment probability truly equals 1, nonparametric identification also lacks support. An empty cell in a finite sample alone cannot prove that the population probability is 1. Quantitative code must not silently fill in 0; it must raise `PositivityViolationError` explicitly. 2. The conditional covariance is $\mathrm{Cov}(T, Y \mid C = 1) = -1/9 \approx -0.1111$. Derivation: given $C = 1$, only the three equally likely states $(0,1)$, $(1,0)$, and $(1,1)$ remain (each has probability $1/3$); $E[T] = 2/3$, $E[Y] = 2/3$, and $E[TY] = 1/3$, so $\mathrm{Cov} = 1/3 - 4/9 = -1/9$ (conditional correlation is as high in magnitude as -0.5). Warning: selection can induce an association if admission follows a collider structure; restricting to index constituents does not itself prove that structure.

43-L05 D_k 16:25; D_k 16:30. The decision must use Version 1 because Version 2 was released at 09:15, after the 09:00 decision, and is future information. Using Version 2 directly creates look-ahead leakage; the PIT filter must exclude it strictly.

43-L06 `2022-03-14 20:05:00 UTC` (daylight-saving time EDT is UTC−4). Comparing only within-day strings (`20:30 < 13:00`) would make ingestion appear later than the signal and cause the data to be discarded incorrectly. Compare full absolute timestamps including the date: `2022-03-14 20:30:00 UTC < 2022-03-15 13:00:00 UTC`. They are 16.5 hours apart, so the point-in-time condition is satisfied.

43-L07 1. Fitting on the full sample lets future test-set terms and document frequencies leak into the training features, causing severe leakage across time. For an OOV term in the test set, the pipeline ignores the unseen feature; a document maps to an all-zero vector only if every term is OOV. The training vocabulary and IDF remain exactly unchanged. Code diagnosis: the first line uses test-set terms and document frequencies in fitting. Replace it with `vectorizer.fit`

`(train_docs)`, then transform each set separately. The input text must still satisfy historical PIT requirements. 2. The fixed lexicon gives this example a unigram score of +2 (it matches only the positive words `"improve"` and `"growth"`); the negation-aware score is −2 (`"not"` and `"no"` trigger sliding windows that reverse the polarity of subsequent positive terms). BoW does not assume statistical independence among terms. The three-word window is demonstrated only for this constructed sentence and is not guaranteed to handle complex negation or sarcasm accurately.

43-L08 Template question: this tests extrapolation across time blocks on the same four-template support, not generalization to new text; the OOV-invariance test is only a preprocessing safeguard. 1. No. An LLM learns from large internet corpora during pretraining, which may include retrospective reports on events from 2022 and therefore knowledge later than the historical signal. This does not prove that this particular output was contaminated, but a pretraining cutoff cannot be equated with historical PIT eligibility. 2. Record the SHA-256 hash of the complete prompt, including the actual input text (a template-only digest cannot check the specific input); changing the input and digest together still passes local recomputation, so retain an independently trusted reference digest. A cryptographic hash is not encryption or immutable evidence. Explicitly record the model status as `not_run / unknown` and the limits on reproducibility, including that zero temperature and a seed do not eliminate cross-hardware drift.

Project 6 —Machine-Learning Forecasts, Costs, and Evidence

Follow a frozen constructed panel through model comparison, a costed ledger, and the limits of the evidence. The reading explains time-grouped validation, metric differences, and common-date IC. The exercises ask you to rebuild calculations and research judgments. Real-data work on P6 remains blocked; constructed results test teaching mechanisms, not markets.

The Research Object

Project 6 is not a contest to find a winning algorithm. It asks when features are available, whether forecasts beat simple baselines, whether rankings form a tradable portfolio, and whether costs or a changing state alter the conclusion. Keep experiment identity explicit and retain negative results.

Trigger. A model has positive mean IC and negative R^2_{OS}. Can you immediately call it effective?

Definition P6.1.1 *(Constructed panel and time process).* The base panel has 40 assets, 1,512 dates, and 46 features: 3 signal features, 40 zero-loading noise features, and three derived price features: rev_5, mom_60_5, and vol_20. A simplified return equation is

$$r_{i,t+1} = \beta_i m_{t+1} + s_t\, 0.0003 \sum_{j=1}^{3} 0.5^{j-1} q_{j,i,t} + \epsilon_{i,t+1}.$$

Here m is a common market shock, q_j carries the constructed signal, and ϵ is idiosyncratic noise. The reversal scenario changes s_t's sign on a preset date. Different q sequences and different noise sequences are independent by construction. Each sequence is a stationary AR(1) over time with $\rho = 0.98$ and marginal variance 1; they are not IID over time. The three price features come from the return/price path, not from an additional independent, equal-variance draw.

P6 and the Chapter 40 interaction experiment

P6 has 46 features, 1,512 dates, and 16 folds. Chapter 40 is separate, with 26 features, 1,260 dates, and 12 folds, a registered primary strength, and a separate fivefold post hoc sensitivity. Do not move its settings, results, or thresholds into P6.

Your turn. P6-L04 Can the P6 experiment with 46 features, 1,512 dates, and 16 folds be merged into the Chapter 40 interaction experiment with 26 features, 1,260 dates, and 12 folds? Explain the design difference and the boundary around the fivefold post hoc comparison.

Your turn. **P6-L06** What does independence mean for the q and noise sequences? Are they IID over time? How are the three price columns generated? Why can the complete 46-dimensional feature set not be described as independent, identically distributed, and homoskedastic?

The Timing Contract

The signal forms at the close on D_k. The label spans the close on D_{k+1} through the close on D_{k+6}, five close-to-close intervals, and matures only after the endpoint close. Execute at the next trading day's close and rebalance every five signal dates. An asset's five-day label is not a daily portfolio holding return.

Proposition P6.2.1 *(Outer and inner folds).* Outer validation is date-grouped expanding-window walk-forward: minimum training 504 dates, test blocks of 63 dates, 16 folds. Exclude rows whose labels have not matured at fit time; keep all assets on a signal date on the same side. The inner loop uses its final two 63-date windows and at least 252 training dates, selecting parameters by mean validation MSE. The outer test set is for evaluation only.

The evaluated-date count

The test windows contain $16 \times 63 = 1,008$ nominal signal dates. The last six labels extend beyond the sample, leaving 1,002 mature-label dates per scenario. A constant forecast has undefined cross-sectional Spearman IC; that date is NaN, not zero.

Ten Prediction Sequences, Eight Main Models

> Version 1 produces zero, train_mean, OLS, Ridge, Lasso, and Elastic Net. Version 2 adds a single tree, random forest, gradient boosting, and PCA + Ridge, for 10 prediction sequences. The main table and figures show eight fitted models and omit the two constant baselines. PCA + Ridge is a dimension-reduced linear model, not a tree.

Model group	Models	Fold-level rule
Linear	OLS, Ridge, Lasso, Elastic Net	Select regularization by inner MSE
Trees	Single tree, RF, HGB	Select parameters on inner date windows
Dimension reduction	StandardScaler + PCA + Ridge	Fit all preprocessing within training folds
Flat baselines	zero, train_mean	Constant forecast; zero return and turnover

Gradient boosting

The fitted estimator is HistGradientBoostingRegressor with early_stopping=False. Its iteration cap is 300; select it with other parameters by mean MSE on the two inner time windows, then refit on all outer-training rows. Never tune on outer-test performance.

Your turn. **P6-L05** What are the six v1 prediction sequences and the four additions in v2? Why does the report say "8 main comparison models" while the frozen run has 10 prediction sequences? Which type of model is PCA + Ridge?

Forecast Metrics and Portfolio Outcomes

Definition P6.4.1 *(Three prediction/portfolio views).*

$$R^2_{\text{OS}} = 1 - \frac{\sum(y - \hat{y})^2}{\sum(y - \hat{y}_{\text{base}})^2}; \qquad IC_t = \text{Spearman}(\hat{y}_{i,t}, y_{i,t}).$$

R^2_{OS} compares asset-level forecast error to a zero or training-mean baseline. Daily IC measures cross-sectional ranking on a signal date. Portfolio Sharpe, net return, drawdown, and turnover depend on weights, execution time, ledger, and cost. Different rankings can all be correct.

Own-date IC versus paired IC

Let D_m be the dates when model m's IC is defined. Its own-date mean is $\bar{IC}_m(D_m)$. To compare with Ridge, use the common set $D_m \cap D_R$:

$$\Delta_m = \bar{IC}_m(D_m \cap D_R) - \bar{IC}_R(D_m \cap D_R).$$

Base Lasso/Elastic Net have 630/882 valid dates of 1,002; reversal Lasso/Elastic Net have 567/819. Common-date means and differences use exactly those date counts. Never subtract a sparse model's own mean from Ridge's 1,002-date mean. NaN is undefined, not zero.

Compressed NW(4) is not four-day HAC

The historical summary dropped missing IC dates and indexed the valid observations consecutively. If actual dates are 0, 2, 3, 5, 8, the fourth lag after compression connects days 0 and 8. It is not a four-trading-day lag. The revised report retains the old diagnostic but does not present it as trading-day inference or compute a new HAC.

Your turn. P6-L01 Can the base Lasso IC of 0.01785759 be compared directly with Ridge's 0.01982995 to conclude that Lasso is weaker?

Your turn. P6-L02 Does compressed NW(4) represent a four-trading-day HAC?

Your turn. **P6-L03** Can Chapter 39 reuse these P6 paired results?

Portfolio Costs and State Changes

> Demean each signal-date forecast across assets and convert it to dollar-neutral target weights with gross exposure 1. Start with 1 million, rebalance every five signal dates, execute at the next close, and charge one-way fees of 0/10/20 bp. Mark NAV daily. Borrow fees, financing spreads, market impact, capacity, and margin costs are not modeled.

Model / scenario	R^2 vs mean	IC	Sharpe 10 bp	Sharpe 20 bp
Elastic Net / base	−0.010%	0.0309	0.81	0.43
Tree / reversal	−0.604%	−0.0120	−1.13	−1.60
PCA + Ridge / base	−0.008%	0.0135	0.25	−0.13
PCA + Ridge / reversal	−0.053%	0.0063	−0.14	−0.52
Ridge / reversal	−0.161%	0.0087	−0.15	−0.53

Elastic Net has good after-cost results in the base construction, but all eight main fitted models have negative 20 bp Sharpe after reversal; the flat baselines have undefined Sharpe. A positive base result cannot erase the reversal outcome.

PCA, Importance, and Stability

Proposition P6.6.1 *(Full-component PCA + Ridge).* With the same StandardScaler, the same Ridge alpha, and whitening off, retaining all 46 principal components is an orthogonal rotation. If $V^\top V = I$, the coordinate change $\theta = V^\top w$ preserves the coefficient norm, $\|\theta\|_2 = \|w\|_2$. Squared error and Ridge penalty remain unchanged, so the full-component pipeline is orthogonally equivalent to Ridge in standardized coordinates. Truncation restricts coefficients to a smaller subspace; it may reduce noise or remove a predictive direction, and is not guaranteed to help.

PCA is not uniformly worse

Relative to the training mean, PCA + Ridge R^2 is $-0.008\%/-0.053\%$ in base/reversal, better than Ridge at $-0.113\%/-0.161\%$. Reversal 10/20 bp Sharpe is $-0.14/-0.52$ for PCA + Ridge, slightly above Ridge's $-0.15/-0.53$, although both are negative. Its IC is 0.0135/0.0063, below Ridge at 0.0198/0.0087. The metrics measure different objects. PCA selected 30 components in 10 folds, 5 in 5, and 15 in 1; this does not establish a stable choice across datasets.

Importance and stability

Grouped permutation is measured on the final 63-date inner-validation window in each fold, with three repeats. Permute the price and all-characteristics groups separately. RF/HGB ablations remove prices, use prices only, or use an inner-loop top 10; the top-10 ablation reuses full-model parameters. Near-zero grouped importance can reflect substitute features, not absence of information. Adjacent-fold rank correlations are low: tree $-0.10/-0.04$, RF 0.01/0.01, and HGB 0.09/0.10 in base/reversal. RF's three-seed prediction correlations are 0.962–0.982, with 10 bp Sharpe 0.44–0.64. HGB is identical across three seeds with early stopping disabled. Stability is not accuracy.

Your turn. **P6-L07** Using R^2 relative to the training mean, IC, and 10/20 bp Sharpe in both scenarios, determine whether PCA + Ridge is uniformly worse than Ridge. Explain why these findings can coexist.

Evidence Strength and Scope

> Across the six main tree comparisons (three tree types × two scenarios), the only $|t| > 2$ is the reversal single tree against Ridge: $\Delta IC = -0.0207$, unadjusted historical NW $t = -2.19$, in the direction of worse performance for the tree. Approximate unadjusted two-sided $p = 0.029$, about 0.17 after Bonferroni over six comparisons; ablations and seeds add more comparisons. This is not confirmatory. All other main comparisons have $|t| < 2$.

Do not invent a protocol threshold

The frozen comparison.claims_rule says to make no economic or alpha claim from constructed data and to retain zero and negative results. It does not define a joint threshold requiring ΔIC NW t≥2 in both scenarios and 20 bp Sharpe above Ridge and Elastic Net. Describe the single-tree result as an unadjusted, descriptive negative result, not a frozen alpha test.

Frozen notebook replay

The I3 P6v2 notebook has 9 cells, 4 code cells. It replays two frozen summaries, date bases, and historical ablation/stability displays, and generates historical_constructed_trees_sharpe.png from frozen summaries into RUN_DIR. Fit is marked SKIPPED; it does not retrain. A8-B is limited-scope accepted pending formal integration; full P6 remains unverified. Real-data work and A10-B data/PIT status remain unverified; empirical research is blocked.

Your turn. **P6-L08** Does the frozen configuration's claims_rule specify a joint threshold requiring "tree-model ΔIC NW t ≥ 2 in both scenarios and 20 bp Sharpe above both Ridge and Elastic Net" ? How should the only main comparison with an absolute t statistic above 2 be described?

Recall Card

Observation	Ask first
Positive IC	Own valid dates or common paired dates?
NaN	Undefined metric, not zero.
NW(4)	Trading days or compressed observations?
Positive Sharpe	Which fee, scenario, drawdown, turnover?
PCA	Full or truncated? Same scaling, alpha, and whitening?
Importance	Substitutes, window length, repeats, post hoc choice?
"Winning" model	Which frozen metric? Multiple tests? Constructed data?

Local reading

The base report and full v2 results are in projects/project06/research_report/README.md and research_report/v2_trees.md. V2_TREES.md is the companion guide; solutions.md holds complete answers for P6-L01–L08 and P6-T01–T04. Configurations, summaries, date-basis JSON, and implementation are available locally. These study notes do not execute models or notebooks or change frozen results.

Project calculations: dates, permutations, coordinates, and cash

A reproducible study puts each input row, loss, and cash movement in the right place. No model is trained here: screen by maturity, diagnose information with joint within-date permutation, then distinguish prediction coordinates from actual holdings.

Your turn. P6-T01 Fit after the close information for D_{10} arrives. Candidate signals are D_2, D_3, D_4, D_5, each with 40 assets, next-close execution and a five-day label. Give each maturity date and the retained row count. Why is randomly splitting assets within a date different from validation on later dates?

Definition P6.9.1 *(Joint within-date permutation).* Draw a separate row order for each date and share it across all group columns. This preserves that date's joint empirical distribution within the group, but may move its relationship with unpermuted columns outside the observed support.

Your turn. P6-T02 Two dates have $q = (1, 2, 3)$ and $(11, 12, 13)$, $z = 2q$, $y = q$, and fixed $f = q$. Labels and predictions are in percentage points; source row indices start at 1. Source orders are $(3, 1, 2)$ and $(2, 1, 3)$. Give the permuted pairs and MSE increase across six rows; write a same-date paired-multiset assertion. Why do separate orders for the two columns fail?

Proposition P6.9.2 *(The full-dimensional orthogonal control).* With the same Scaler, alpha, input rows, and intercept convention, without whitening, $Z = XV$ with full-dimensional orthogonal V gives identical Ridge predictions. Truncated PCA has no such guarantee.

Proof. Set $\beta = Vb$. Predictions satisfy $X\beta =$ _____; orthogonality gives the penalty norm _____.[fill in: Explain why the coordinate map is one-to-one and why whitening changes the meaning of the same alpha.]

Your turn. P6-T03 $X^\top X = \mathrm{diag}(2, 8)$, $X^\top y = (2, 4)^\top$, $\alpha = 2$, and $V = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Find $\hat{\beta}, \hat{b}, V\hat{b}$. What is removed if only the first column of V is retained?

A target is an instruction. Actual weights divide marked position values by current NAV. Price drift and fees change the numerator or denominator. Following cash event by event separates predictive error from implementation.

Your turn. **P6-T04** Start with 1,000 cash and A/B prices of 100. Buy five A shares and short five B shares, paying 10 bp one-sided fees from cash without resizing dollar targets. Next prices are 110/90. Find first post-fee cash/NAV, next pre-trade NAV and actual weights. Reset targets to positive/negative 0.5 of pre-trade NAV; calculate total traded notional, fee, and post-trade NAV. Allow fractional shares; omit financing, dividends, and impact.

Project 6: Questions and Complete Answers

P6-L01 · Own-date IC and common-date pairing

Question: Can the base Lasso IC of 0.01785759 be compared directly with Ridge's 0.01982995 to conclude that Lasso is weaker?
Answer: No. Lasso's mean is over 630 dates with defined IC; Ridge's mean is over all 1,002. On the common 630 dates, Lasso is 0.01785759, Ridge is 0.01613330, and the difference is +0.00172429. It is descriptive and has no valid inferential t statistic in this display. NaN is undefined, not zero.

P6-L02 · Compressed NW(4)

Question: Does compressed NW(4) represent a four-trading-day HAC?
Answer: No. With actual valid dates 0, 2, 3, 5, 8, compression makes the fourth lag connect dates 0 and 8, eight trading days apart. It is not a four-trading-day HAC. No new HAC was calculated.

P6-L03 · P6 and Chapter 39

Question: Can Chapter 39 reuse these P6 paired results?
Answer: No. The row-level forecasts required for Chapter 39 are unavailable. P6 has a different configuration and run identity, so it cannot fill the gap. New training cannot create historical predictions. Keep the Chapter 39 paired quantity marked as unrecomputable.

P6-L04 · P6 and Chapter 40

Question: Can the P6 experiment with 46 features, 1,512 dates, and 16 folds be merged into the Chapter 40 interaction experiment with 26 features, 1,260 dates, and 12 folds? Explain the design difference and the boundary around the fivefold post hoc comparison.
Answer: No. The generating process, sample, and split differ. Chapter 40 has a registered primary strength and a separate fivefold post hoc sensitivity; both belong to its experiment. Neither the interaction structure nor the post hoc result becomes a P6 feature, result, or threshold. Methods may be taught side by side only if experiment identity remains explicit.

P6-L05 · Prediction sequences and main-table model count

Question: What are the six v1 prediction sequences and the four additions in v2? Why does the report say "8 main comparison models" while the frozen run has 10 prediction sequences? Which type of model is PCA + Ridge?
Answer: Version 1 has zero, train_mean, OLS, Ridge, Lasso, and Elastic Net. Version 2 adds a single tree, RF, HGB, and PCA + Ridge, for 10 sequences. The table shows eight fitted models and leaves out the two constant flat-portfolio baselines. PCA + Ridge standardizes, applies PCA, then fits Ridge; it is a dimension-reduced linear model, not a tree.

P6-L06 · Feature process and independence

Question: What does independence mean for the q and noise sequences? Are they IID over time? How are the three price columns generated? Why can the complete 46-dimensional feature set not be described as independent, identically distributed, and homoskedastic?
Answer: Different constructed q/noise sequences are independent by the generator, but each is an AR(1) over time with $\rho = 0.98$ and marginal variance 1, so they are not IID. The price columns rev_5, mom_60_5, and vol_20 derive from the return/price path with common market, q-signal, and idiosyncratic shocks. The 43 characteristic sequences and three derived price features have different structures; the 46 columns are not draws from one independent, identically distributed, equal-variance population.

P6-L07 · PCA and Ridge

Question: Using R^2 relative to the training mean, IC, and 10/20 bp Sharpe in both scenarios, determine whether PCA + Ridge is uniformly worse than Ridge. Explain why these findings can coexist.

Answer: No. PCA + Ridge R^2 versus the training mean is $-0.008\%/-0.053\%$ in base/reversal, slightly better than Ridge at $-0.113\%/-0.161\%$. In reversal its 10/20 bp Sharpe is $-0.14/-0.52$, slightly higher than Ridge's $-0.15/-0.53$, though both are negative. Its IC is 0.0135/0.0063, lower than Ridge's 0.0198/0.0087. Forecast error, rank correlation, and portfolio results measure different objects.

P6-L08 · Protocol and multiple comparisons

Question: Does the frozen configuration's claims_rule specify a joint threshold requiring "tree-model ΔIC NW t ≥ 2 in both scenarios and 20 bp Sharpe above both Ridge and Elastic Net" ? How should the only main comparison with an absolute t statistic above 2 be described?

Answer: No. claims_rule prohibits economic or alpha claims from constructed data and requires retaining zero and negative results. The only $|t| > 2$ among the six main comparisons is the reversal single tree against Ridge: $\Delta IC = -0.0207$, unadjusted NW $t = -2.19$, in the direction of worse tree performance. Approximate unadjusted $p = 0.029$, about 0.17 after Bonferroni over six. Since ablations and seeds add comparisons, describe this as an unadjusted descriptive result from constructed data, not a confirmatory significant finding.

Complete answers to the project calculations

P6-T01

Signals D_2, D_3, D_4, D_5 execute on D_3, D_4, D_5, D_6 and mature on D_8, D_9, D_{10}, D_{11}. The first three dates are eligible: keep $3 \times 40 = 120$ rows. Equality requires the close information to have arrived. Splitting assets within a date shares the daily shock and is not validation on later dates; recheck maturity at each inner fitting time.

P6-T02

First date: $(3, 6), (1, 2), (2, 4)$; second: $(12, 24), (11, 22), (13, 26)$. With labels unchanged, errors are $(2, -1, -1, 1, -1, 0)$, SSE is 8, and MSE increases by $4/3$ squared percentage points. Directly check `Counter(permuted[d]) == Counter(original[d])`, each column's multiset, and $z = 2q$; see the standard-library code in this project's solutions. Global cross-date orders and separate column orders change the wrong structure. A fixed model's unused column has exactly zero importance; a fitted finite-sample model need not.

P6-T03

$X\beta = XVb = Zb$ and $\|\beta\|^2 = \|b\|^2$. Full orthogonality makes the map invertible, preserving squared loss and the same-alpha penalty. Whitening adds directional rescaling and breaks this control. $\hat{\beta} = (0.5, 0.4)^\top$, $\hat{b} = (0.4, -0.5)^\top$, and $V\hat{b} = \hat{\beta}$. Retaining only the first column sets $\beta_1 = 0$, giving $(0, 0.4)^\top$. This may reduce noise or lose signal; use in-training selection and independent testing rather than a variance guarantee. No P6 training was added.

P6-T04

First fee: $1000 \times 0.001 = 1$, giving cash/NAV 999. Next positions are 550 and -450 with cash 999, so NAV is 1099 and weights are $550/1099$ and $-450/1099$, approximately 0.500455 and -0.409463. Targets become $+549.5, -549.5$: sell 0.5 dollars of A and short another 99.5 dollars of B. Total notional 100, fee 0.1, cash $999 + 0.5 + 99.5 - 0.1 = 1098.9$, net position value 0, and NAV 1098.9. Absolute post-fee weights are $549.5/1098.9 \approx 0.500046$, not exactly 0.5. Event-by-event cash updates and position-value drift determine rebalancing notional; Chapter 41's fixed-weight cost proxy cannot replace these identities.

Chapter 44 —Stochastic Calculus and Derivatives

Begin with replication under two states, then use information and quadratic variation to establish continuous replication. Run each simulation under the specified measure, and reconcile both shares and cash in every hedge. All numerical values in this volume are SIMULATED / CONSTRUCTED.

Trigger. Before calculating, identify the payoff, measure, time to maturity, and units. After completing each prompt, cover the answer and redo the key equalities.

Your turn. 44-L01 Mapping: 44-E01, 44-P01: Brownian scaling and the scope of the proof. Over one year, a fair coin is tossed for N steps, with increment $\xi_i/\sqrt{N}$ and variance $1/N$ per step. The terminal variance is _____; if the increment is changed to ξ_i/N, the terminal variance is _____. For a Brownian increment $\Delta W \sim N(0,h)$, the difference quotient has distribution _____. Explain why this rules out a finite derivative in probability at a fixed time, and why it is not a complete proof of nowhere differentiability. For equal steps with $T=1, N=4$, the expected sum of squared increments is _____, and its variance is _____. What condition is needed to replace finite QV by T?

Your turn. 44-L02 Mapping: 44-E02, 44-C01: information and martingales. Let $X_0 = 0, X_1 = \xi, X_2 = -\xi$, where ξ is fair ± 1; the mean at all three times is 0. Given $\mathcal{F}_1$, $E[X_2 \mid \mathcal{F}_1] =$_____, so this _____ a martingale. State the three conditions in the definition involving adaptedness, integrability, and conditional expectation. If every left-point integrand value uses future W_T, the discrete sum is _____ and its expectation is _____. Why can a numerical array shape check not prove adaptedness?

Your turn. **44-L03** Mapping: 44-E03, 44-P01: where does the second-order term come from? For the constructed sequence $W = [0, .2, .1, .4]$, first calculate QV = _____ and the left-point sum = _____. Starting from $a(b-a) = (b^2 - a^2 - (b-a)^2)/2$, telescope the sum and write the general finite expression including W_0. For Brownian motion starting at 0, the limit is _____. In Itô's formula for $f = x^2$, fill in the term missing from the ordinary chain rule; explain the bookkeeping meaning of $(dW)^2 = dt$.

Your turn. **44-L04** Mapping: 44-E04, 44-P02: GBM and error. Under P, the total-return drift including dividends is $\mu = .08$, with $q = .02, \sigma = .2$. The stock-price drift $\beta =$ _____; for $f = \log S$, $f' = 1/S, f'' = -1/S^2$, and the log drift is _____. Fill in the explicit solution and $E[S_T]$ and $\text{Var}(S_T)$. With the same dW, define $D_m = S_T^E - S_T^{exact}$: the strong terminal RMSE = _____, weak mean difference = _____, and paired SE = _____. Must the weak sample difference be monotone across grids? Why or why not?

Your turn. 44-L05 Mapping: 44-E05: P/Q and continuous dividends. Let $\mu = .08, r = .05, q = .02, \sigma = .2, T = 1$; μ always denotes the total-return drift including dividends. $\lambda = (\mu - r)/\sigma =$ _____. The density is $Z_t = \exp($ _____ $)$, and $W_t^{\mathbb{Q}} = W_t^{\mathbb{P}} +$ _____; after substitution, the Q stock-price drift is _____. How do we assess the Novikov expectation $e^{\lambda^2 T/2}$ in this finite, constant-coefficient model? The drift of $e^{-rt}S_t$ is _____. Give one correct martingale process that includes cash dividends and one reinvestment process, and state the required moment condition.

Your turn. 44-L06 Mapping: 44-E06, 44-P03: tree replication. Let $S_0 = K = 100, S_u = 120, S_d = 80, q = 0$, with bank growth 1.05; the call pays 20/0 in the two states. Solve $120\Delta + 1.05A = 20$ and $80\Delta + 1.05A = 0$ to obtain $\Delta =$ _____, $A =$ _____, and price = _____. Risk-neutral $p =$ _____; how can you check the discounted expectation? For CRR with dividends, let $a = e^{(r-q)dt}$; write p and the admissible interval. Does changing the real-world up probability to .3 change the replication price? Explain why p must not be clipped when it falls outside its admissible range.

Your turn. **44-L07** Full coverage of 44-E07, partial coverage of 44-E08 (PDE and Feynman–Kac): from self-financing to the PDE. The stock holding is Δ, cash is A, and total value is $V = \Delta S + A$. The self-financing return is $dV = \Delta(dS + qSdt) + rAdt$; Itô gives $dV = [V_t + (\mu - q)SV_S + \sigma^2 S^2 V_{SS}/2]dt + \sigma S V_S dW_P$. Matching the noise gives $\Delta =$ _____; write the two drift terms to be equated, cancel μSV_S, and obtain the PDE _____. The call terminal payoff is _____. What condition is needed for the discounted Q conditional-expectation representation? Explain why a local martingale does not automatically preserve its expectation.

Your turn. **44-L08** Mapping: 44-E09, 44-P04: units of the Greeks. Base case: $S = K = 100, T = 1, r = .05, q = .02, \sigma = .2$, $d1 = .25, d2 = .05$. Call Delta = $e^{-qT}\Phi(d1)$; Vega = $Se^{-qT}\phi(d1)\sqrt{T} = 37.9011575100$. The first-order currency amount for a 1 pp volatility move is _____. The Theta convention is _____, so which sign is needed for a central difference in remaining maturity? Give the second-difference formula for Gamma. At expiry with $S = K$, what are the left and right Deltas? How can a very small step amplify rounding error?

Your turn. 44-L09 Mapping: 44-E10, 44-P04: SE for Monte Carlo with correlated samples. The M independent Z_i generate antithetic payoffs Y_i^+, Y_i^-. The independent observations are $X_i =$ _____, the price estimate = _____, and SE = _____. Fill in the covariance term missing from the flattened algorithm's $\mathrm{Var}(X)$. Given that `y, ya` are discounted payoffs generated from the same z and $-z$, $M = \mathrm{len}(y)$, and `np` is NumPy, diagnose the following incorrect code and state the correct independent sampling unit:

```
se_wrong = np.r_[y, ya].std(ddof=1) / np.sqrt(2*M)
```

Why does one BS price falling within a 95% approximate CI not prove the algorithm is correct? Which error is removed by sampling exact GBM terminal values, and which error remains? How should random $M = 1$ be reported?

Your turn. 44-L10 Mapping: 44-P05, 44-C02: the ledger and research judgment. There are 12 steps on a P path, stock-price drift $\mu - q = .06$, Q pricing drift $r - q = .03$, and one-way fee = .001. Given old cash A, old share count Δ, and the left-point price S, cash just before the end-of-interval rebalance is _____. How do the purchase price and fee for the current change in shares $\delta\Delta$ enter cash? Is the expiry liquidation charged as well? In a frozen simulation, no-fee error SD falls from 1.982914 at 12 steps to .432230 at 252 steps; with fees, average cash fees rise from .226679 to .611734, and mean error changes from −.332425 to −.669815. Which model-internal conclusions do these results support? Give one verifiable condition that could refute "higher frequency in live trading must be better."

From information to weighted quadratic variation, then reconcile cash and the time scale. The next two are new calculations; L01--L10 are unchanged.

Your turn. 44-L11 Mapping: new 44-E11: why left-point information matters. The four equally likely terminal states are $(+,+),(+,-),(-,+),(-,-)$. The values of $M_2 = \xi + \eta$ are _____; given F_1, the conditional means in the two groups are _____. What measurability condition is represented by having the same value within each group? For a deterministic partition, let C_i be $\mathcal{F}_{t_i}$-measurable with $|C_i| \le K$, and let $\varepsilon_i = (\Delta W_i)^2 - \Delta t_i$. Write $E[\varepsilon_i \mid \mathcal{F}_{t_i}]$ and $E[\varepsilon_i^2 \mid \mathcal{F}_{t_i}]$; for $i < j$, with respect to which information set do you condition the cross term? Complete

$$E[(\sum C_i \varepsilon_i)^2] \le _____.$$

For $T = 1, N = 4, K = 2$, the upper bound is _____. Why can this still not replace a proof controlling the general Itô remainder?

Your turn. 44-L12 Mapping: new 44-E12: verifying the solution and the cash account. With zero-fee instantaneous rebalancing, $S = 100$, shares change from .6 → .5, and A=−50: the change in stock market value is _____, the change in cash is _____, and total value before/after is _____. For a smooth semimartingale Delta, $d(\Delta S) = \Delta dS + Sd\Delta + d[\Delta, S]$; fill in the cash equation so the rebalancing terms cancel:

$$dA = _____.$$

For $F(t, w) = S_0 e^{(\beta - \sigma^2/2)t + \sigma w}$, write the three partial derivatives and simplify $F_t + F_{ww}/2$ to verify GBM. Why is it always positive? With calendar time $t = .25$ and expiry 1, API T = _____ years; the units of λ are _____.

Chapter 44: Complete Answers and Recalculations

> Answers appear on pages separate from their prompts. Exact unrounded values are in Section 8 of the chapter body and the frozen experiment; displayed six-decimal values are not required to meet floating-point assertion tolerances.

44-L01

The two terminal variances are 1 and 1/N, respectively. The difference quotient is $N(0, 1/h)$. For any fixed finite M, the probability that it falls in $[-M, M]$ tends to 0; it is not tight and cannot converge in probability to a finite value. Almost-sure nowhere differentiability requires controlling all times simultaneously with a standard theorem; this calculation only gives a fixed-time argument. For N=4, $E[QV] = 4(.25) = 1$, and the variance is $4 \cdot 2(.25)^2 = .5$. A deterministic partition with maximum step size tending to 0 gives $\mathrm{Var}(QV) \leq 2T|\Pi| \to 0$, so $QV \to T$ in L^2; on a finite grid it generally does not equal T.

44-L02

The conditional expectation is $-\xi \neq \xi$, so this is not a martingale. A martingale requires adaptedness, integrability at every time, and $E[M_t \mid \mathcal{F}_s] = M_s$. An Itô integrand also needs, for example, progressive measurability or predictability and $E\int H^2 dt < \infty$. With future W_T copied at every left point, the sum is $W_T \sum \Delta W = W_T^2$, with expectation T and second moment $3T^2$; the Itô isometry's right-hand side is T^2. Array shape only shows that multiplication is compatible; it cannot tell when information becomes available. Changing a future suffix without changing the past checks dependence in the given realization, but finite tests do not prove adaptedness for an arbitrary process.

44-L03

The increments are $.2, -.1, .3$, so QV = .14 and the left-point sum is $.2(-.1) + .1(.3) = .01$. The general finite expression is $(W_T^2 - W_0^2 - QV)/2$, obtained by telescoping the squared endpoints. For Brownian motion starting at 0, the limit is $(W_T^2 - T)/2$. Itô's formula for x^2 gives $d(W^2) = 2W\,dW + dt$; the second derivative 2 times $\frac{1}{2}(dW)^2$ leaves dt. The bookkeeping rule represents the integral limit of quadratic variation; it does not say that an individual finite $(\Delta W)^2$ equals dt.

44-L04

$\beta = .06$, and the log drift is .04. The explicit solution is $S_t = S_0 \exp[(\beta - \sigma^2/2)t + \sigma W_t]$. The exponential moments of a normal variable give $E[S_t] = S_0 e^{\beta t}$ and $E[S_t^2] = S_0^2 e^{(2\beta+\sigma^2)t}$; subtracting the squared mean gives variance $S_0^2 e^{2\beta t}(e^{\sigma^2 t} - 1)$. Strong RMSE is $\sqrt{M^{-1} \sum D_m^2}$, weak mean difference is $\bar{D}$, and SE = $s_D/\sqrt{M}$. Sampling error can make the weak sample difference nonmonotone; in this chapter the exact and Euler population means are $S_0 e^{\beta T}$ and $S_0(1 + \beta dt)^N$, respectively. Path RMSE with different noise is not strong discretization error.

44-L05

$\lambda = .15$, $Z_t = \exp(-.15W_P(t) - .01125t)$, and $W_Q = W_P + .15t$. The Novikov expectation $e^{.01125T}$ is a finite constant for finite T. Substituting $dW_P = dW_Q - .15dt$ makes the P stock-price drift $.06 - .2(.15) = .03$ under Q. Under Q, the drift of the discounted ex-dividend price is $-qe^{-rt}S$. The correct total-return process with cash dividends is

$$G_t = e^{-rt}S_t + \int_0^t e^{-ru}qS_u du, \qquad dG_t = \sigma e^{-rt}S_t dW_Q.$$

Reinvesting dividends gives e^{qt} shares and discounted wealth $e^{-(r-q)t}S_t$, whose drift is zero. On a finite horizon, the GBM second moment is finite, so the relevant stochastic integral is square-integrable and the process is a true martingale, not merely a local martingale. When $\sigma = 0$, the density formula with division by σ does not apply.

44-L06

Subtracting the equations gives $\Delta = .5$. The down-state equation gives $A = -40/1.05 = -38.0952380952$, and the initial price is $50 - 40/1.05 = 11.9047619048$. In the up state the portfolio is $60 - 40 = 20$; in the down state it is $40 - 40 = 0$. $p = (1.05 - .8)/(.4) = .625$, and the discounted expectation $.625(20)/1.05$ gives the same price. The real-world up probability .3 does not change the replication cost of these two state payoffs. For a CRR tree with reinvested dividends, the number of shares at the right endpoint is multiplied by e^{qdt}. Rearranging the initial cost gives $p = (e^{(r-q)dt} - d)/(u - d)$, with the strict admissibility condition $d < e^{(r-q)dt} < u$. An out-of-range p means the tree assumptions do not satisfy no-arbitrage; clipping it changes the original replication relation. Reject the tree or choose an explicitly valid grid rather than silently altering the probability.

44-L07

Matching the random terms gives $\Delta = V_S$. The drift-matching equation is

$$V_t + (\mu - q)SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} = \mu SV_S + rV - rSV_S.$$

After canceling the true drift, the PDE is $V_t + (r-q)SV_S + \sigma^2 S^2 V_{SS}/2 - rV = 0$, with call terminal payoff $(S-K)^+$. Cash A pays for rebalancing; $Sd\Delta$ must not be counted as an external contribution. Applying Itô's formula to discounted V under Q, the PDE removes the drift. In this chapter, positive S, constant-coefficient GBM, finite T, a call or put with linear growth, $C^{1,2}$ regularity for $\tau > 0$, bounded Delta, and the GBM second moment ensure that the stochastic integral is square-integrable, so conditional expectation is valid. Taking the terminal limit gives $V(t,S) = E_Q[e^{-r(T-t)}h(S_T) \mid \mathcal{F}_t]$; an arbitrary local martingale does not automatically preserve its mean.

44-L08

The first-order currency amount for 1 pp is $.01 \times 37.9011575100 = .3790115751$. Theta = $dV/dt = -dV/d\tau$, so the central difference in maturity must be negated. The Gamma difference is $[V(S + h_S) - 2V(S) + V(S - h_S)]/h_S^2$. At expiry, the left and right call Deltas at the kink are 0 and 1; there is no ordinary unique derivative. An implementation that rejects all Greeks at T=0 or zero volatility is defining API scope, not claiming that every derivative away from the kink fails to exist. Approximation error competes with rounding; Gamma also divides by the square of the step. In the main simulation, relative step 10^{-4} gives error about -1.2×10^{-10}, while 10^{-8} gives error about .01658. Decimal places alone do not establish accuracy.

44-L09

$X_i = (Y_i^+ + Y_i^-)/2$, price estimate $\bar{X}$, sample variance $s_X^2 = \sum(X_i - \bar{X})^2/(M-1)$, and SE = $s_X/\sqrt{M}$; the 95% normal approximation is $\bar{X} \pm 1.95996398454\, SE$.

$$\mathrm{Var}(X) = \tfrac{1}{4}[\mathrm{Var}(Y^+) + \mathrm{Var}(Y^-) + 2\,\mathrm{Cov}(Y^+, Y^-)].$$

The code above flattens correlated paths. Pair them first, then calculate the standard error from the M independent pair averages:

```
pair = (y + ya) / 2
se = pair.std(ddof=1) / np.sqrt(M)
```

Treating the flattened array as 2M independent observations omits the covariance. The two paths within a pair are not independent; independence across pairs and finite variance support the approximation. Coverage on one run is random and does not prove correctness; also check the analytic value and independent formulas and boundaries. Exact Q terminal GBM sampling removes time-discretization bias but leaves sampling error. With random M=1, SE cannot be estimated, so reject it. For a deterministic payoff, SE may be reported directly as 0.

44-L10

Old shares earn interval dividends at the left-point price, while old cash earns interest: $A^{pre} = Ae^{rdt} + \Delta S(e^{qdt} - 1)$. This is a left-point cash-dividend approximation; the continuous-model amount is $\int q\Delta S_u du$, so an exact terminal value does not make the entire ledger exact. After rebalancing, $A' = A^{pre} - \delta\Delta S_{now} - fee|\delta\Delta|S_{now}$, new shares are $\Delta' = \Delta + \delta\Delta$, and portfolio value is $A' + \Delta' S_{now}$. Initial share purchases and expiry liquidation both incur fees; terminal shares are 0, and the error is liquidation cash minus the payoff. The main simulation supports the model-internal comparison that error SD falls as frequency rises without fees, while cash costs increase and mean error becomes more negative with fees. A verifiable case in which live-trading costs from quote delays, jumps, or high-frequency fees push losses beyond a risk limit could refute "higher frequency must be better in live trading." This requires permitted data, timing, and a separation between training and evaluation; simulation cannot replace empirical evidence. The complete reference implementation and row-by-row assertions are in 44-P05 of the chapter body; checking only final cash totals does not validate the share, dividend, and financing sequence.

44-L11 (new, complete answer)

The four values of M_2 are 2, 0, 0, and -2. The conditional mean in the positive-ξ group is $(2+0)/2 = 1$; in the negative-ξ group it is $(0-2)/2 = -1$. These equal ξ. M_2 is not $\mathscr{F}_1$-measurable because its values differ within each group; the conditional expectation is constant within each group and is measurable. The centered squared increment is $\varepsilon_i = (\Delta W_i)^2 - \Delta t_i$. The future Brownian increment is independent of the left-point information, so $E[\varepsilon_i \mid \mathscr{F}_{t_i}] = 0$ and $E[\varepsilon_i^2 \mid \mathscr{F}_{t_i}] = 2(\Delta t_i)^2$. For $i < j$, condition on $\mathscr{F}_{t_j}$: $C_i\varepsilon_i C_j$ is known, and $E[\varepsilon_j \mid \mathscr{F}_{t_j}] = 0$, so the cross term vanishes.

$$\begin{aligned} E[(\sum C_i\varepsilon_i)^2] &= 2\sum E[C_i^2](\Delta t_i)^2 \\ &\le 2K^2T|\Pi| \to 0. \end{aligned}$$

The numerical bound is 2. This bounds expected squared error, not pathwise error. Bounded left-point weights may depend on the past; they need not be independent of one another. A general Itô proof still requires Riemann-sum convergence, approximation, localization, and remainder control; this argument does not establish the result for arbitrary random partitions.

44-L12 (new, complete answer)

Stock value goes from 60 to 50, cash goes from -50 to -40, and total value is 10 both before and after. The sale transfers 10 within the portfolio. The cash expansion

$$dA = rAdt + q\Delta Sdt - Sd\Delta - d[\Delta, S]$$

cancels the two rebalancing terms in the product formula, giving $dV = \Delta dS + q\Delta Sdt + rAdt$. This is the expansion for the smooth semimartingale Delta in this chapter; a general strategy is defined by the self-financing return integral. $F_t = (\beta - \sigma^2/2)F$, $F_w = \sigma F$, and $F_{ww} = \sigma^2 F$. Itô's drift is βF and its random coefficient is σF, with initial value S_0; a positive initial value times an exponential remains positive. API T = $\tau = .75$ years. The units of λ are year$^{-1/2}$, so λW and $\lambda^2 t$ are dimensionless; calendar Theta remains $-V_\tau$.

Chapter 45 —Production Quant Pipelines

Gate One: The Data Gate

> The data gate is the pipeline's first validation step. Before any feature engineering or computation begins, it checks the input schema, numerical finiteness, and units against the data contract.

In the same quote row, 100 and 10000 may both be finite positive numbers, yet their units determine whether they can represent the same price. Check USD/share before calculating momentum to separate "mathematically computable" from "permitted by the domain." Historical prices are endpoints of the lookback window; the current quote is the input to this decision. The two endpoints serve different purposes. Missing values must not be filled with zero automatically: rejecting the input preserves its failure identity, while zero filling can silently alter the strategy. The teaching entry point has no credentials and does not fetch online prices, so readers can trace each input through the normalized JSON and lineage hash.

Trigger. Triggered when the data-ingestion module receives a market, financial, or alternative-data slice.

Core Concept and Data Contract

Definition 45.1 *(Data Completeness and Finiteness Constraints).* Let the input table $\mathcal{D}$ contain the field set $\mathcal{F}$. A compliant input must satisfy:

(i) Schema completeness: $\mathcal{F}_{\text{required}} \subseteq \mathcal{F}$; neither the permanent security identifier `permanent_security_id` nor any time-series timestamp may be missing.

(ii) Numerical finiteness: for the spot-equity price field P, require $P \in \mathbb{R}^{+}$ and _____; never allow NaN or $\pm\infty$.

(iii) Explicit and consistent units: every monetary value must declare its unit (for example, _____); implicit FX assumptions are prohibited.

Recall: What error will arise downstream if spot-equity data contain a nonpositive price?

Reading Bridge: Package Identity and the Scope of Reading

A source-layout package keeps its core under `src/quant_course`; the CLI, Notebook, and tests all call the same module. PEP 517 defines the build frontend/backend interface, while PEP 518 declares build dependencies; neither requires a directory named src, which is a layout convention. A cold kernel must import the package again rather than rely on old variables. The setuptools backend and source-package discovery configured in `pyproject.toml` answer different questions. This batch did not reinstall the package.

Four rows with four fields and a hypothetical eight bytes per field take 128 bytes when read in full. Projection pushdown reads only the price column, or 32 bytes; if statistics let the engine skip the two old rows in one row group, the data portion is 16 bytes, plus metadata. This explains how columnar Parquet and the DuckDB engine can reduce reads; it is not a measurement of file size or performance. The main example still reads CSV/JSON when optional packages are missing; a future faster read of the revision table would still violate the rules.

Raw prices of 100 and 10^{-320} are both finite and positive, but a division-based momentum calculation can overflow. Features must also be finite after calculation. On rejection, retain FEATURE_VALIDATION_FAILED/STOPPED and zero fills; do not clip the result into a positive signal. Executions and marks must also cover the same asset set as observations. Stress simple returns must be finite and ≥ -1; positive scenarios are allowed, and a loss may be negative.

Data-Gate Decision Policy

If a data record violates Definition 45.1, the data gate rejects it and terminates the current pipeline. It does not silently fill with a mean or carry values forward.

Your turn 45-L01 [Maps to body: 45-P01, 45-E01] **Why this blocks downstream work:** If missing fields or NaN values pass through, later calculations may propagate invalid values; the precise scope depends on the operator and error handling. This example rejects the input against its contract first. It does not assume what an optimizer that has not been implemented would necessarily do to its weights.
Code diagnosis: Analyze the problem with this data-cleaning snippet:

```
df['price'] = df['price'].fillna(method='ffill').fillna(0.0)
```

Explain why this code violates explicit validation, then write a self-contained set of compliant assertions:

Example 45.2 *(Rejecting Nonpositive Prices and Invalid Units).* During the observation window on 2026-09-01, asset A is quoted at 100.0 USD/share and asset B at 50.0 USD/share. If a transmission error sends asset B's price as −50.0 or mislabels its unit as GBX/share, the data gate raises an exception and rejects it.

Your turn 45-L02 [Maps to body: 45-P01] **Research judgment (domain-contract boundary):** An analyst argues, "Whenever a negative price is detected, just take its absolute value $|P|$ and continue." Is that reasonable for spot equities and physically delivered commodity futures? Why?

Gate Two: The Timing Gate

> The timing gate uses a strict temporal partial order to physically separate past decisions from future outcomes and prevent look-ahead bias.

Observation time is not signal time. In this example, a constructed quote occurs and is ingested at 16:00 on September 1; the signal forms at 16:05. Two simulated fills occur at 09:00 and 09:01 on September 2. The simulator saves the next day's price only to produce a retrospective ledger;

the feature layer still reads observations only. Freshness also depends on event age: a stale quote does not become fresh just because it was written to an as-of snapshot moments ago. Returns start at each fill time, so the earlier position cannot claim price movement from before it was purchased. A label can enter drift monitoring only after its interval ends and it is mature.

One UTC Timeline: Inputs, Holding Period, and Label

09-01 16:00	Quote event; available, ingested, and as_of are all no later than 16:05.
09-01 16:05	The signal reads only the version available then; no position is held yet.
09-02 09:00/09:01	Trades start the holding intervals for A and B, respectively.
09-02 16:00	The mark closes the ledger; the label interval ends and meets the maturity threshold.
09-02 17:00	Only if it becomes available and is ingested now may it update monitoring; at 16:30 it is still unavailable.

This connects L03–L05 and L17: temporal ordering still requires a historical-version check. Maturity and actual availability are two separate conditions; a price rise before the fill is not a holding-period return.

Trigger. Triggered before feature extraction and signal calculation.

Point-in-Time Causal Partial Order

Definition 45.3 *(Point-in-Time Partial-Order Condition).* Suppose each input record has timestamp tuple $(t_{\text{event}}, t_{\text{available}}, t_{\text{ingested}}, t_{\text{as_of}}, t_{\text{signal}}, t_{\text{trade}})$. A necessary temporal condition for the timing gate to pass is:

$$\max(t_{\text{available}}, t_{\text{ingested}}, t_{\text{as_of}}) \leq _____ < t_{\text{trade}}$$

where $t_{\text{as_of}}$ is the business snapshot's cutoff time.

Further thought: Is satisfying the timing inequality sufficient to prove that data have no future revisions? How should backdated corrections be guarded against?

Your turn 45-L03 [Maps to body: 45-E01] **Why this blocks downstream work:** If $t_{\text{available}} > t_{\text{signal}}$, the strategy uses data in the backtest that had not yet been released. This produces performance that cannot be reproduced in live trading.
Fill in the mathematical expressions for the temporal partial-order chain:

1. For the quote event that has already occurred in this example, event time t_{event} and external availability time $t_{\text{available}}$ (do not generalize this to announcements of future-effective events): _____ .
2. Signal-generation time t_{signal} and order-matching time t_{trade} must satisfy: _____ .

Algebraic Decomposition of Holding-Period Return

For asset i, the actual post-fill holding interval is the left-open, right-closed interval $\mathcal{I}_{\text{holding}} = (t_{\text{trade},i}, t_{\text{mark}}]$. Let the lag-period growth rate be $g = P_{\text{trade}}/P_{\text{signal}} - 1$, and the holding-period return be $r = P_{\text{mark}}/P_{\text{trade}} - 1$. The difference between the illusory return measured from t_{signal} and the actual return is:

$$R_{\text{illusory}} - r = _____$$

Your turn 45-L04 [Maps to body: 45-E01, 45-P01] **Code diagnosis (conditional look-ahead leakage):** Analyze this join:

```
pd.merge_asof(market_df,fund_df,on='date')
```

If `fund_df`'s `date` is the quarter-end date (2026-03-31), but the actual disclosure date is 2026-04-28, explain why this join conditionally leaks future information when matching a trade on 2026-04-01:

Your turn 45-L05 [Maps to body: 45-E01] **Research judgment:** Asset A fills at 09:00Z the next day; asset B fills at 09:01Z; both are settled at 16:00Z. Their holding intervals differ by one minute. When calculating intraday volatility and risk exposure, can you simply assume their holding times are identical? Why?

Gate Three: The Execution Gate and Idempotent State Machine

> In environments where network timeouts and scheduler retries are possible, the execution gate uses a persistent ledger to guarantee idempotency: on a rerun it submits only the unfilled remainder and never resends excess orders.

The fixed targets are ten shares of A and twenty of B. Under the full-fill conditions in this example, once A's ten shares have filled, a rerun produces a zero remainder; if B has not filled, its remainder is twenty shares. Deduplication identity binds the run key, input configuration, and code fingerprint; the system does not infer success from "the directory seems to exist." The risk denominator for a new target must also account for purchase cash and commissions: including the new market value in NAV while forgetting the cash outflow inflates capital and hides leverage. A stop can prevent later simulated fills, but cannot erase A's fill that has already happened. The exception event and ledger must remain.

Trigger. Triggered when trade instructions are generated and are about to be submitted for simulated or live matching.

Incremental Order Operator and Persistent State

Definition 45.4 *(Incremental Order-Generation Operator).* Let the current target holdings be $\mathbf{Q}^* \in \mathbb{R}^M_{\geq 0}$, and let the cumulative fills recorded in the persistent state database be $\mathbf{Q}_{\text{filled}} \in \mathbb{R}^M_{\geq 0}$. The actual order increment is:

$$\Delta\mathbf{Q} = _____$$

Recall: If a scheduled job is retried twice in succession after a timeout, what will a system without state-based deduplication do?

Your turn 45-L06 [Maps to body: 45-E03, 45-P02] **Why this blocks downstream work:** Without deduplication based on persistent state, a network timeout or duplicate scheduled trigger can execute the same batch of buy instructions multiple times. Holdings can multiply rapidly and breach the margin-control limit.
State-machine calculation: The target share vector is $\mathbf{Q}^* = (10, 20)$. After asset A (10 shares) fills, the system experiences a controlled interruption; SQLite records $\mathbf{Q}_{\text{filled}} = (10, 0)$. After restart:

1. Asset A's incremental order is $\Delta Q_A = \max(0, 10 - 10) =$ _____.
2. Asset B's incremental order is $\Delta Q_B = \max(0, 20 - 0) =$ _____.

Reading Bridge: Two SQL Queries, One Logical Task

The actual `fills` table uses the text key `fill_id`, with columns `asset`, `signed_quantity`, `fill_price`, and `commission`. A costs $10 \times 100.1 + 1.001 = 1002.001$ USD in cash; B costs $20 \times 50.05 + 1.001 = 1002.001$ USD. An SQL sum grouped by run key is 2004.002, leaving 7995.998 USD in cash. Both `COUNT(DISTINCTfill_id)` and the row count are 2; the number of assets is not generally the number of fills. A unique key cannot replace validation of the recovery payload or ordering.

The same logical run started at 09:00:03 and retried at 09:00:05 still has one key. If a real broker API times out, the result can only be classified as UNKNOWN: query the original order identity or remain stopped; do not assume no fill and resend. This batch made no API calls; the failure is a controlled simulation triggered on the first or second cumulative actual fill. A failure-name is unchanged if B fills first. If B is the only target, recovery after its first fill can proceed without adding an order; a failure before a second fill has no applicable target and preflight rejects it.

Persistent Transaction Controls

This single-process core commits SQLite fills and then writes an audit event; it cannot claim that both are committed in one transaction. `BEGIN IMMEDIATE` is an optional write-transaction mode, not a multiprocess lock implemented in this batch. If the same `run_key` arrives with a mismatching input fingerprint, the execution gate rejects it directly.

Your turn 45-L07 [Maps to body: 45-E03, 45-P02] **Code diagnosis:** Analyze the logic defect in this execution loop:

```
forsym,targetintargets.items():execute_order(sym,target)
```

Explain why this code cannot prevent duplicate fills on a rerun, then write self-contained code that queries state and calculates the remainder:

Your turn 45-L08 [Maps to body: 45-E05] **Research judgment (emergency kill switch):** If, within the first ten minutes after the market opens on a given day, asset A's simulated fill price deviates by more than 15% from its previous close, and the portfolio's instantaneous cumulative loss reaches 80% of its preset limit, what should the execution gate do? To what state should it transition?

Gate Four: Reconciliation and Financial Conservation

> The reconciliation gate uses double-entry accounting principles to close the loop on cash outflows, embedded costs, explicit commissions, and end-of-period NAV generated by discrete trades.

A's fill price of 100.1 already embeds a total one-dollar spread cost over ten shares, for notional value 1001; then a 1.001 commission is charged, so cash falls by 1002.001. B has twice the quantity and half the price, giving the same outflow. At the end, A's rise contributes twenty dollars and B's decline contributes negative twenty dollars, so gross PnL is zero; net PnL is still negative 4.002. Cash of 7995.998 plus 2000 in holdings equals NAV of 9995.998. List these figures independently before reading the program summary; this makes it possible to spot double-counting even when the same calculation function is internally consistent.

Units Ladder: One Ledger for L09--L12

10 share $\times$ 100.100 USD/share = 1001.000 USD. Only notional plus commission is deducted from cash; the spread is already embedded in the fill price. Gross and net PnL, cash, and NAV are all in USD. Net return is $-4.002/10000 = -0.0004002 = -0.04002\%$, a dimensionless ratio. Check the 10^{-6} USD amount tolerance and the 10^{-6} share quantity tolerance separately; they cannot be interchanged.

Trigger. Triggered after trade matching and before end-of-period valuation and report generation.

Reading Bridge: Reconciliation Events Are Not Platform Logs

The SQLite event fields in this example are event_time, run_key, event_name, and details_json; the event timestamp is separate from the quote timestamp. A failed cash, quantity, or NAV check is saved as STOPPED; a terminal discrepancy is saved as TERMINAL_RECONCILIATION_FAILED. No report is published, the original two fills remain, and a retry with the same key is rejected. The amount tolerance is 10^{-6} USD; the share-conservation tolerance is 10^{-6} share. Their units cannot be exchanged.

A counter accumulates attempts or unique fills; a retry may increase the attempt count while the unique-fill count stays unchanged. A gauge is a mutable current value such as cash or position. A histogram bins latencies of 0.02, 0.07, and 0.20 seconds into 0.05, 0.10, and infinity buckets; cumulative counts are 1, 2, and 3, and the sum is 0.29 seconds. Bucket counts are not an exact P99. This batch did not deploy Prometheus or a logging service. The CLI emits JSON, and audit facts are stored in SQLite.

Cost Breakdown and Cash-Conservation Identity

Under assumptions A1–A5 (zero initial position, fully filled buys, one currency, zero interest on borrowing, and zero market impact): the buy fill price is $P_{\text{fill}} = P_{\text{mid}}(1 + \phi)$, and explicit commission is C. Cash is reduced by $N + C$ (notional plus commission), while gross return is calculated at the mid price:

$$\text{NAV}_t - \text{Cash}_0 \equiv _____ - _____$$

Your turn 45-L09 [Maps to body: 45-E02, 45-P03] **Why this blocks downstream work:** If the reconciliation gate detects an excessive cash residual, the trading ledger and valuation model have a major accounting inconsistency (such as an omitted commission, reversed sign, or phantom holding). Any return report published at that point would contain false financial data.

Hand calculation: Given $\text{Cash}_0 = 10000.000$ USD:

- Buy 10 shares of asset A: mid price 100, half-spread 10 bps, commission 10 bps. Fill price = _____, cash outflow = _____, remaining cash = _____.
- Buy 20 shares of asset B: mid price 50, half-spread 10 bps, commission 10 bps. Fill price = _____, cash outflow = _____, ending cash = _____.

Your turn 45-L10 [Maps to body: 45-E02, 45-P03] **Research judgment (double-counting the spread):** Suppose a program deducts cash as follows: `cash-=(shares *fill_price+spread_cost+commission)`. Derive the resulting end-of-period NAV bias. What amount is the bias exactly equal to? State whether the erroneous NAV is too high or too low, and write both the signed difference NetPnL $-$ ($\text{NAV}_{\text{err}} - \text{Cash}_0$) and its absolute residual. If the spread is mistakenly deducted on both trades, what are both amounts in USD?

Your turn 45-L11 [Maps to body: 45-E02, 45-P03] **Gross and net return decomposition (constructed hand calculation):** The end-of-period mark prices are $P_A = 102, P_B = 49$.

1. Asset A gross return: $10 \times (102 - 100) =$ _____ USD.
2. Asset B gross return: $20 \times (49 - 50) =$ _____ USD.
3. Portfolio gross PnL = _____ USD.
4. Total friction cost = _____ USD.
5. Portfolio net PnL = _____ USD.
6. Ending total NAV = _____ USD.

Your turn 45-L12 [Maps to body: 45-P03] **Financial tolerances for floating-point arithmetic:** Why does this example use unit-specific tolerances for binary floating-point reconciliation, $|\text{Residual}| \leq 10^{-6}$ USD (amount tolerance) and $|\Delta Q| \leq 10^{-6}$ share (quantity tolerance), instead of relying on Python's exact equality comparison `==`? Why can a hand calculation using a Decimal constructed from a decimal string use an exact assertion? What operator tests identity?

Gate Five: Recovery and Crash Repair

> A single-machine operating system does not provide a two-phase commit spanning SQLite and the filesystem. The recovery gate uses a separate staging directory and atomic rename to avoid producing a partially written artifact after an interruption, while preserving failed evidence for diagnosis.

Fill status is in SQLite; output files are in a directory, and the two do not share a transaction.

If an interruption occurs after the first fill, recovery first checks the stored fill identity, time, quantity, and fees, then rebuilds cash and positions. If it occurs after the directory is published but before COMPLETED is recorded, the recovery gate distinguishes three publication states: for an unpublished run, start a new attempt in staging; for a valid published run, record the COMPLETED terminal state without filling again; for a damaged publication, reject it strictly and preserve the damaged evidence rather than silently overwriting it on retry. These checks cover controlled Python exceptions and local-directory publication. They do not claim guarantees for actual power loss, broker retries, or multi-host concurrency.

Trigger. Triggered when computation artifacts are written to disk, a pipeline terminal state is recorded, or the process restarts after a crash.

Atomic Rename and Self-Healing State Machine

All computation artifacts are first written to a separate staging directory with a run sequence number, `staging_<run_key>_<attempt>`. After validation, the low-level system call `os.rename(staging, final)` performs the atomic replacement. The published target is handled through three distinct paths: unpublished, valid publication with a missing terminal registration, or damaged publication that is explicitly rejected. Failed staging directories and damaged artifacts must be preserved for traceability; blind deletion or overwrite on rerun is prohibited.

> **Your turn** 45-L13 [Maps to body: 45-E04, 45-P04] **Why this blocks downstream work:** If a downstream system consumes an output file before it is fully written, automated archiving and portfolio scheduling can fail with a JSON parsing error and interrupt the daily batch.
> **Code diagnosis:** Explain why atomic publication across mounted filesystems raises `EXDEV` on Linux:
>
> ```
> os.replace('/tmp/staging_run1','/data/output/run1')
> ```

Check Persistent Evidence First, Then Choose a Recovery Action (L06, L14, L18)

State	Evidence	Action
Unpublished after first fill	Same input identity; fills=(10,0)	Add twenty shares of B; do not rebuy A
Unpublished after later fills	Fills recorded; staging not published	Reconcile the remainder and artifacts, then publish
Published and valid	Artifact hash matches the database expectation	Record terminal state only; do not fill again
Published but damaged	Partial artifact, hash mismatch, and database record	Reject and preserve the evidence

This is local evidence under controlled single-machine failures, not proof of a cross-resource transaction or power-loss behavior. If a real network fill is unknown, query the original order or stop; "not recorded locally" does not mean "not filled externally." This table does not promise exactly-once execution at a broker.

Your turn 45-L14 [Maps to body: 45-E04, 45-P04] **Failure-simulation walkthrough (three publication branches):** After asset A (10 shares) is recorded, the system experiences a controlled interruption. After restart, asset B (20 shares) fills. Reconciliation passes, but another interruption occurs after `os.rename` and before the database commits `COMPLETED`. On the next restart, how should the recovery gate handle each case: unpublished, published and valid, or published but damaged?

Your turn 45-L15 [Maps to body: 45-P04] **Research judgment:** An engineer proposes, "If a crash occurs, just clear the output directory and rerun everything; it is simpler." Why is this unacceptable in a production system with actual fills and external state?

Architecture Extension: Secret Isolation, Monitoring, and Distributed-System Boundaries

> A production-grade pipeline must manage credentials, system metrics, and model-lifecycle degradation monitoring carefully, while recognizing the physical consistency limits of distributed systems.

A model alert is the start of an investigation. Changes in input KS and PSI describe differences in marginal feature distributions only; changes in mature residuals also require distinguishing feature errors, data delays, and a change in the economic relationship. Page–Hinkley (PH)-style cumulative monitoring distinguishes a fixed baseline from a running-mean variant, and residual monitoring can update online only after labels mature. Before rollback, check compatibility between model and transformation versions, the input fields expected by the old model, and existing positions and cash; stop new orders first if needed. Secrets also have a lifecycle: temporary mounts, least privilege, redaction, and rotation go further than "switching to environment variables." Docker, MLflow, monitoring, and scheduling are discussed here only as boundaries. The research technical protocol has been accepted, but permission and authorization to execute against the target are still absent; this constructed loss example cannot be used to judge real alpha.

Trigger. Triggered when the pipeline is connected to a monitoring system, deployed in a production container, or managed across model versions.

Reading Bridge: Container Entrypoint and Version Rollback

The Docker reference file starts with a Python image in FROM, sets /app in WORKDIR, installs hypothetical image dependencies in RUN, copies the specified src/capstone/code via COPY, and uses ENTRYPOINT to run `capstone/run_offline.py`. The build context is the course root; `capstone/.dockerignore` is not the root ignore file for that context. The image was not built or run, and an unlocked tag and dependencies cannot prove byte-for-byte reproducibil-

ity. Build ARG/ENV must not carry secrets; a runtime secret mount still needs least privilege and redaction. This batch contains no credentials.

Experiment tracking associates input/config/code hashes, state, and output; both successful net PnL of −4.002 and failure state STOPPED are retained. Before rollback, check the feature schema, transformation version, cash, and positions; do not delete old fills. MLflow, a real paper-trading interface, and automatic retraining were not run.

Credential Isolation, Feature Drift, and Distributed Consistency

Reading Bridge: Hand-Calculate Drift First, Then Check the Maturity Clock

The Kolmogorov–Smirnov (KS) D is the largest difference between two empirical CDFs. For a baseline of 1, 2, 3, 4 and new observations of 2, 3, 4, 5, at x=1 the CDF values are 1/4 and 0, and at x=4 they are 1 and 3/4, so D=0.25, a dimensionless value. This calculates D only. With four rows, ties, and possible financial dependence, it does not justify the classic p-value for independent continuous samples.

The Population Stability Index (PSI) freezes bins first. Baseline counts are 3, 4, 3; new counts are 2, 5, 3. The proportions are B=(0.3,0.4,0.3) and A=(0.2,0.5,0.3); the three terms in $\sum(A-B)\ln(A/B)$ are 0.0405465, 0.0223144, and 0, totaling 0.0628609, a dimensionless value. Handling of empty bins must be specified in advance, not filled ad hoc; 0.1 and 0.25 are not universal alert theorems.

The fixed-baseline Page–Hinkley (PH)-style rule is $S_0 = M_0 = 0$, $S_t = S_{t-1} + e_t - \mu_0 - \delta$, and $M_t = \min(M_{t-1}, S_t)$; alert when $S_t - M_t > h$. Errors $e = |y - \hat{y}|$ and μ_0, δ, h are returns in decimal form. With values 0.10, 0.01, and 0.05, errors of 0.10/0.15/0.18 give S values of −0.01, 0.03, 0.10 and S-M values of 0, 0.04, 0.11, so an alert occurs only on the third mature label. A cumulative/running mean first updates $\bar{e}_t = \bar{e}_{t-1} + (e_t - \bar{e}_{t-1})/t$, then uses current $\bar{e}_t$ to update that step's increment, $V_t = V_{t-1} + e_t - \bar{e}_t - \delta$ ($\bar{e}_0 = V_0 = 0$); earlier increments are not backfilled. This differs from a frozen baseline and from a fixed-window rolling mean. River also has forgetting and two-sided settings; it was not run in this batch.

The prediction is made at 16:05 on September 1. Its label matures only after the interval ends at 16:00 on September 2 and the label is available; if it is not ingested until 17:00, it cannot update the residual at 16:30. After maturity, monitoring can update online or replay offline, but it cannot use a future label early. An alert does not prove alpha decay; check data and transformations first, then decide whether to stop orders or roll back to a compatible version.

> **Your turn** 45-L16 [Maps to body: 45-E05] **Secret-credential management:** Under The Twelve-Factor App, how should live-trading API keys and database passwords be injected into the system? If a configuration dictionary must be written to a log, what defensive measures should be taken?

Your turn 45-L17 [Maps to body: 45-E06, 45-C02] **Marginal feature drift, covariate shift, and concept drift:** Suppose the marginal distribution of a core factor changes sharply, the Kolmogorov–Smirnov test has a p-value below 0.001, and the Population Stability Index (PSI) is > 0.25. Why does this not directly imply alpha decay? Under strict covariate shift, which conditional distribution must remain invariant? In a Page–Hinkley (PH) test, how does each step's increment update under the fixed baseline and cumulative/running-mean variants, and why can the final mean not be used to rewrite past increments? Let $e_1 = 0.20$, $\mu_0 = 0.10$, $\delta = 0.01$, and $h = 0.05$, with a zero initial value: $PH_1 =$ _____; does it raise an alert? For the sequence $(0.10, 0.10, 0.10, 0.30)$, what are the cumulative mean and the mean over the most recent two periods? The two-period window is included only to distinguish terminology; implementing a third PH variant is not required. When are labels both mature and actually available for an online update?

Your turn 45-L18 [Maps to body: 45-C01] **Distributed double-spend protection:** In a microservice architecture spanning multiple servers, why can SAGA or 2PC still not guarantee exactly-once external fills mathematically if the broker gateway does not support idempotent order IDs?

Loom Exercise Solutions and Derivations (Solutions to 45-L01--L18)

> This section contains complete derivations, diagnostic analyses, and self-contained reference answers for all active-recall exercises 45-L01 through 45-L18 in the Chapter 45 Loom notes. The answers remain traceable to the body exercises (45-E / 45-P / 45-C) through the stated question mappings.

45-L01 [Maps to: 45-P01, 45-E01]: Data Integrity and the Risk of Silent Filling

Problem analysis: The snippet below has serious engineering defects:

```
df['price'] = df['price'].fillna(method='ffill').fillna(0.0)
```

1. Forward fill (ffill) can carry a stale price into the current decision window during a long trading halt or data gap, creating a falsely stable series.
2. Filling the remaining values with 0.0 introduces a nonpositive price. A later log-return calculation $\ln(P_t/P_{t-1})$ will encounter division by zero and a mathematical-domain error.
3. It violates the production-pipeline principle of failing explicitly on dirty data and hides systemic failures in upstream data sources.

Fill in the definition: prices must be finite and strictly positive; the price unit is USD/share and the quantity unit is share. The snippet below checks numeric values only. See the core of body 45-P01 for validation of the full field set, units, timestamps, and freshness. **Numerical assertion logic:**

```
import math

def validate_prices(prices):
    for p in prices:
        assert isinstance(p, (int, float))
        assert not isinstance(p, bool)
        assert math.isfinite(p) and p > 0
validate_prices([100.0, 50.0])
for value in [None, True, float("nan"), float("inf"), 0, -1]:
    try:
        validate_prices([value])
    except AssertionError:
        pass
    else:
        raise AssertionError("invalid price accepted")
```

45-L02 [Maps to: 45-P01]: Negative Prices and Asset-Class Boundaries

Research judgment: The proposed rule is unreasonable:

1. Spot equities have limited-liability characteristics, and their theoretical and market price lower bound is 0. A negative price indicates an upstream parsing or transmission error. Taking its absolute value disguises bad data as valid.

2. For physically delivered commodity futures with storage and delivery costs, a negative price can be a real economic outcome. Taking its absolute value would turn it positive and completely reverse the position's value direction. The system should parameterize data contracts by asset class and distinguish the physical limits of financial instruments.

45-L03 [Maps to: 45-E01]: Point-in-Time Partial Order and Necessary Timing Conditions

Fill-in answers:

1. Partial-order chain: $\max(t_{\text{available}}, t_{\text{ingested}}, t_{\text{as_of}}) \leq t_{\text{signal}} < t_{\text{trade}}$.
2. Event time and availability time: $t_{\text{event}} \leq t_{\text{available}}$ (this refers to a quote event that has already occurred; an announcement of a future-effective event may be published in advance, so this total ordering is not imposed on it).
3. Signal time and trade time: $t_{\text{signal}} < t_{\text{trade}}$ (there must be a strict time window for algorithmic decision computation and order submission).

The blank in the timing definition is t_{signal}. The holding-period decomposition is $g(1+r)$, since $(1+g)(1+r)-1-r = g+gr$. Both g and r are simple returns in decimal form; a rise outside the traded holding period cannot be counted in strategy PnL.

45-L04 [Maps to: 45-E01, 45-P01]: A Conditional Look-Ahead Leakage Counterexample

Code diagnosis: Joining directly on the fiscal-period end date is conditionally look-ahead leakage. For example, a company's first-quarter reporting period ends on 2026-03-31, but the report is not publicly disclosed until 2026-04-28. If the strategy matches on period-end when trading on 2026-04-01, it uses information that has not been released 27 days early. Fix the join by using the actual public timestamp, `available_at`.

45-L05 [Maps to: 45-E01]: Unequal Discrete Holding Periods

Research judgment: You cannot simply assume the holding times are identical. Asset A fills at 09:00Z and settles at 16:00Z, so its economic holding period is 7 hours. Asset B fills at 09:01Z and settles at 16:00Z, so its holding period is 6 hours and 59 minutes. In high-frequency trading or an extremely volatile market, that minute may contain a key macroeconomic release. Rigorous risk measurement must calculate each exposure duration and realized variance using the actual fill time $(t_{\text{trade},i}, t_{\text{mark}}]$.

The configured share targets are finite real numbers and may be zero when there is no positive signal. The values 10 and 20 in this example are integer choices; no broker lot size is checked. Recovery of the already recorded first fill does not retrigger after_first_fill. A B-first sequence and a B-only target use the same cumulative fill count. An independent terminal reconciliation failure also transitions to STOPPED; the fills remain and no output is published.

45-L06 [Maps to: 45-E03, 45-P02]: Incremental Operator State Calculation

Derivation: The incremental operator is $\Delta\mathbf{Q} = \max(\mathbf{0}, \mathbf{Q}^* - \mathbf{Q}_{\text{filled}})$.

1. Asset A: target 10 shares, already filled 10 shares, $\Delta Q_A = \max(0, 10-10) = 0$ shares. Skip the order.

2. Asset B: target 20 shares, filled 0 shares, $\Delta Q_B = \max(0, 20 - 0) = 20$ shares. Submit a buy order for 20 shares.

After rerun, asset A's increment is 0 and asset B is topped up by 20 shares, so cumulative holdings converge exactly to the target vector $(10, 20)$.

45-L07 [Maps to: 45-E03, 45-P02]: Duplicate-Order Defect in a Stateless Execution Loop

Code diagnosis: The loop submits each target directly without comparing it with the persistent fill ledger. On a network timeout and retry, or after a process interruption and restart, it will submit the full quantity again for assets that already filled, doubling the holdings. **Compliant state query and remainder calculation:**

```
def compute_delta_orders(targets, filled_map):
    deltas = {}
    for sym, target in targets.items():
        filled = filled_map.get(sym, 0)
        delta = max(0, target - filled)
        if delta > 0:
            deltas[sym] = delta
    return deltas
assert compute_delta_orders({'A': 10, 'B': 20}, {'A': 10}) == {'B': 20}
```

45-L08 [Maps to: 45-E05]: Emergency Kill-Switch Action

Response process: The following is a conceptual response for an external deployment. This batch has no trading gateway, in-flight orders, or message sending; its actual behavior is to write STOPPED and an event, and reject new simulated fills. 1. Immediately send a cancellation request to the trading gateway for all open, unfilled orders; 2. Stop the strategy loop and reject any subsequent order generation; 3. Persist the run state in SQLite as `STOPPED` and write an alert event to the log; 4. Notify a human to investigate and prohibit automatic restart until the issue is cleared.

45-L09 [Maps to: 45-E02, 45-P03]: Core Numerical Hand Check

Exact hand calculation (constructed example): Initial cash is $\text{Cash}_0 = 10000.000$ USD.

1. Asset A (10 shares):
   - Fill price: $P_{\text{fill},A} = 100.0 \times (1 + 0.0010) = 100.100$ USD/share.
   - Notional amount: $N_A = 10 \times 100.100 = 1001.000$ USD.
   - Explicit commission: $C_A = 1001.000 \times 0.0010 = 1.001$ USD.
   - Cash deduction: $\Delta\text{Cash}_A = -(1001.000 + 1.001) = -1002.001$ USD.
   - Cash balance after purchase: $= 10000.000 - 1002.001 = 8997.999$ USD.
2. Asset B (20 shares):
   - Fill price: $P_{\text{fill},B} = 50.0 \times (1 + 0.0010) = 50.050$ USD/share.
   - Notional amount: $N_B = 20 \times 50.050 = 1001.000$ USD.
   - Explicit commission: $C_B = 1001.000 \times 0.0010 = 1.001$ USD.
   - Cash deduction: $\Delta\text{Cash}_B = -(1001.000 + 1.001) = -1002.001$ USD.
   - Ending cash balance: $= 8997.999 - 1002.001 = 7995.998$ USD.

45-L10 [Maps to: 45-E02, 45-P03]: Deriving the Effect of Double-Counting the Spread

Derivation: The notional amount at the fill price already includes the difference between the buy fill and the mid price:

$$N = \Delta Q \cdot P_{\text{fill}} = \Delta Q \cdot P_{\text{mid}} + \Delta Q \cdot (P_{\text{fill}} - P_{\text{mid}}) = \Delta Q \cdot P_{\text{mid}} + S$$

If S is deducted again, cash becomes $\text{Cash}_{\text{err}} = \text{Cash}_{\text{correct}} - S$. End-of-period total net asset value is $\text{NAV}_{\text{err}} = \text{Cash}_{\text{err}} + V = \text{NAV}_{\text{correct}} - S$. Thus the calculated change in net asset value is $\text{NAV}_{\text{err}} - \text{Cash}_0 = \text{NetPnL} - S$, so NAV is understated by S USD. The signed reconciliation difference is

$$D_{\text{signed}} = \text{NetPnL} - (\text{NAV}_{\text{err}} - \text{Cash}_0) = +S,$$

and the reported residual is $|D_{\text{signed}}| = S$, which does not carry direction. Cash and NAV are both too low. If both trades are charged twice, $S = 1 + 1 = 2.000$ USD, erroneous NAV is 9993.998 USD, the signed difference is +2.000 USD, and the absolute residual is 2.000 USD. This matches body E02; the negative sign of the cash outflow is not confused with the sign of the reconciliation difference.

45-L11 [Maps to: 45-E02, 45-P03]: Gross and Net Return and the NAV Closure

Calculation results:

1. Asset A gross return: $10 \times (102.0 - 100.0) = +20.000$ USD.
2. Asset B gross return: $20 \times (49.0 - 50.0) = -20.000$ USD.
3. Portfolio gross PnL = $+20.000 - 20.000 = 0.000$ USD.
4. Total friction cost: spread costs $(1.000 + 1.000)$+ commission costs $(1.001 + 1.001) = 4.002$ USD.
5. Portfolio net PnL = $0.000 - 4.002 = -4.002$ USD.
6. Ending market value of positions: $V = 10 \times 102.0 + 20 \times 49.0 = 2000.000$ USD.
7. Ending total NAV = $7995.998 + 2000.000 = 9995.998$ USD.
8. Reconciliation identity: $\text{NAV} - \text{Cash}_0 = 9995.998 - 10000.000 = -4.002$ USD. The identity holds.

45-L12 [Maps to: 45-P03]: Floating-Point Reconciliation Tolerances

Explanation: Computers store double-precision floating-point values in binary under the IEEE 754 standard. Decimal fractions can be truncated slightly when converted; for example, in Python, $0.1 + 0.2 - 0.3 \approx 5.55 \times 10^{-17} \neq 0$. A direct `==` check can therefore treat a correct ledger as unreconciled because of a rounding tail. Use small financial tolerances instead, as defined for this core: amount tolerance $|\text{Residual}| \leq 10^{-6}$ USD and quantity tolerance $|\Delta Q| \leq 10^{-6}$ share, with each tested in its own unit.

`==` is an equality comparison; `is` is an identity comparison. A Decimal constructed from a decimal string can be checked exactly at the precision used in this hand calculation. The amount tolerance in this example is not a universal guarantee for every account scale.

45-L13 [Maps to: 45-E04, 45-P04]: Atomic Rename Failure Across Mount Points

Explanation: The atomicity of POSIX `rename`/`os.replace` relies on changing a directory-entry pointer within one filesystem. If the source path (for example, the `/tmp` mount) and target path (for example, `/data`) are on different filesystems, their inodes cannot be shared across devices, so the system call raises `EXDEV`; an atomic replacement cannot be performed.

45-L14 [Maps to: 45-E04, 45-P04]: Three Publication Branches and Crash-Recovery Judgment

Three recovery paths: When scanning the target directory and database state, the recovery gate distinguishes these three execution paths:

1. **Unpublished branch (target output directory does not exist):** No valid publication has completed; this does not imply that only the first fill exists. First validate the same-identity persistent SQLite fills: if only A10 exists, fill B20; if A10 and B20 are both stored, add zero fills. Reconstruct or validate artifacts from those stored fills, open a new staging directory `staging_<run_key>_attempt2`, complete matching and reconciliation as needed, then call `os.rename` and record `COMPLETED`.

2. **Valid-publication branch (target output directory exists and its self-contained artifacts and hashes agree):** The artifacts were published atomically, but terminal-state registration was interrupted. Recovery checks the SQLite `run_key`, input-configuration fingerprint, and independently persisted artifact hash. Once they match, it records `COMPLETED` directly without any further trade matching, preserving idempotency.

3. **Damaged-publication branch (target output directory exists but artifacts are incomplete or the fingerprint has been altered):** Recovery must raise an explicit rejection and preserve the damaged artifact for human review. It must not silently overwrite the artifact or create a new attempt and match trades again, which could conceal a systemic failure.

45-L15 [Maps to: 45-P04]: Risks of a Blind Full Rerun

Reasoning: A production system is connected to external state and funded accounts. A full rerun erases the system's memory of what happened in the outside world. If the previous run already filled 10 shares in the market, a full rerun assumes nothing filled and submits another buy order for 10 shares, doubling the position. A production system must treat persistent state as its source of truth.

45-L16 [Maps to: 45-E05]: Configuration and Secret Boundaries

Reference judgment: Deployment configuration should be separated from code, but environment variables alone do not ensure secret safety. In a real system, a controlled credential service or runtime secret mount can provide least-privilege credentials. Logs, SQLite, configuration snapshots, and prompts must not store secret values. Exception messages and nested objects also need controls; masking sensitive keys alone does not prove that nothing leaks. Docker build ARG/ENV values can appear in the image or build records and must not carry build secrets. Temporary build-secret mounts and runtime credentials serve different stages. This example configures or reads no credentials.

45-L17 [Maps to: 45-E06, 45-C02]: Statistical Drift Tests and Lifecycle Monitoring

Method distinctions and response protocol:

1. **Full names and properties of the tests:**
   - **Kolmogorov–Smirnov Test (KS test):** Tests differences in a univariate continuous distribution using the supremum of the difference between empirical cumulative distribution functions, $D = \sup_x |F_{\text{curr}}(x) - F_{\text{ref}}(x)|$.
   - **Population Stability Index (PSI):** Measures overall feature deviation through binned relative entropy, $\text{PSI} = \sum(p_i - q_i)\ln(p_i/q_i)$.
   - **Page–Hinkley Test (PH test):** The one-sided fixed-baseline rule in this exercise freezes μ_0: $U_0 = 0$, $U_t = U_{t-1} + e_t - \mu_0 - \delta$, and $PH_t = U_t - \min_{0\le i\le t} U_i$. Set $U_t = S_t$ and $M_t = \min_{0\le i\le t} U_i$ to obtain the S/M recurrence in the body. Errors and μ_0, δ, h are returns in decimal form. First-point counterexample: $0.20 - 0.10 - 0.01 = 0.09$; $\min(U_0, U_1) = 0$, so $PH_1 = 0.09 > 0.05$ and an alert is raised. If the old-style minimum omits U_0, it gives $0.09 - 0.09 = 0$ and misses the alert. A cumulative/running mean (Running/Cumulative Mean) uses $\bar{e}_t = \bar{e}_{t-1} + (e_t - \bar{e}_{t-1})/t$, then updates $V_t = V_{t-1} + e_t - \bar{e}_t - \delta$ ($\bar{e}_0 = V_0 = 0$). The current mean enters only the current increment; each historical increment keeps the mean used at that time. Do not substitute the final $\bar{e}_t$ into every historical term. This differs from a fixed-window rolling mean. For $(0.10, 0.10, 0.10, 0.30)$, the cumulative mean is $0.60/4 = 0.15$, while the most-recent-two-period mean is $0.40/2 = 0.20$. The window is a terminology counterexample only; this exercise does not ask for another algorithm or make an unconditional claim about relative sensitivity.
2. **Covariate shift versus alpha decay:** KS and PSI describe changes in the marginal distribution $P(X)$ of input features. Strict covariate shift additionally requires $P(Y \mid X)$ to remain invariant. A marginal change alone neither implies nor rules out a change in the conditional relationship, and it does not directly establish alpha decay. Financial time series are strongly autocorrelated and violate the i.i.d. assumption of classical statistical procedures; empirical thresholds such as $\text{PSI} > 0.25$ are engineering warning references, not statistical or causal guarantees.
3. **Online updates with mature labels:** In time-series prediction, the forecast is made at time t, but the realized-return label matures only at the end of the holding interval, $t+\Delta t_{\text{label}}$. Before then the forward residual does not physically exist. Wait for both label maturity and actual availability and ingestion before sending the residual to the online state machine. Offline replay is also possible; a future label cannot be used early.
4. **Tiered response:** Level 1 warning (reduce weight to 50%) → Level 2 circuit breaker (switch to the Golden Model baseline and stop opening new positions) → Level 3 review (offline sample slicing and feature-ablation analysis).

45-L18 [Maps to: 45-C01]: Physical Limits of Distributed Consistency

Boundary analysis: Even with SAGA or 2PC, if the external broker does not deduplicate by a unique order-submission key (`client_order_id`), then during a network partition or timeout the local system cannot know whether the order was matched at the exchange. Blindly sending a compensating retry can cause a duplicate fill; cancelling can fail because the order

may already have filled. A distributed quant system therefore depends on broker-side idempotent deduplication and active order-status queries. A local two-phase protocol alone cannot guarantee exactly-once execution.

Capstone: from admissible input to an interpretable net conclusion

> The spine: distinguish engineering from research, then connect eligibility, signals, a continuous self-financing ledger and paired inference. Every numerical exercise below is an explicitly constructed hand calculation, not market research.

Two interfaces

Begin by naming the economic object in each input. AssetA and AssetB are constructed engineering assets: prices use USD/share, quantities use share, and clocks resolve seconds. The ten research industries are provider portfolio categories with monthly simple total returns. Their names are not permanent security identifiers. Identity, rejection and recovery ideas transfer between layers; a normalized wealth index cannot be passed to the share matcher as a quote.

Your turn.

CAP-L01

The engineering ledger ends at NAV 9995.998 USD with costs of 4.002 USD. Can these fill the French research net monthly difference? State the input and units of each interface, and name the missing bridge.

Rights and technical gates

The technical protocol closes calendar gate G02 and statistical/cost gate G04. A well-defined method does not establish permission for the activity: G01 still lacks the actual actor, purpose and corresponding rights evidence; G05 remains not_authorized. Public access does not supply that evidence. Unknown historical PIT and prior exposure are separate states that engineering tests cannot clear.

Your turn.

CAP-L02

Someone proposes downloading the CSV and running all 13 paths now because the protocol is accepted, then resolving rights later. Identify the missing qualifications and state what the report can currently deliver and cannot fill.

Eligibility, maturity and intervals

A YYYYMM key labels an interval rather than a trade timestamp. Returns occupy half-open months $[B_t, B_{t+1})$, with assumed cutoff $S_t = B_t - 1$ second and virtual allocation marker $T_t = B_t$. This order is an assumption, not evidence of historical publication. Label end, actual availability and ingestion must all precede fitting or error updating. Purging asks whether intervals overlap, a separate question.

Your turn.

CAP-L03

A constructed January label ends at February 1, 00:00, but becomes available and ingested on February 2. May it enter a fit on February 1 at noon? Does the adjacent February interval overlap January? Is a one-month signal skip an embargo?

Eleven compounded terms

The main signal multiplies eleven wealth factors before holding month t; it does not sum eleven returns. The general window is $[t-s-L, t-s-1]$, where L counts terms and s is skip. Main values $L = 11, s = 1$ give $t - 12$ through $t - 2$. Correct endpoints do not establish contemporaneous publication; calendar arithmetic and information eligibility remain distinct.

Your turn.

CAP-L04

Give the main window and skipped month for January 2026; give the L17/skip1 and L11/skip2 windows for January 1990. Using only constructed returns of +10% and -10%, show why compounding cannot be replaced by a sum.

Stable ranks and thirteen paths

The full cross-section sorts by descending score, then ascending fixed manifest index to break ties. Main top3 weights are 1/3 each, with zero elsewhere. There is no training or optimization. Nine strategy paths cover the main specification and single-factor sensitivities; four equal-weight cost paths permit matched comparisons. Fixing them constrains future choices but cannot turn exposed history into a blind holdout.

Your turn.

CAP-L05

Durbl and Manuf tie for third place, with manifest order NoDur, Durbl, Manuf. Which enters? List all 9+4 paths and explain why no Cartesian product of L and k is added.

Drift before target trades

Rebalancing starts from actual amounts x_i carried by the prior ledger, rather than its old target weights. Normalization requires a positive denominator. Post-cost targets are $w_i V_{post}$, so trades are $w_i V_{post} - x_i$. Setting targets from pre-cost NAV and deducting costs afterwards generally misses the desired post-cost weights.

Your turn.

CAP-L06

Constructed example: start with 0.5 in each asset; monthly returns are +20% and -20%, with no other cost that month. What are x, NAV and drifted weights at the next boundary? Why are trades nonzero if the next target is again 50/50?

The self-financing fixed point

Take finite nonnegative drifted amounts, nonnegative weights summing to one, $V_{pre} > 0$ and $0 \le c < 1$. Let $A(v) = \sum_i |w_i v - x_i|$. The fee equation is $v + cA(v) = V_{pre}$. Since $|A(v_2) - A(v_1)| \le |v_2 - v_1|$, its left side increases by at least $(1-c)(v_2 - v_1) > 0$. Thus there is at most one solution on this domain; existence and feasibility follow the Chapter 29 and ledger conditions.

Your turn.

CAP-L07

Continue CAP-L06 with x=(.6,.4), w=(.5,.5), c=.001. Assume each target amount lies between .4 and .6. Solve Vpost, signed trades, absolute traded amount and fee; verify the targets and assumed signs.

Entry, exit and continuous phases

At first entry from cash 1, $x = 0$ and absolute traded amount equals V_{post}, giving $V_{post} = 1/(1+c)$. The final month earns asset returns before liquidation multiplies wealth by $(1-c)$. Intermediate phase boundaries neither liquidate nor re-enter. Dividing a phase plot by its opening NAV changes display only. The first main-sample denominator carries August 2015 ending NAV; the final row includes August 2026 liquidation.

Your turn.

CAP-L08

Derive the entry formula and the 440-month product identity. Why does restarting from cash 1 and charging entry cost in September 2015 produce a different 132-month result? Give the 176/132/132 dates.

□□□□□

Rate c and NAV loss C

Research rate c applies to the sum of absolute buys and sells; 10bps=.001. It is an all-in scenario, not a calibrated commission or an annual fee. NAV loss $C = F/V_{pre}$ has a different denominator. With $C = .001$ and gross allocation return 2%, net return is $(1-C)(1+.02)-1$; the drag participates in subsequent wealth growth rather than being a mechanical 10bp subtraction.

Your turn.

CAP-L09

Absolute traded amount is 200.10 USD and c=.001. What is the fee? May it be halved or charged an additional spread? In a separate construction, C=.001 and gross return is 2%; find net return and drag, and explain why c and C are not interchangeable.

□□□□□

Paired HAC and covariance correction

The experimental unit is one paired difference $d_t = r_{M0,t} - r_{EW10,t}$ per month, not ten independent industry replications. Main values are n=132, intercept-only k=1 and lag6, with the same n denominator for every autocovariance. Bartlett weights for $j = 1..6$ are $1 - j/7$. Correction $q = n/(n-k)$ applies to covariance. Since $X^T X = n$ in this design, $\mathrm{SE}(\hat{\mu}) = \sqrt{\widehat{\Omega}/131}$.

Your turn.

CAP-L10

Write gamma and Omega, then derive corrected mean variance. For a hand check only, use constructed d=(.01,.03,.05,.07), n4 and L0; find the mean, residual sum of squares, mean variance and SE.

□□□□□

Intervals, undefined cases and RF once

The main direction is $H_1 : \mu_d > 0$. The one-sided 95% lower bound uses $t_{.95,131}$; a two-sided 95% interval uses $t_{.975,131}$. These are approximations under dependence, not exact coverage guarantees. Same-month RF cancels between the two main returns, so d is not reduced by RF again. FF3 attribution instead takes net total return less the matching FF3 RF once, with an intercept and three published spreads, k=4.

Your turn.

CAP-L11

One main month is missing but 131 rows remain: may row offsets stand in for monthly lags? If SE=0, may p=0 be reported? May attribution reuse k1 correction or subtract RF again from Mkt-RF?

Fixed circular block bootstrap

The bootstrap preserves local monthly dependence in the main 132 paired differences: primary block12, 5000 replicates and seed20261012; blocks6/24 are planned interval sensitivities. Initialize one dedicated random.Random(seed) generator once before the 5000-replication loop. All replications consume that same evolving state; call randrange(n) for each required block, without reseeding inside a replication. Circular blocks are concatenated until ceil(n/block) blocks cover n, then truncated. Quantiles interpolate at $(B-1)p$. An unfavorable HAC interval does not justify reporting bootstrap alone.

Your turn.

CAP-L12

Without sampling or executing code, take constructed n5, block3 and starts4 and2 on indices0..4. Give the concatenated indices truncated to five. Four sorted replicate means are0,1,2,3: find the interpolated p=.25 quantile. Which p values define the main one- and two-sided bounds?

A complete report and its failure branches

The 18 report items connect input, action, output and interpretation. Sources, dictionary, PIT and cleaning establish admissible input; features, signals, portfolio and costs establish a net ledger. Validation, performance, risk, attribution, robustness and sensitivity support interpretation. Overfitting, limitations and conclusion bound the answer. Generic prompts about training, neutralization and grids are historical guidance, not execution requirements of this French protocol.

Your turn.

CAP-L13

For report items8,9,11 and14, name one required artifact and one rejection/error branch. Which phases should item14 use? May it add restarted first/second-half experiments?

Exposure and risk evidence

P3 already displayed the full January1990-August2026 history and benchmark results; P1/P2 add partial exposure, and other access is unknown. Fixing future rules constrains further choice but cannot remove hindsight. Industry total returns lack security-level events, order books, volume and executable ETF data. They cannot estimate real capacity or establish that the underlying market is free of survivorship bias.

Your turn.

CAP-L14

Why is locked reporting not a blind holdout? Without evidence that DSR/PBO apply, how should item16 be filled? What can a uniform constructed -10% shock, top3 weight chart and monthly tail statistics explain, and what cannot they establish?

Answering the final question

The main endpoint is the mean paired monthly net difference, not the highest Sharpe or most attractive wealth curve. Statistical evidence and the economic threshold address distinct questions: a positive one-sided lower bound supports the prespecified positive direction;10bps/month is an illustrative economic design threshold. Negative, zero, crossing-zero and cost-eroded gross advantages all deserve reporting. A gate-blocked run produces no market effect estimate.

Your turn.

CAP-L15

Write an acceptable conclusion for each: currently unexecuted; a future positive mean with interval crossing zero; future net mean<=0; future lower bound>0 but mean below10bps/month. Which permits claiming proof of slow information response or permanent strategy failure?

Capstone complete solutions

> Attempt the exercises before checking each stable ID. These answers explain the protocol and constructed arithmetic; they supply no market results.

CAP-L01: Two interfaces

No. Engineering consumes two-asset prices, quantities and execution/mark clocks, then produces a monetary ledger. Research consumes ten-industry monthly total returns and weights, with normalized holding amounts, then produces monthly net returns. The missing bridge is the monthly-return continuous-ledger adapter; it is unimplemented and execution is unauthorized. Engineering PASS supports the constructed engineering contract and supplies no French target-market observations.

CAP-L02: Rights and technical gates

G01 needs rights evidence for the actual actor and use; G05 needs explicit execution authorization. Strict historical PIT is also unproven. The report can deliver the design, fields, clocks, formulas, planned outputs and failure interpretations. It cannot download/read target CSV or ZIP data, train, execute paths or target bootstrap, or fill net means, Sharpe ratios, intervals or a winner. full_preregistration, market_verified and execution_authorized remain false.

CAP-L03: Eligibility, maturity and intervals

It cannot enter the fit: the interval has ended, but availability and ingestion follow the fit. Adjacent half-open intervals share only a boundary, so this pair does not require overlap purging; the maturity gate still fails. Skip excludes the most recent signal month and is not an embargo. The forward-only design has future-side embargo=0; with unknown publication history, the maturity gap is pending_available_history rather than an assumed month.

CAP-L04: Eleven compounded terms

The January 2026 main window is January-November 2025, skipping December. For January 1990, L17/skip1 runs July 1988-November 1989, 17 terms; L11/skip2 runs December 1988-October 1989, 11 terms, skipping November and December. Constructed wealth is $1.1(0.9) = 0.99$, a return of -0.01; the sum is zero and omits the -0.01 cross term. No market returns enter this calculation.

CAP-L05: Stable ranks and thirteen paths

Durbl enters because its manifest index comes first. The nine strategies are M0, C0, C25, C50, L5, L17, K2, K4 and S2; benchmarks are EW0, EW10, EW25 and EW50. M0 is 11/1/3/10bps. C changes cost only, L lookback only, K count only, and S2 skip only. Each benchmark holds all ten industries at 10%. A cross-grid adds unregistered paths and selection freedom; richer plots do not authorize it. Optimization trials remain zero.

CAP-L06: Drift before target trades

$x = (0.6, 0.4)$, $V_{pre} = 1$, and drifted weights are $(0.6, 0.4)$. Prices changed the realized proportions although the old targets were equal. The next rebalance must sell the first asset and buy the

second. Each post-cost target is $0.5V_{post}$, not an amount of 0.5 before solving costs. At zero NAV the next weights and return ratios are undefined and must be rejected rather than replaced with equal weights.

CAP-L07: The self-financing fixed point

On the assumed sign region, absolute traded amount is $(.6 - .5v) + (.5v - .4) = .2$. Thus $v = 1 - .001(.2) = .9998$, with targets of .4999 each, inside the assumed interval. Signed trades are $(-.1001, +.0999)$, summing in absolute value to .2000; fee is .0002. Sale proceeds .1001 less purchases .0999 exactly pay the .0002 fee, leaving zero cash. Targets .9998 plus costs .0002 equal initial NAV 1. These trades are normalized monetary amounts, not shares.

CAP-L08: Entry, exit and continuous phases

At entry $v = 1-cv$, hence $v(1+c) = 1$. In ordinary months $1+r_t = (V_{post,t}/V_{pre,t})(1+R_{alloc,t})$; the last month also multiplies by $(1-c)$. Consecutive denominators telescope, yielding $\prod_{t=1}^{440}(1+r_t) = V_{terminal}/1$. Development is January 1990-August 2004, 176 months; validation September 2004-August 2015, 132; main reporting September 2015-August 2026, 132. A cash restart changes drifted positions and fees, creating an additional path. The main segment product must equal terminal cash divided by August 2015 ending NAV.

CAP-L09: Rate c and NAV loss C

The fee is .20010 USD, with no halving and no additional spread charge. In the separate example net return is $(.999)(1.02) - 1 = .01898$, and drag is $.02 - .01898 = .00102$, or 10.2bps. The denominator for c is traded amount; that for C is pre-trade NAV. Conversion needs the traded-amount/NAV ratio. Chapter 45 embeds a half-spread and adds explicit commission in a different share-based engineering example; it does not recalibrate the main research c.

CAP-L10: Paired HAC and covariance correction

$\hat{\gamma}_j = n^{-1}\sum_{t=j+1}^{n} u_t u_{t-j}$ and $\widehat{\Omega} = \hat{\gamma}_0 + 2\sum_{j=1}^{6}(1-j/7)\hat{\gamma}_j$. Corrected variance is $q\Omega/n = \Omega/(n-1)$. Do not multiply SE by q again or replace denominator n by n-j. The constructed mean is .04, residuals $(-.03, -.01, .01, .03)$ and squared sum .002. At L0, Omega=.0005, mean variance $.0005/3 = 1/6000$, and SE is approximately .01290994. This equals sample variance/n and uses none of the 132 target-market rows.

CAP-L11: Intervals, undefined cases and RF once

No. A missing month requires rejected/undefined status with an explanation; dropping it and compressing the calendar or silently reducing L changes the contract. Zero/nonfinite SE, n<=L and insufficient identification must not create artificial significance. Only valid finite SE supports $t = \hat{\mu}/SE$ and the stated approximate intervals. Attribution uses $n/(n-4)$; y subtracts RF once, while published Mkt-RF, SMB and HML spreads receive no further subtraction. Alpha is model-dependent retrospective attribution, not automatic evidence of predictive skill.

CAP-L12: Fixed circular block bootstrap

The first block is4,0,1 and the second2,3,4; truncation gives4,0,1,2,3. Position $(4-1).25 = .75$ lies between0 and1, giving quantile .75. Main p values are .05 for a one-sided lower bound and

.025/.975 for a two-sided interval. Main n132/block12 requires11 blocks. This n5 hand example explains mechanics without replacing the specification, running target data or adding a selection family. Report HAC, main bootstrap and block6/24 intervals even when they disagree.

CAP-L13: A complete report and its failure branches

Item8 needs a continuous ledger of pre/post/end NAV, drift, targets, trades and fees; reject budget failure or missed post-cost targets. Item9 needs absolute traded amount/fee and a gross-to-net bridge in declared units; halving or double-charging spread is wrong. Item11 needs the main132 paired differences, HAC/fixed-bootstrap intervals and status; missing months or degenerate SE cannot be patched with p values. Item14 needs all planned cost/skip sensitivities, the three fixed phases and block6/24 intervals; omitting negative results or choosing favorable intervals is wrong. Slice the continuous176/132/132 phases; do not add half-sample cash restarts.

CAP-L14: Exposure and risk evidence

History is exposed or of unknown access, so it is not unused data and full_preregistration=false. Item16 records known exposure, unknown access,13 planned paths and zero optimization. Mark DSR/PBO not_applicable/not_evaluated with reasons rather than inventing values. The constructed shock checks a risk formula; weights show industry concentration; tail statistics describe adverse observations at the stated monthly frequency. None establishes real liquidity, ADV capacity, intraday tails or causal mechanism, or closes the historical P6/39 evidence gaps.

CAP-L15: Answering the final question

Currently: the design and technical protocol are accepted, execution is blocked, and no target-market result permits judging $\mu_d > 0$. A positive mean crossing zero gives insufficient evidence under the stated sample/dependence assumptions. A nonpositive mean shows no sample net advantage for this fixed specification. A positive lower bound with mean below10bps supports the direction but misses the design economic threshold. None identifies slow-response causality or proves permanent ineffectiveness. Future conclusions retain costs, risks, scope and all planned sensitivities; blocked is never a zero effect.